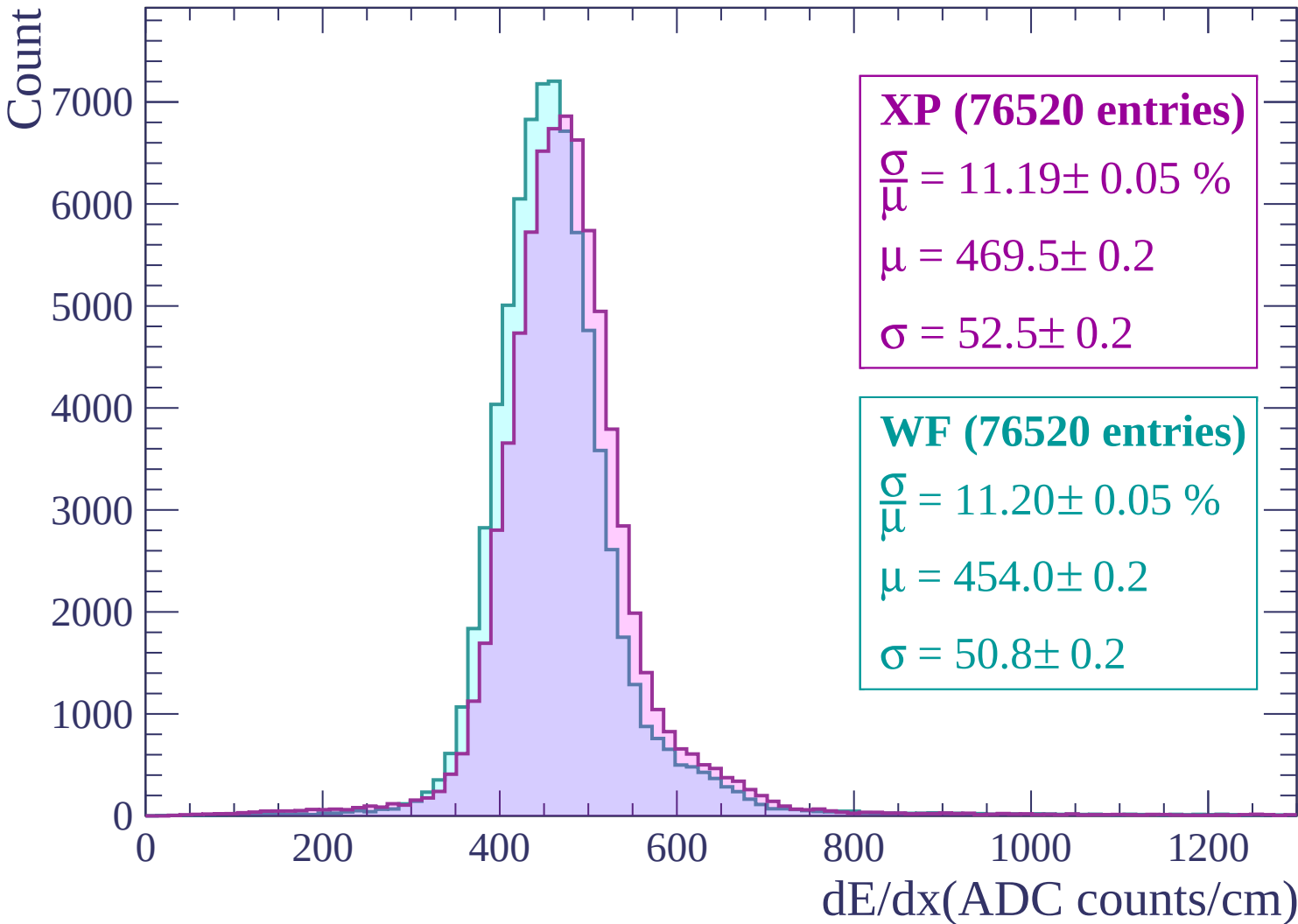
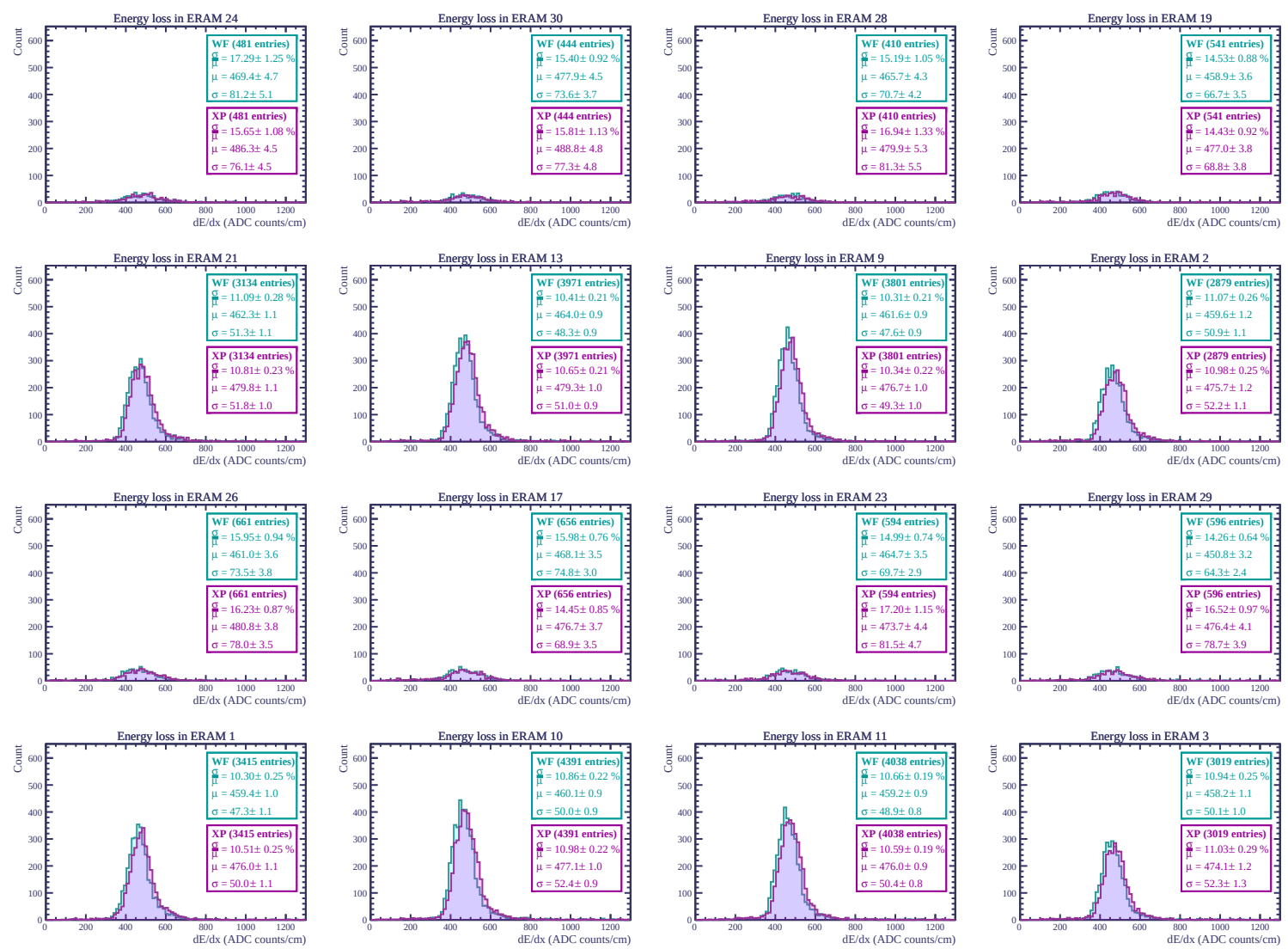
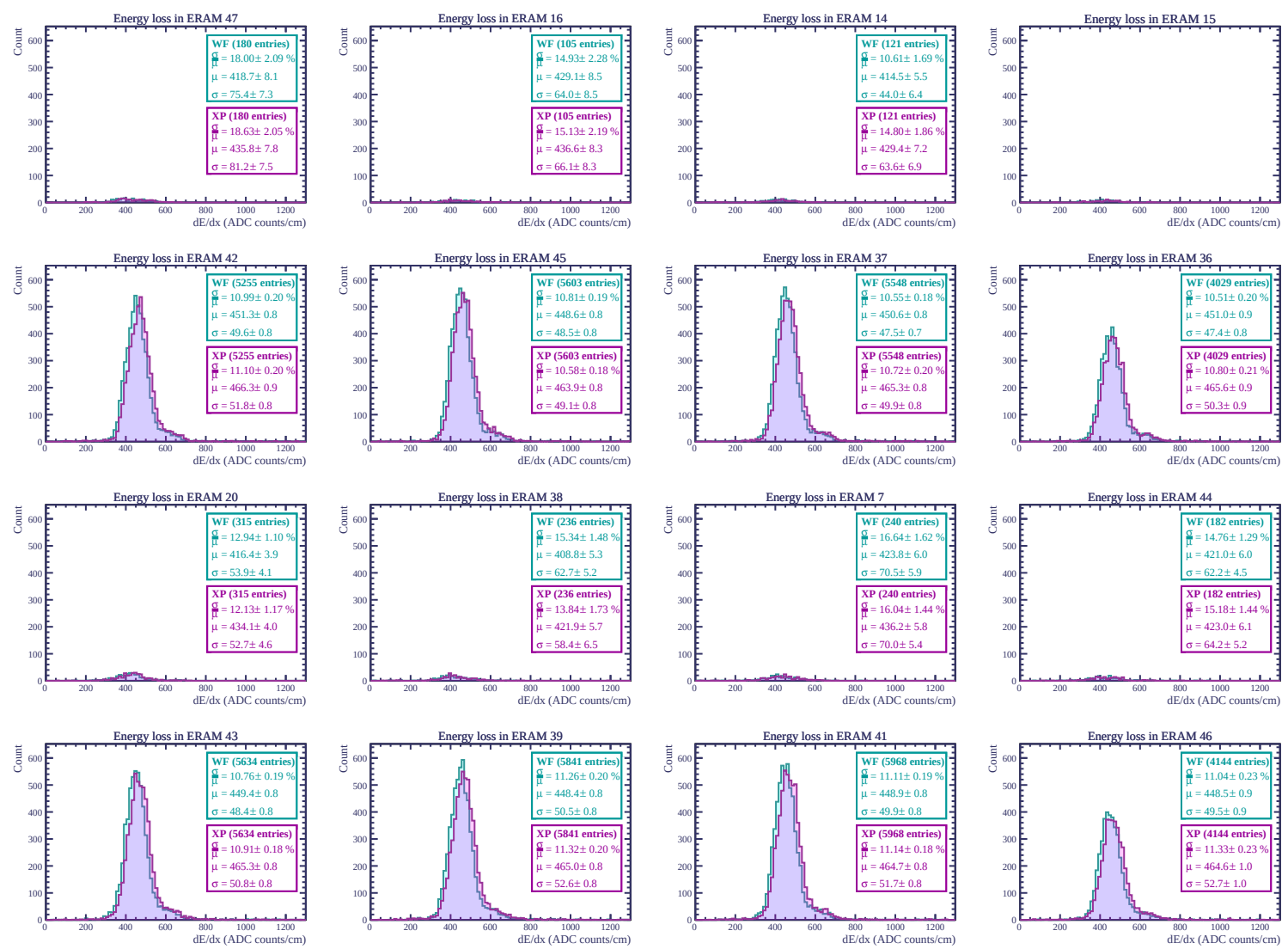
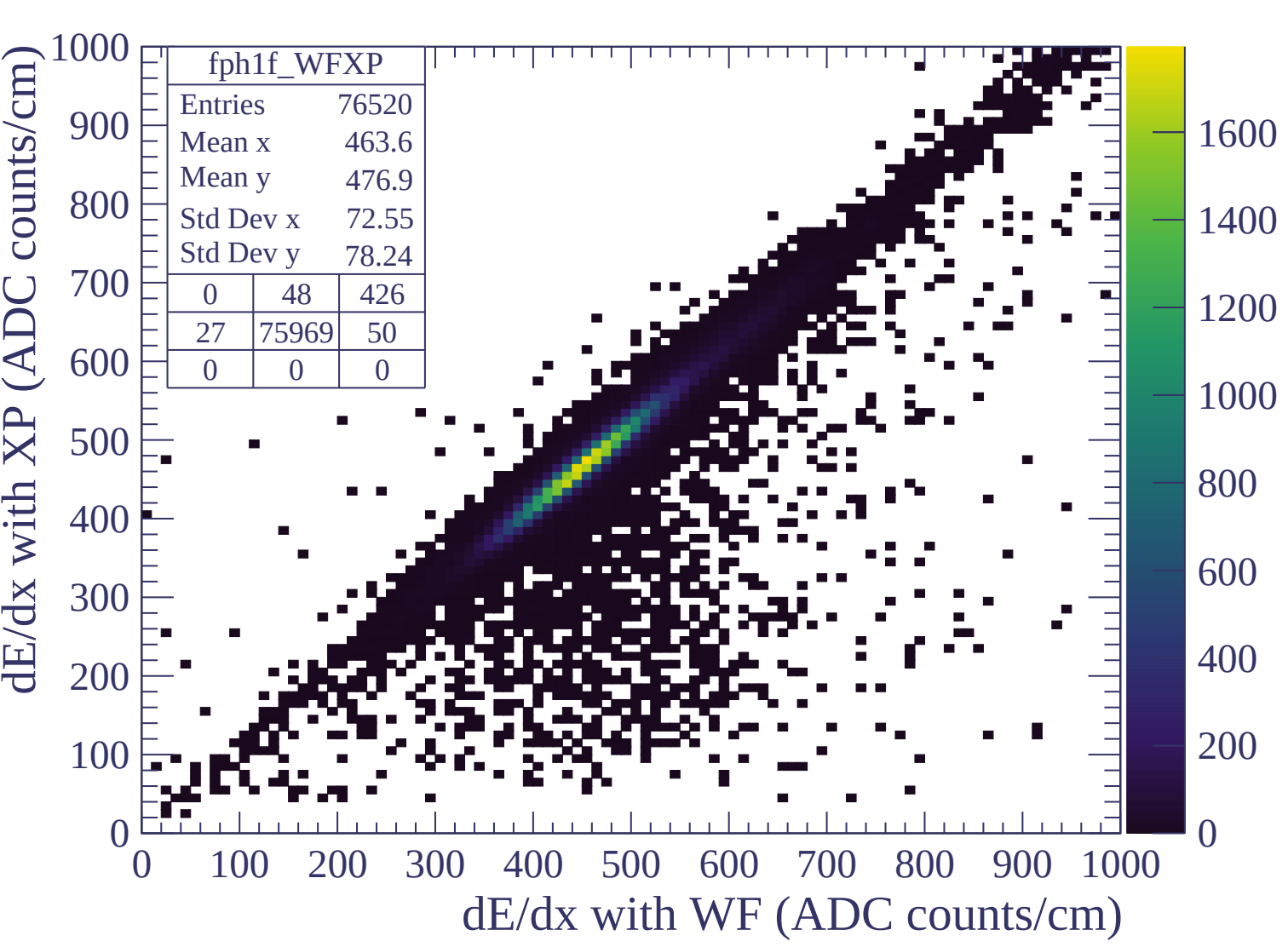
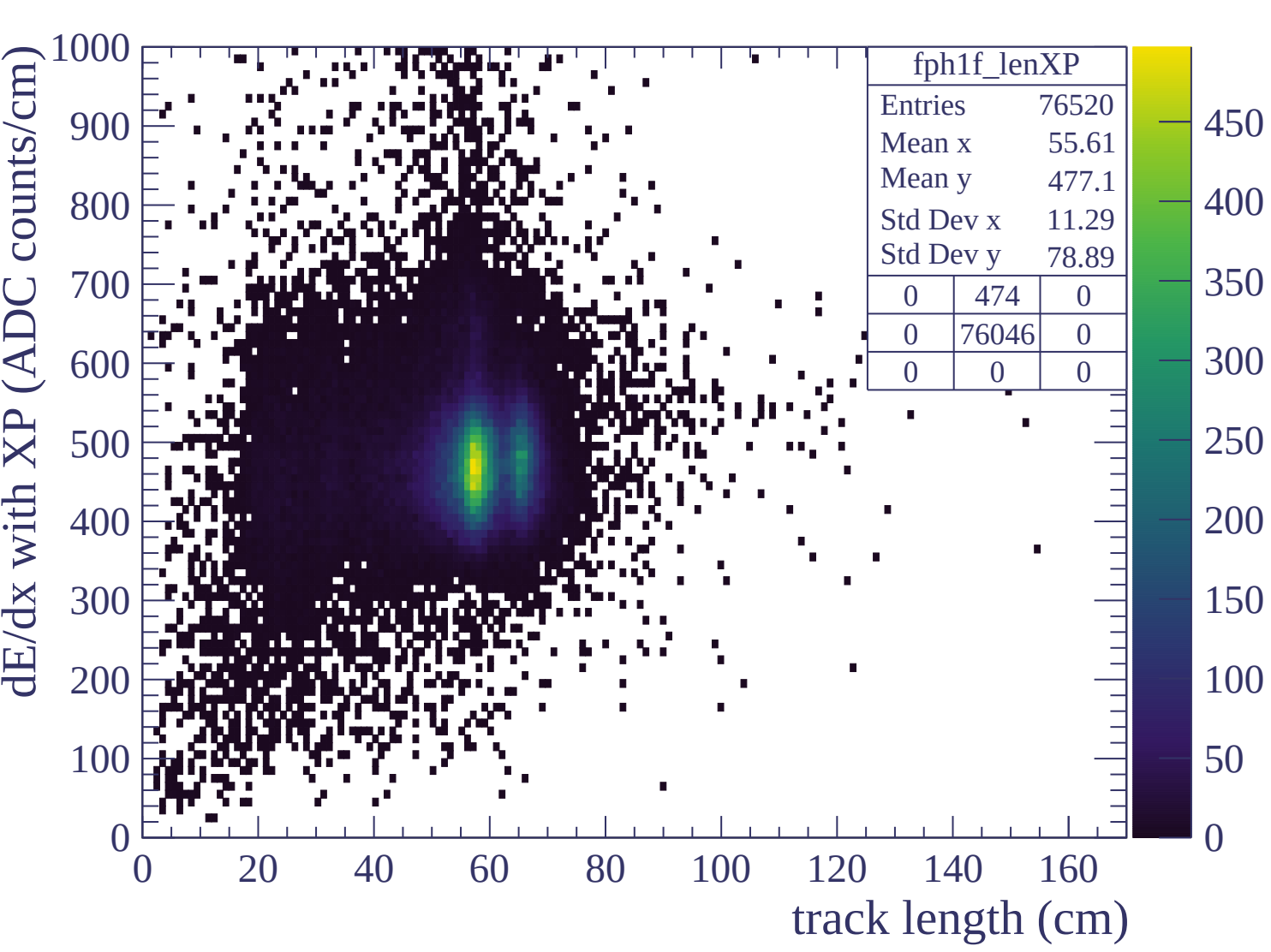


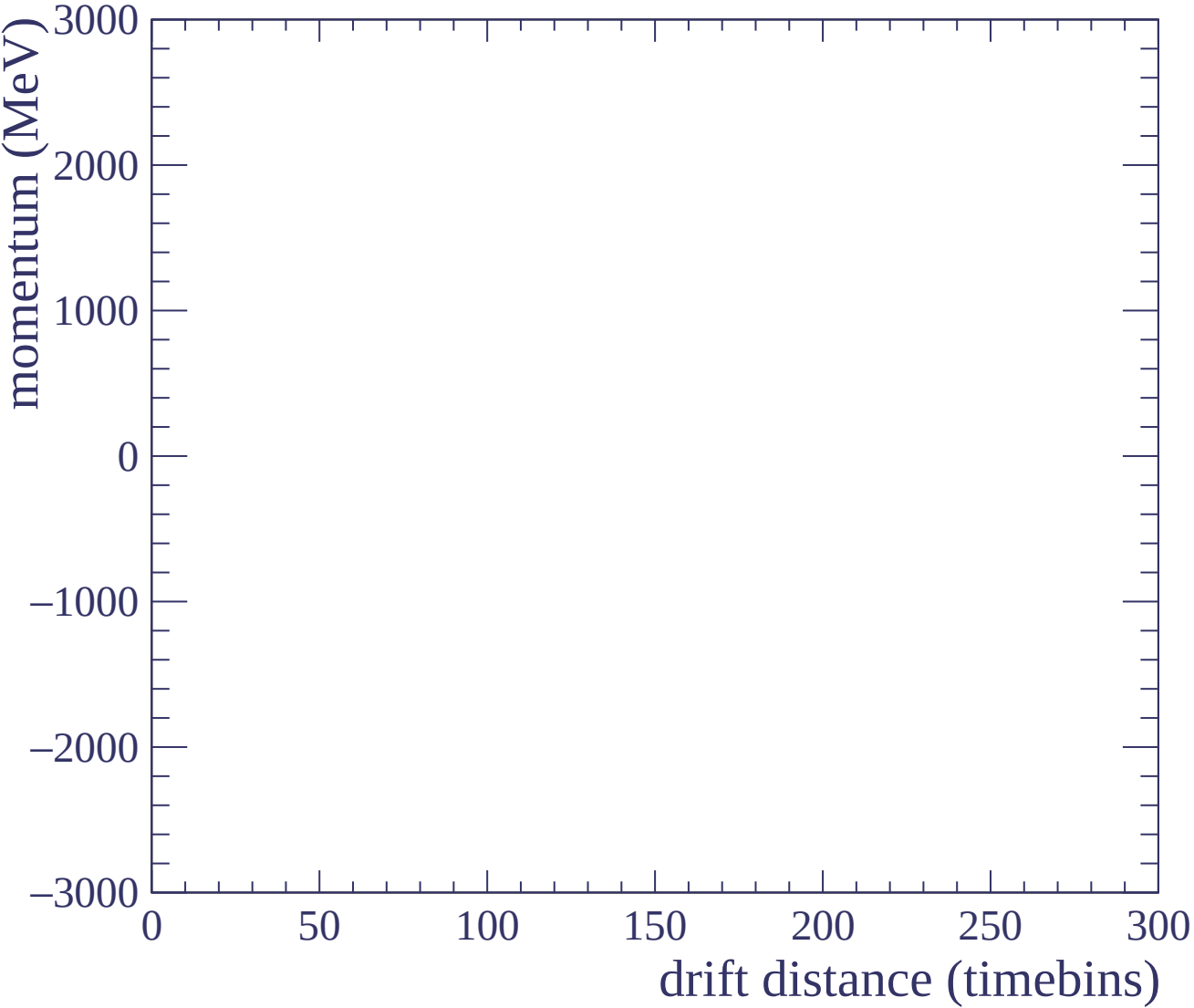


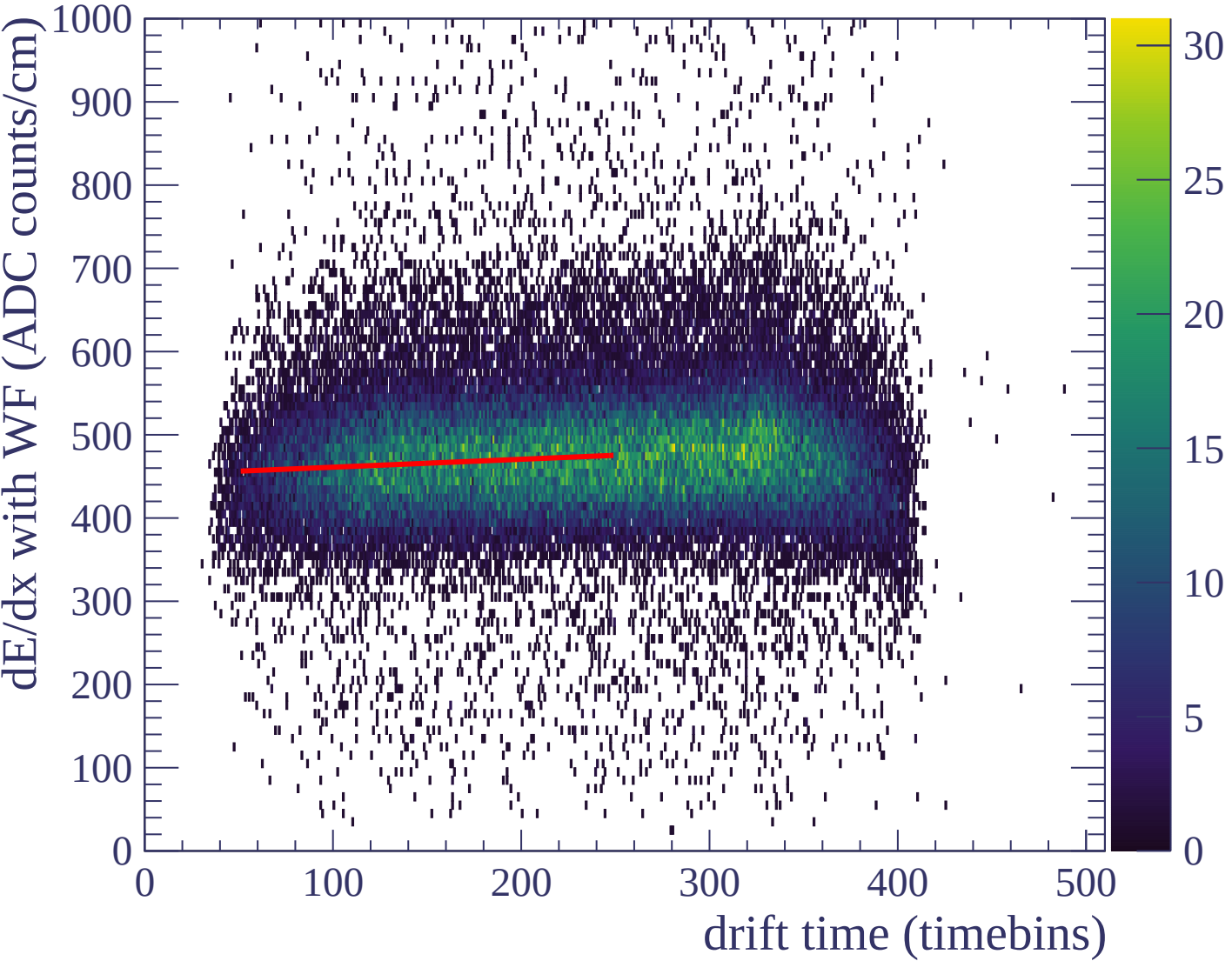


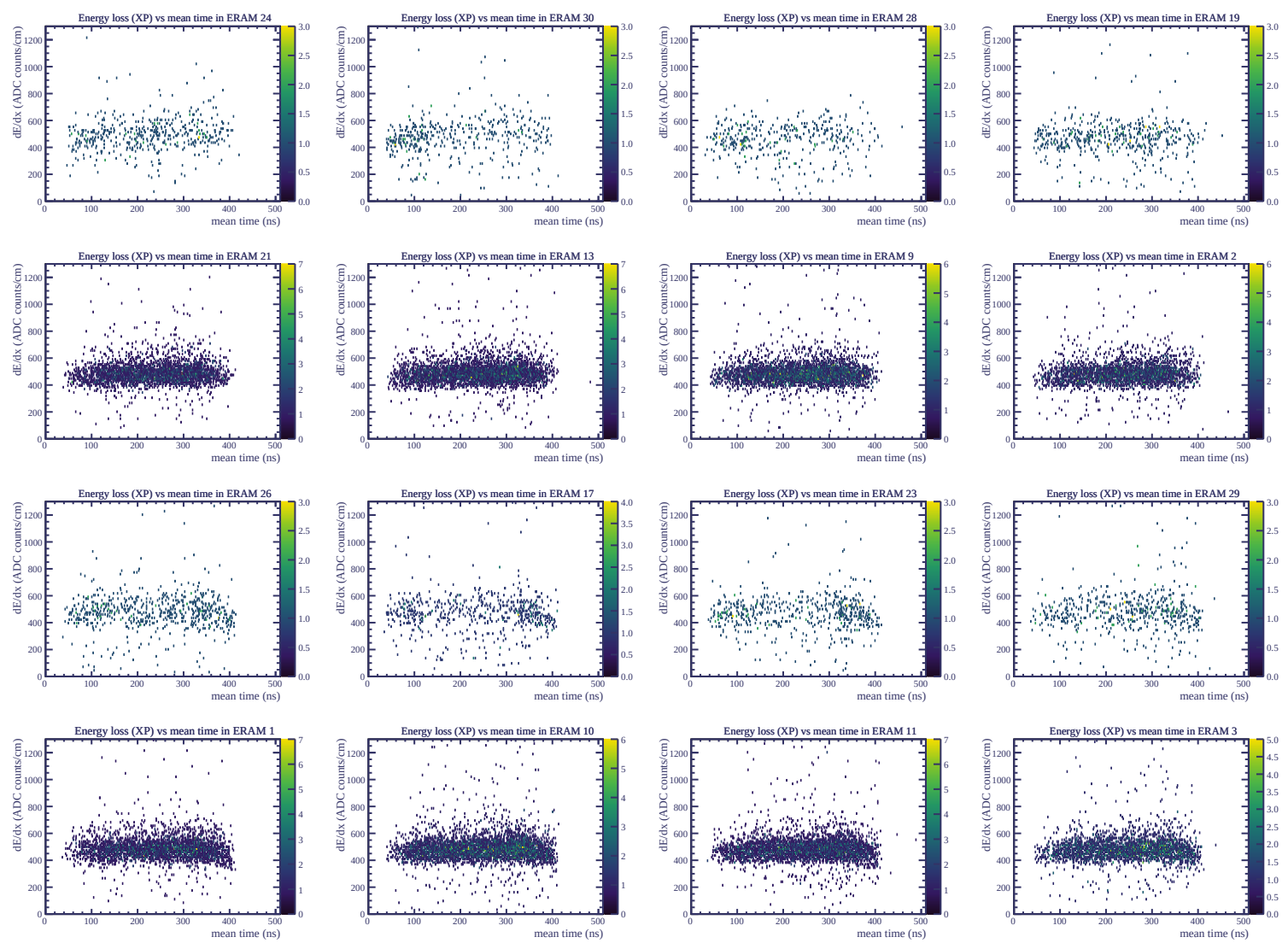


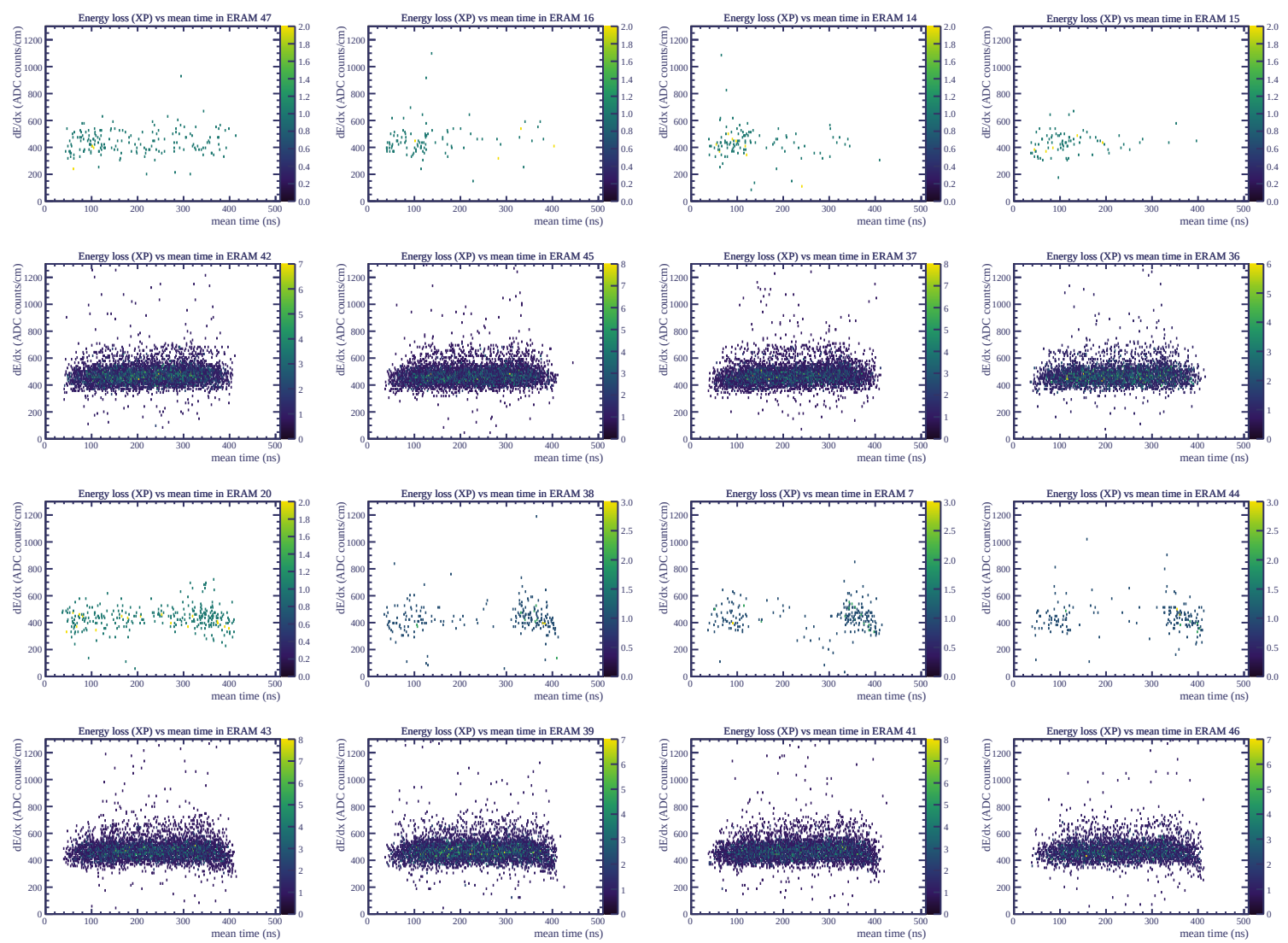


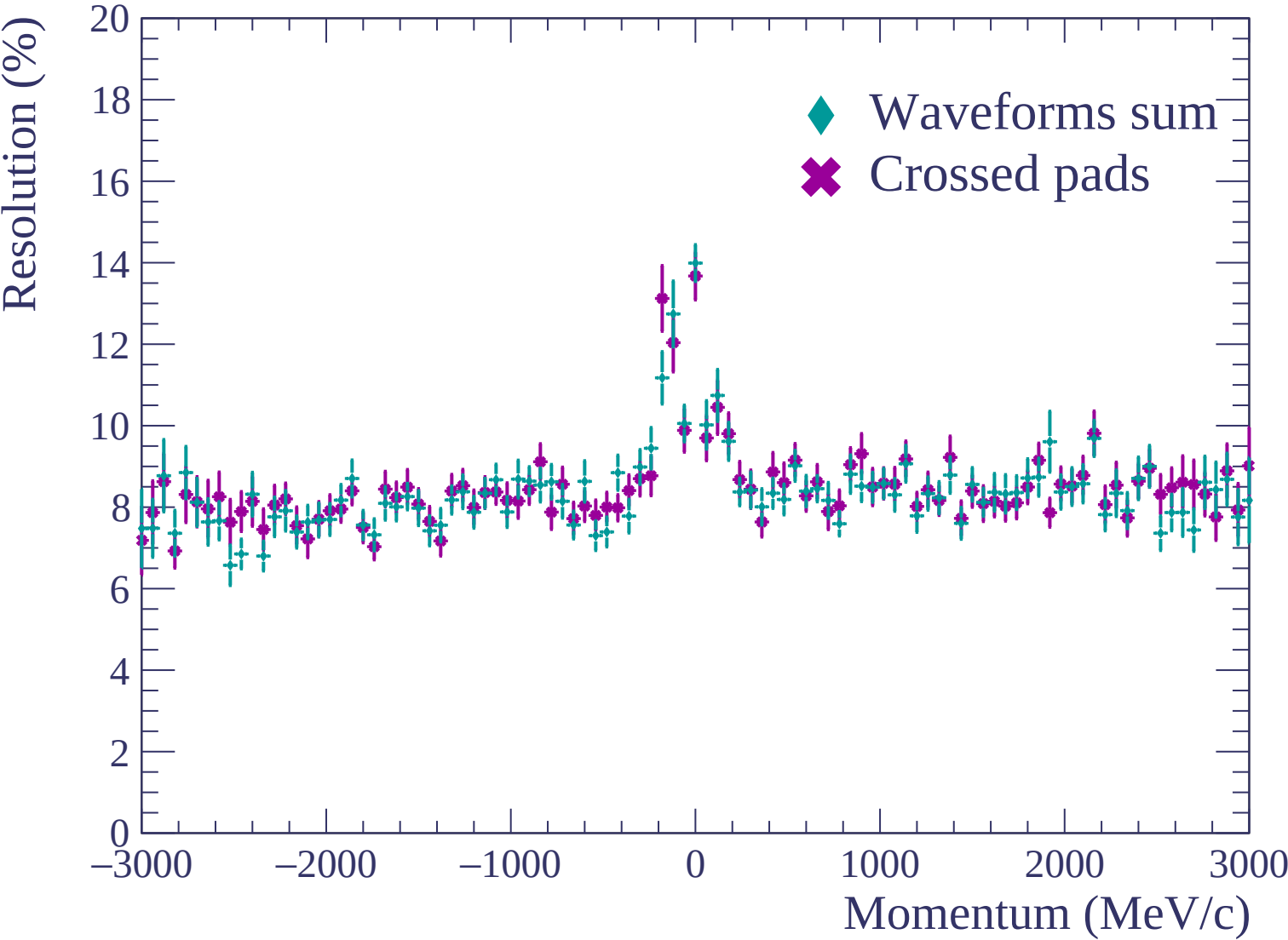


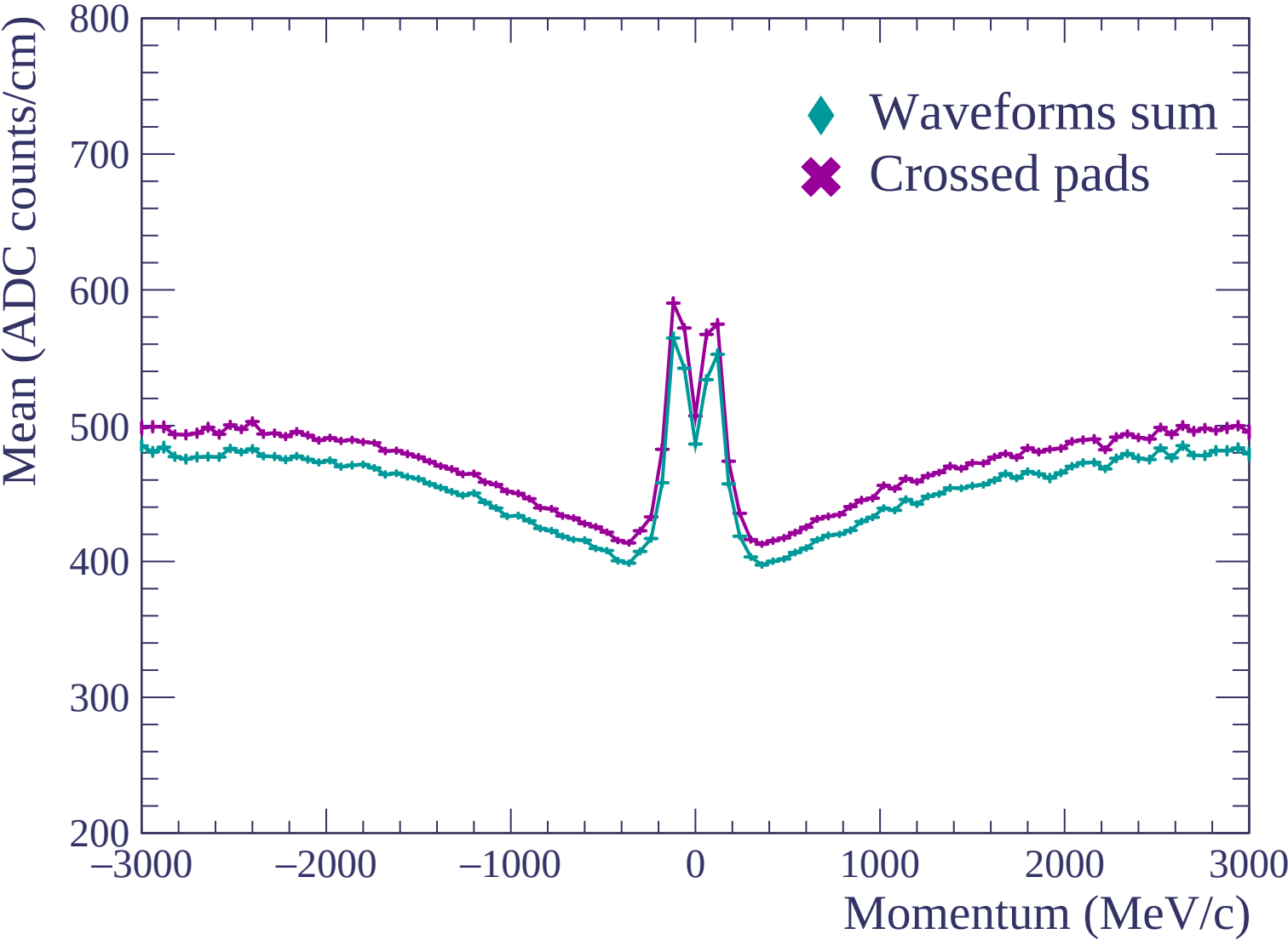


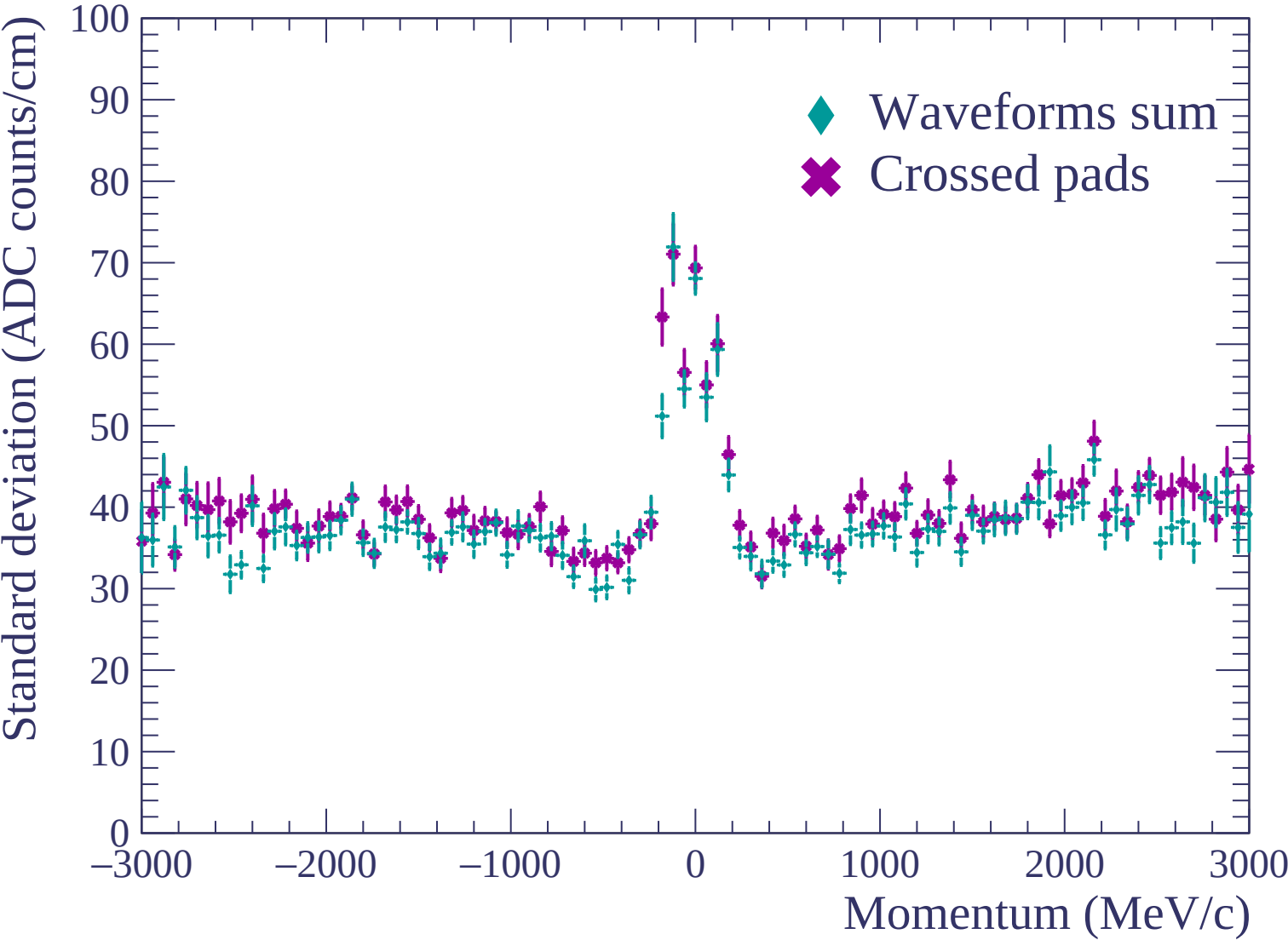


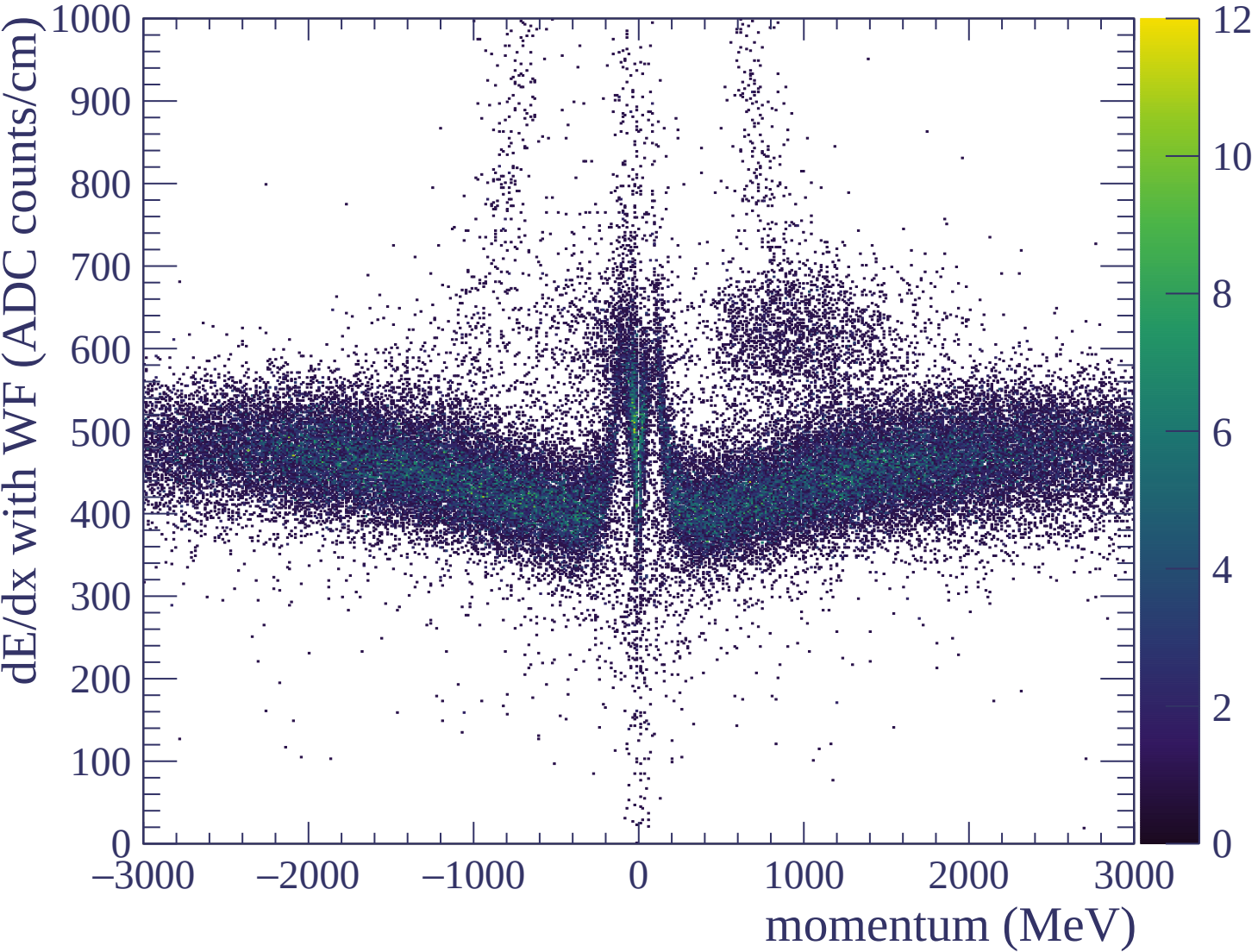


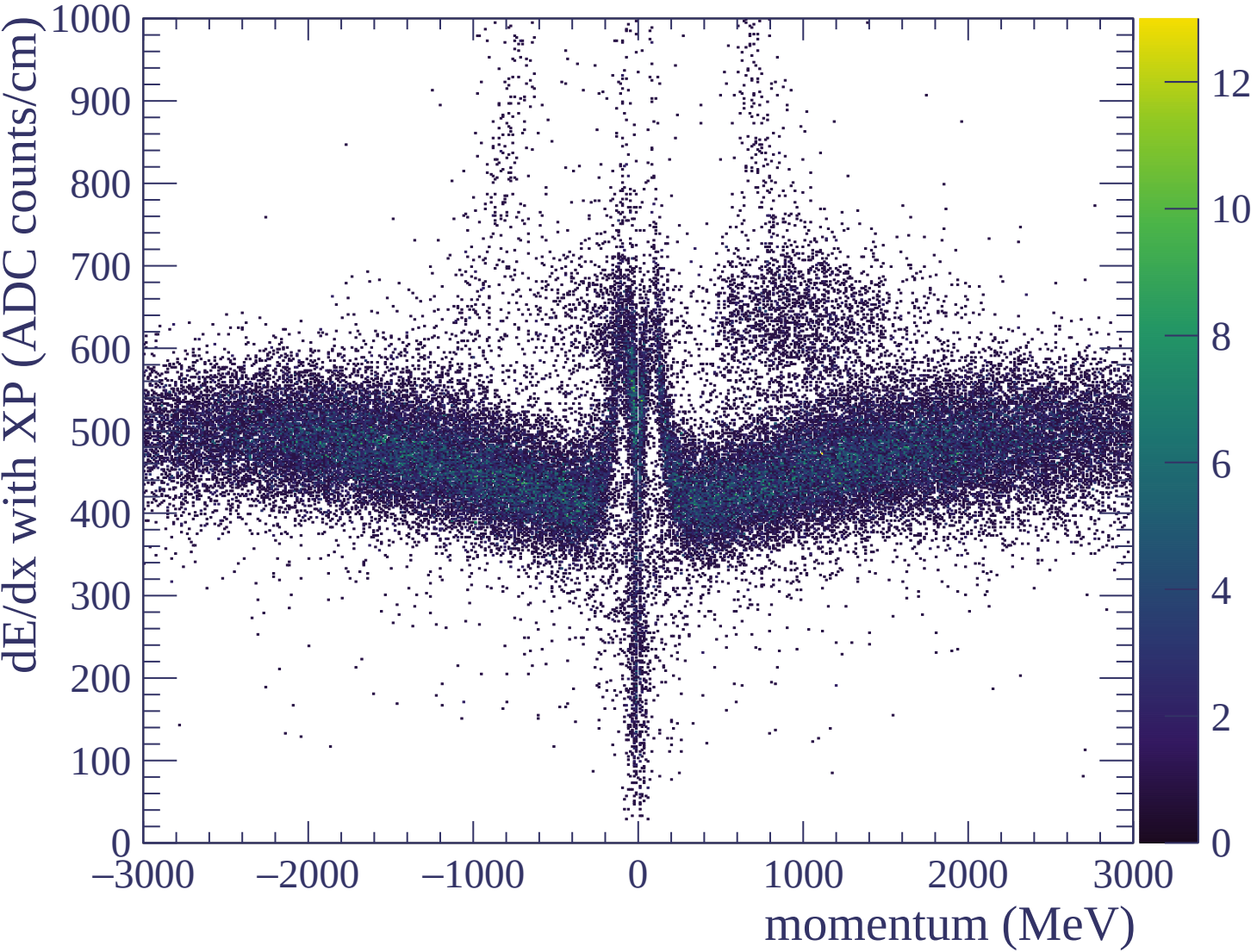


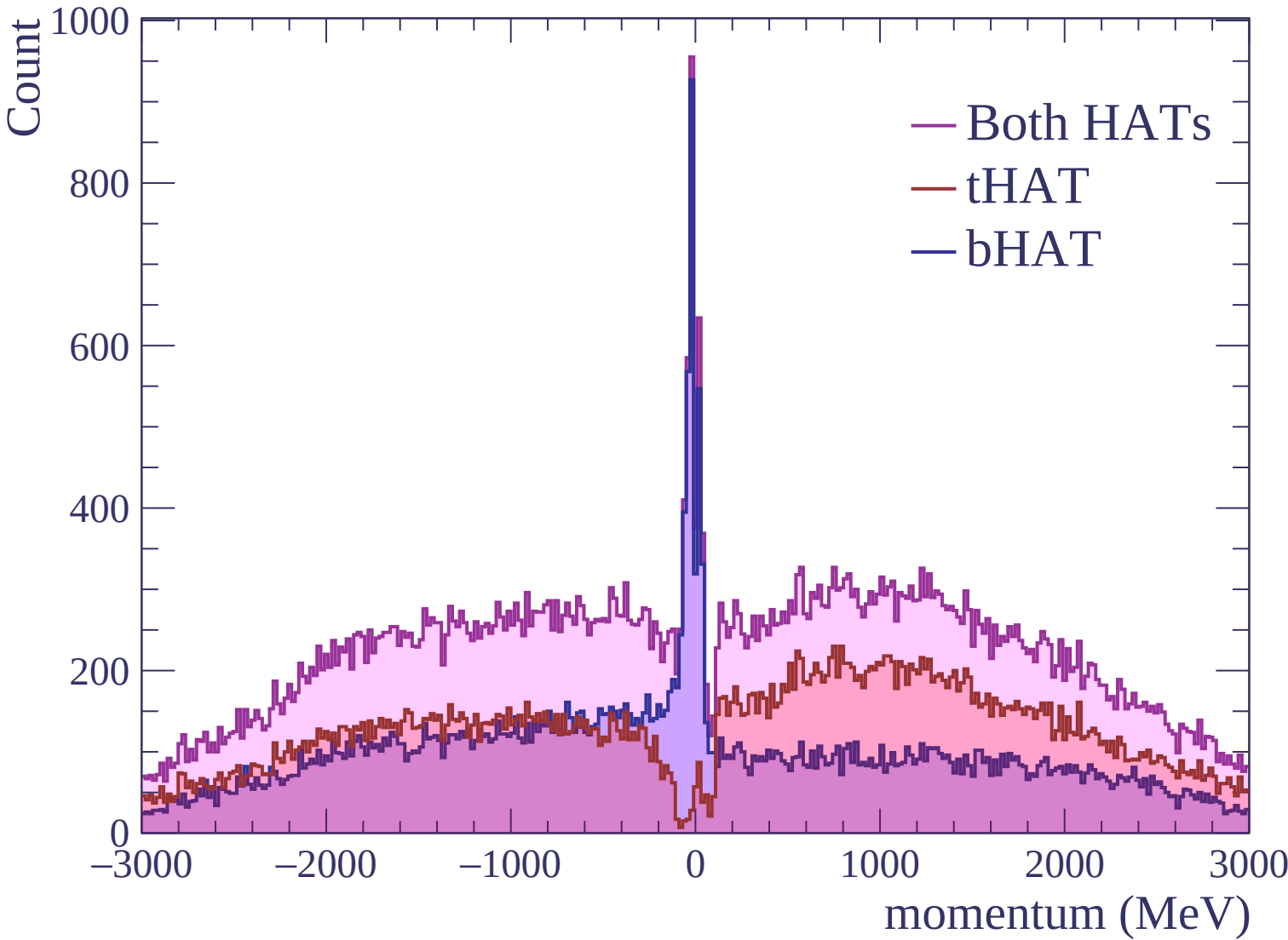


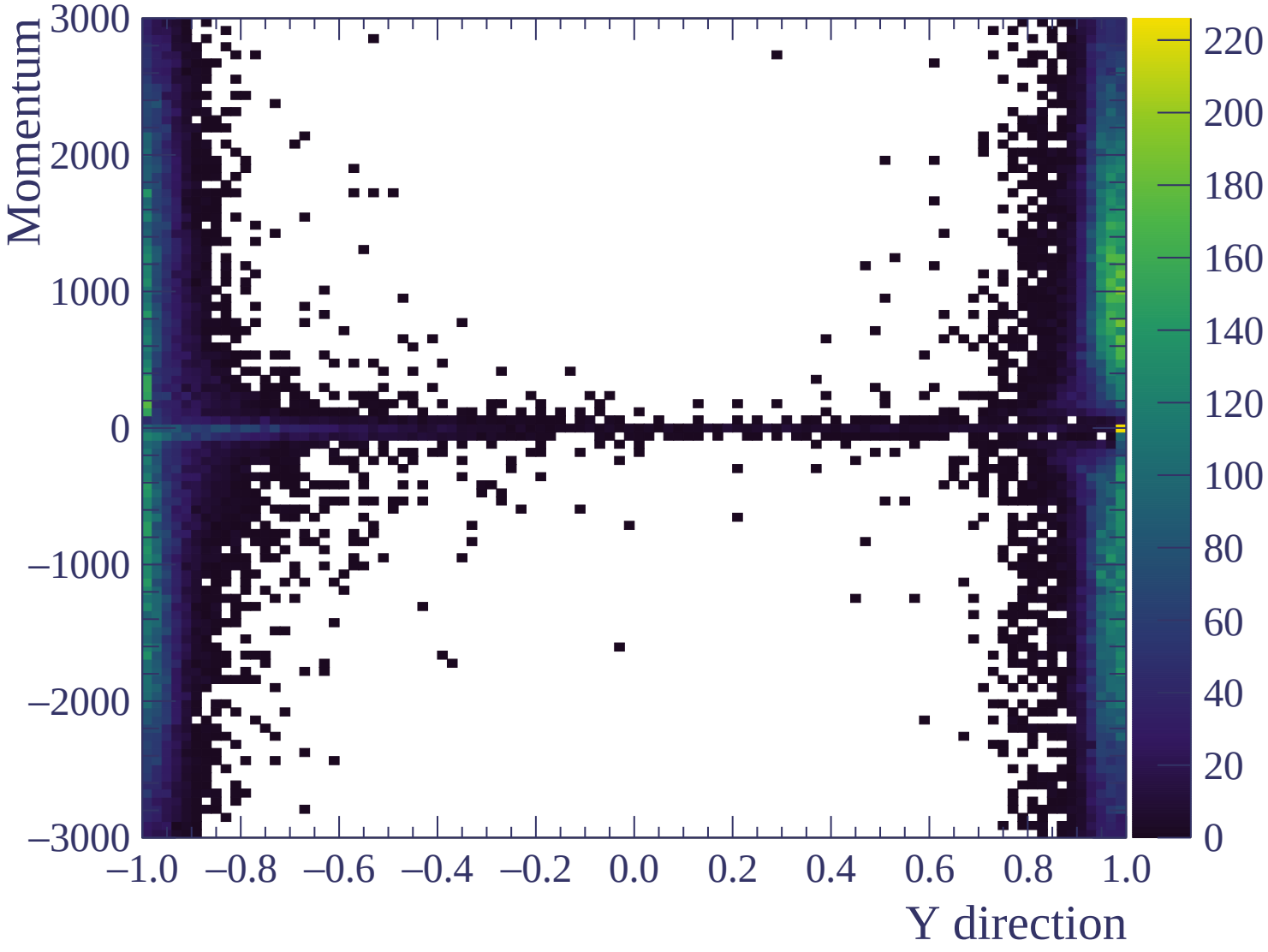


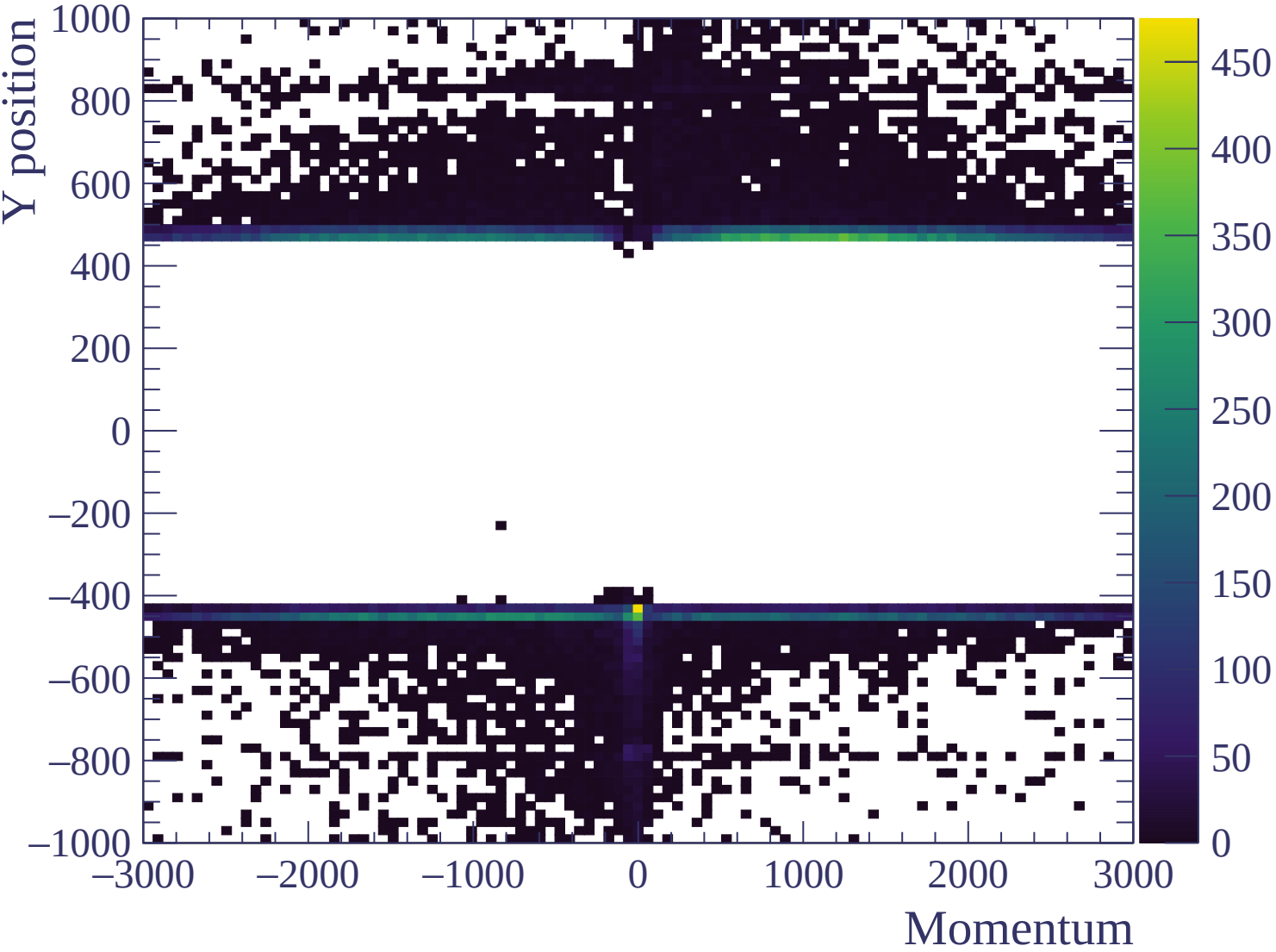


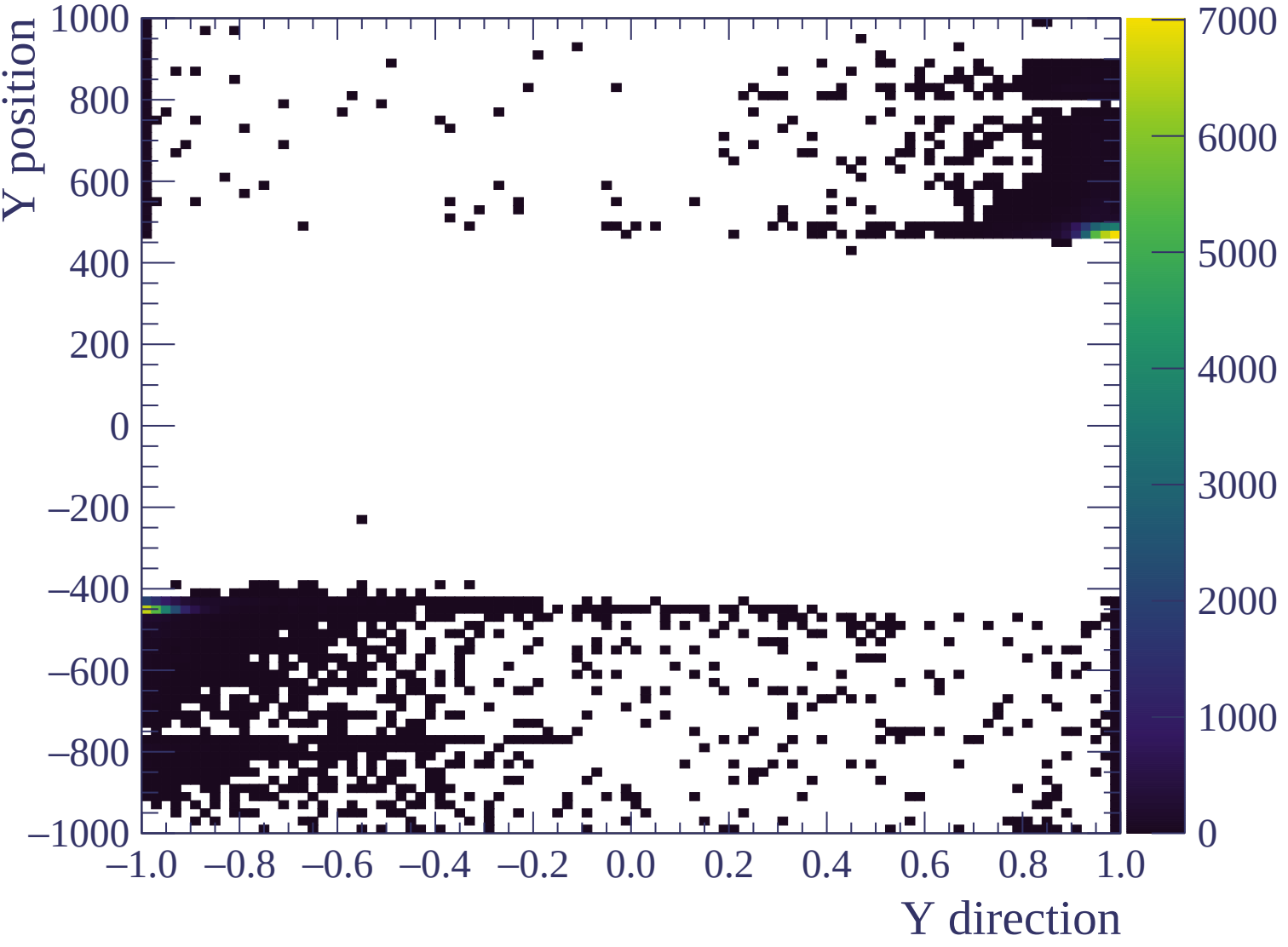


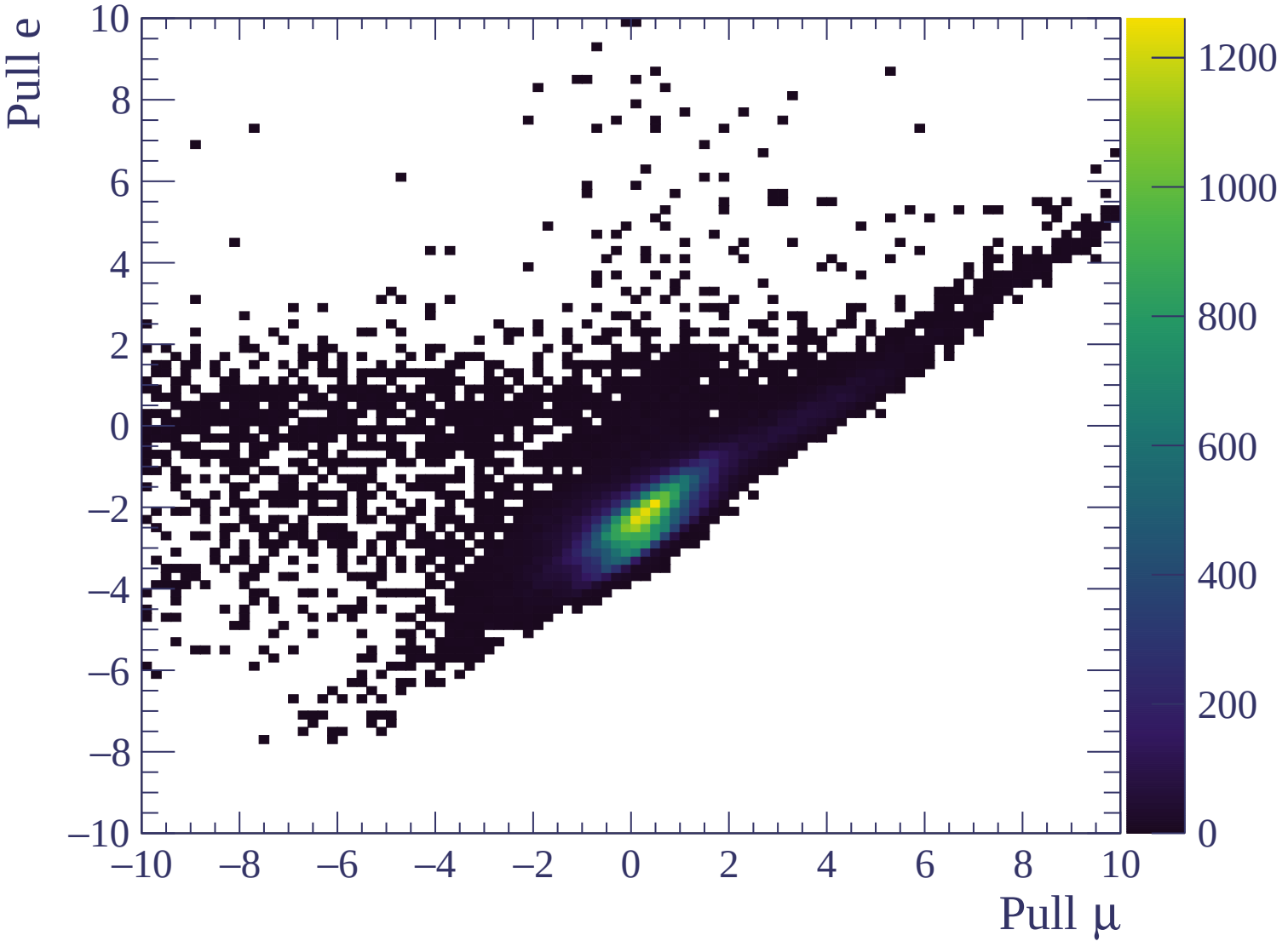






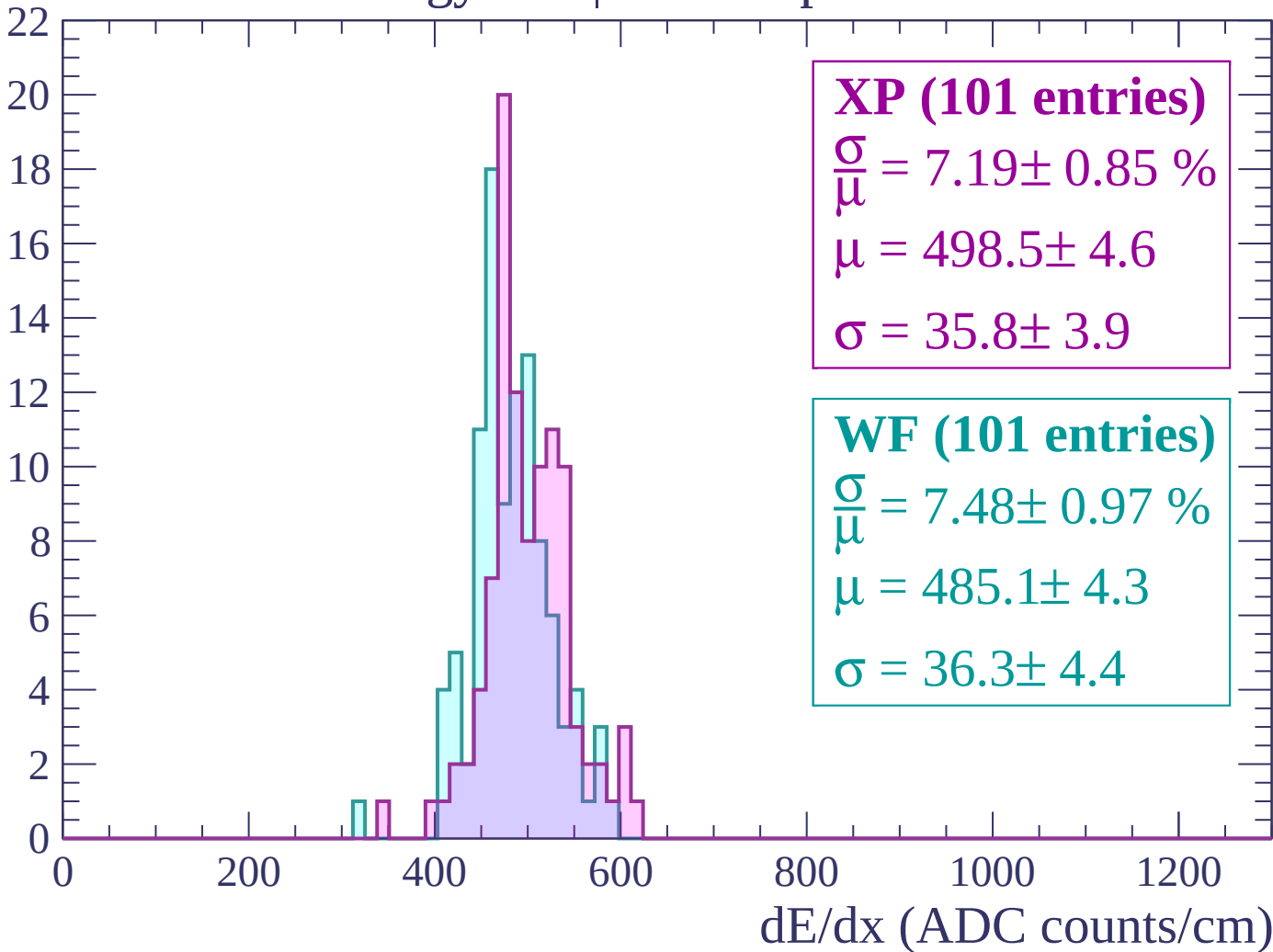






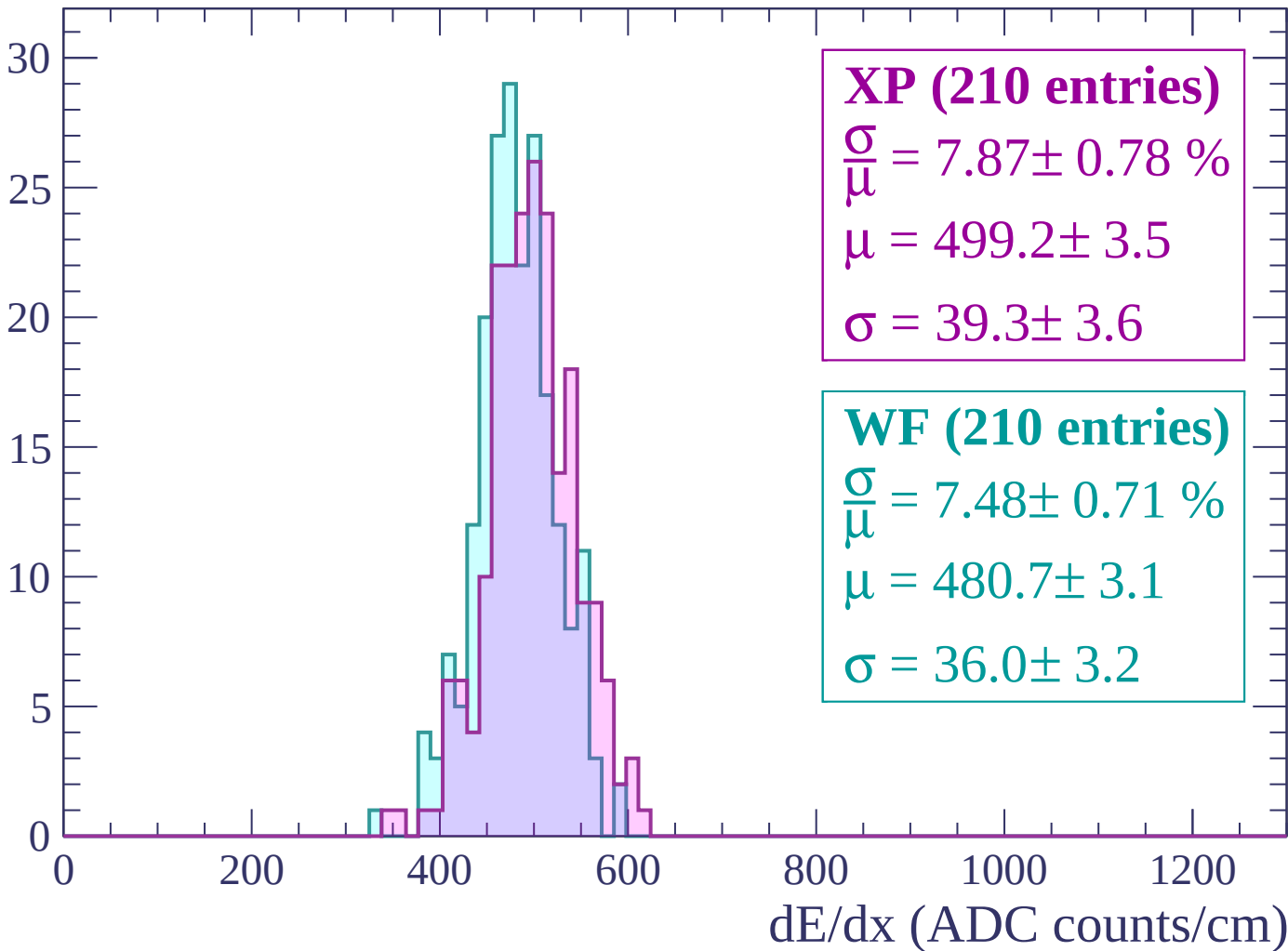
Energy loss | $-3000 < p < -2940$

Count



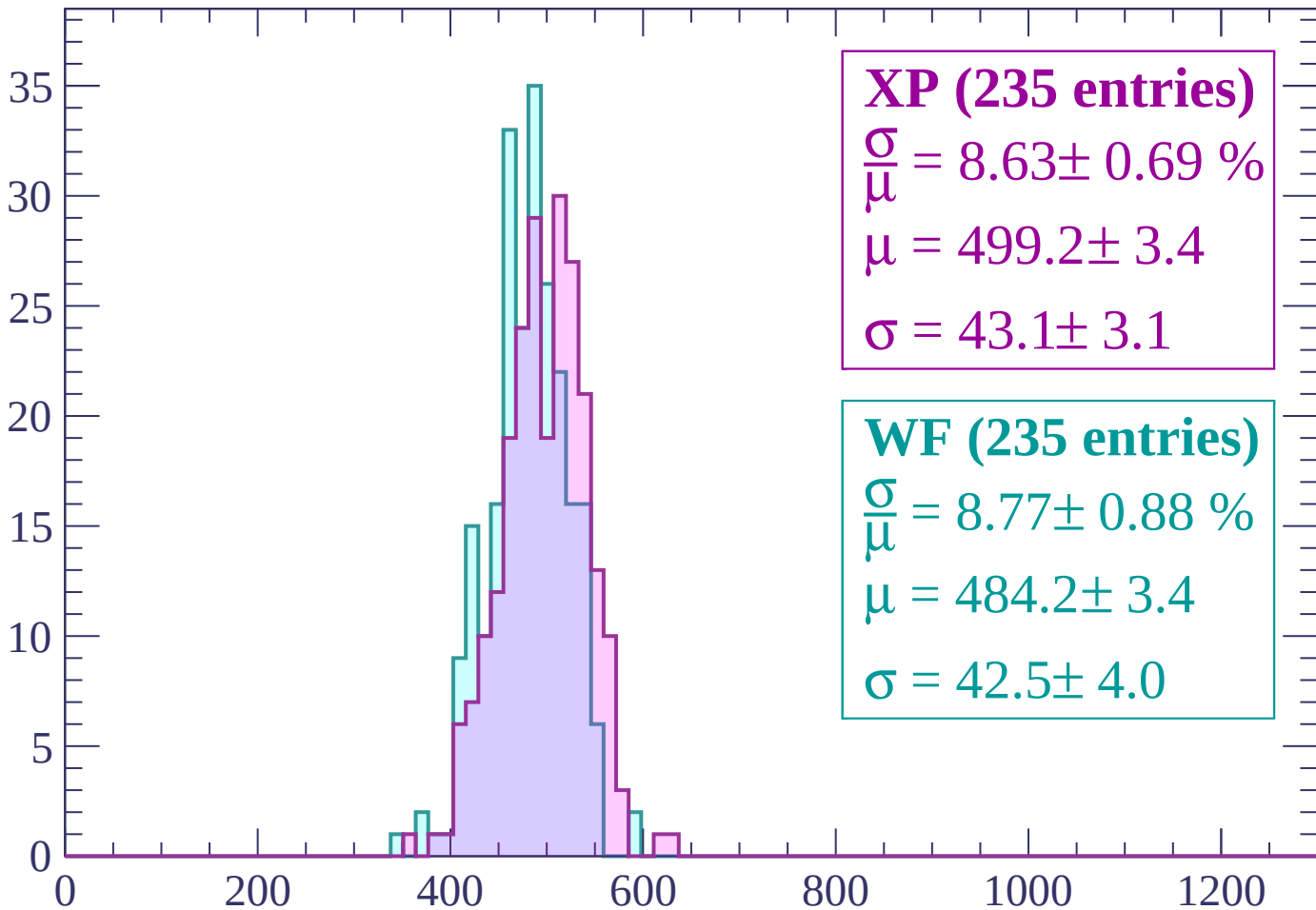
Energy loss | $-2940 < p < -2880$

Count



Energy loss | $-2880 < p < -2820$

Count



XP (235 entries)

$$\frac{\sigma}{\mu} = 8.63 \pm 0.69 \%$$

$$\mu = 499.2 \pm 3.4$$

$$\sigma = 43.1 \pm 3.1$$

WF (235 entries)

$$\frac{\sigma}{\mu} = 8.77 \pm 0.88 \%$$

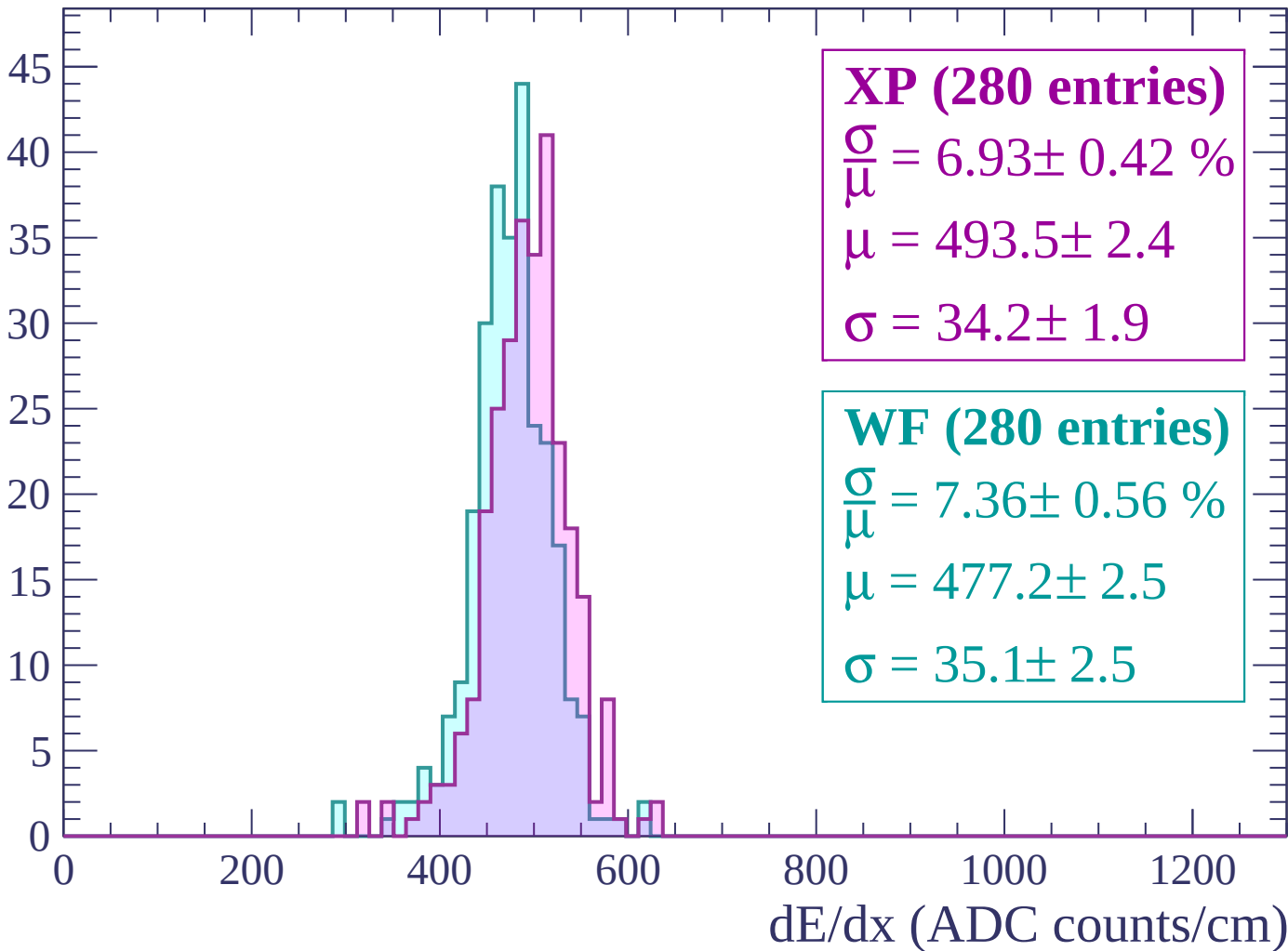
$$\mu = 484.2 \pm 3.4$$

$$\sigma = 42.5 \pm 4.0$$

dE/dx (ADC counts/cm)

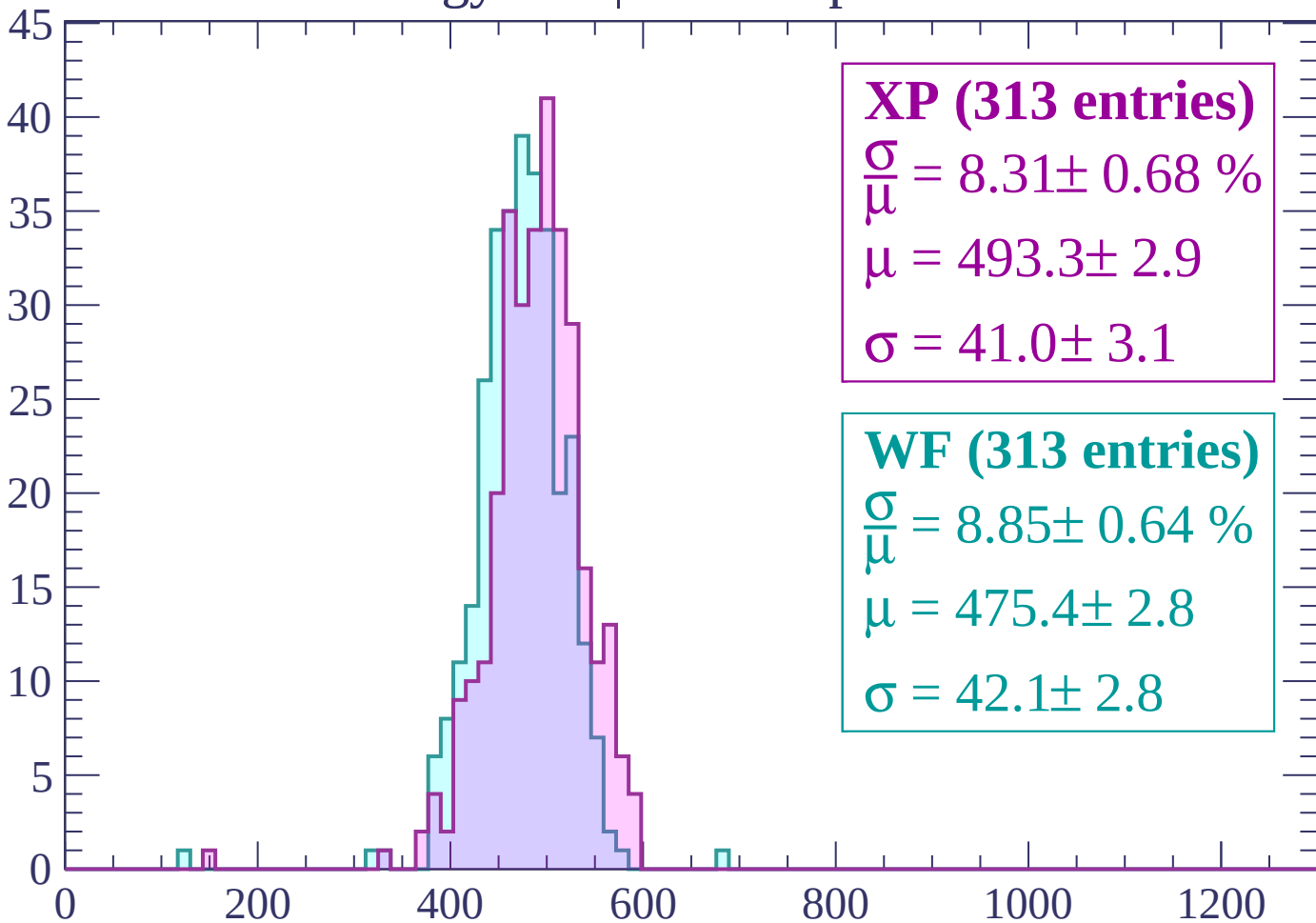
Energy loss | $-2820 < p < -2760$

Count



Energy loss | $-2760 < p < -2700$

Count



XP (313 entries)

$$\frac{\sigma}{\mu} = 8.31 \pm 0.68 \%$$

$$\mu = 493.3 \pm 2.9$$

$$\sigma = 41.0 \pm 3.1$$

WF (313 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.64 \%$$

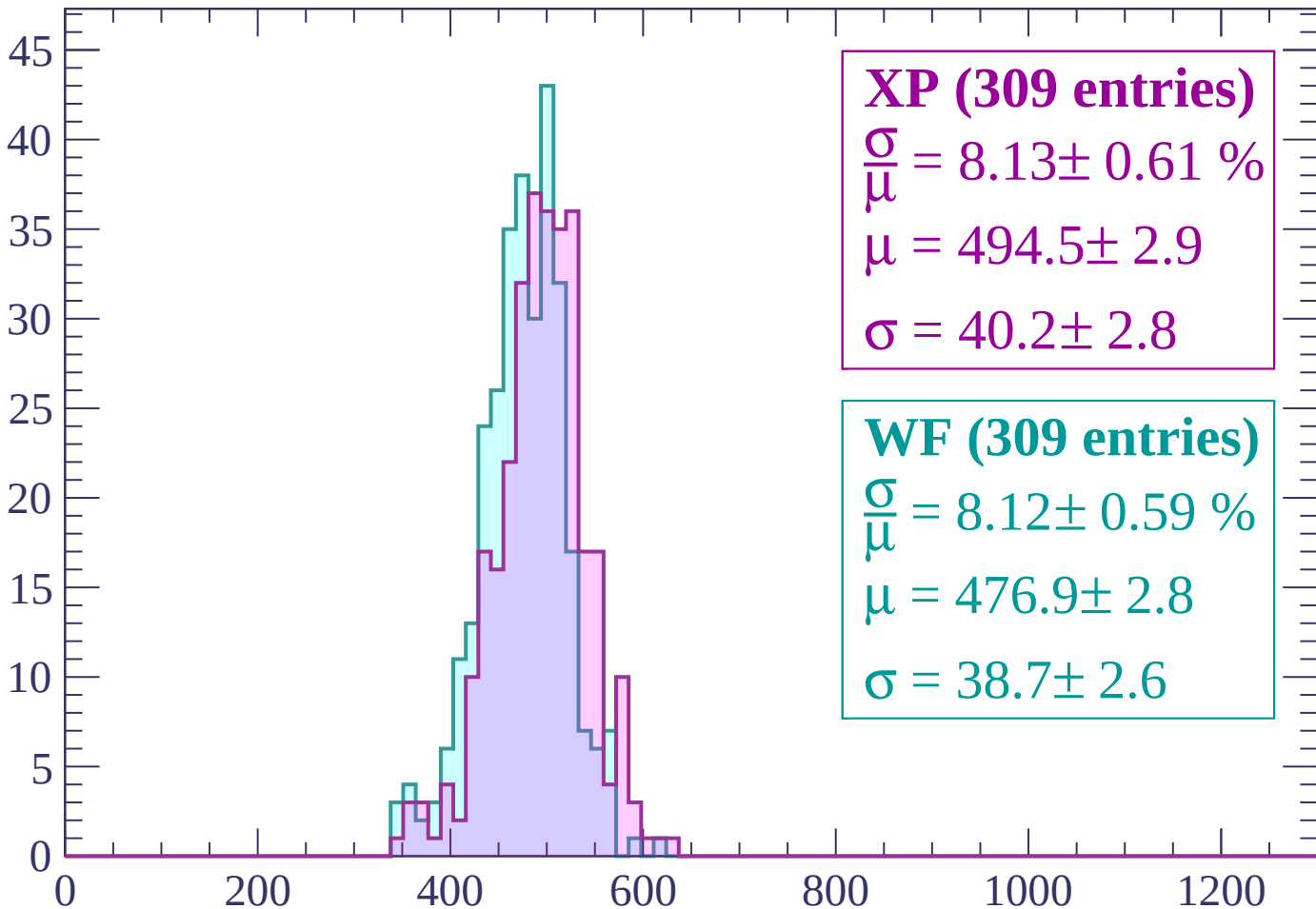
$$\mu = 475.4 \pm 2.8$$

$$\sigma = 42.1 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $-2700 < p < -2640$

Count



XP (309 entries)

$$\frac{\sigma}{\mu} = 8.13 \pm 0.61 \%$$

$$\mu = 494.5 \pm 2.9$$

$$\sigma = 40.2 \pm 2.8$$

WF (309 entries)

$$\frac{\sigma}{\mu} = 8.12 \pm 0.59 \%$$

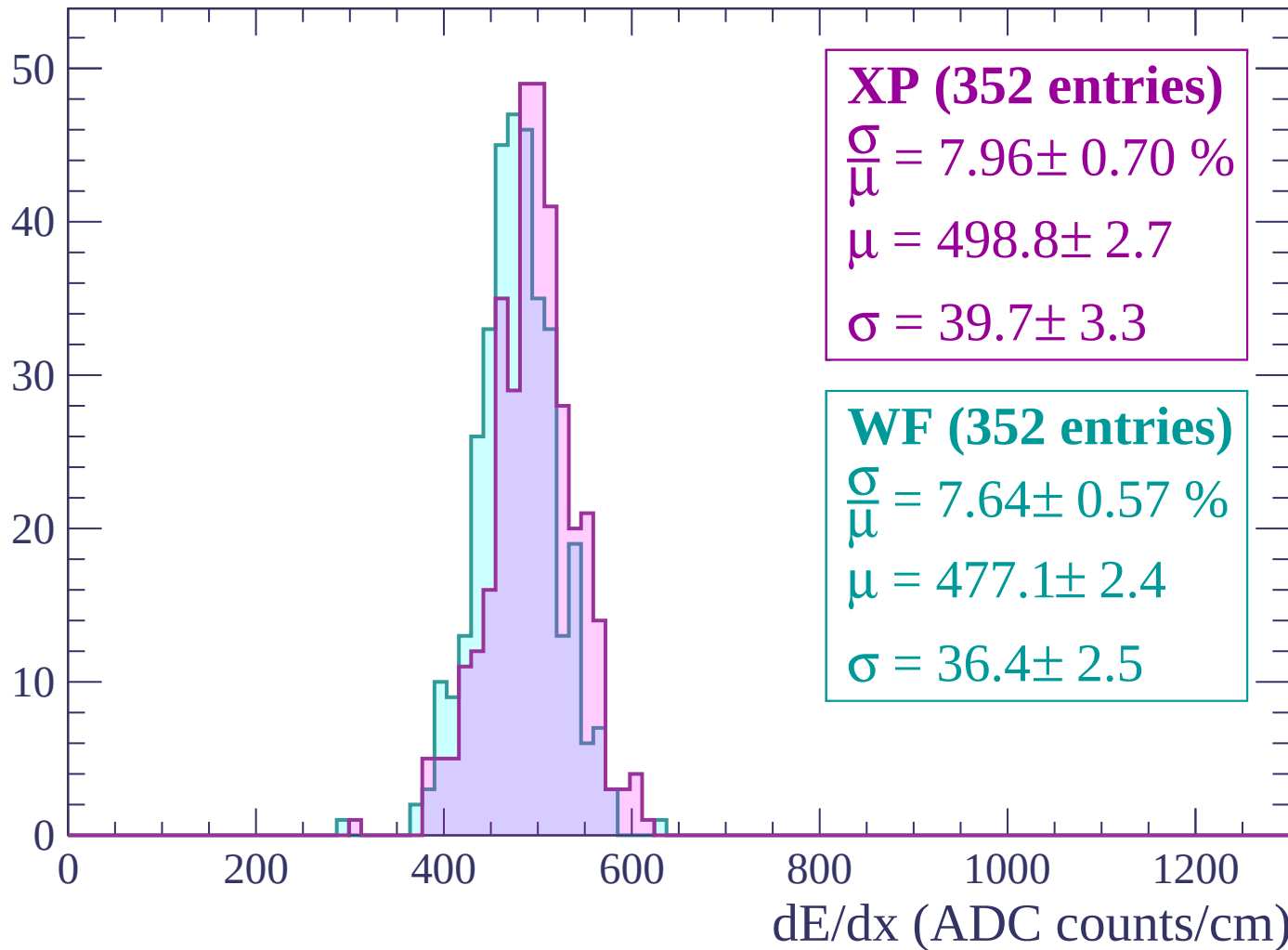
$$\mu = 476.9 \pm 2.8$$

$$\sigma = 38.7 \pm 2.6$$

dE/dx (ADC counts/cm)

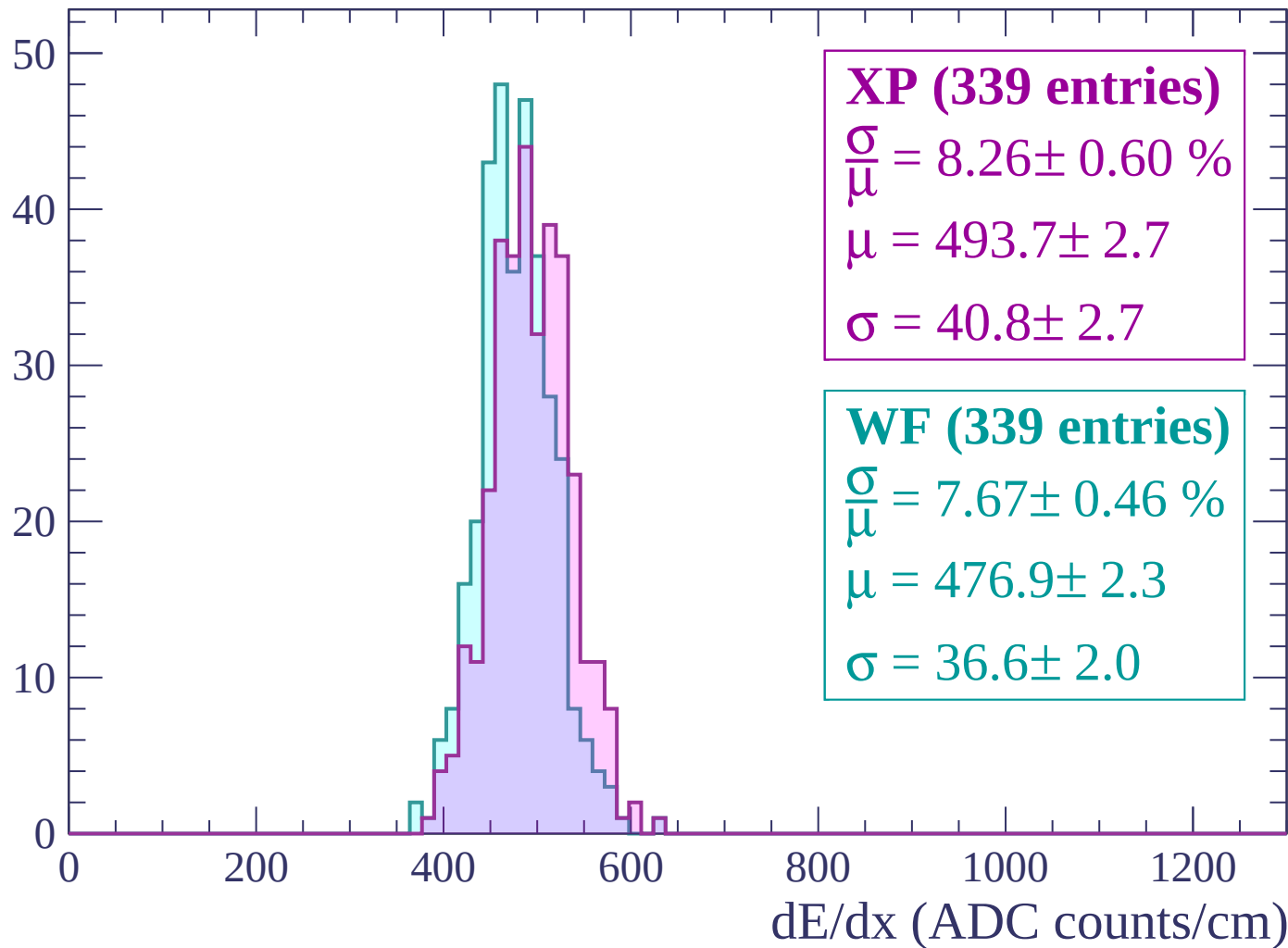
Energy loss | $-2640 < p < -2580$

Count



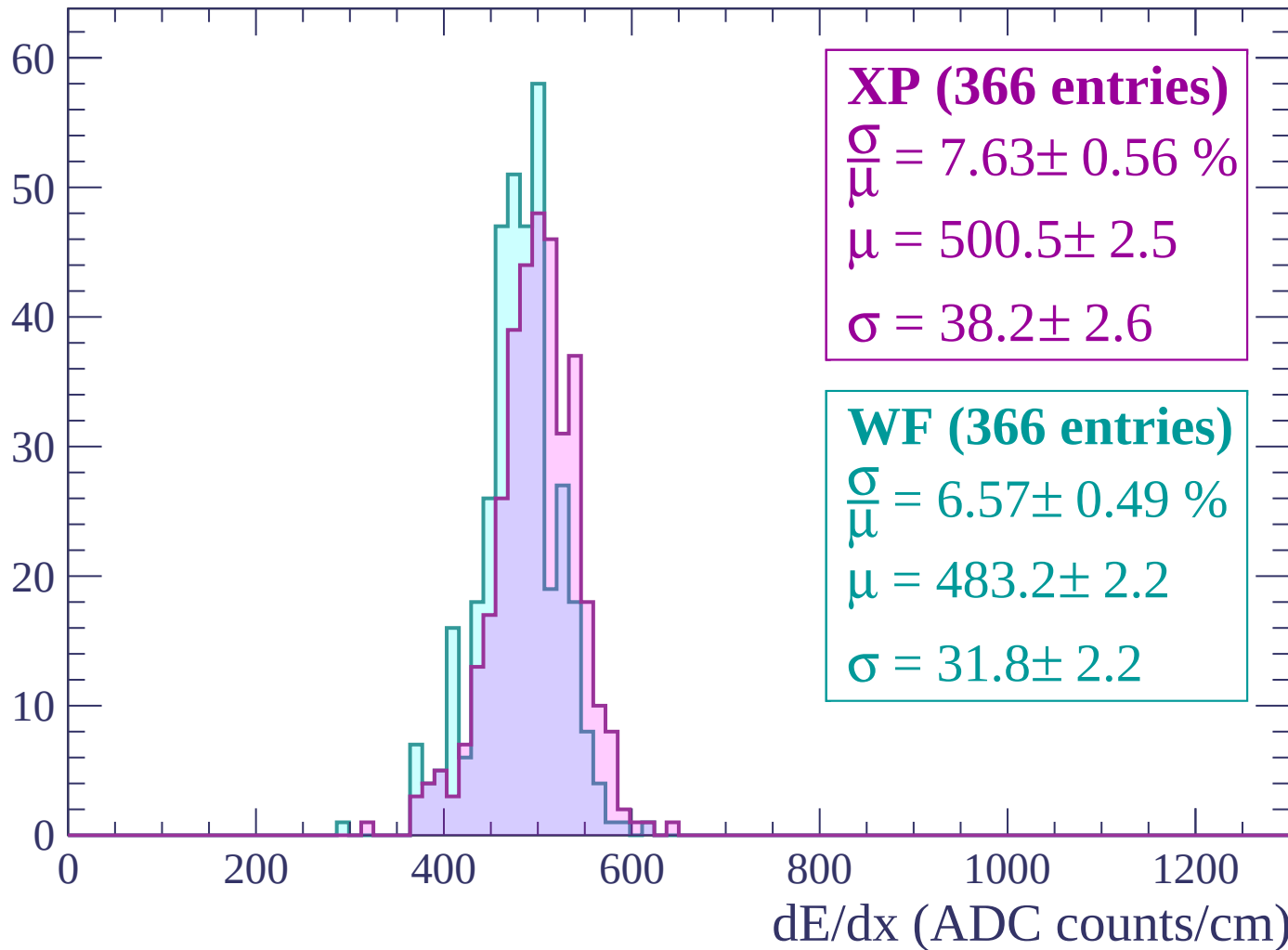
Energy loss | $-2580 < p < -2520$

Count



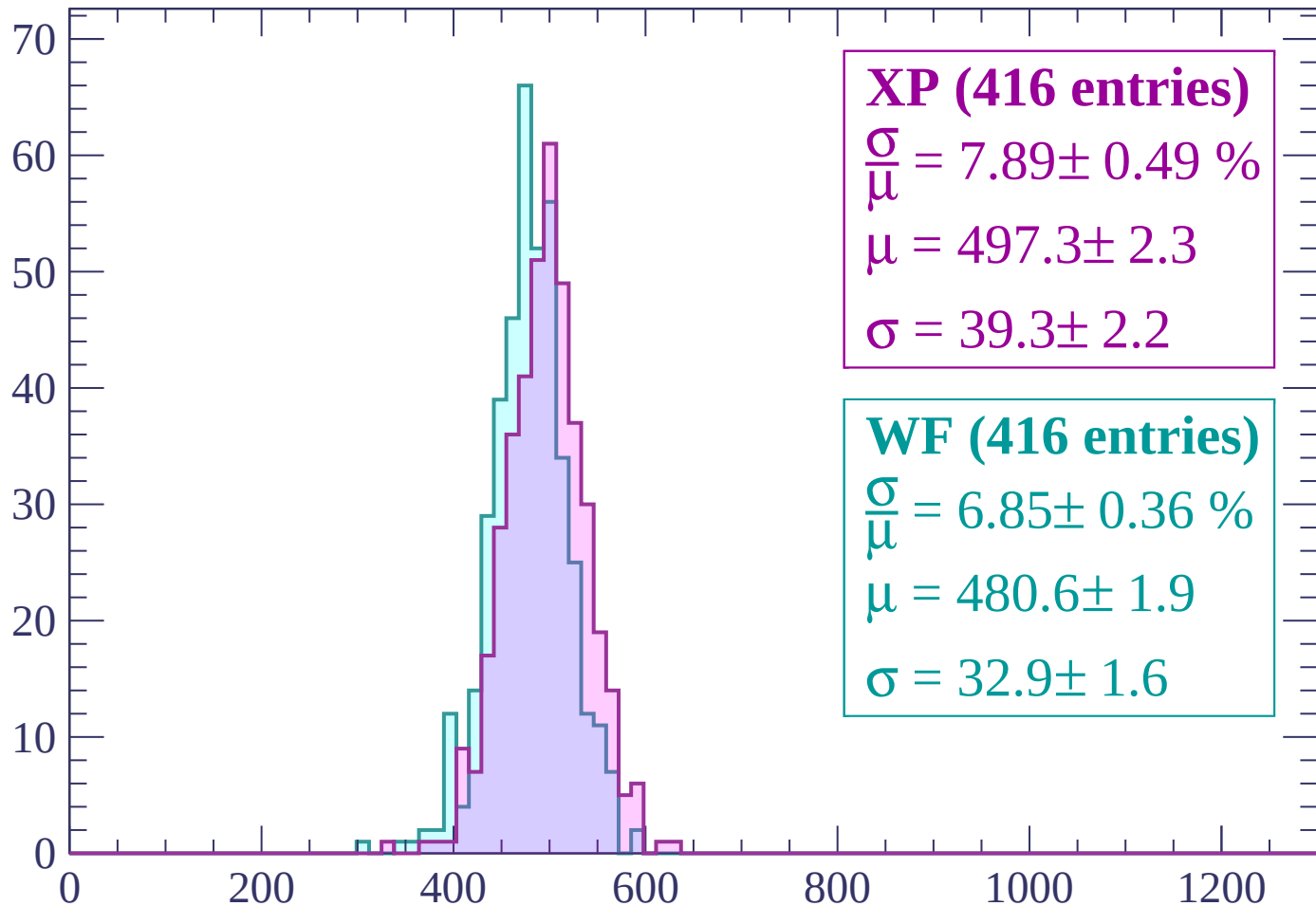
Energy loss | $-2520 < p < -2460$

Count



Energy loss | $-2460 < p < -2400$

Count



XP (416 entries)

$$\frac{\sigma}{\mu} = 7.89 \pm 0.49 \%$$

$$\mu = 497.3 \pm 2.3$$

$$\sigma = 39.3 \pm 2.2$$

WF (416 entries)

$$\frac{\sigma}{\mu} = 6.85 \pm 0.36 \%$$

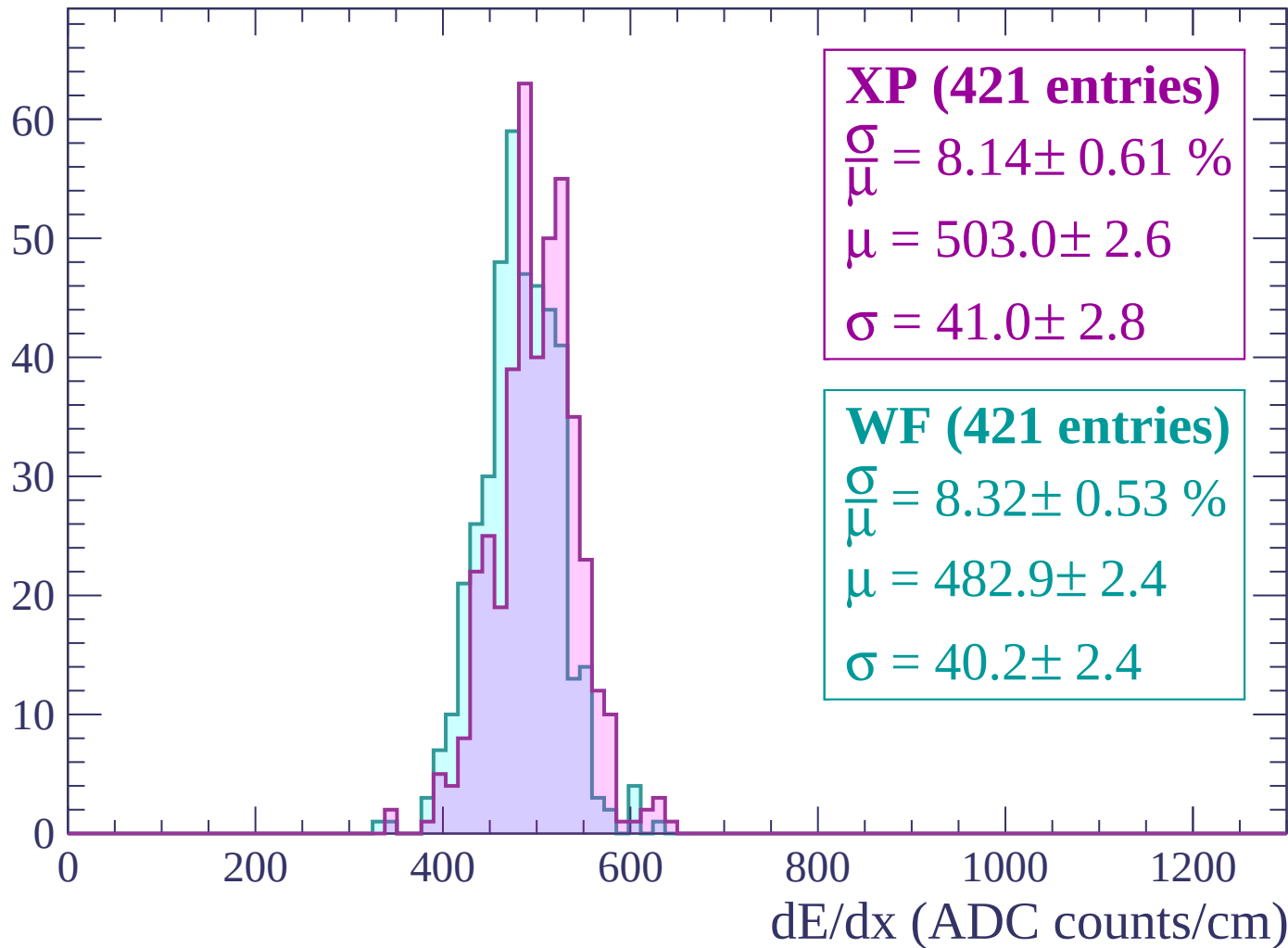
$$\mu = 480.6 \pm 1.9$$

$$\sigma = 32.9 \pm 1.6$$

dE/dx (ADC counts/cm)

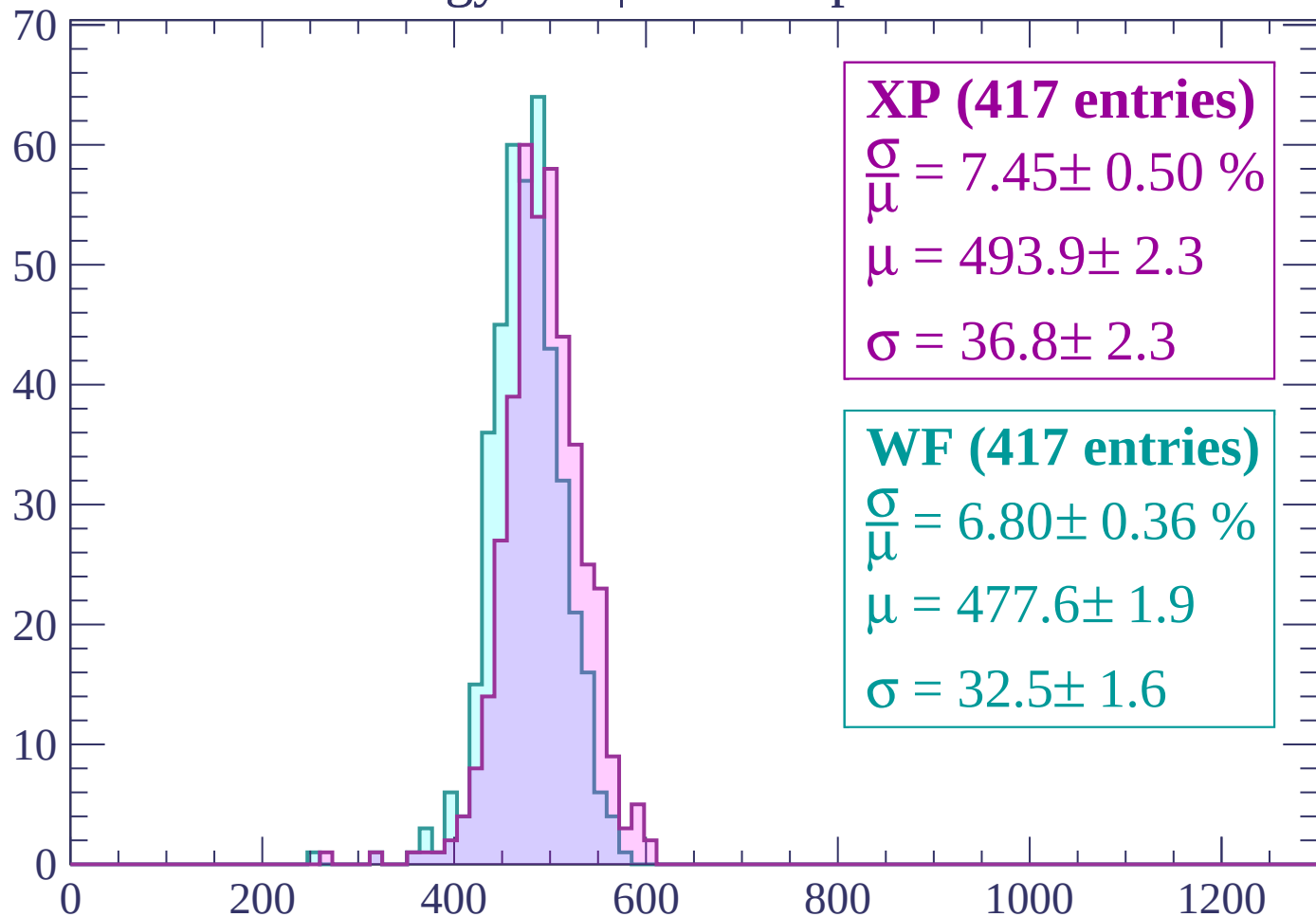
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (417 entries)

$$\frac{\sigma}{\mu} = 7.45 \pm 0.50 \%$$

$$\mu = 493.9 \pm 2.3$$

$$\sigma = 36.8 \pm 2.3$$

WF (417 entries)

$$\frac{\sigma}{\mu} = 6.80 \pm 0.36 \%$$

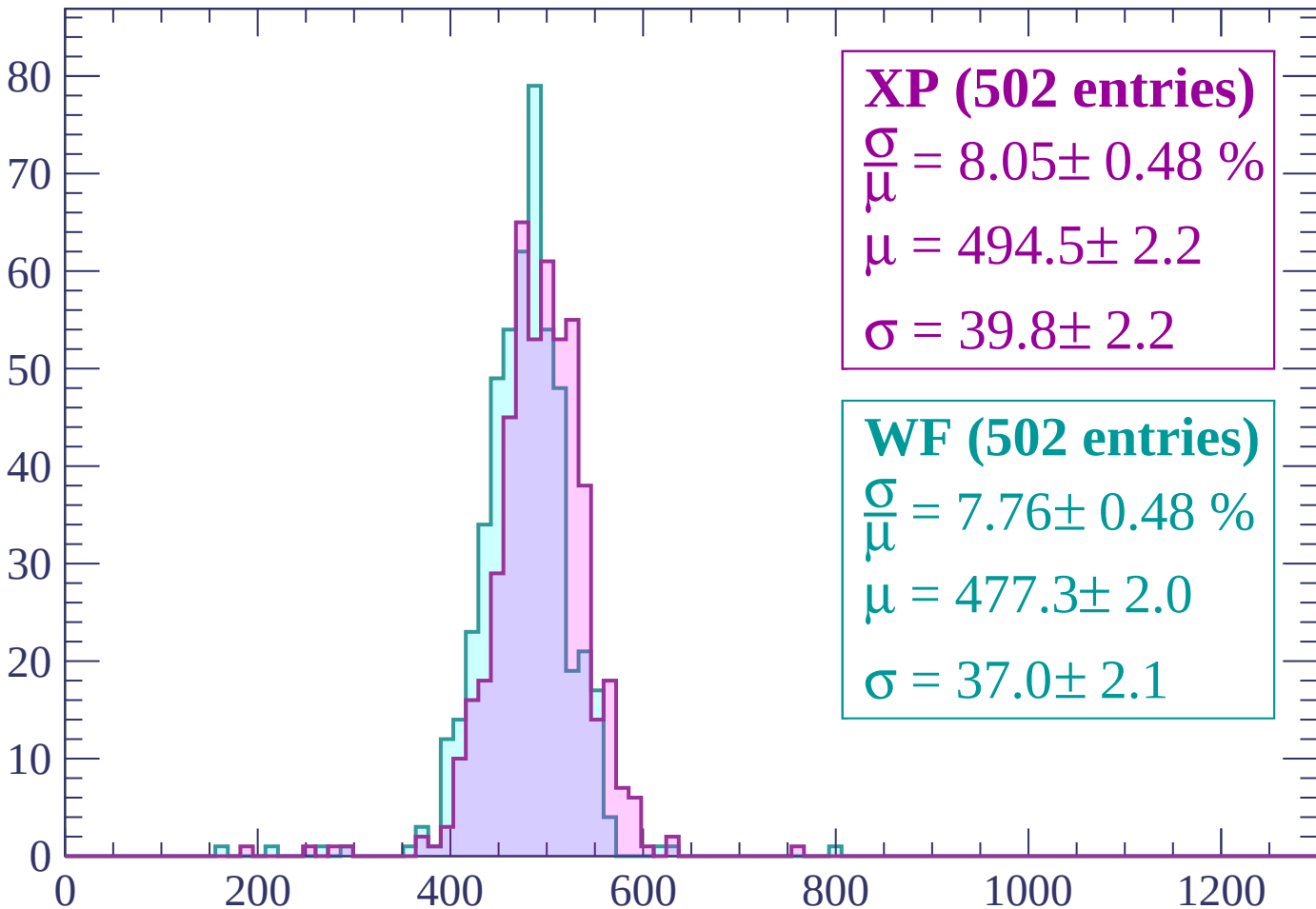
$$\mu = 477.6 \pm 1.9$$

$$\sigma = 32.5 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-2280 < p < -2220$

Count



XP (502 entries)

$$\frac{\sigma}{\mu} = 8.05 \pm 0.48 \%$$

$$\mu = 494.5 \pm 2.2$$

$$\sigma = 39.8 \pm 2.2$$

WF (502 entries)

$$\frac{\sigma}{\mu} = 7.76 \pm 0.48 \%$$

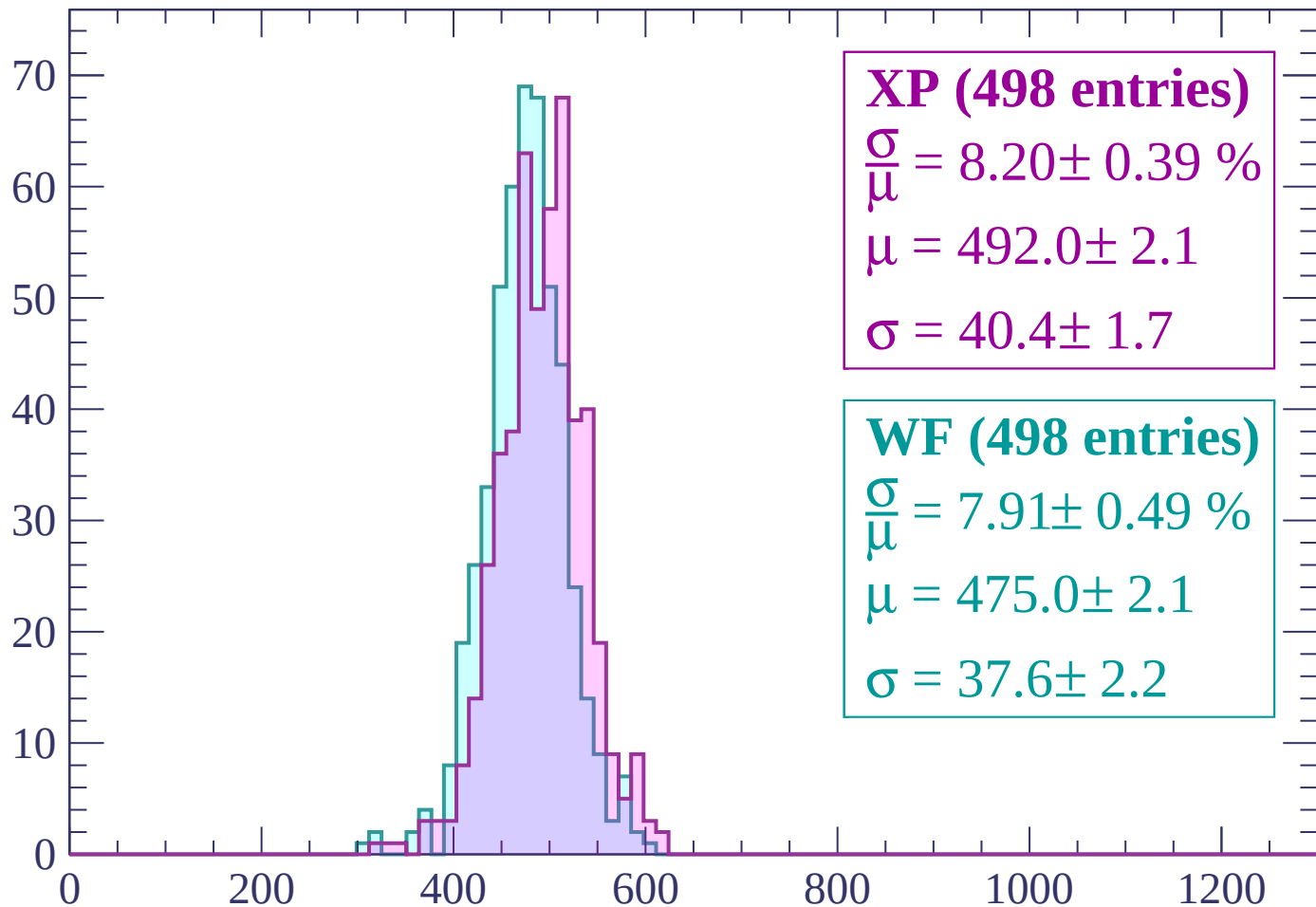
$$\mu = 477.3 \pm 2.0$$

$$\sigma = 37.0 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-2220 < p < -2160$

Count



XP (498 entries)

$\frac{\sigma}{\mu} = 8.20 \pm 0.39 \%$

$\mu = 492.0 \pm 2.1$

$\sigma = 40.4 \pm 1.7$

WF (498 entries)

$\frac{\sigma}{\mu} = 7.91 \pm 0.49 \%$

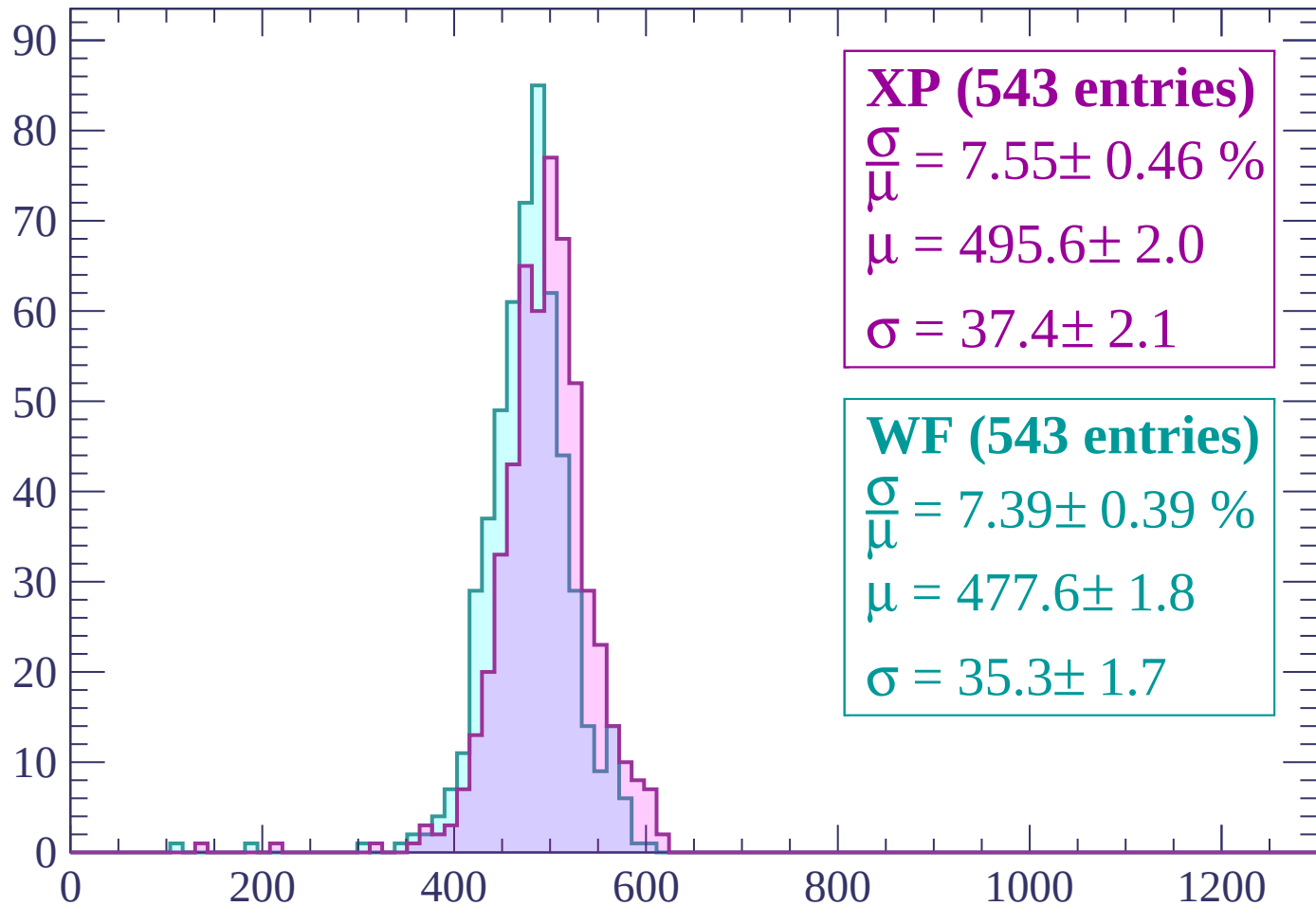
$\mu = 475.0 \pm 2.1$

$\sigma = 37.6 \pm 2.2$

dE/dx (ADC counts/cm)

Energy loss | $-2160 < p < -2100$

Count



XP (543 entries)

$$\frac{\sigma}{\mu} = 7.55 \pm 0.46 \%$$

$$\mu = 495.6 \pm 2.0$$

$$\sigma = 37.4 \pm 2.1$$

WF (543 entries)

$$\frac{\sigma}{\mu} = 7.39 \pm 0.39 \%$$

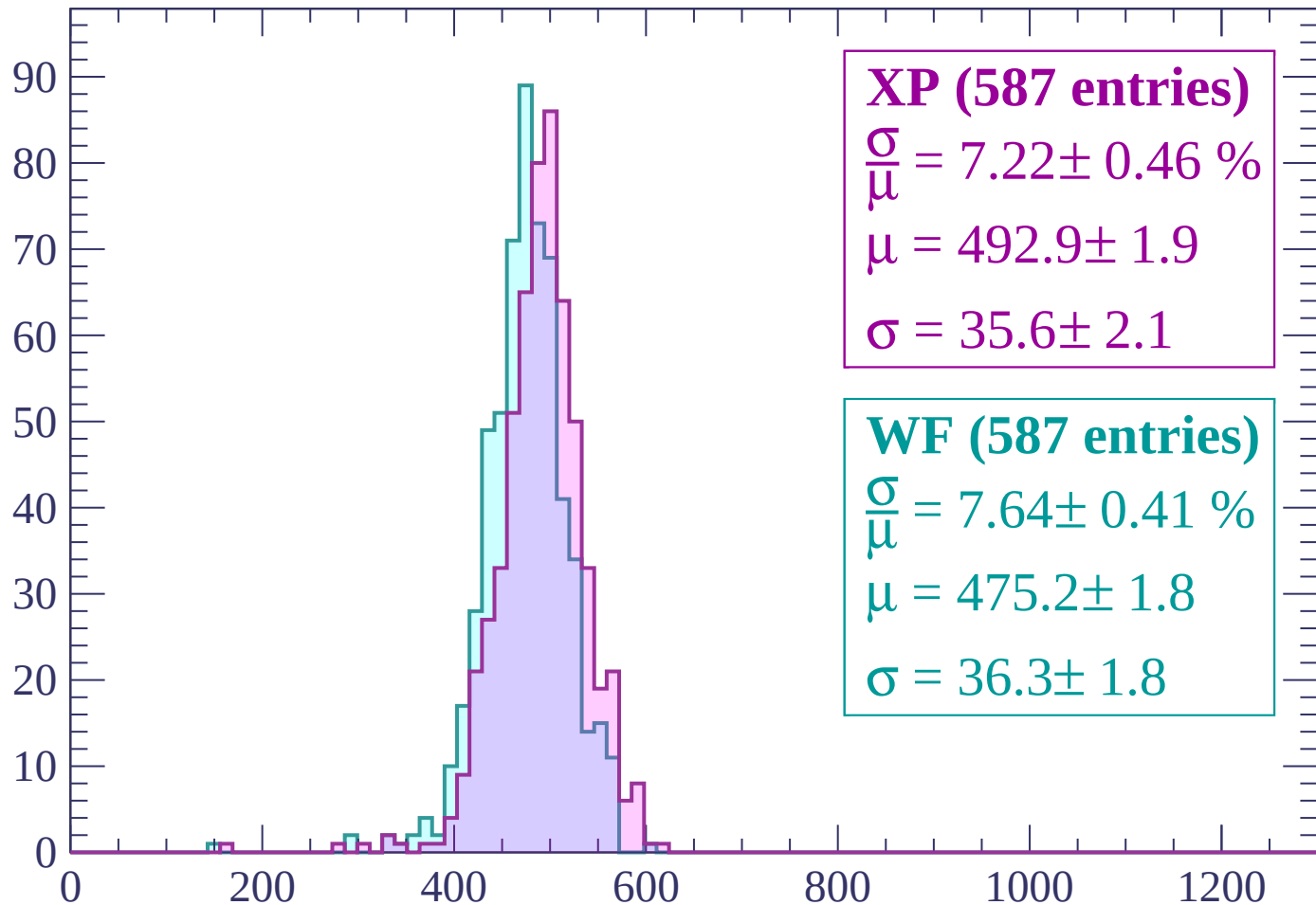
$$\mu = 477.6 \pm 1.8$$

$$\sigma = 35.3 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count



XP (587 entries)

$$\frac{\sigma}{\mu} = 7.22 \pm 0.46 \%$$

$$\mu = 492.9 \pm 1.9$$

$$\sigma = 35.6 \pm 2.1$$

WF (587 entries)

$$\frac{\sigma}{\mu} = 7.64 \pm 0.41 \%$$

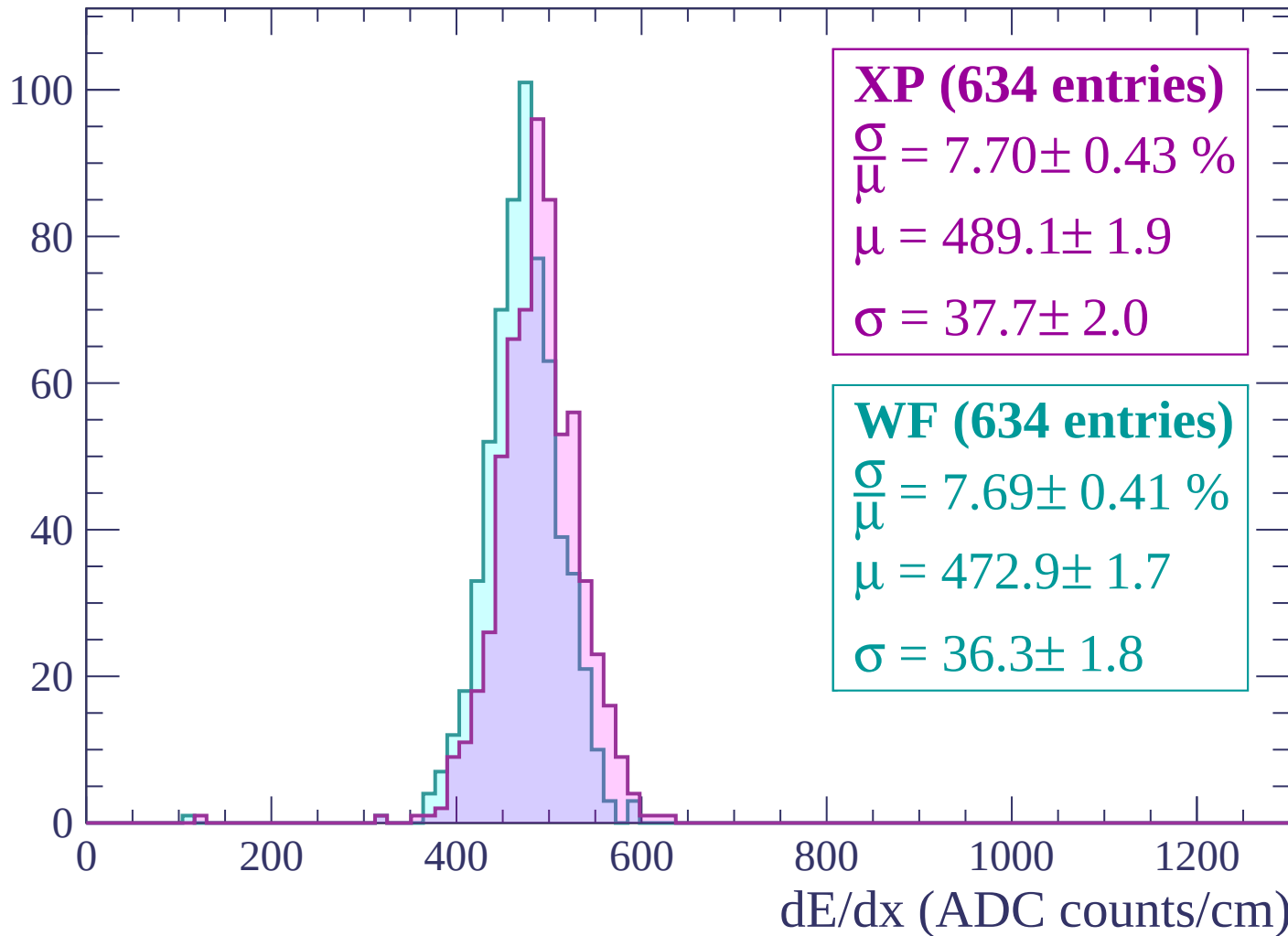
$$\mu = 475.2 \pm 1.8$$

$$\sigma = 36.3 \pm 1.8$$

dE/dx (ADC counts/cm)

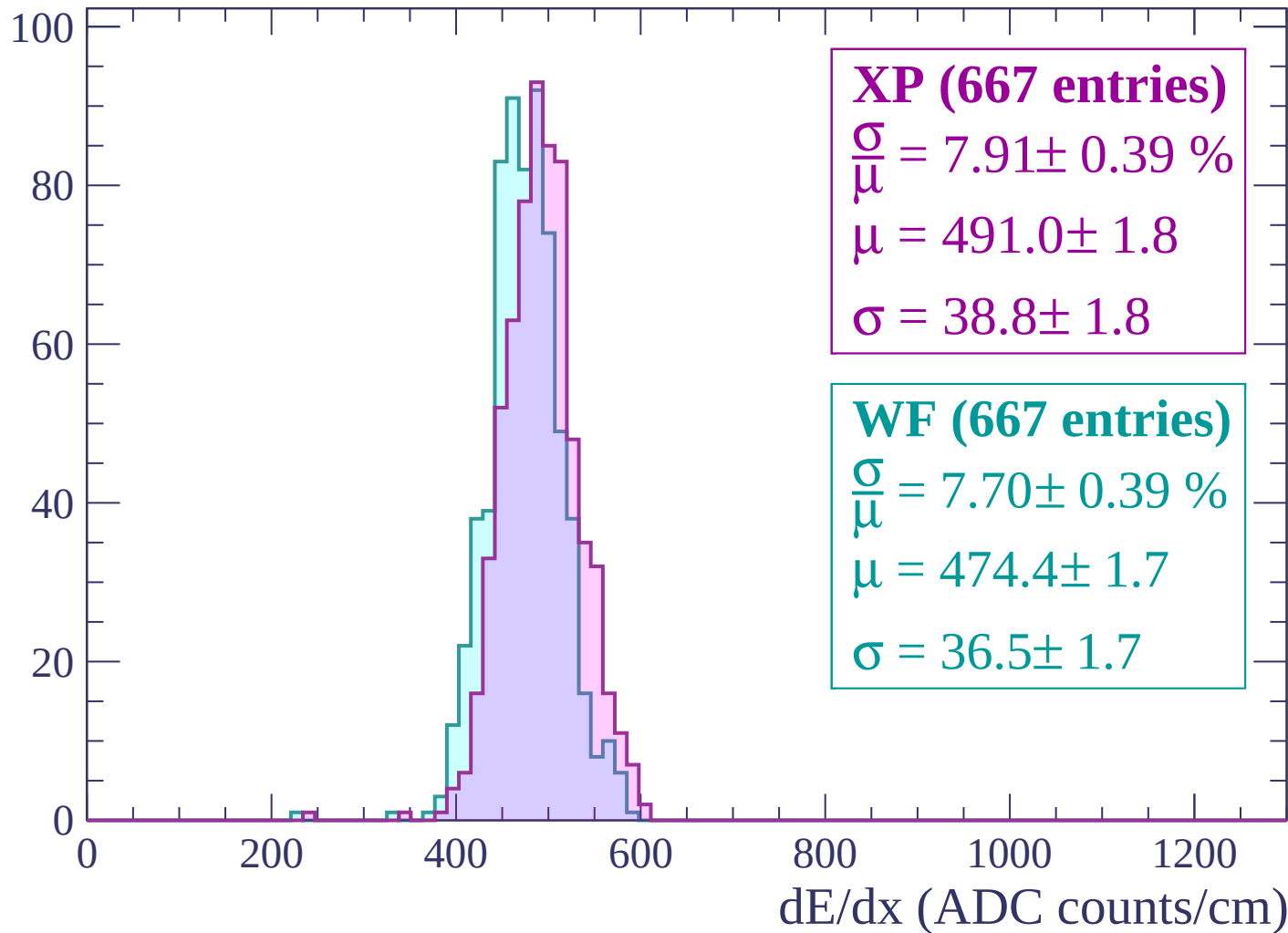
Energy loss | -2040 < p < -1980

Count



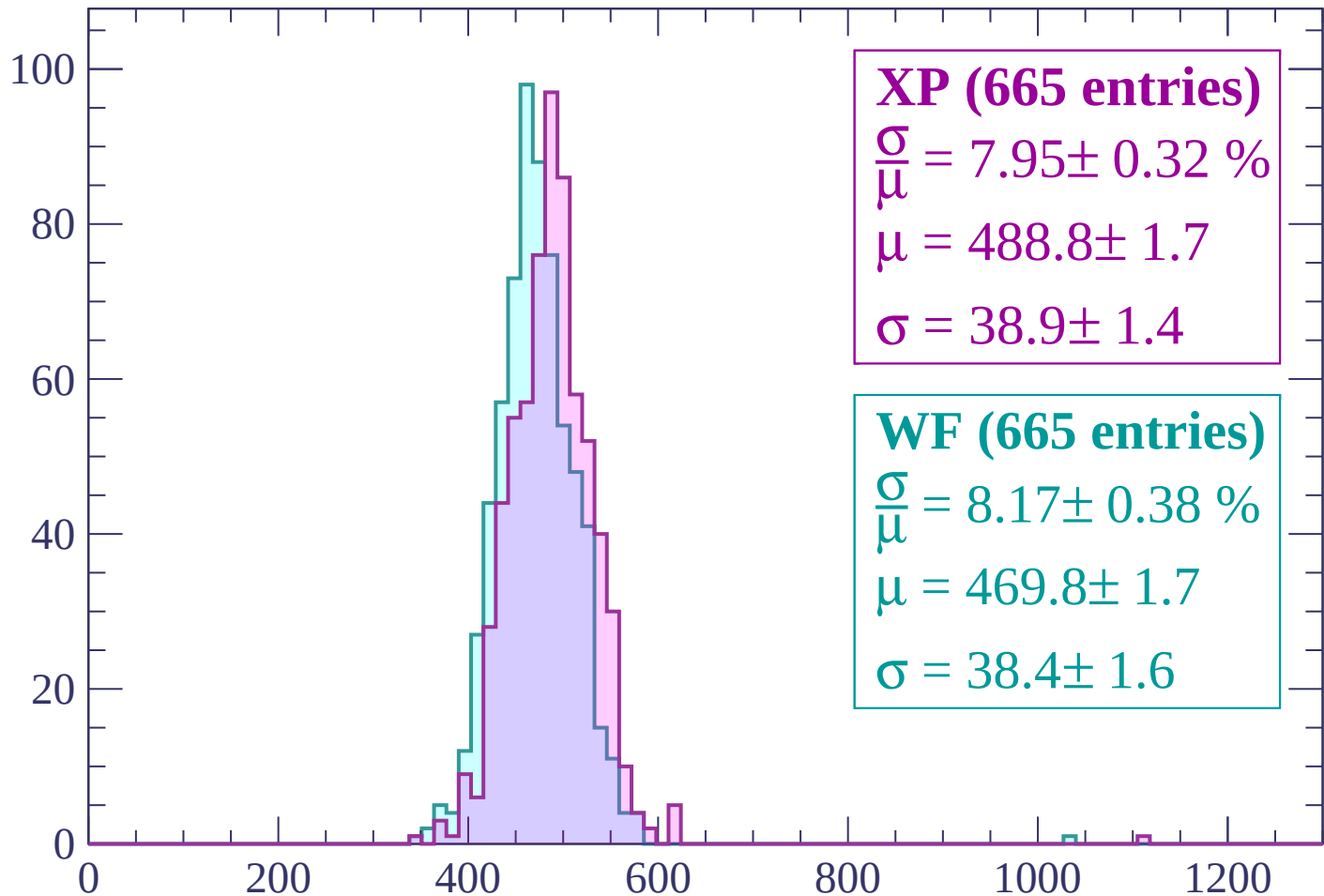
Energy loss | -1980 < p < -1920

Count



Energy loss | -1920 < p < -1860

Count



XP (665 entries)

$$\frac{\sigma}{\mu} = 7.95 \pm 0.32 \%$$

$$\mu = 488.8 \pm 1.7$$

$$\sigma = 38.9 \pm 1.4$$

WF (665 entries)

$$\frac{\sigma}{\mu} = 8.17 \pm 0.38 \%$$

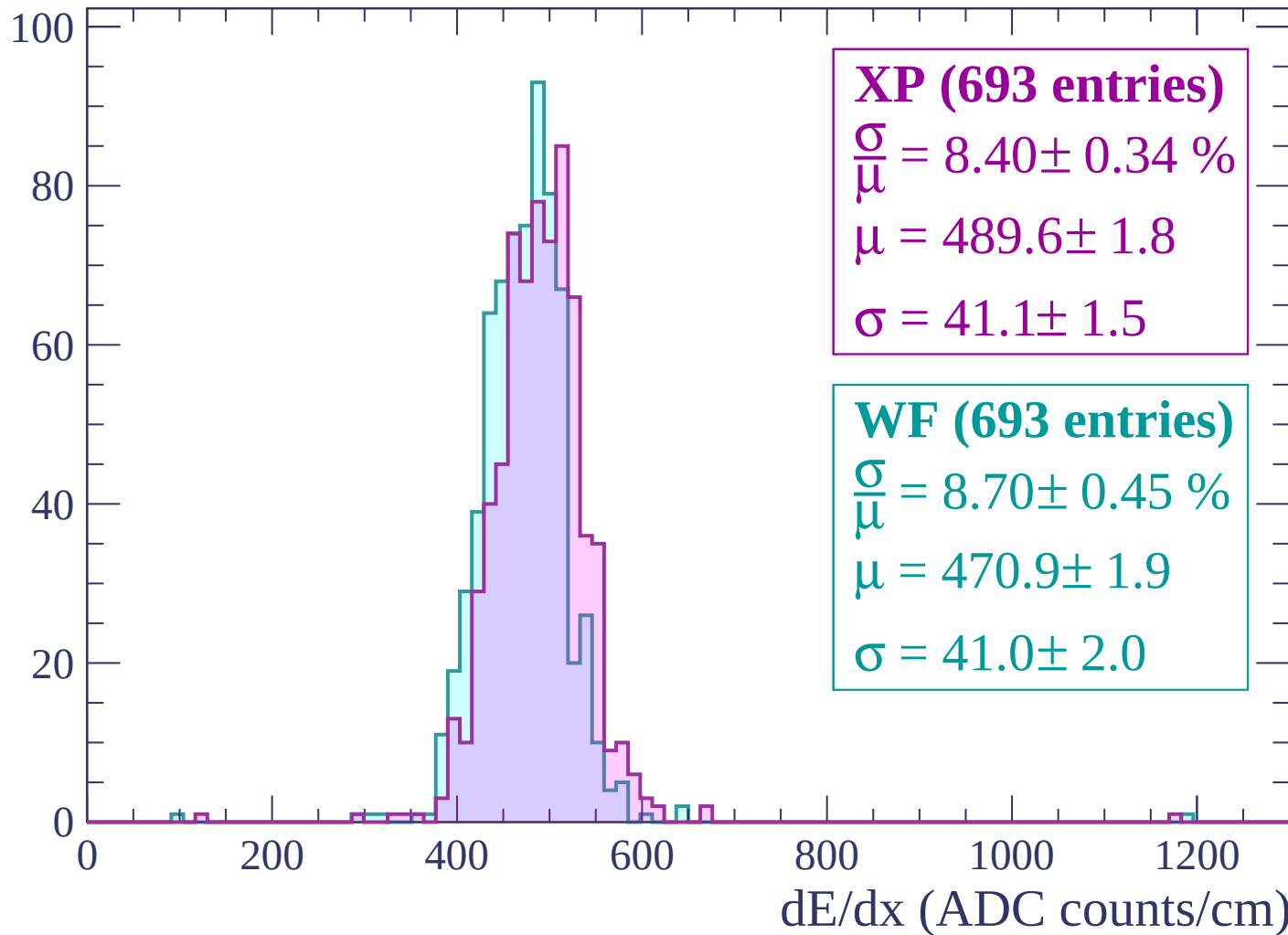
$$\mu = 469.8 \pm 1.7$$

$$\sigma = 38.4 \pm 1.6$$

dE/dx (ADC counts/cm)

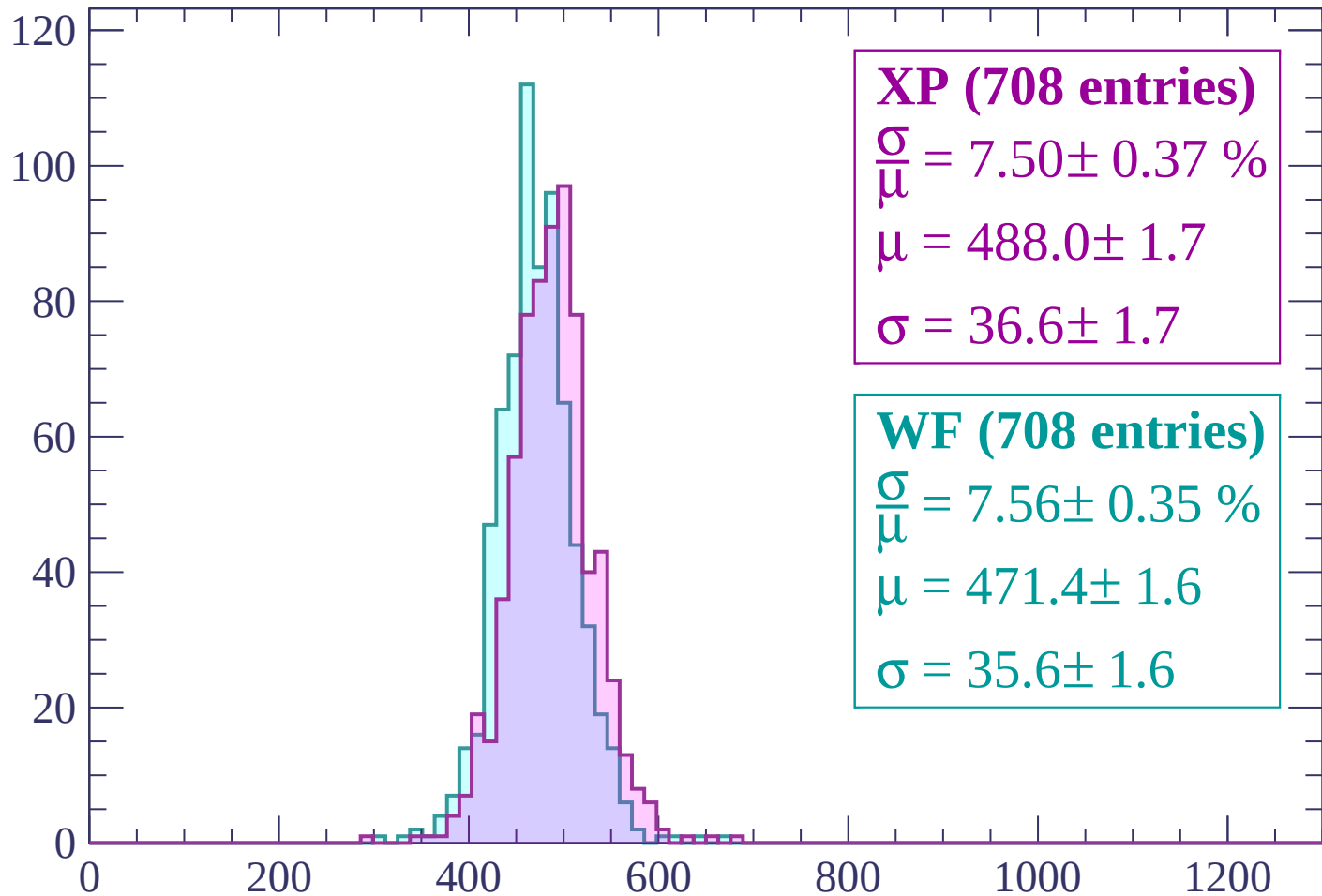
Energy loss | $-1860 < p < -1800$

Count



Energy loss | $-1800 < p < -1740$

Count



XP (708 entries)

$$\frac{\sigma}{\mu} = 7.50 \pm 0.37 \%$$

$$\mu = 488.0 \pm 1.7$$

$$\sigma = 36.6 \pm 1.7$$

WF (708 entries)

$$\frac{\sigma}{\mu} = 7.56 \pm 0.35 \%$$

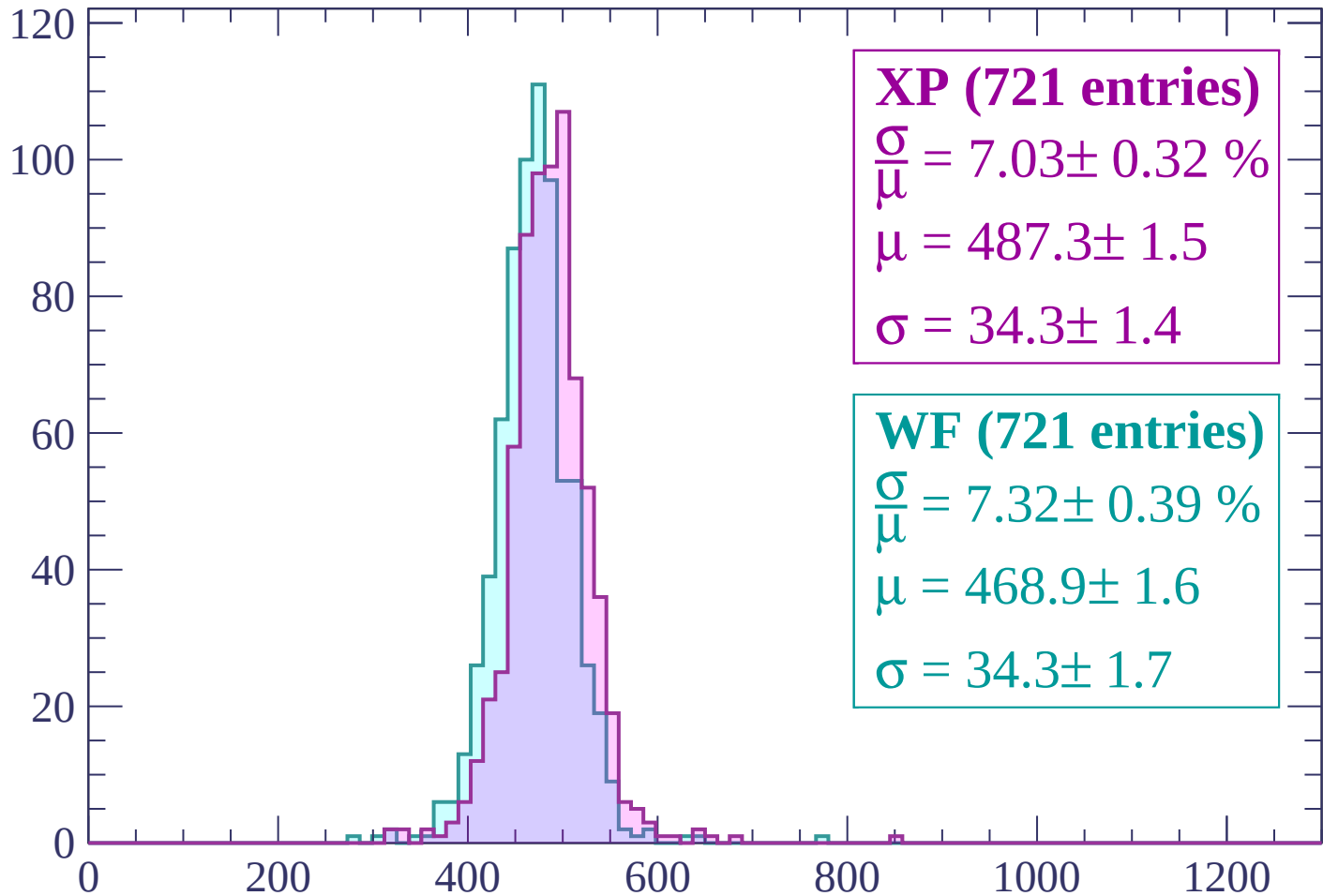
$$\mu = 471.4 \pm 1.6$$

$$\sigma = 35.6 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (721 entries)

$\frac{\sigma}{\mu} = 7.03 \pm 0.32 \%$

$\mu = 487.3 \pm 1.5$

$\sigma = 34.3 \pm 1.4$

WF (721 entries)

$\frac{\sigma}{\mu} = 7.32 \pm 0.39 \%$

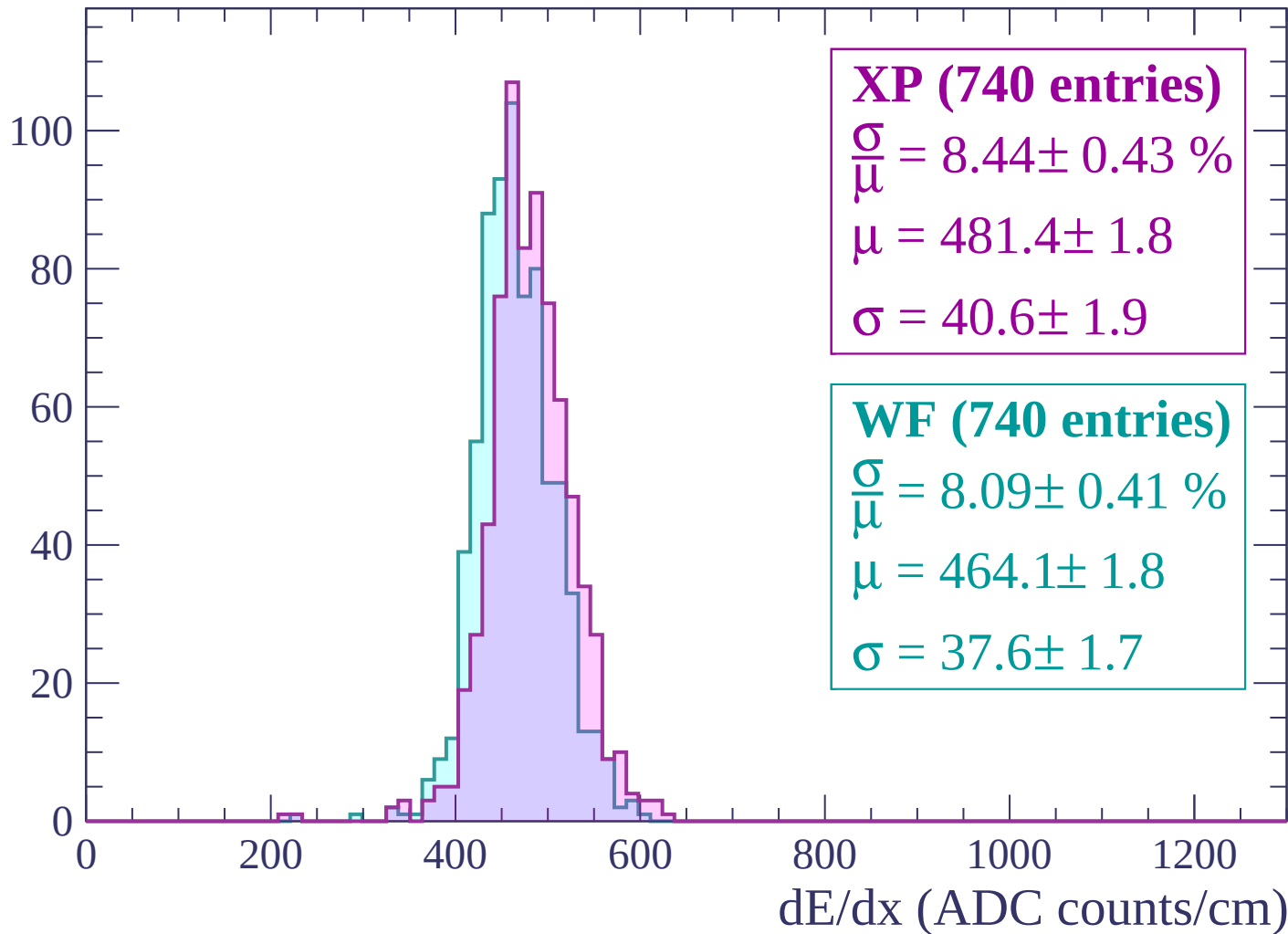
$\mu = 468.9 \pm 1.6$

$\sigma = 34.3 \pm 1.7$

dE/dx (ADC counts/cm)

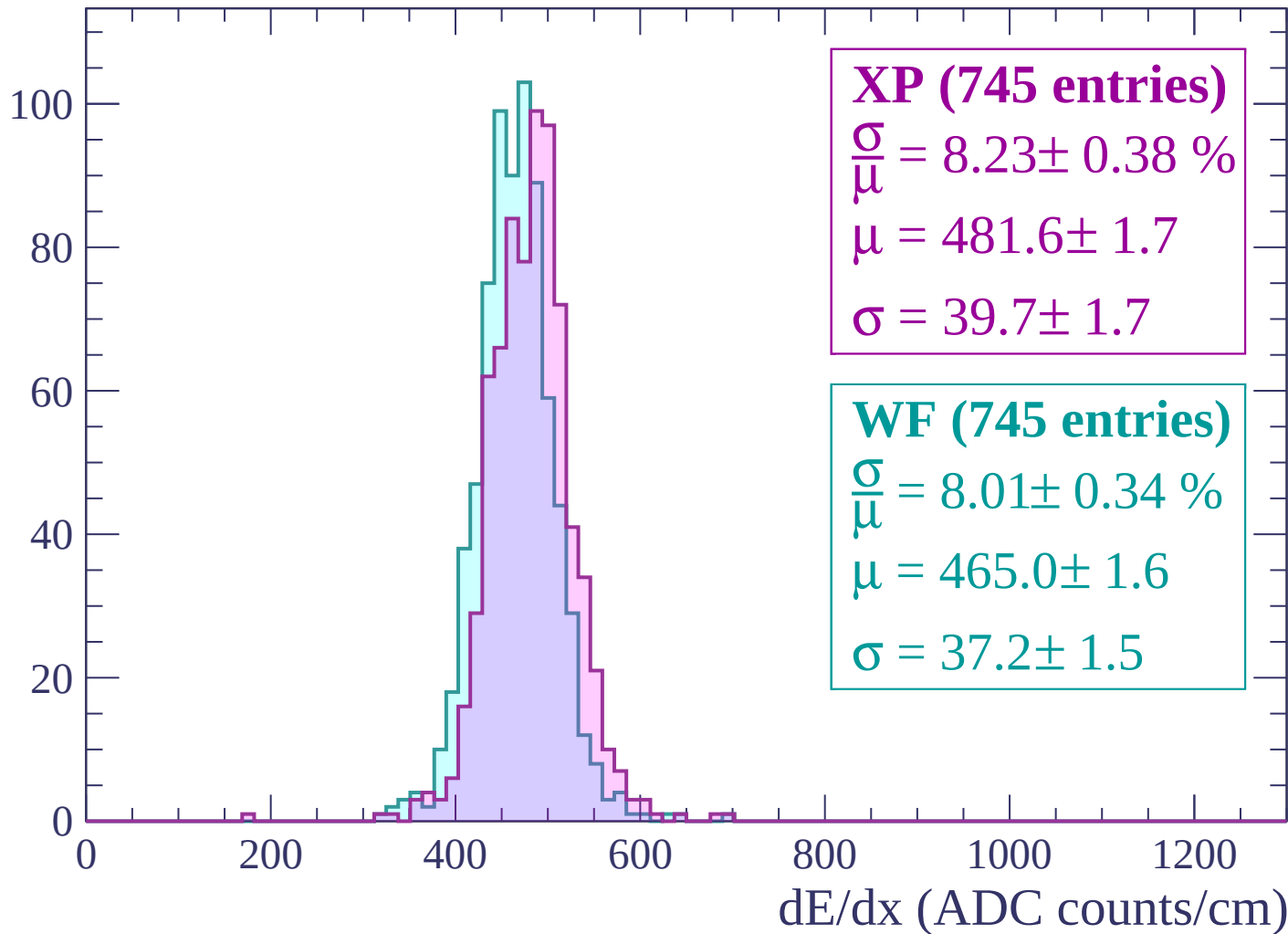
Energy loss | $-1680 < p < -1620$

Count



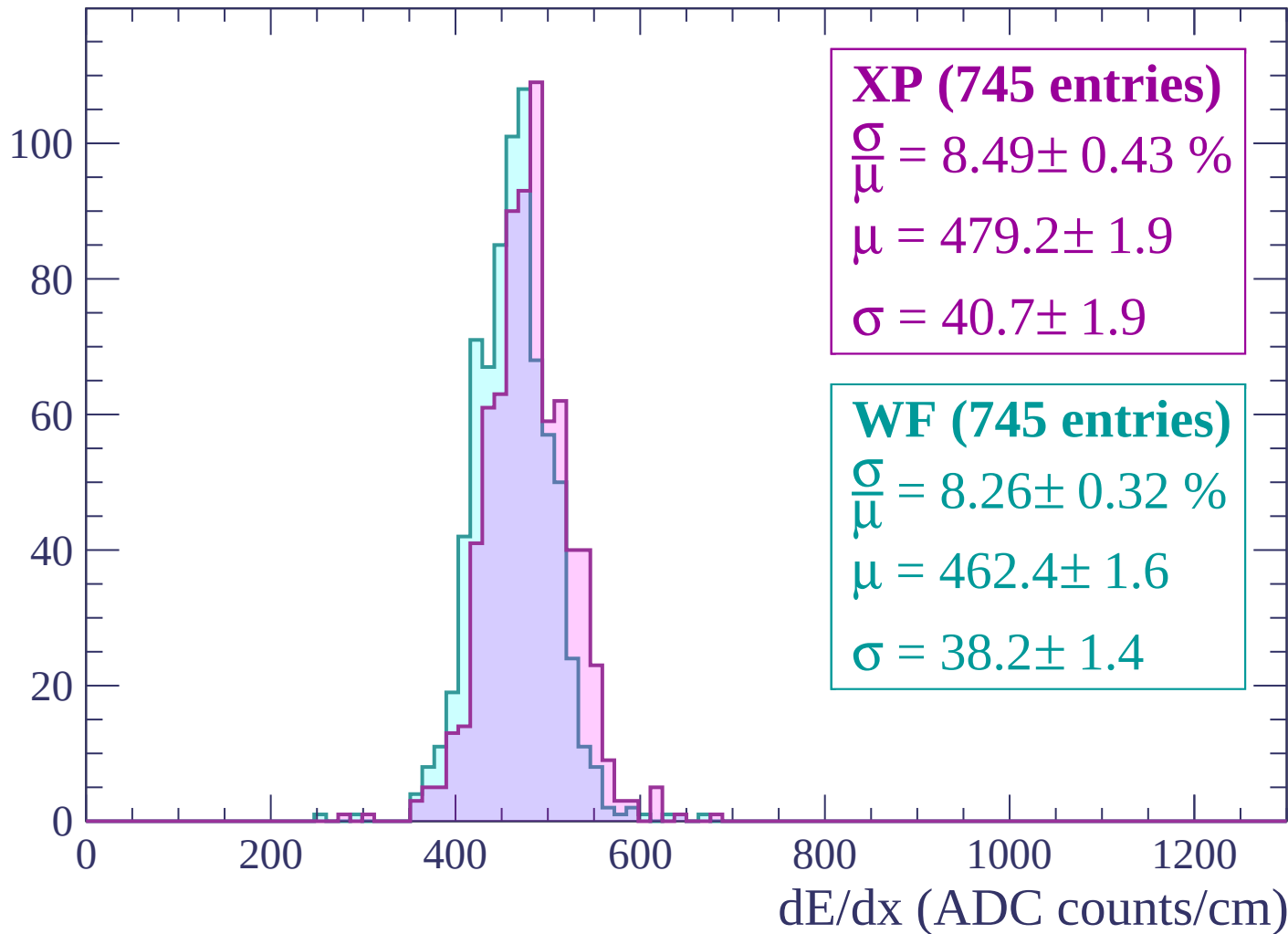
Energy loss | -1620 < p < -1560

Count



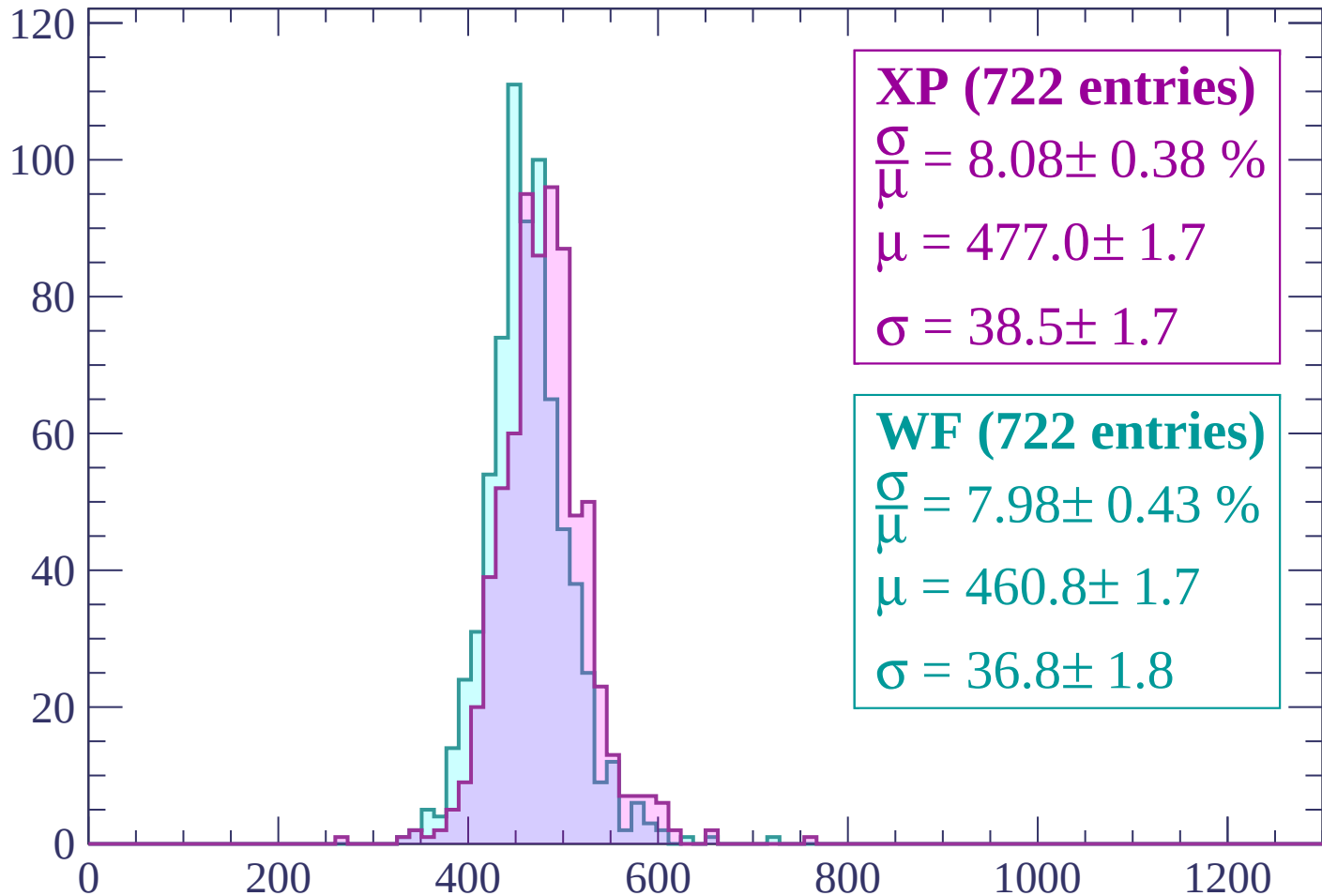
Energy loss | $-1560 < p < -1500$

Count



Energy loss | $-1500 < p < -1440$

Count



XP (722 entries)

$$\frac{\sigma}{\mu} = 8.08 \pm 0.38 \%$$

$$\mu = 477.0 \pm 1.7$$

$$\sigma = 38.5 \pm 1.7$$

WF (722 entries)

$$\frac{\sigma}{\mu} = 7.98 \pm 0.43 \%$$

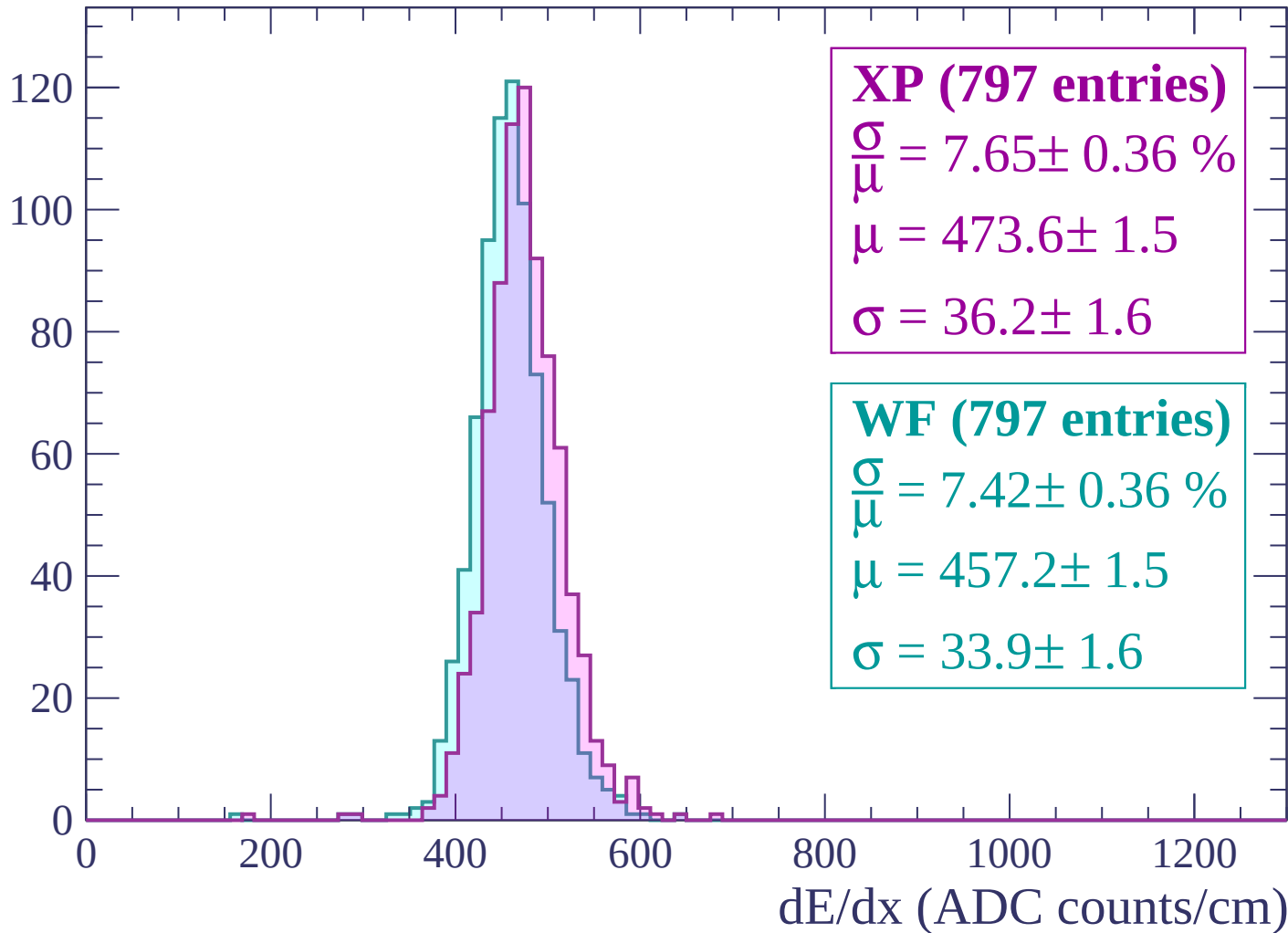
$$\mu = 460.8 \pm 1.7$$

$$\sigma = 36.8 \pm 1.8$$

dE/dx (ADC counts/cm)

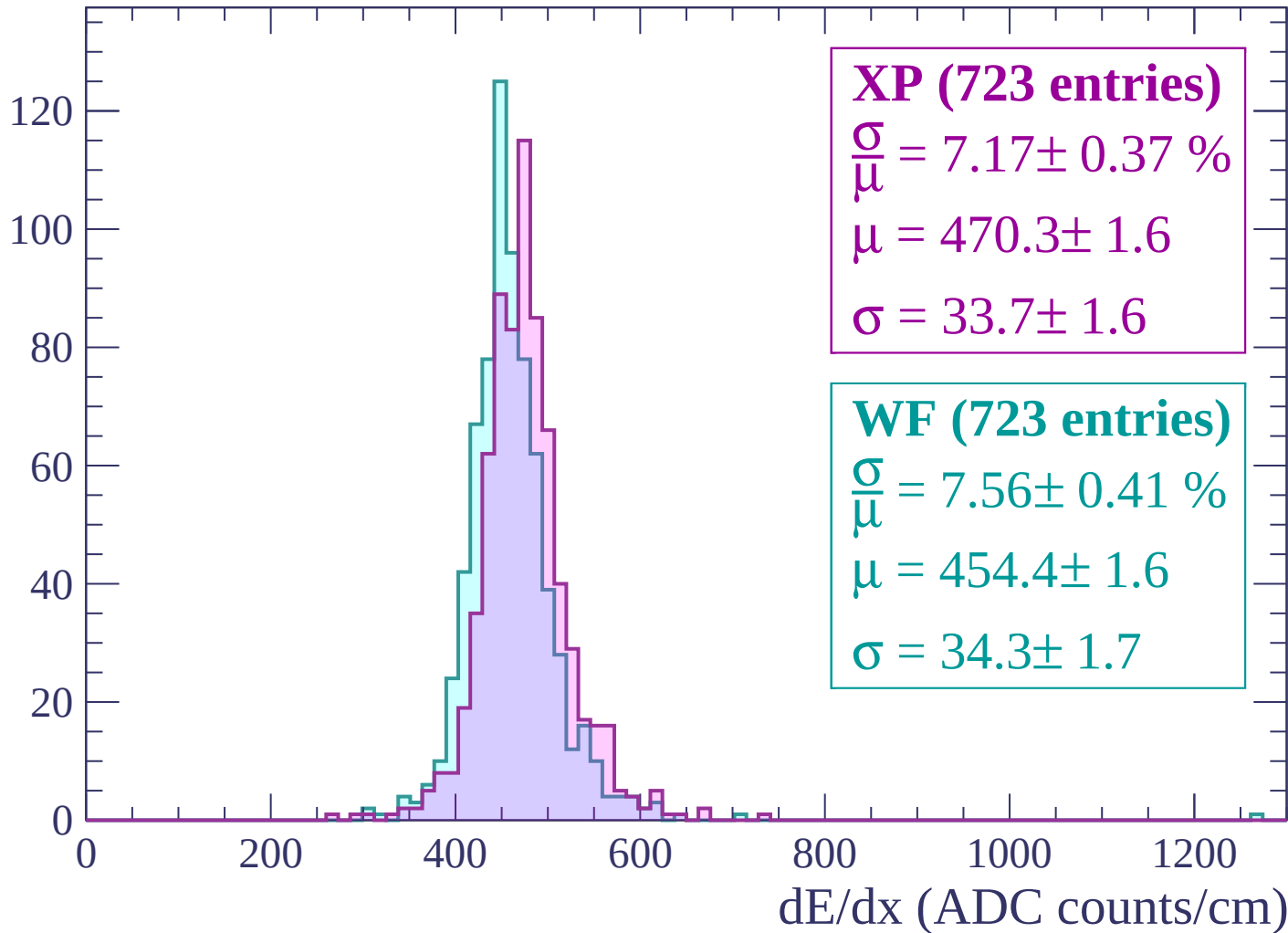
Energy loss | -1440 < p < -1380

Count



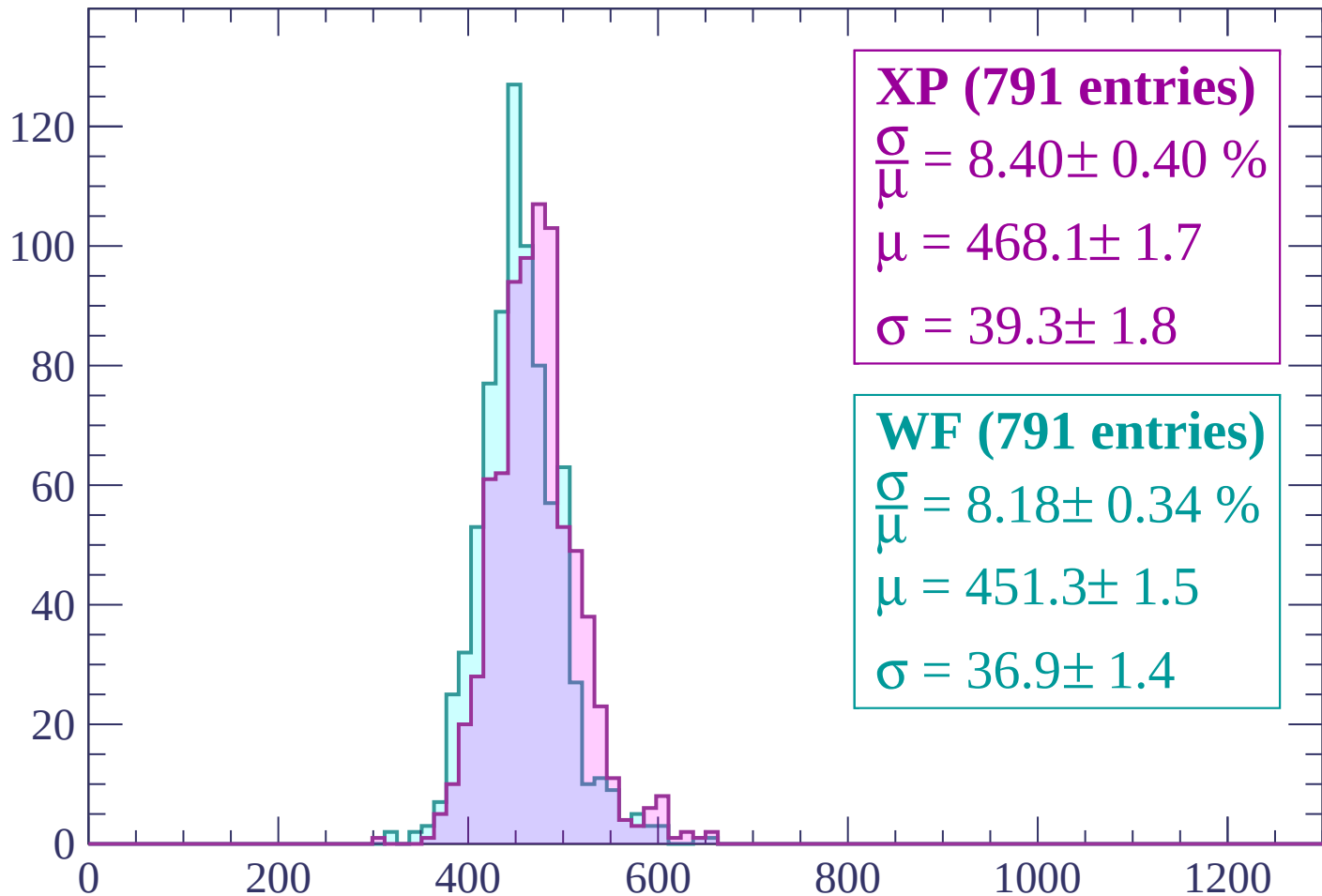
Energy loss | $-1380 < p < -1320$

Count



Energy loss | $-1320 < p < -1260$

Count



XP (791 entries)

$\frac{\sigma}{\mu} = 8.40 \pm 0.40 \%$

$\mu = 468.1 \pm 1.7$

$\sigma = 39.3 \pm 1.8$

WF (791 entries)

$\frac{\sigma}{\mu} = 8.18 \pm 0.34 \%$

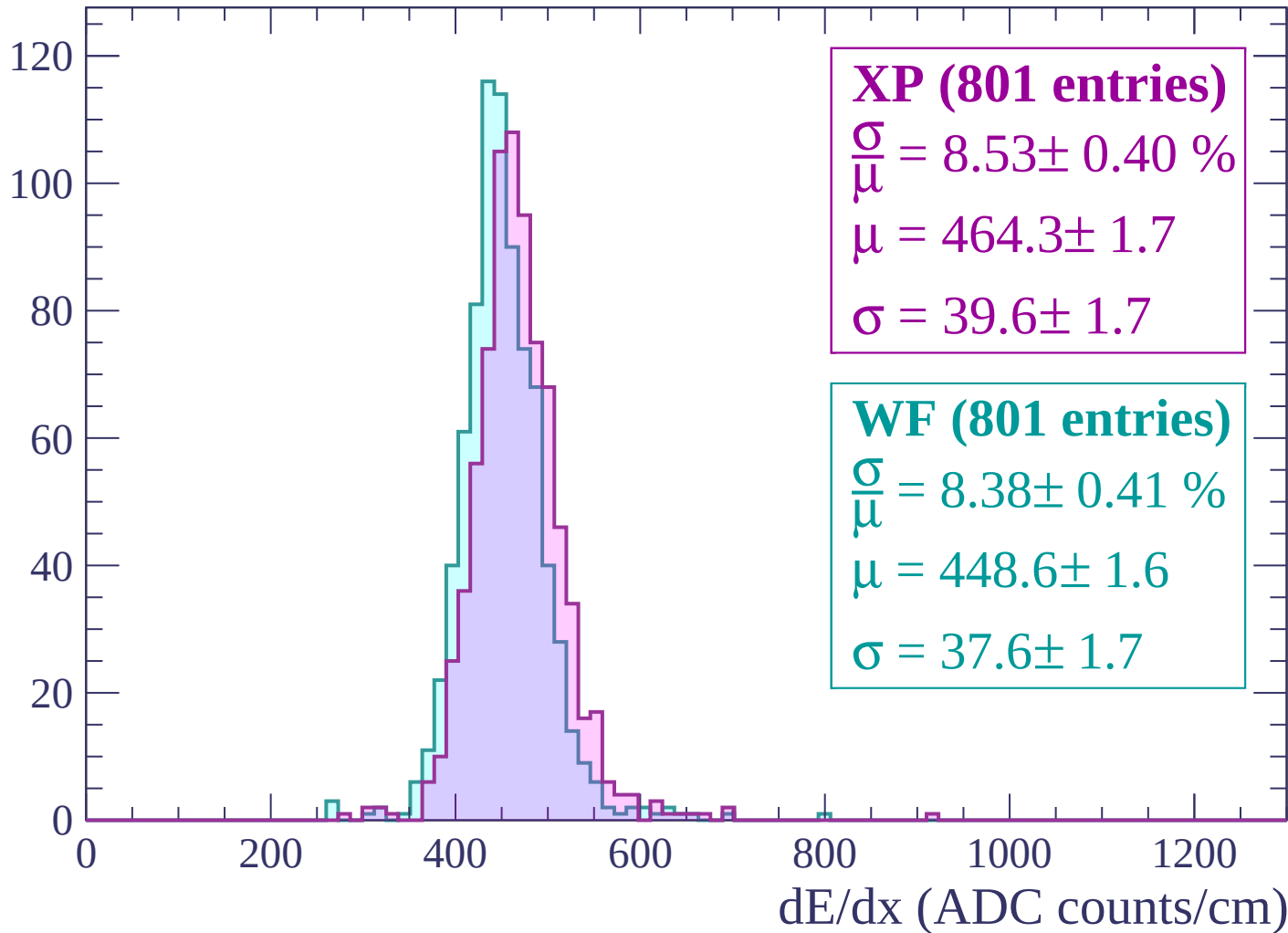
$\mu = 451.3 \pm 1.5$

$\sigma = 36.9 \pm 1.4$

dE/dx (ADC counts/cm)

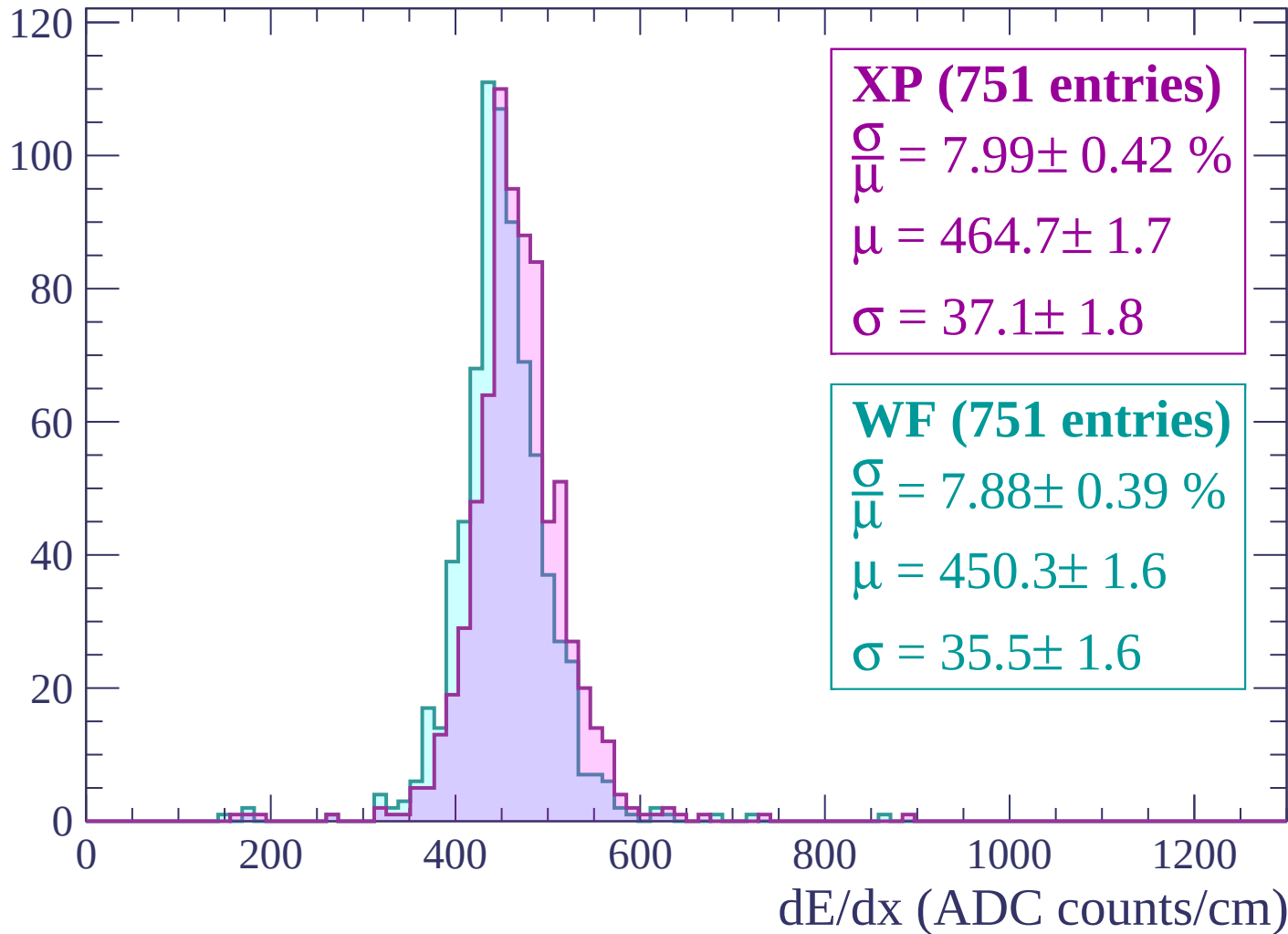
Energy loss | $-1260 < p < -1200$

Count



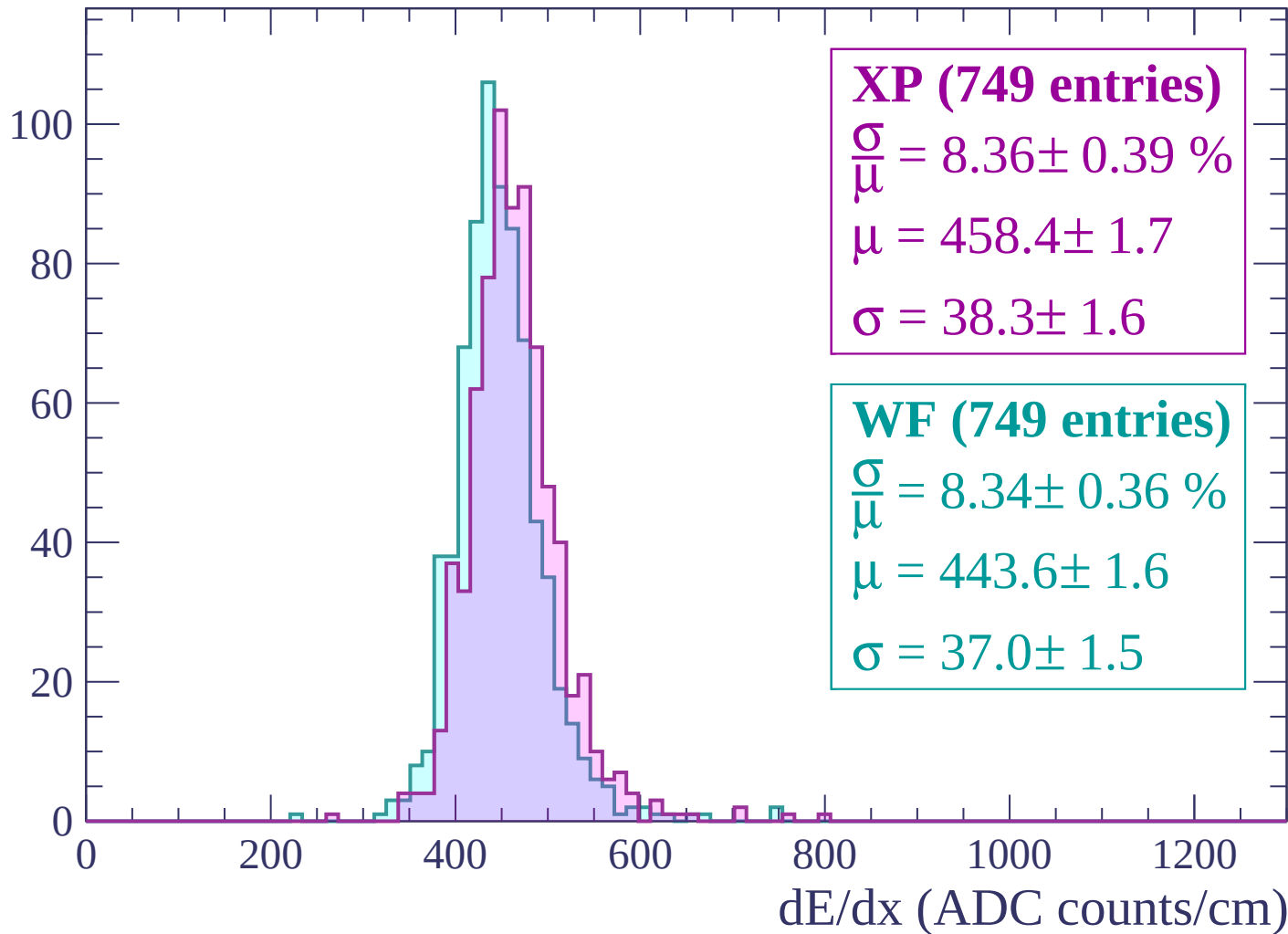
Energy loss | $-1200 < p < -1140$

Count



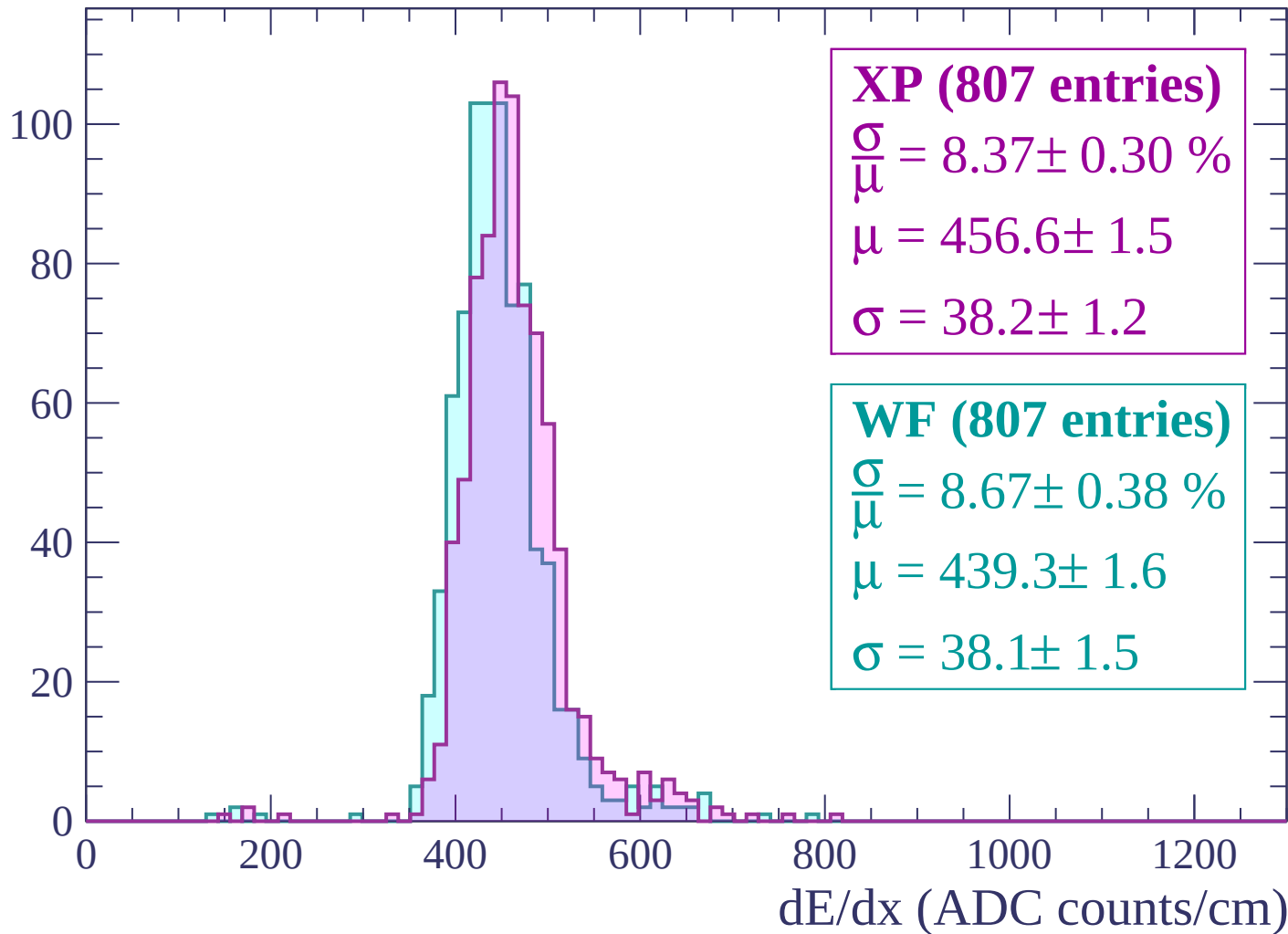
Energy loss | $-1140 < p < -1080$

Count



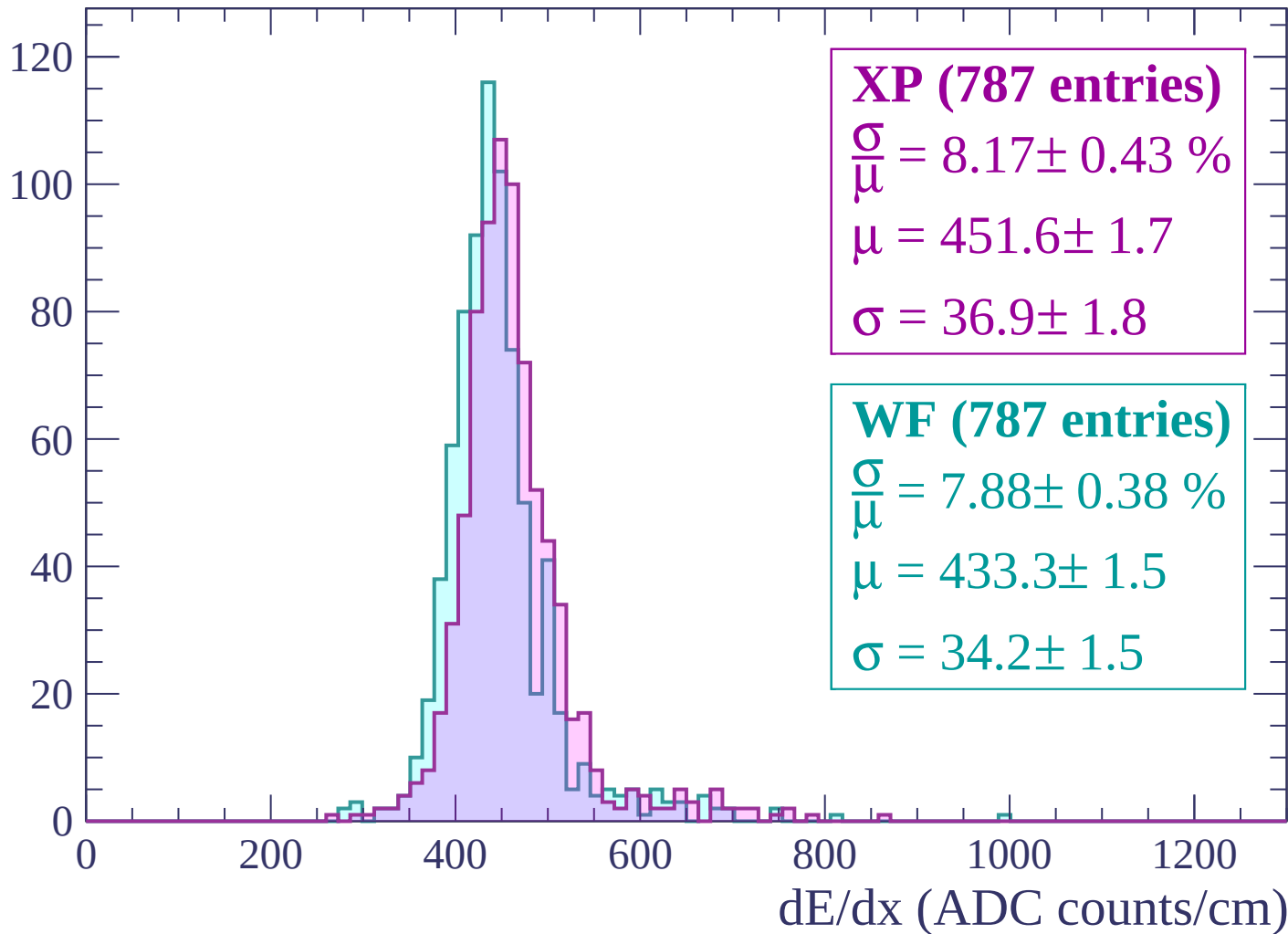
Energy loss | -1080 < p < -1020

Count



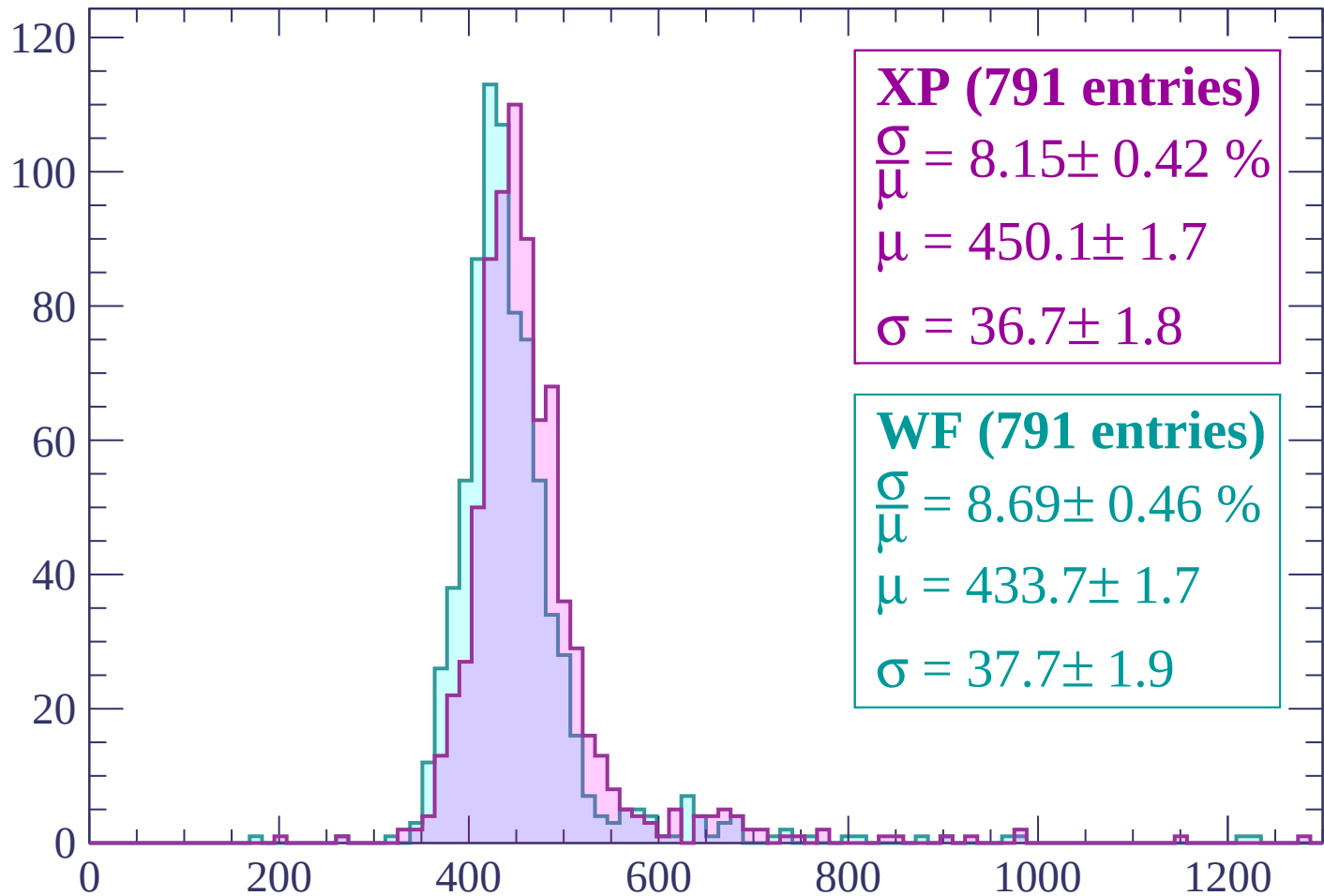
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP (791 entries)

$$\frac{\sigma}{\mu} = 8.15 \pm 0.42 \%$$

$$\mu = 450.1 \pm 1.7$$

$$\sigma = 36.7 \pm 1.8$$

WF (791 entries)

$$\frac{\sigma}{\mu} = 8.69 \pm 0.46 \%$$

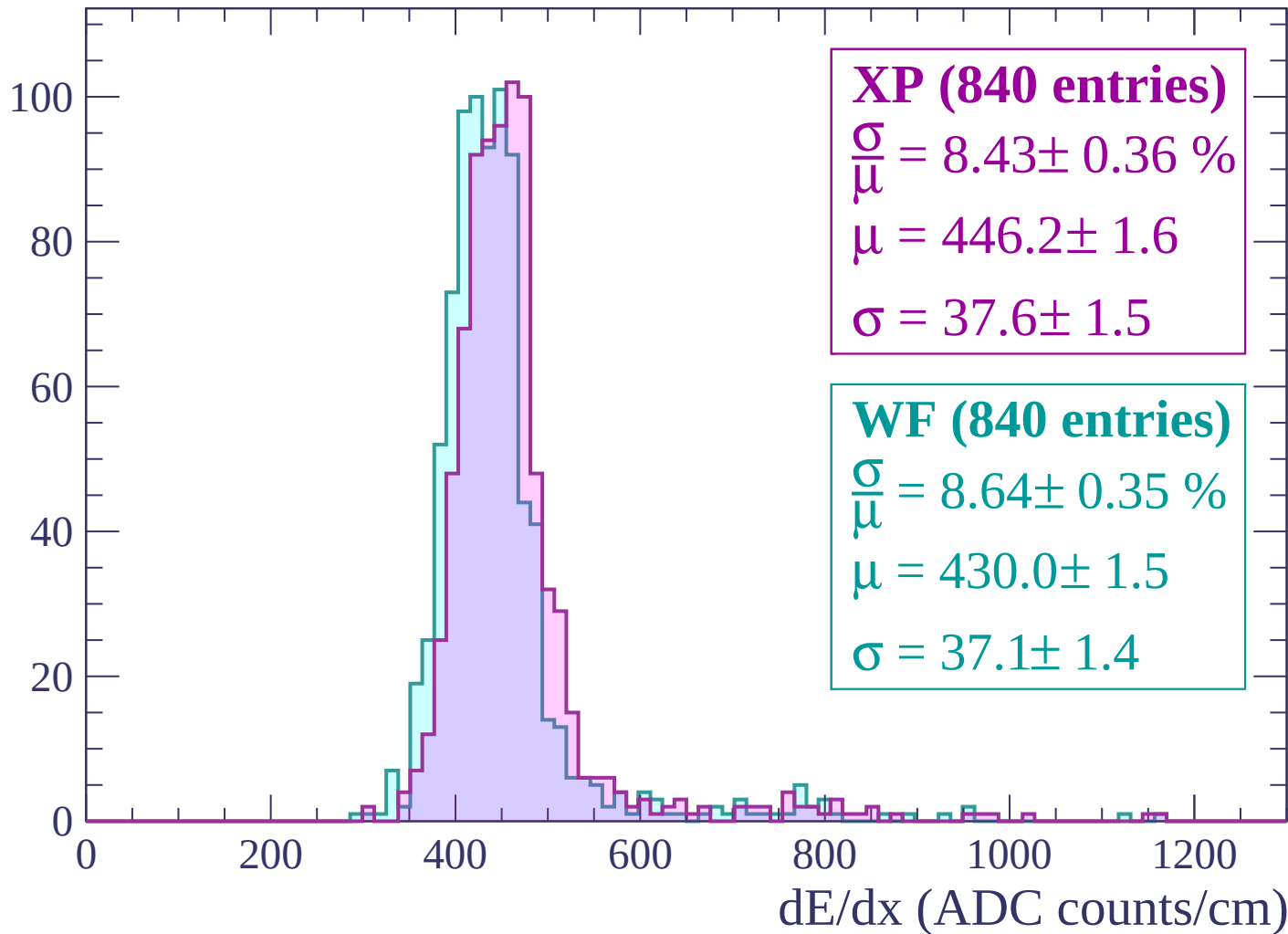
$$\mu = 433.7 \pm 1.7$$

$$\sigma = 37.7 \pm 1.9$$

dE/dx (ADC counts/cm)

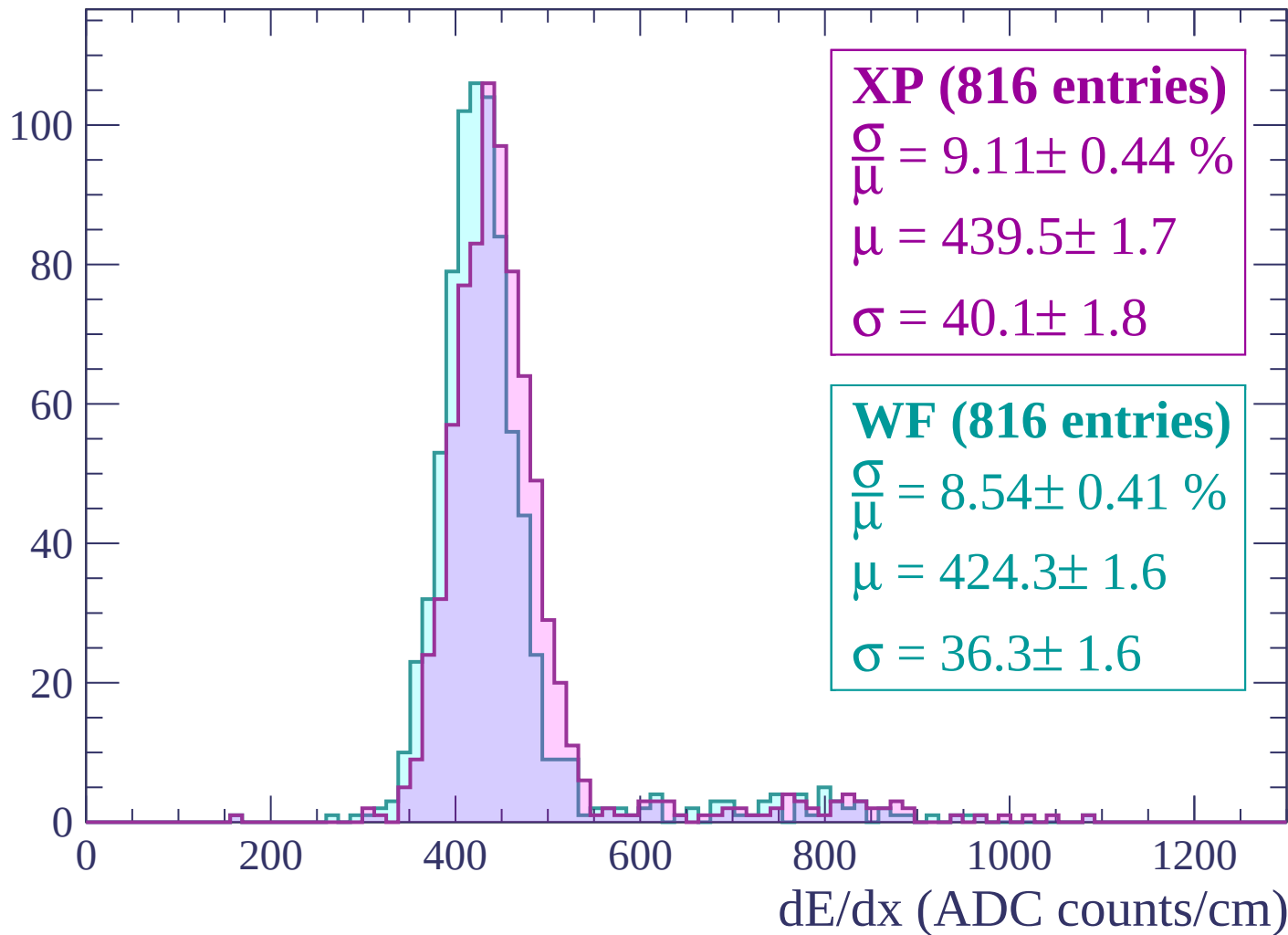
Energy loss | $-900 < p < -840$

Count



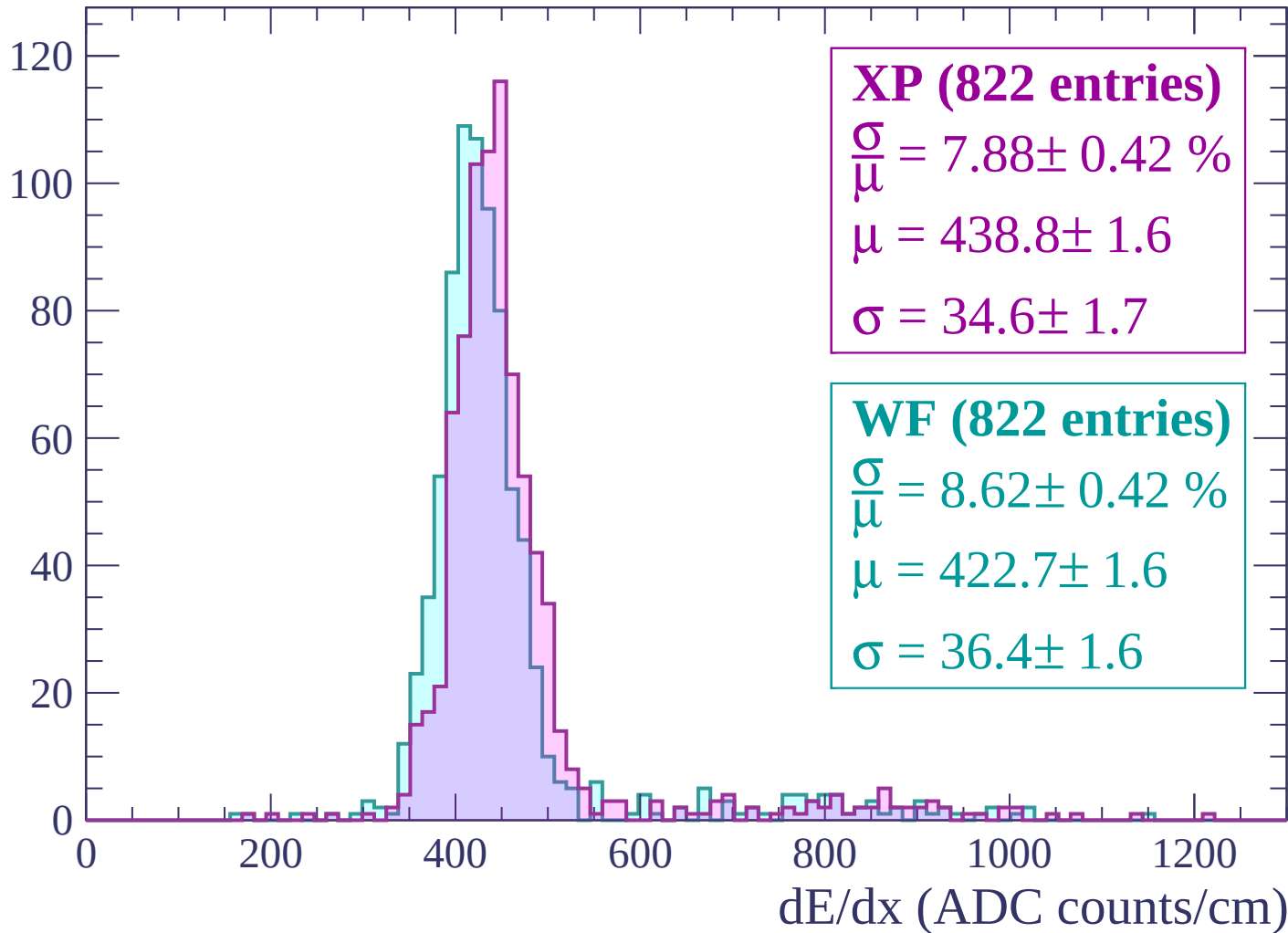
Energy loss | $-840 < p < -780$

Count



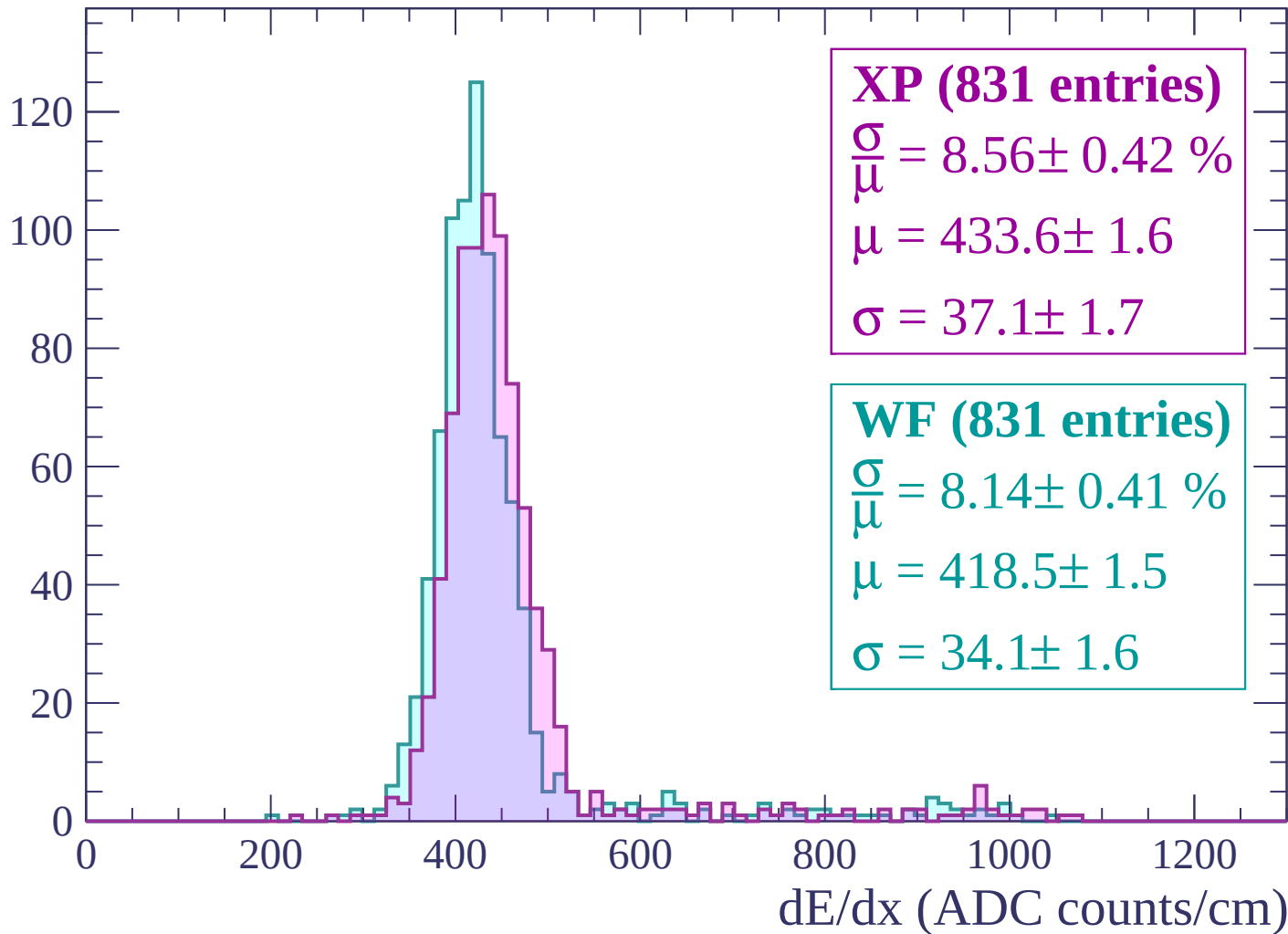
Energy loss | $-780 < p < -720$

Count



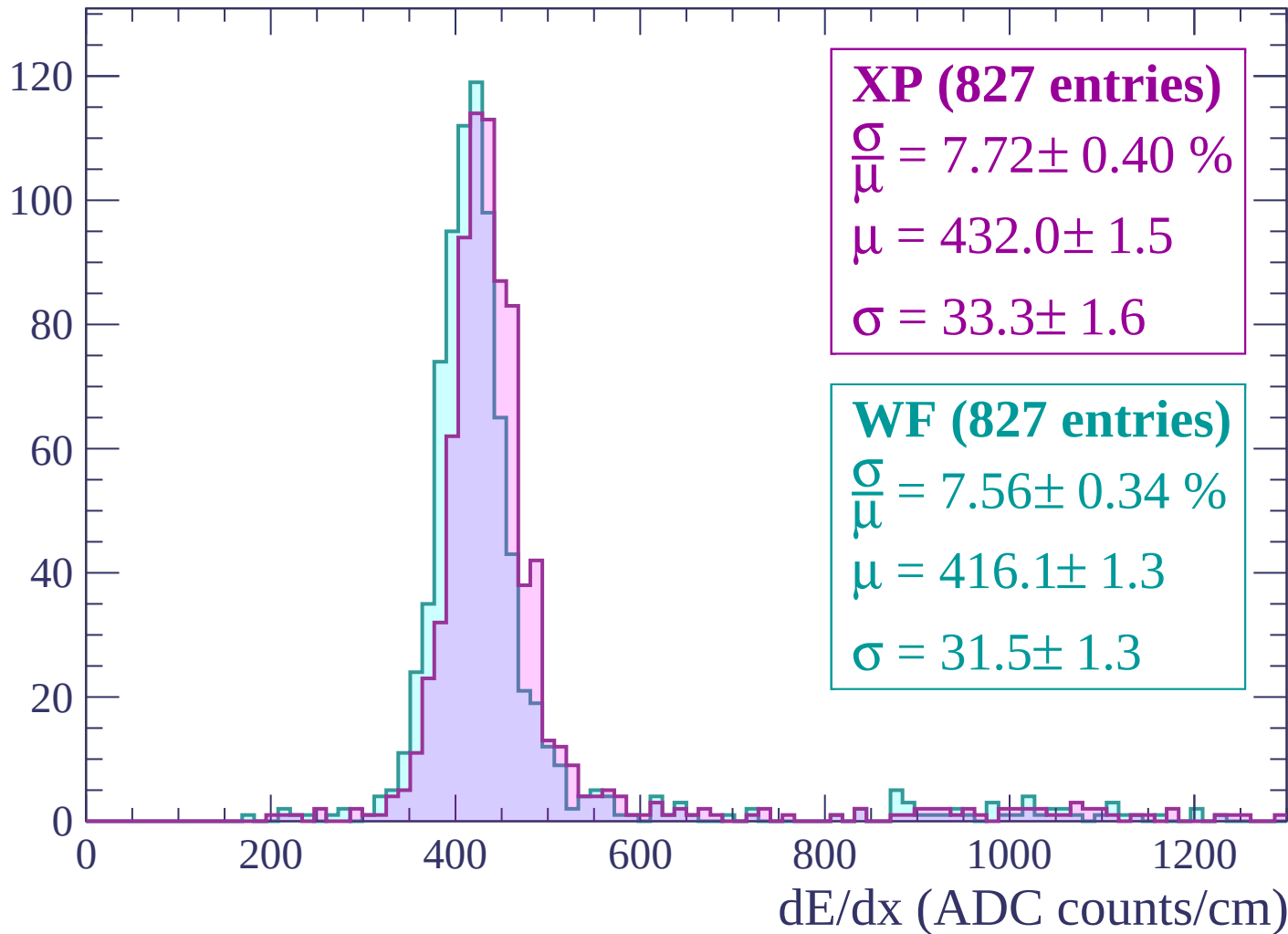
Energy loss | $-720 < p < -660$

Count



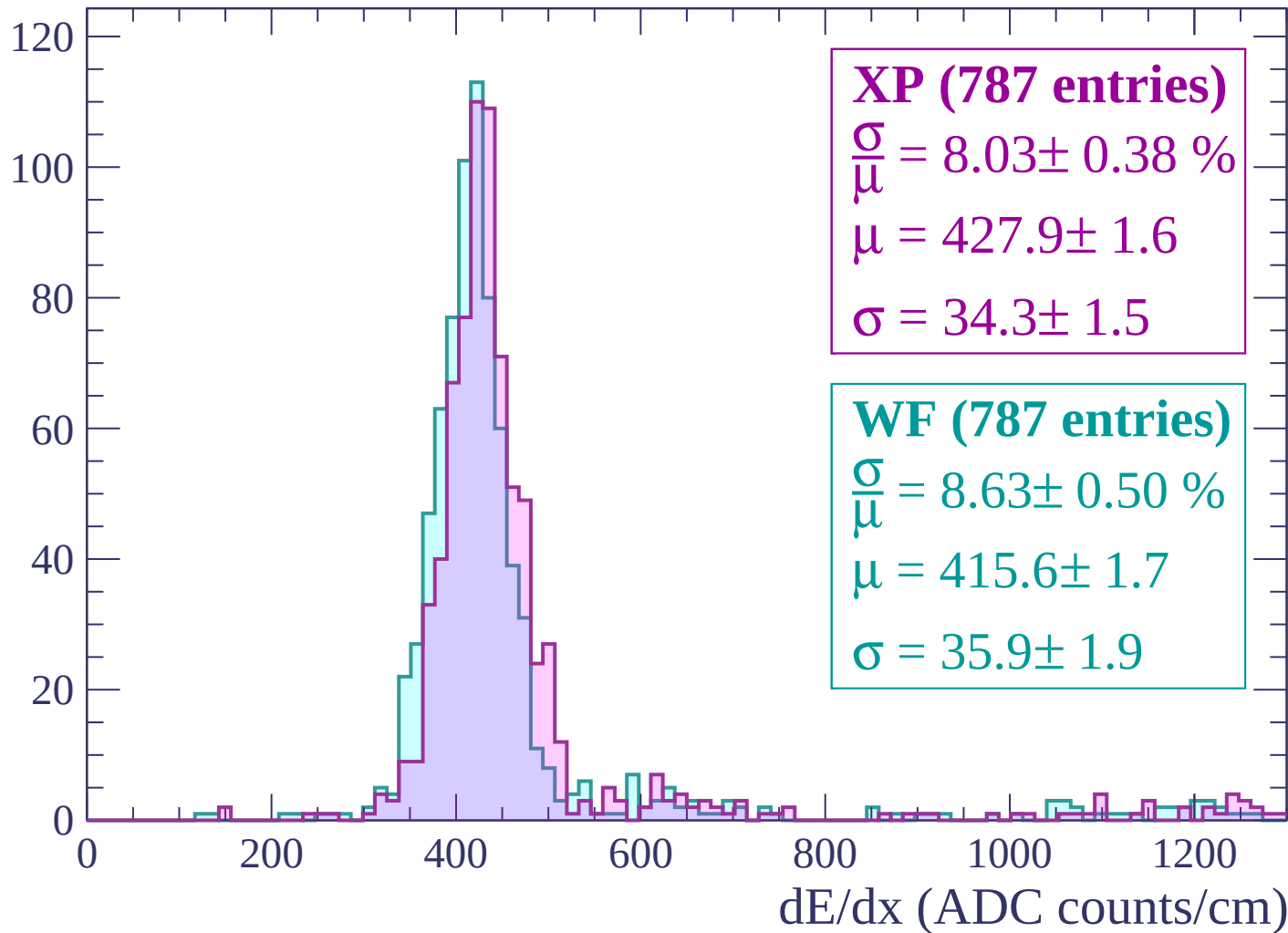
Energy loss | $-660 < p < -600$

Count



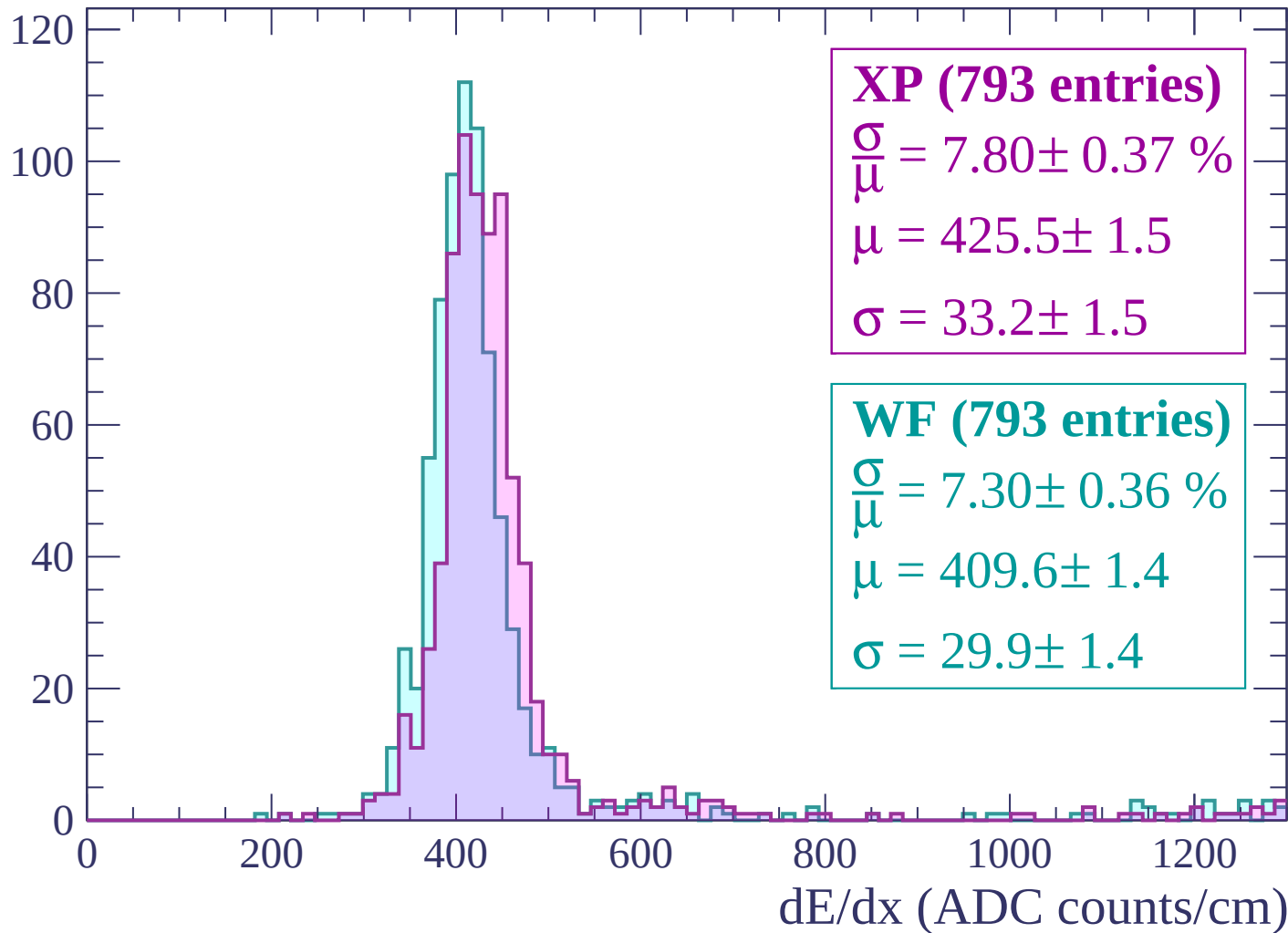
Energy loss | $-600 < p < -540$

Count



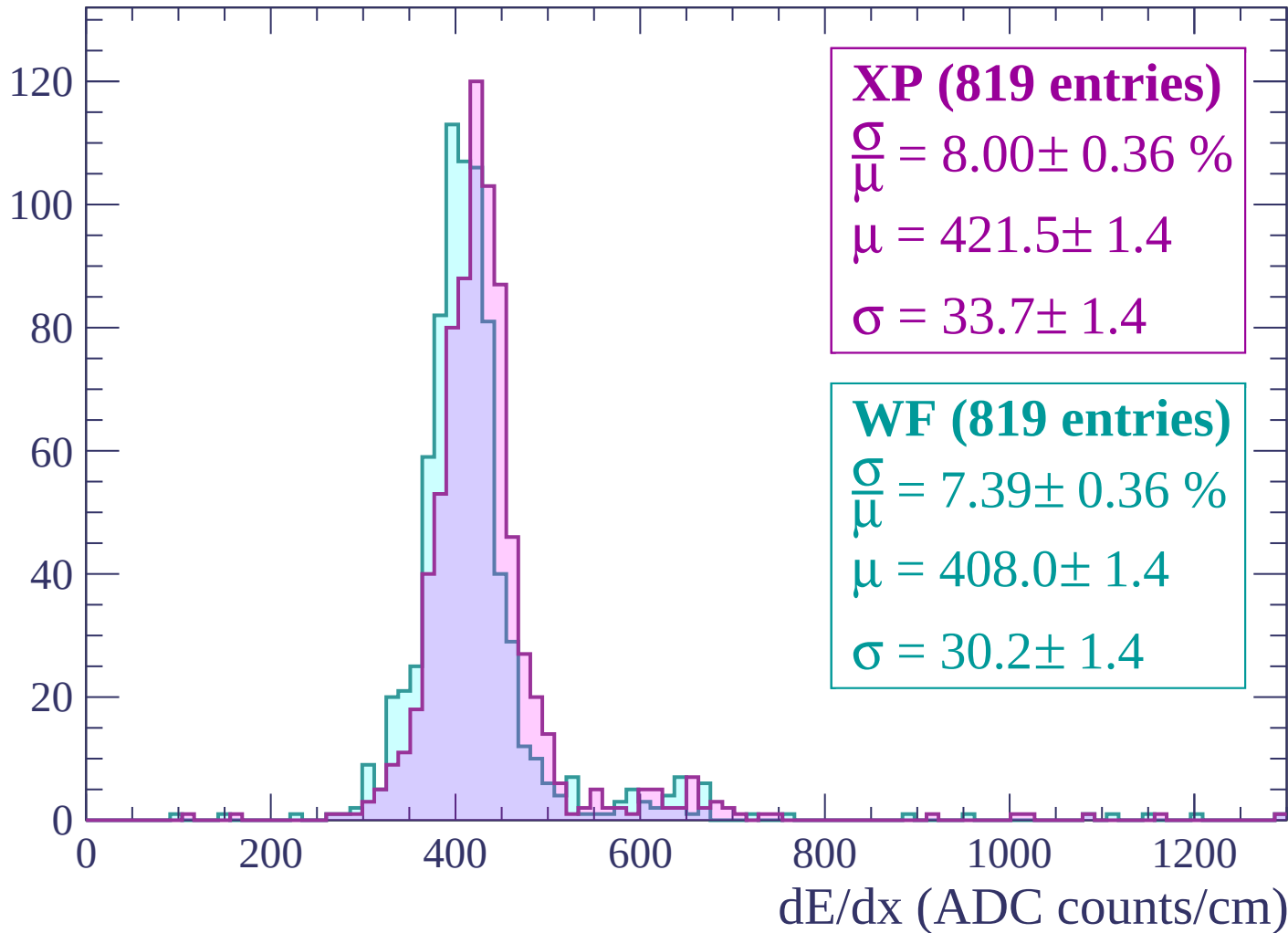
Energy loss | $-540 < p < -480$

Count



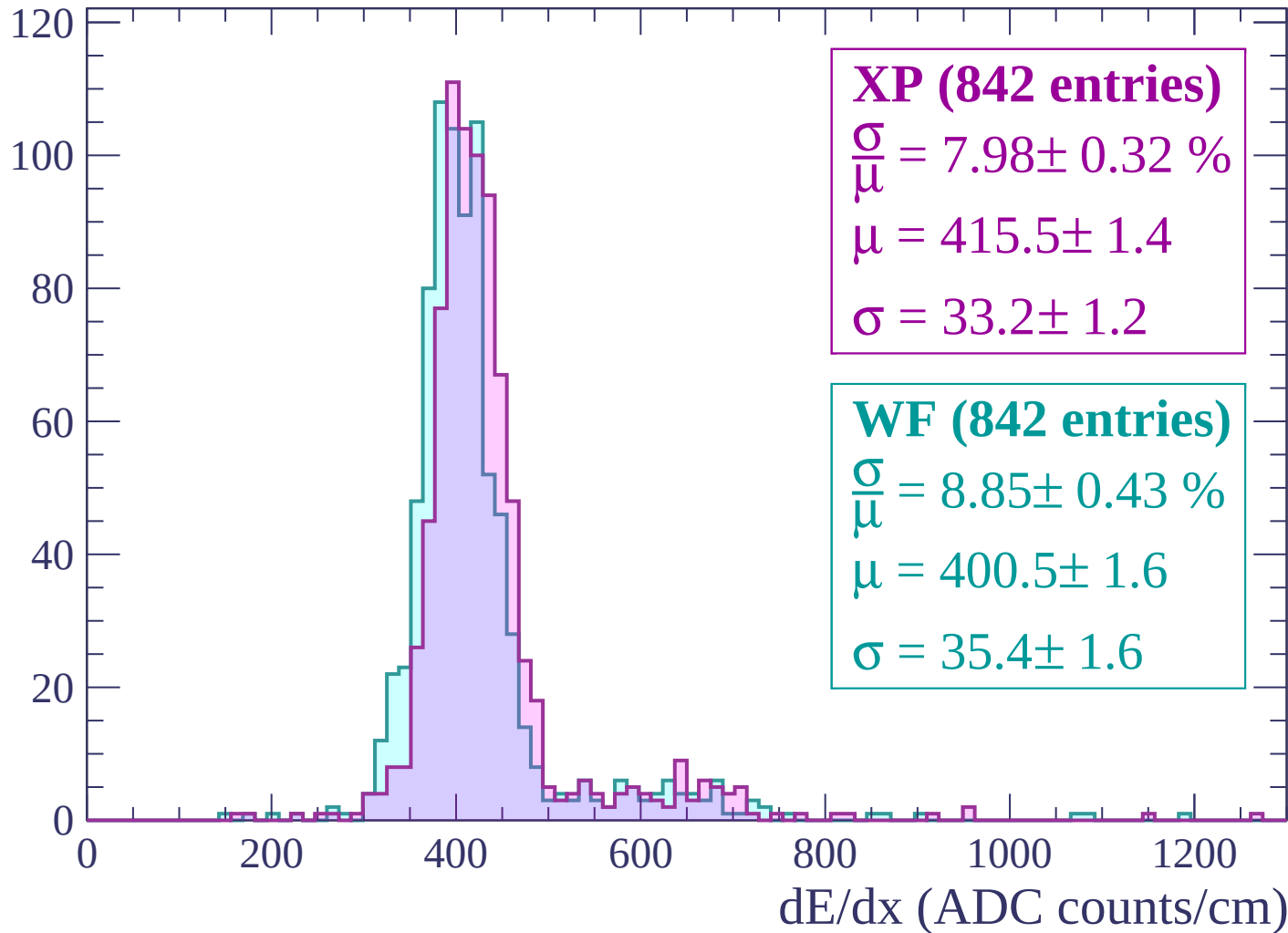
Energy loss | $-480 < p < -420$

Count



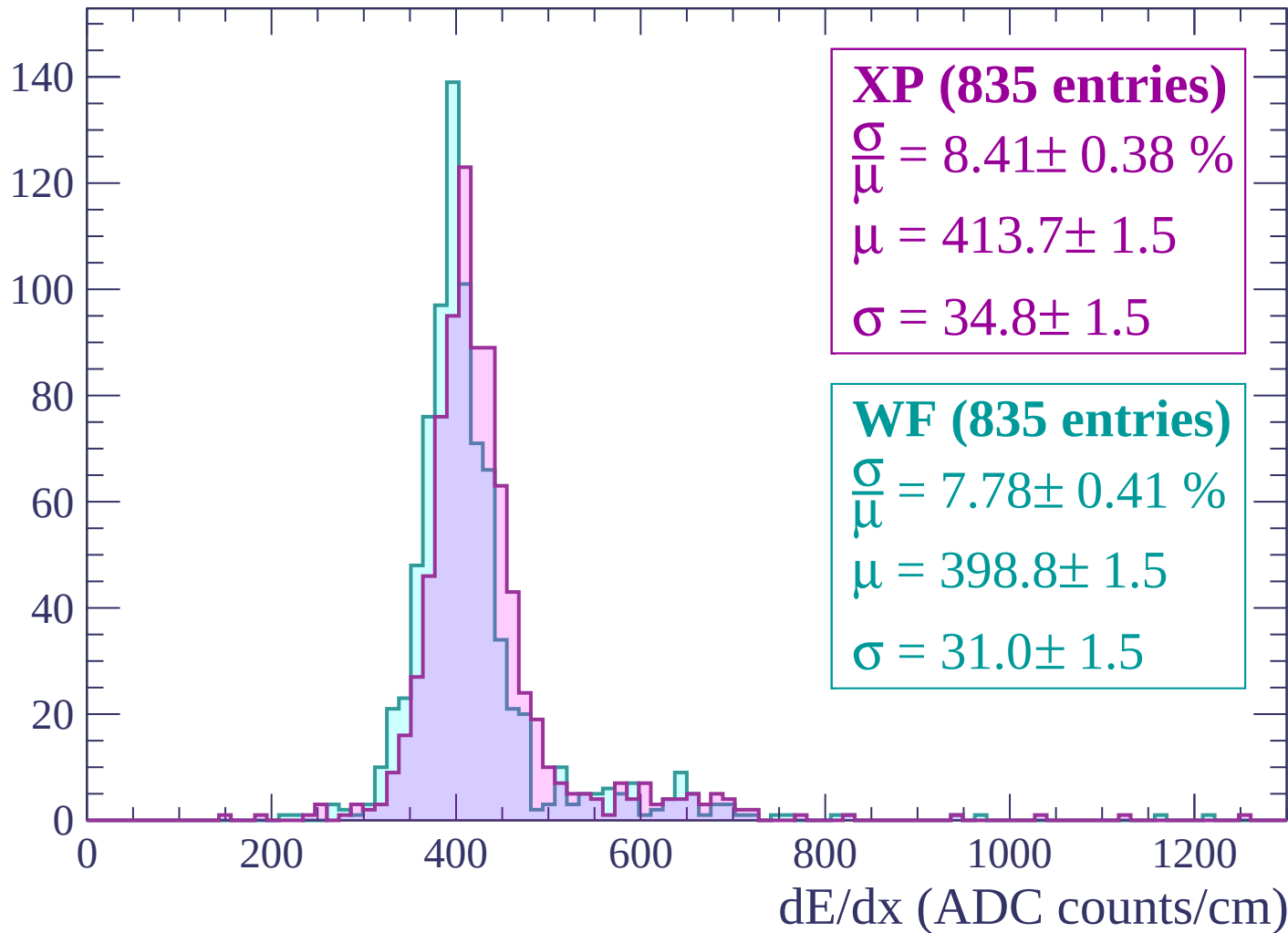
Energy loss | $-420 < p < -360$

Count



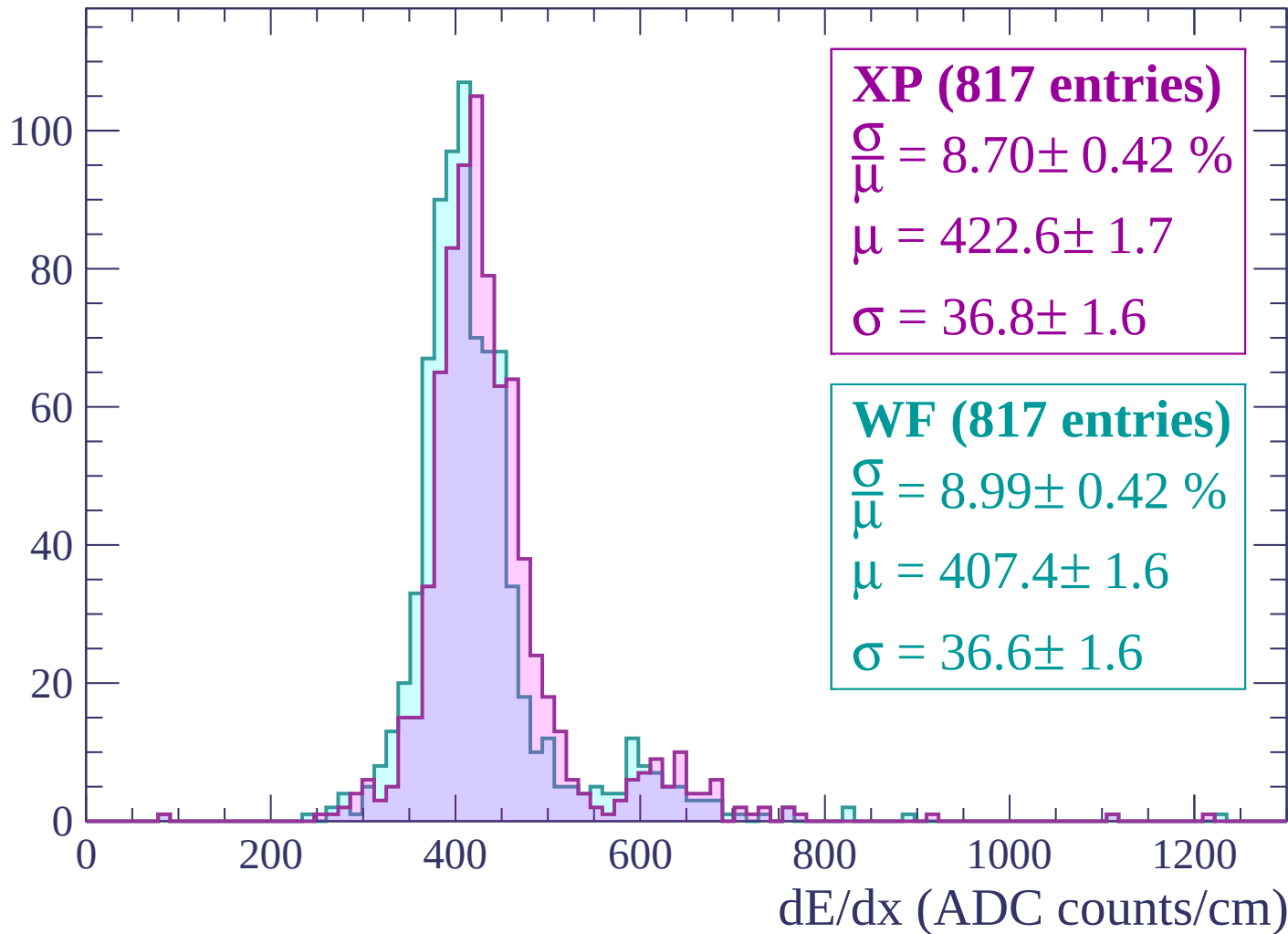
Energy loss | $-360 < p < -300$

Count



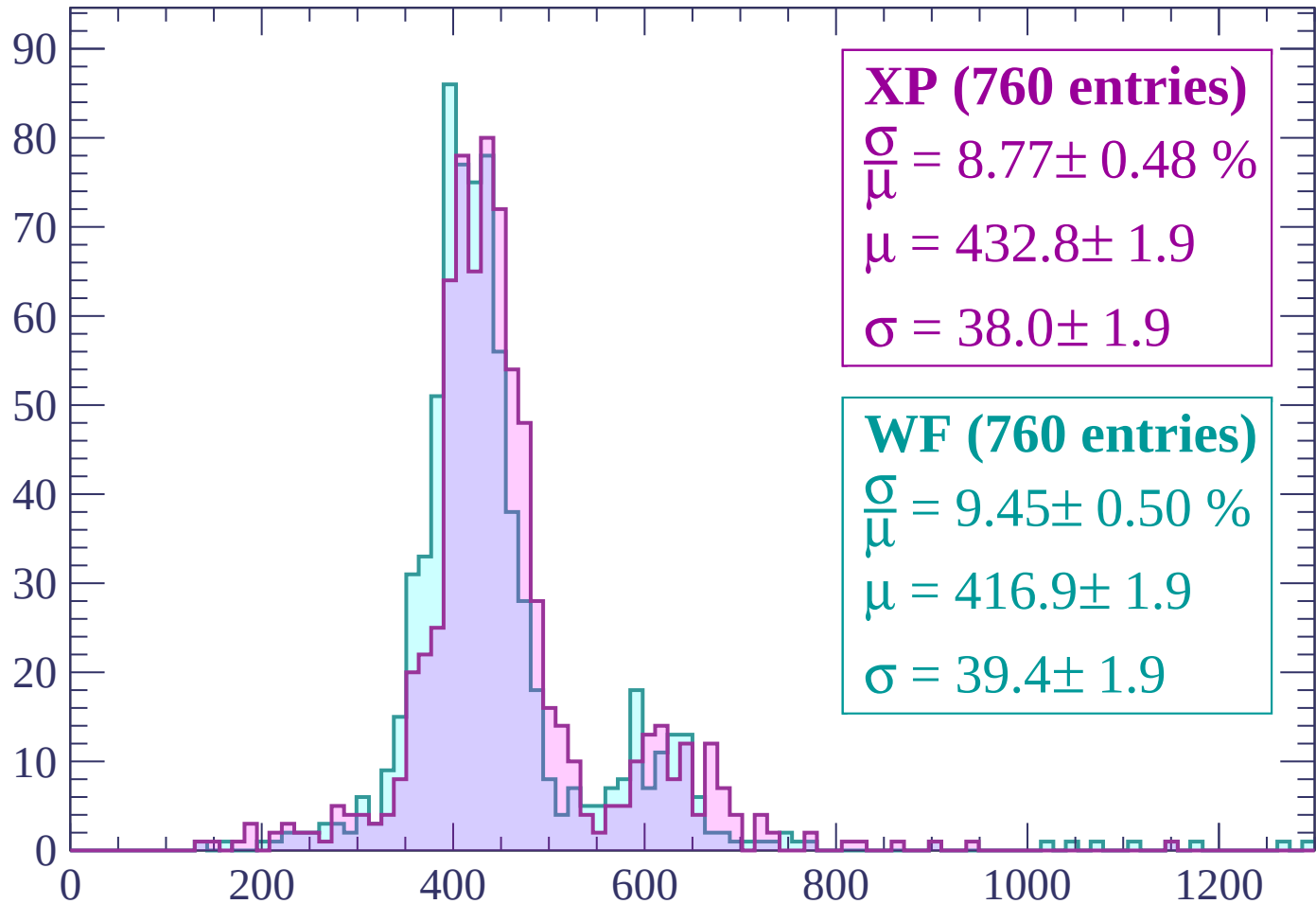
Energy loss | $-300 < p < -240$

Count



Energy loss | $-240 < p < -180$

Count



XP (760 entries)

$\frac{\sigma}{\mu} = 8.77 \pm 0.48 \%$

$\mu = 432.8 \pm 1.9$

$\sigma = 38.0 \pm 1.9$

WF (760 entries)

$\frac{\sigma}{\mu} = 9.45 \pm 0.50 \%$

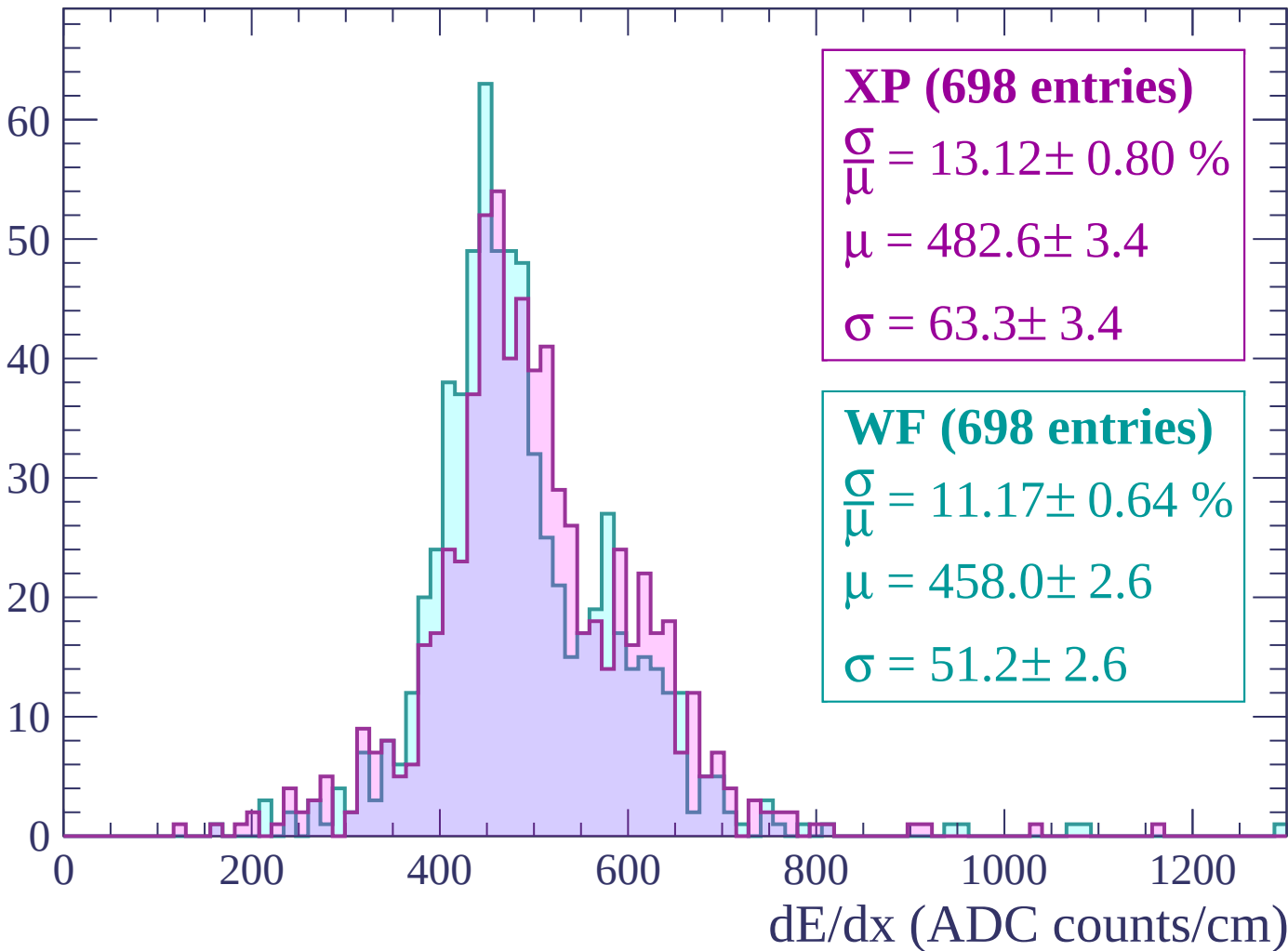
$\mu = 416.9 \pm 1.9$

$\sigma = 39.4 \pm 1.9$

dE/dx (ADC counts/cm)

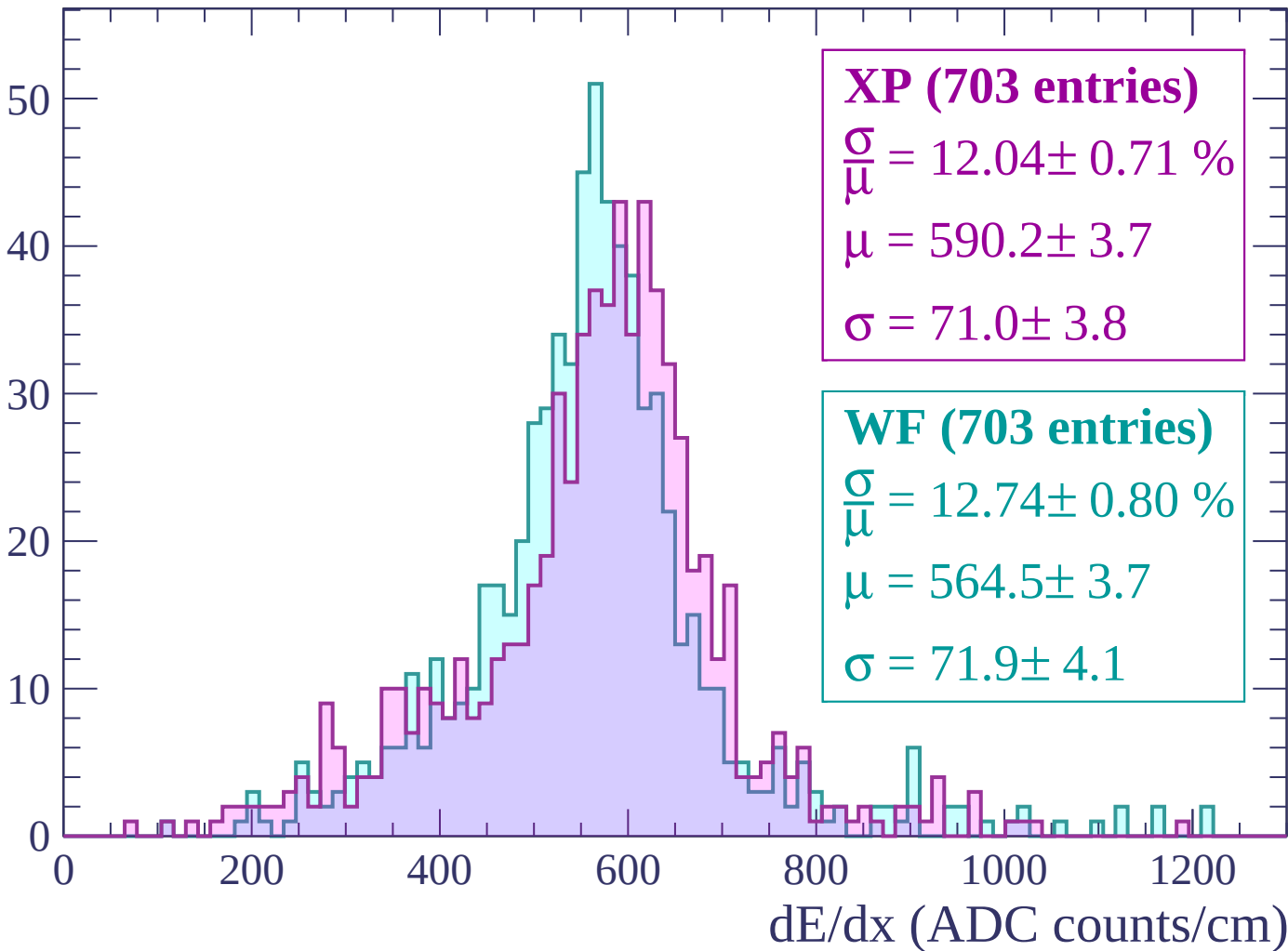
Energy loss | $-180 < p < -120$

Count



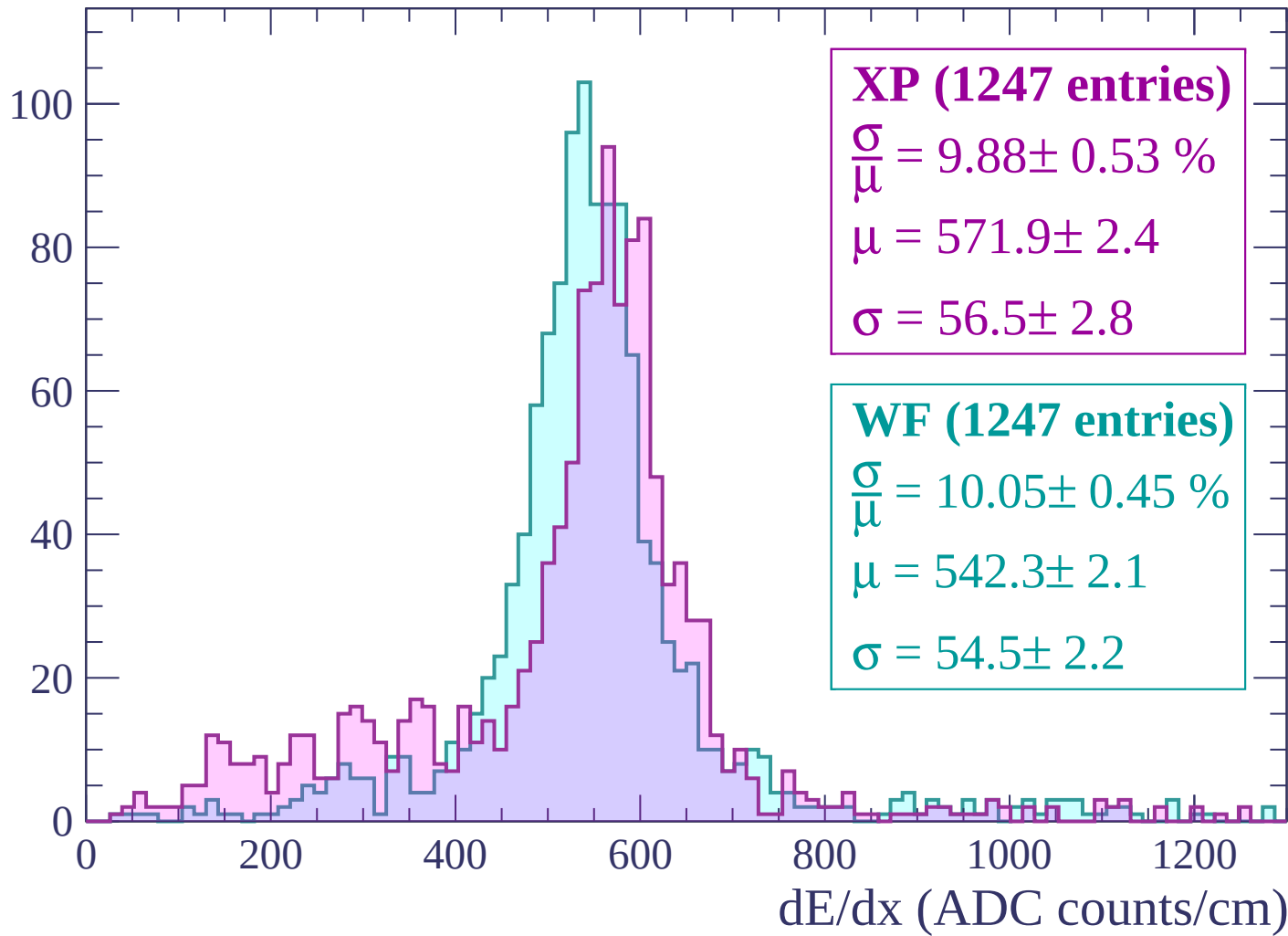
Energy loss | $-120 < p < -60$

Count



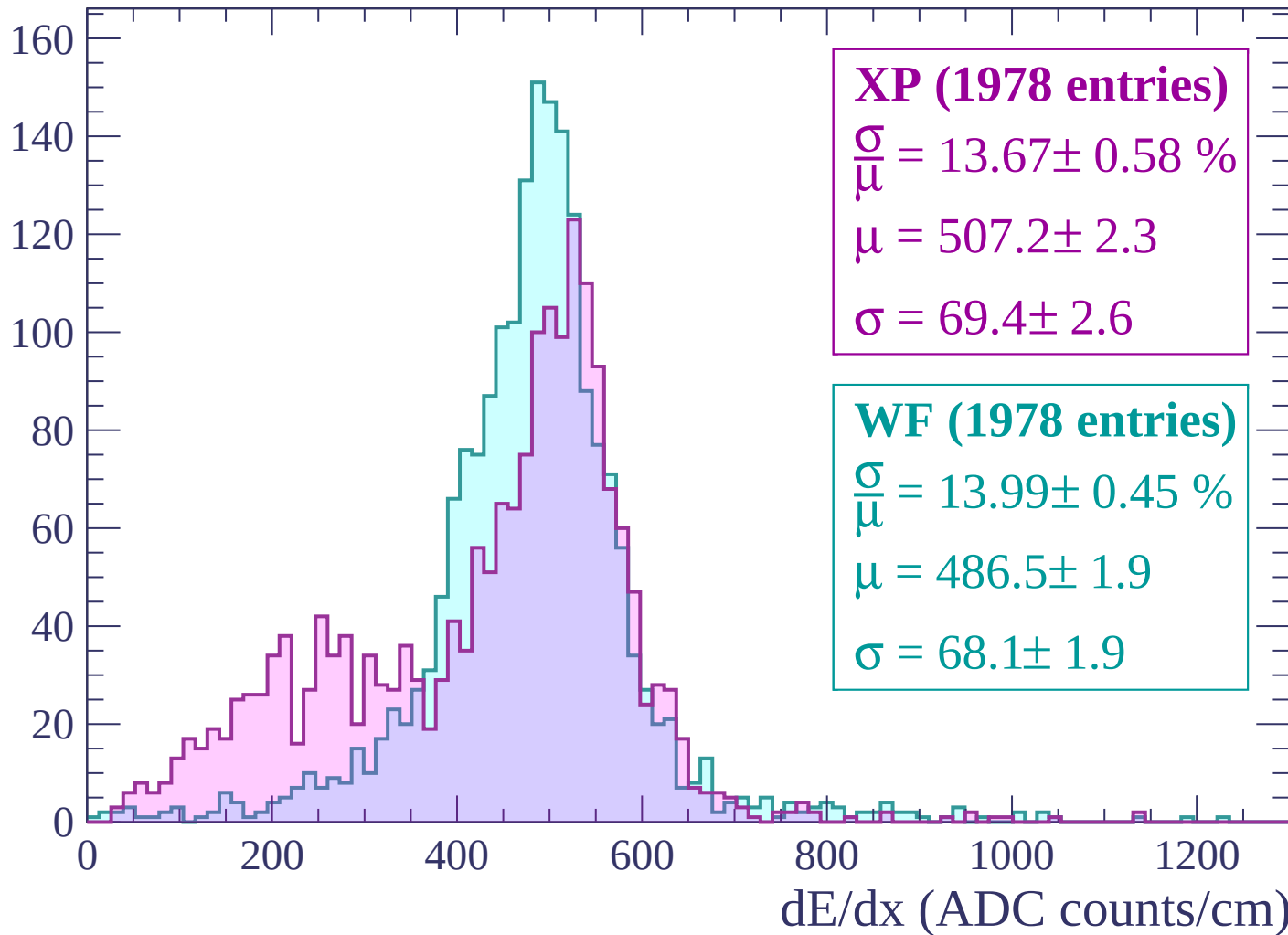
Energy loss | $-60 < p < 0$

Count



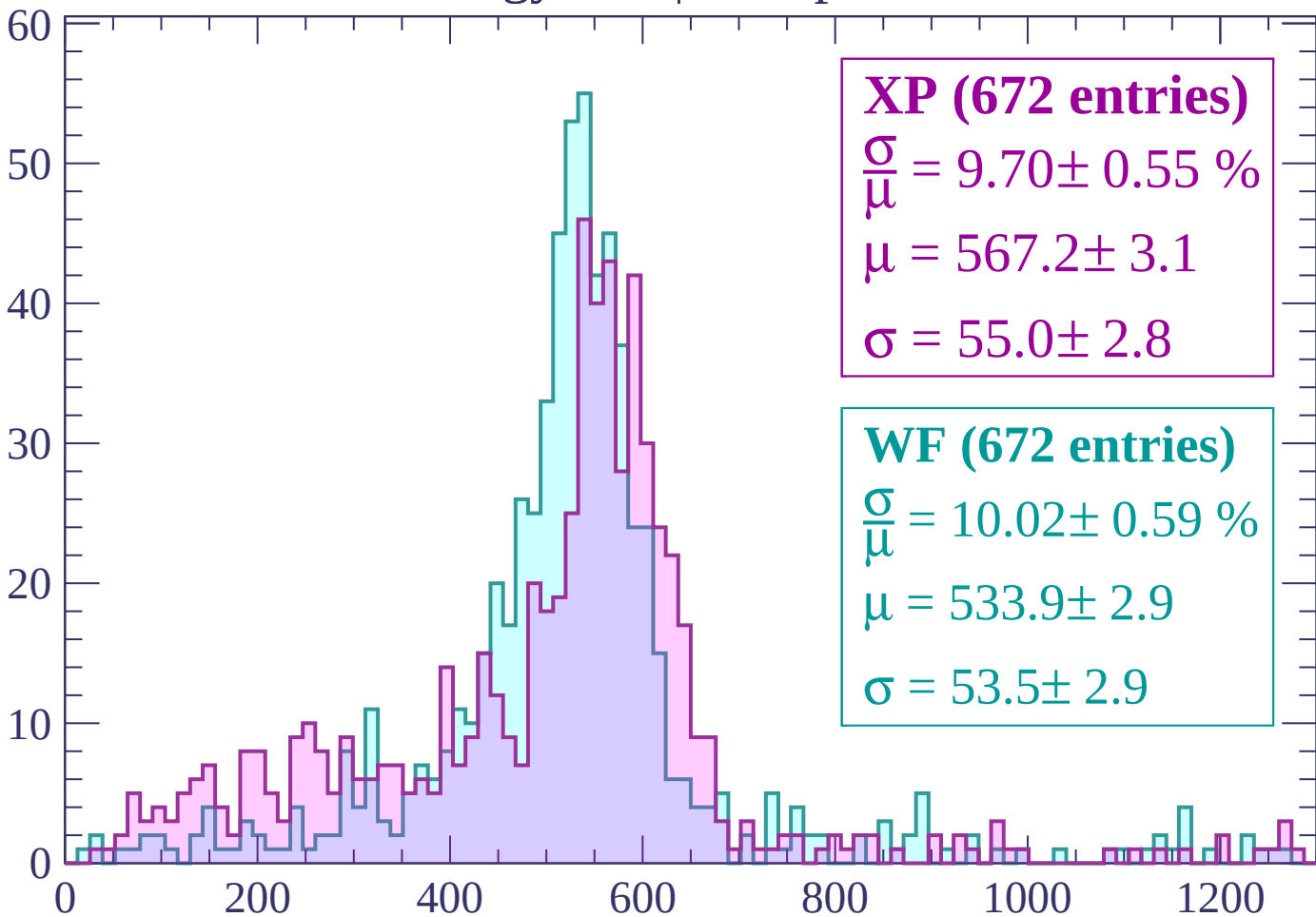
Energy loss | $0 < p < 60$

Count



Energy loss | $60 < p < 120$

Count



XP (672 entries)

$\frac{\sigma}{\mu} = 9.70 \pm 0.55 \%$

$\mu = 567.2 \pm 3.1$

$\sigma = 55.0 \pm 2.8$

WF (672 entries)

$\frac{\sigma}{\mu} = 10.02 \pm 0.59 \%$

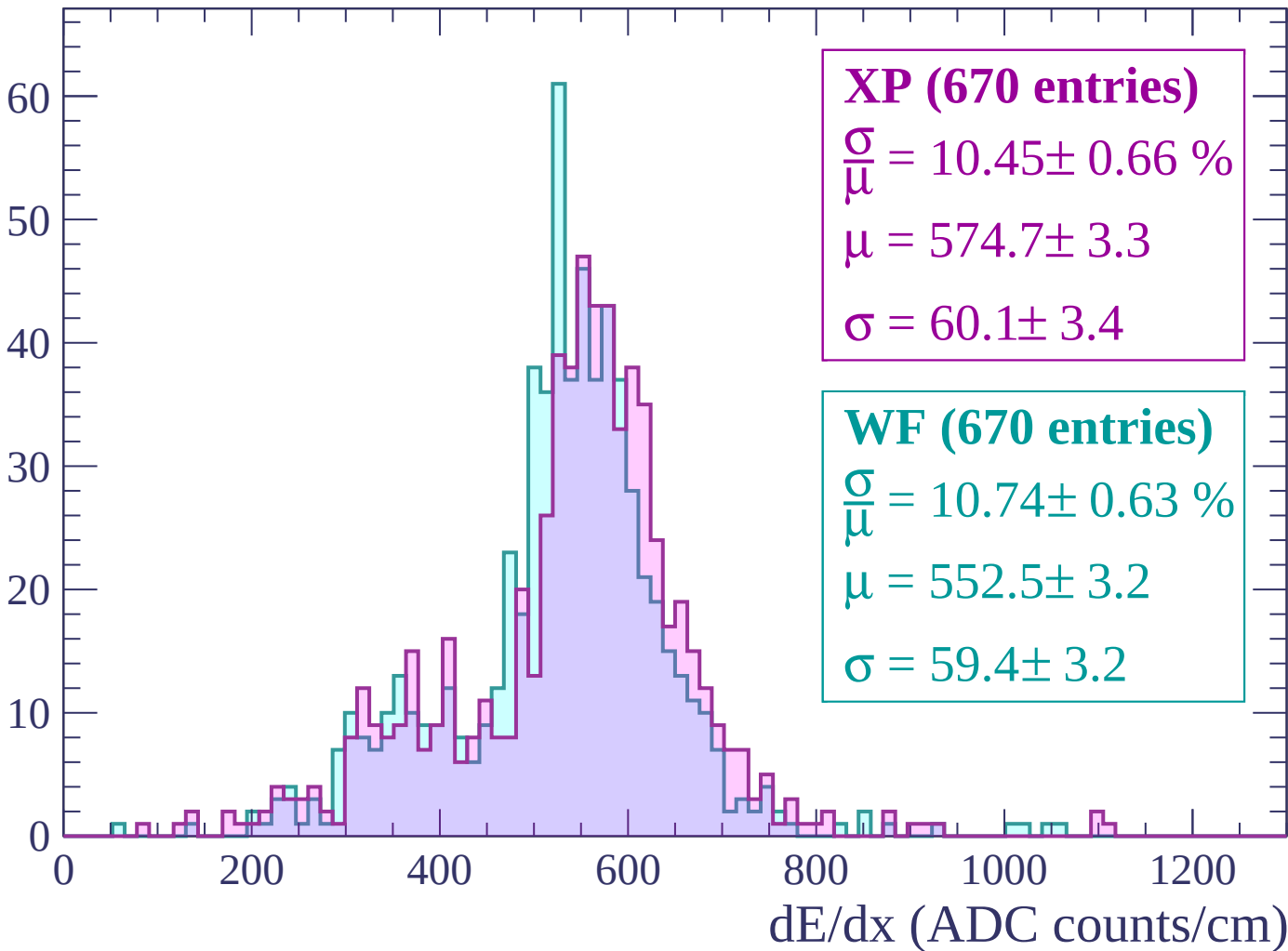
$\mu = 533.9 \pm 2.9$

$\sigma = 53.5 \pm 2.9$

dE/dx (ADC counts/cm)

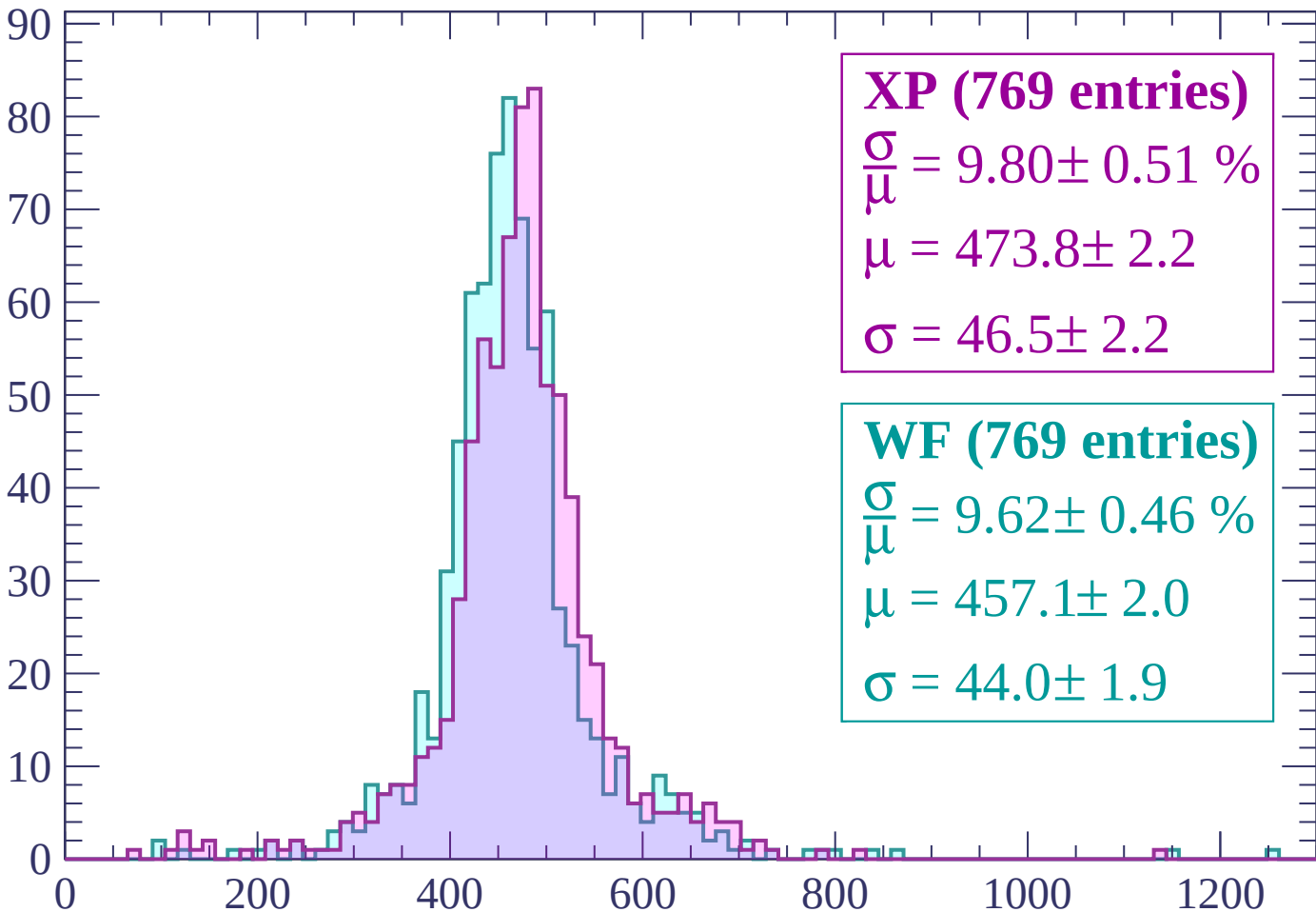
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (769 entries)

$$\frac{\sigma}{\mu} = 9.80 \pm 0.51 \%$$

$$\mu = 473.8 \pm 2.2$$

$$\sigma = 46.5 \pm 2.2$$

WF (769 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.46 \%$$

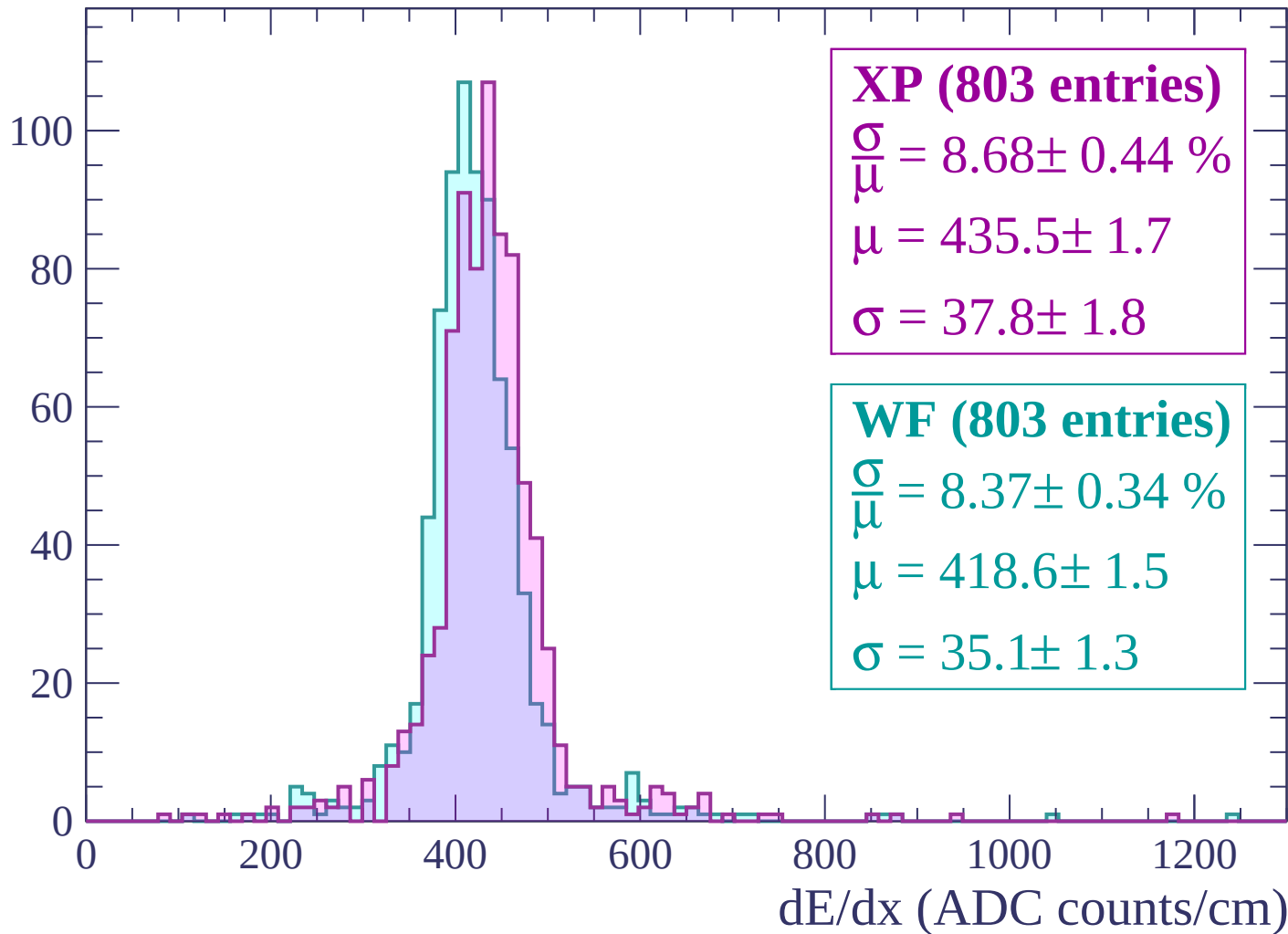
$$\mu = 457.1 \pm 2.0$$

$$\sigma = 44.0 \pm 1.9$$

dE/dx (ADC counts/cm)

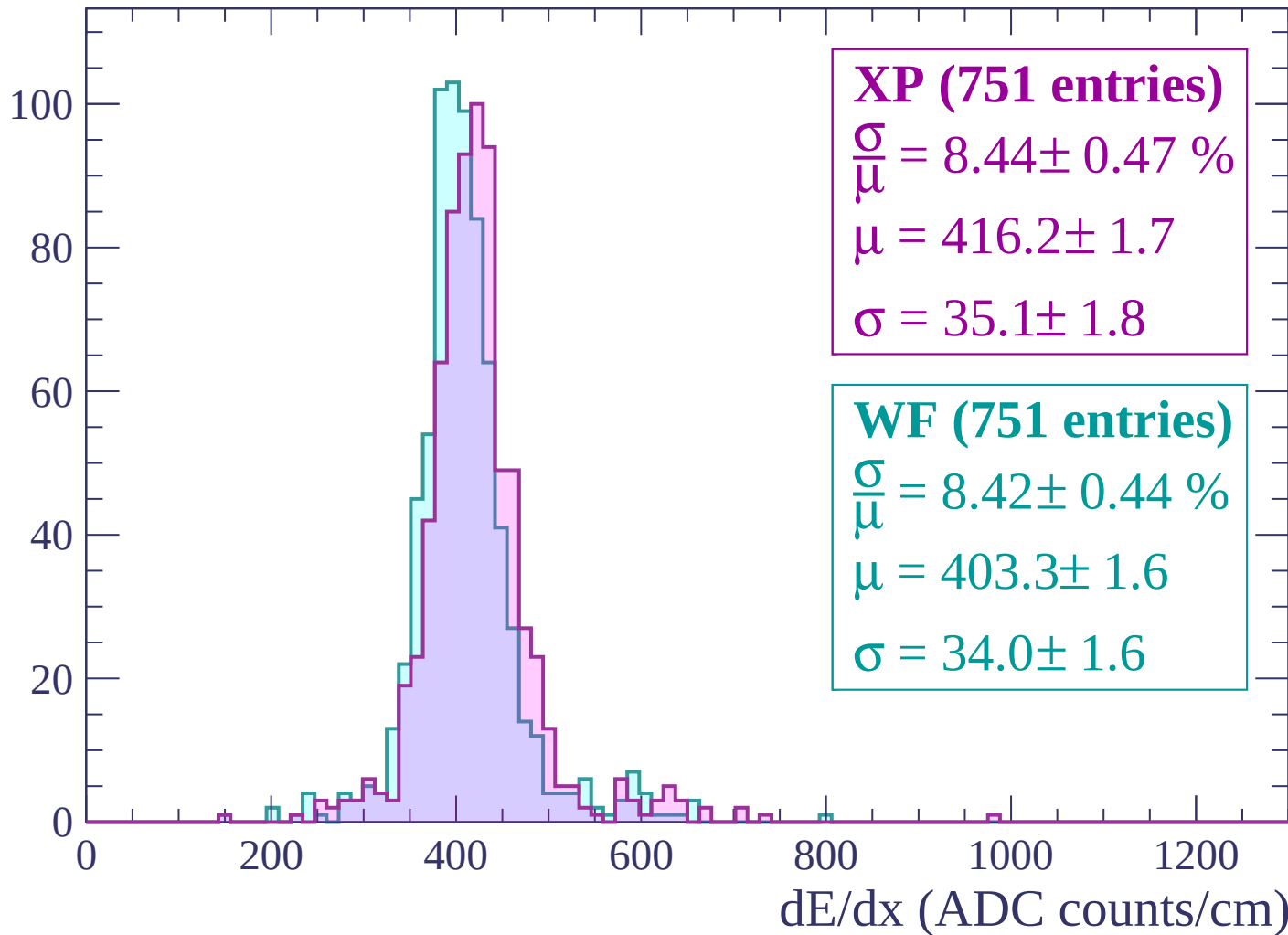
Energy loss | $240 < p < 300$

Count



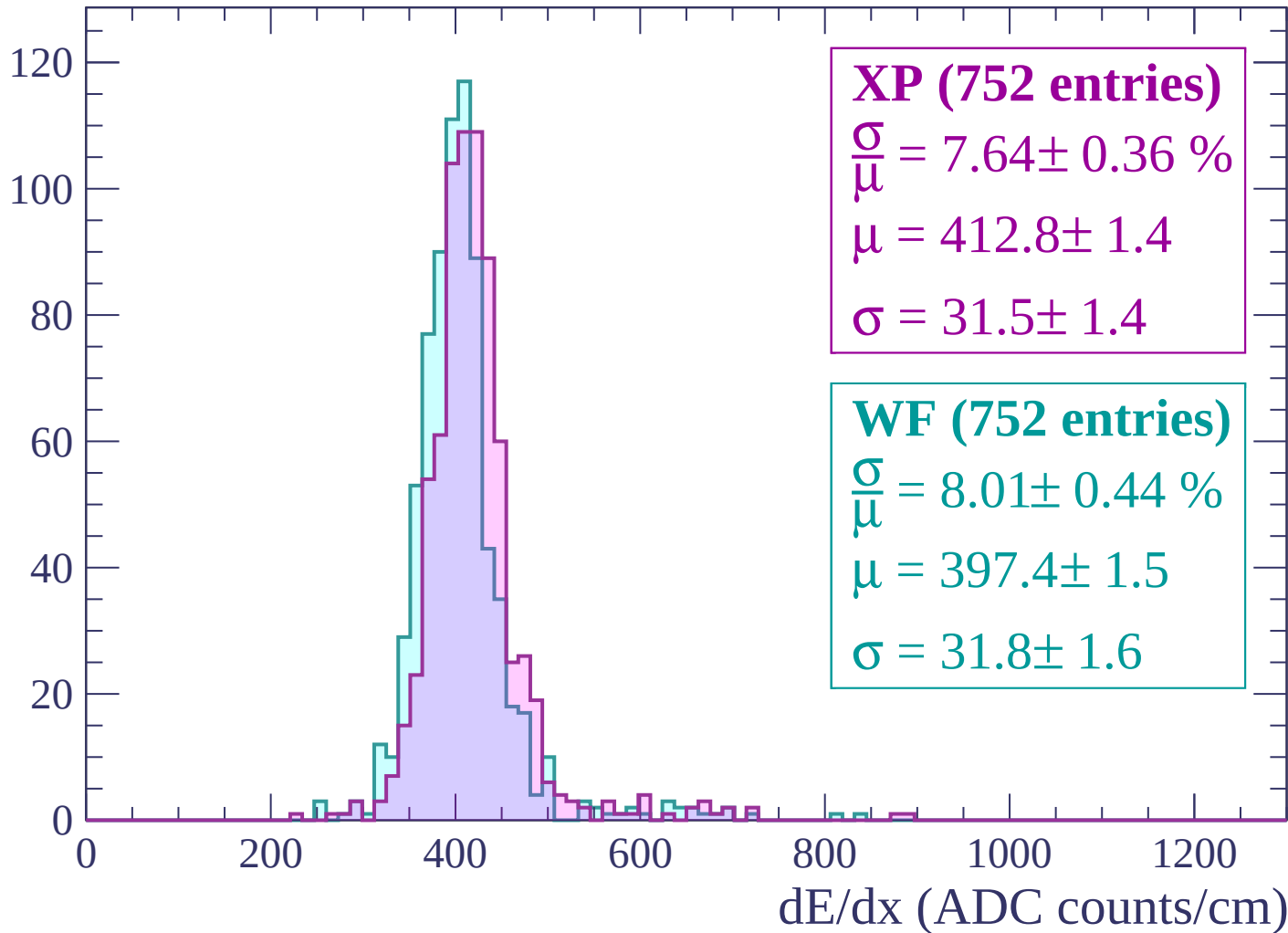
Energy loss | $300 < p < 360$

Count



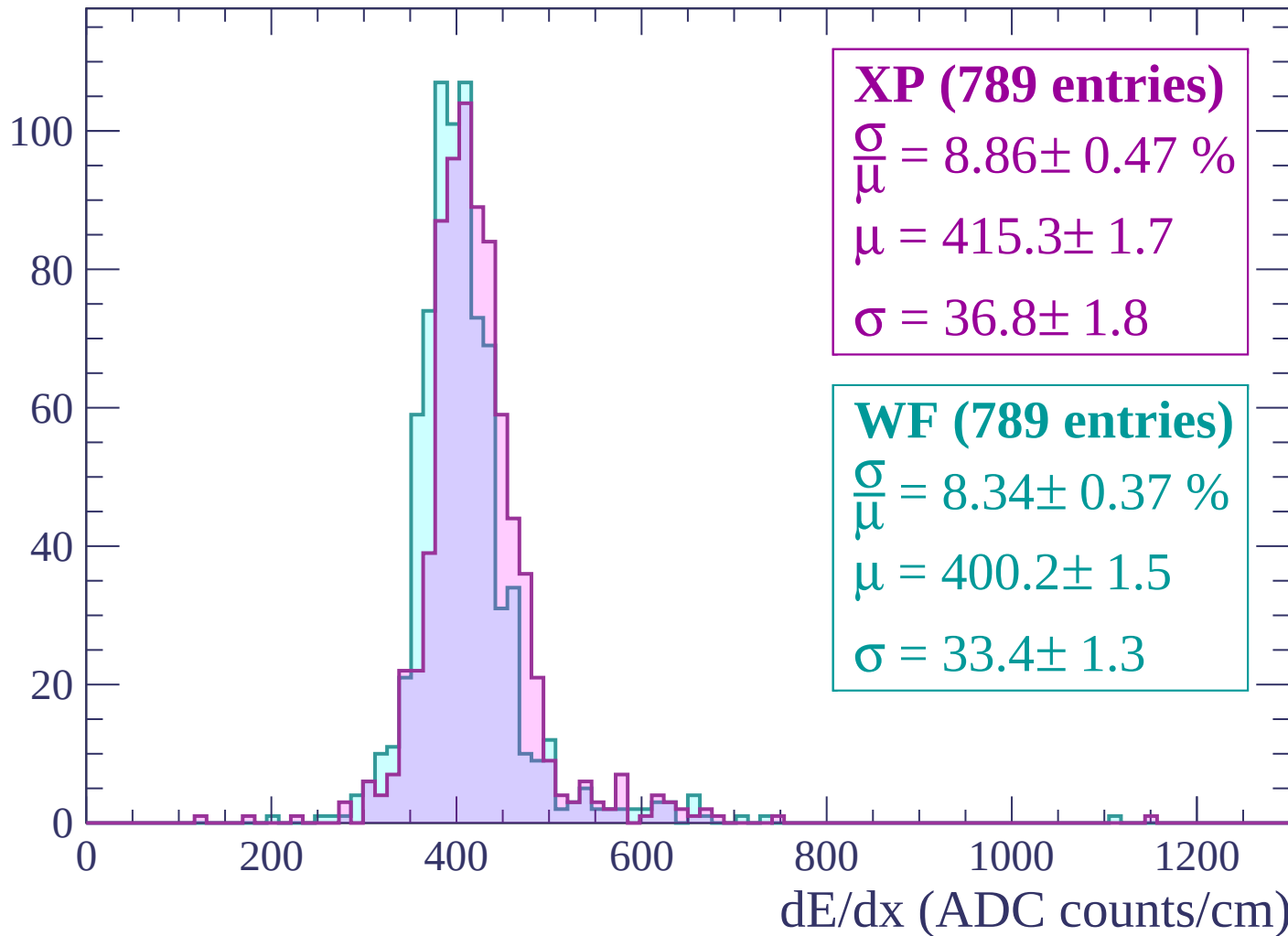
Energy loss | $360 < p < 420$

Count



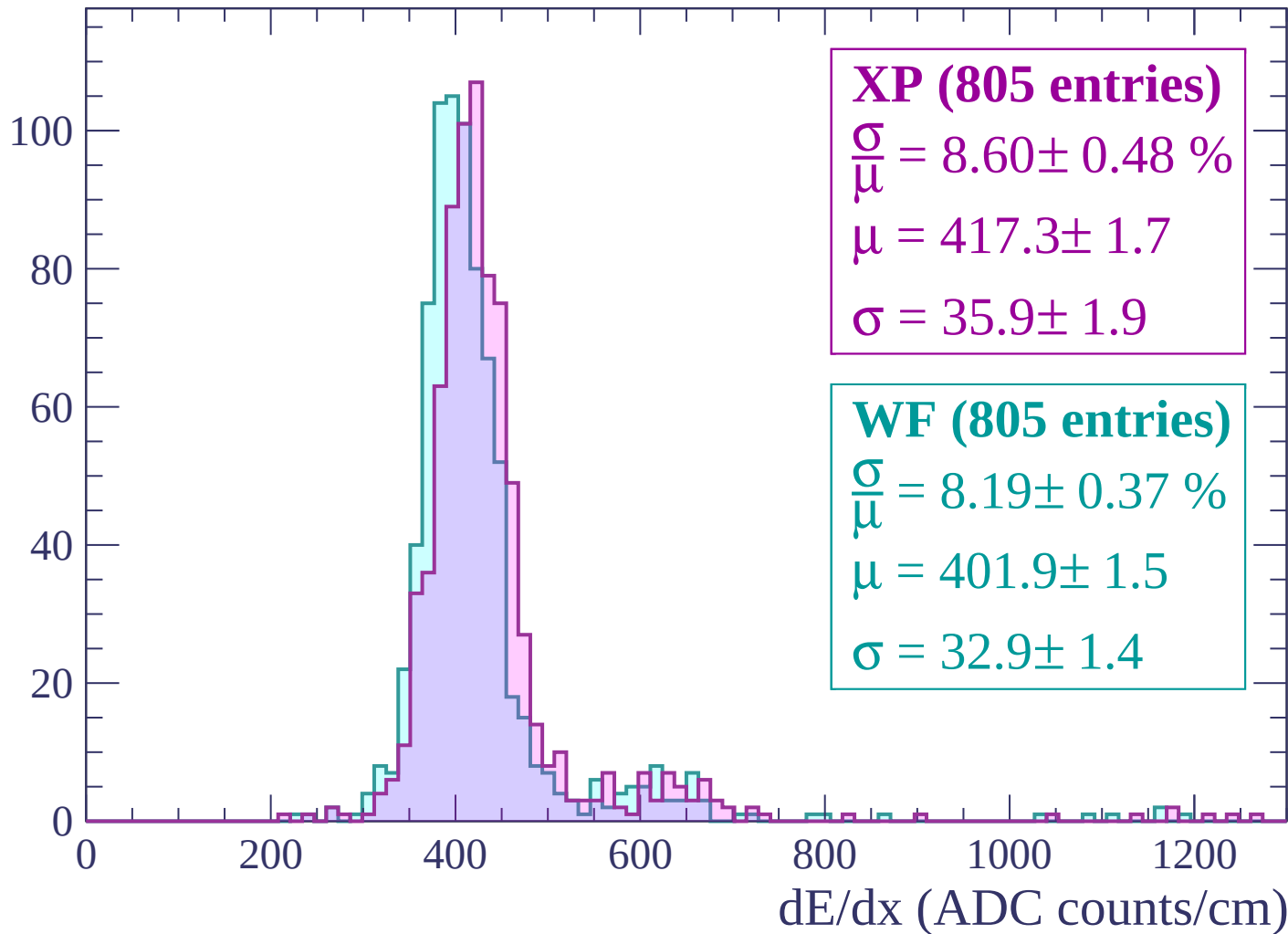
Energy loss | $420 < p < 480$

Count



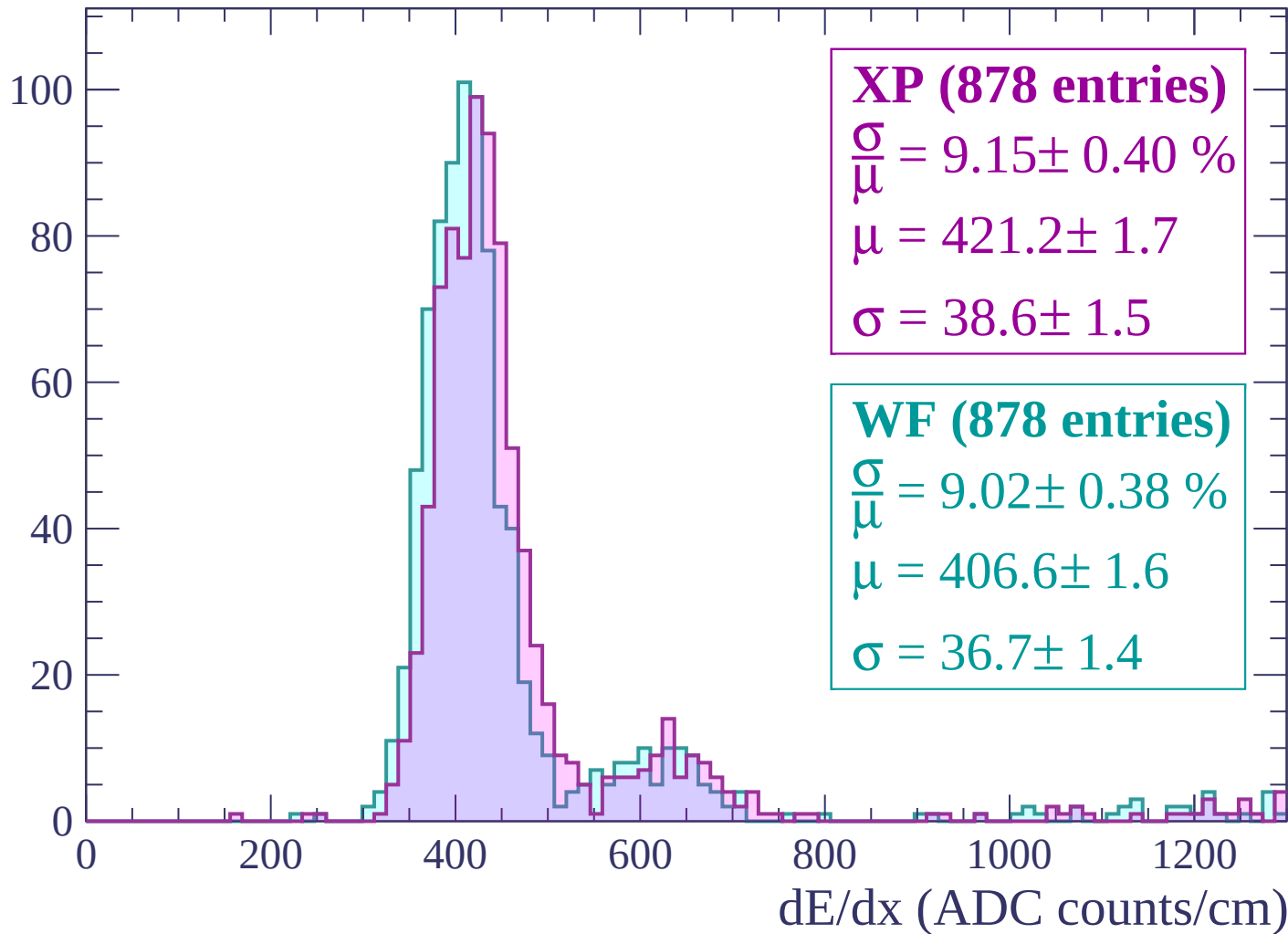
Energy loss | $480 < p < 540$

Count



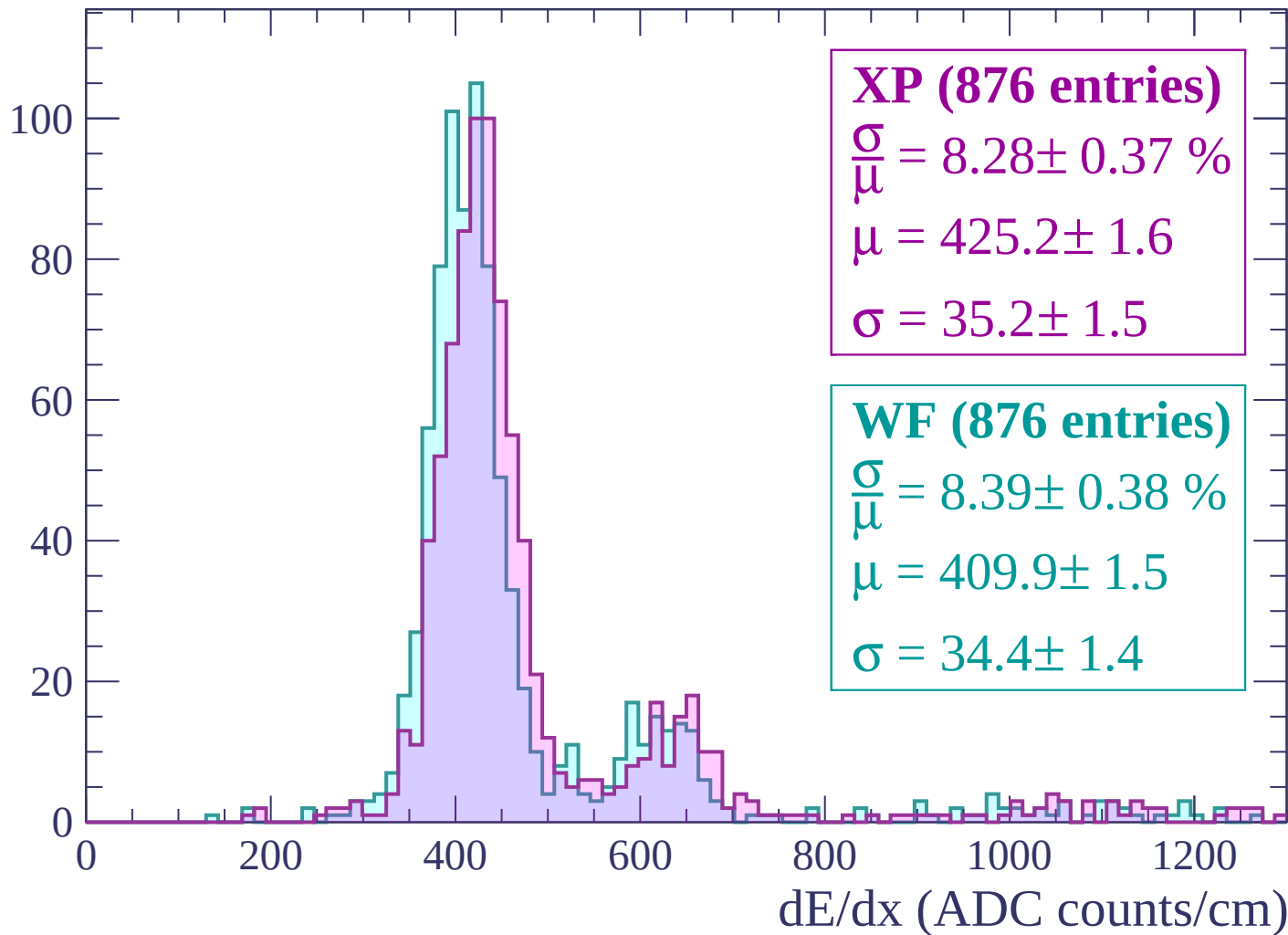
Energy loss | $540 < p < 600$

Count



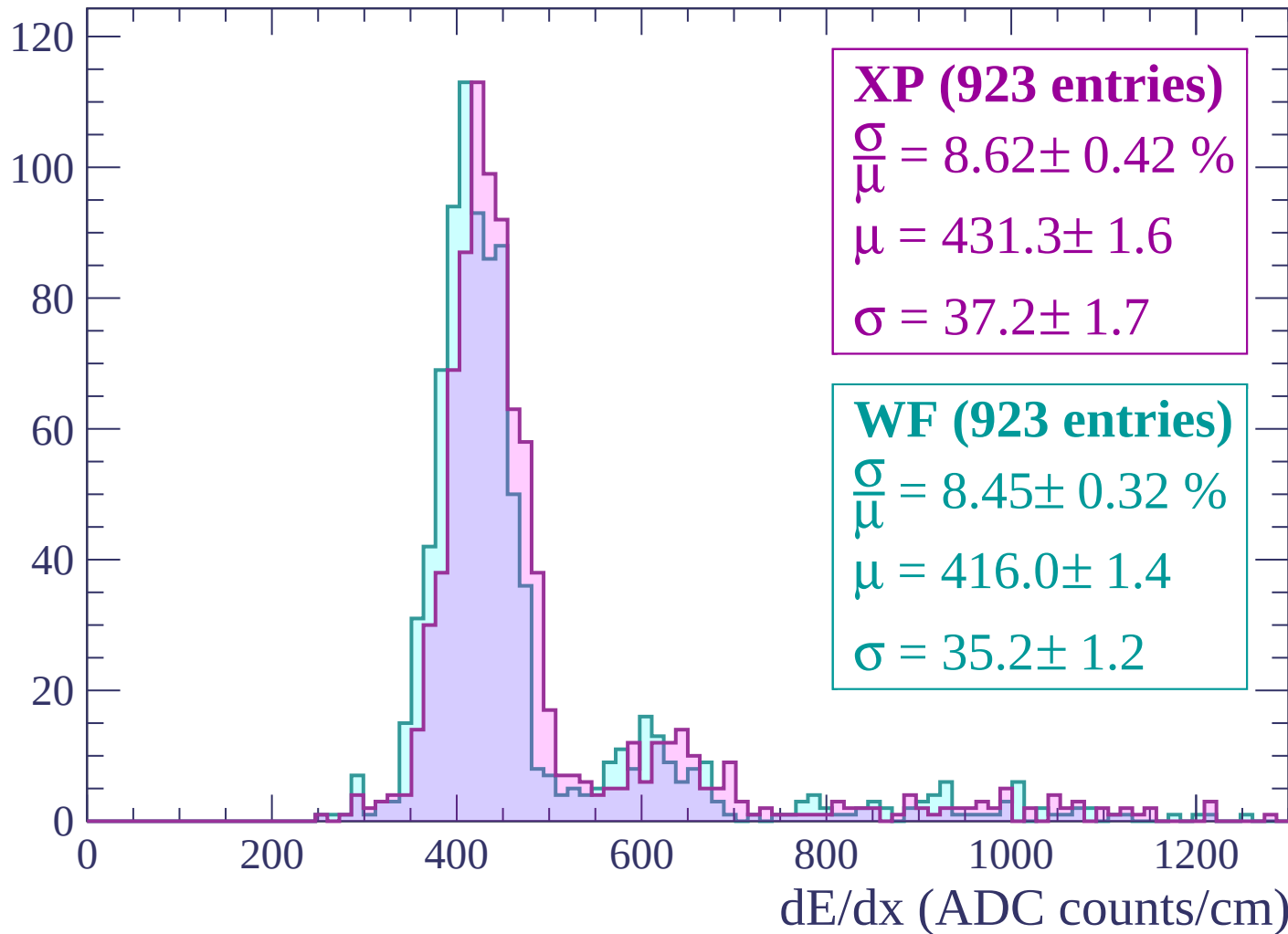
Energy loss | $600 < p < 660$

Count



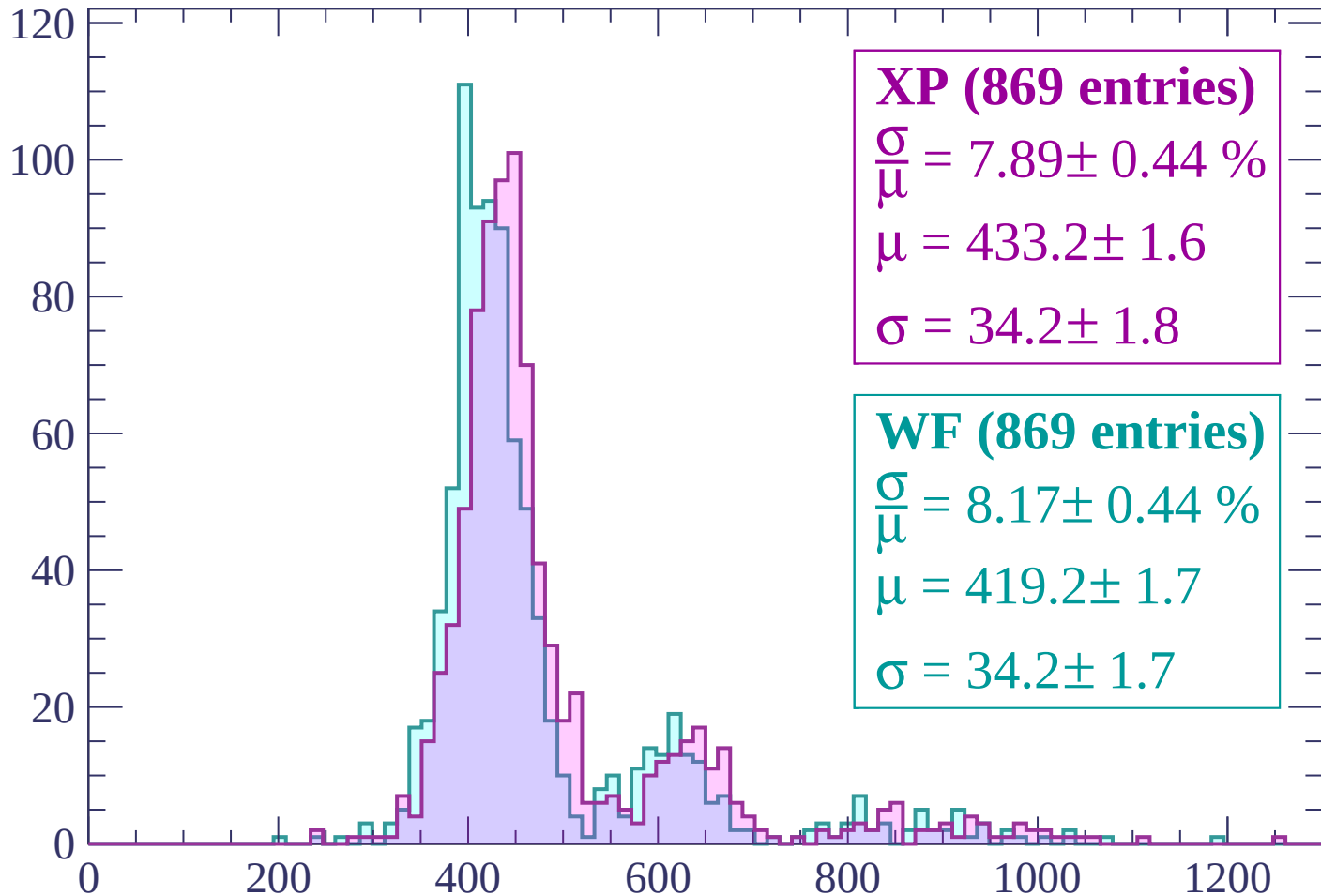
Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count



XP (869 entries)

$$\frac{\sigma}{\mu} = 7.89 \pm 0.44 \%$$

$$\mu = 433.2 \pm 1.6$$

$$\sigma = 34.2 \pm 1.8$$

WF (869 entries)

$$\frac{\sigma}{\mu} = 8.17 \pm 0.44 \%$$

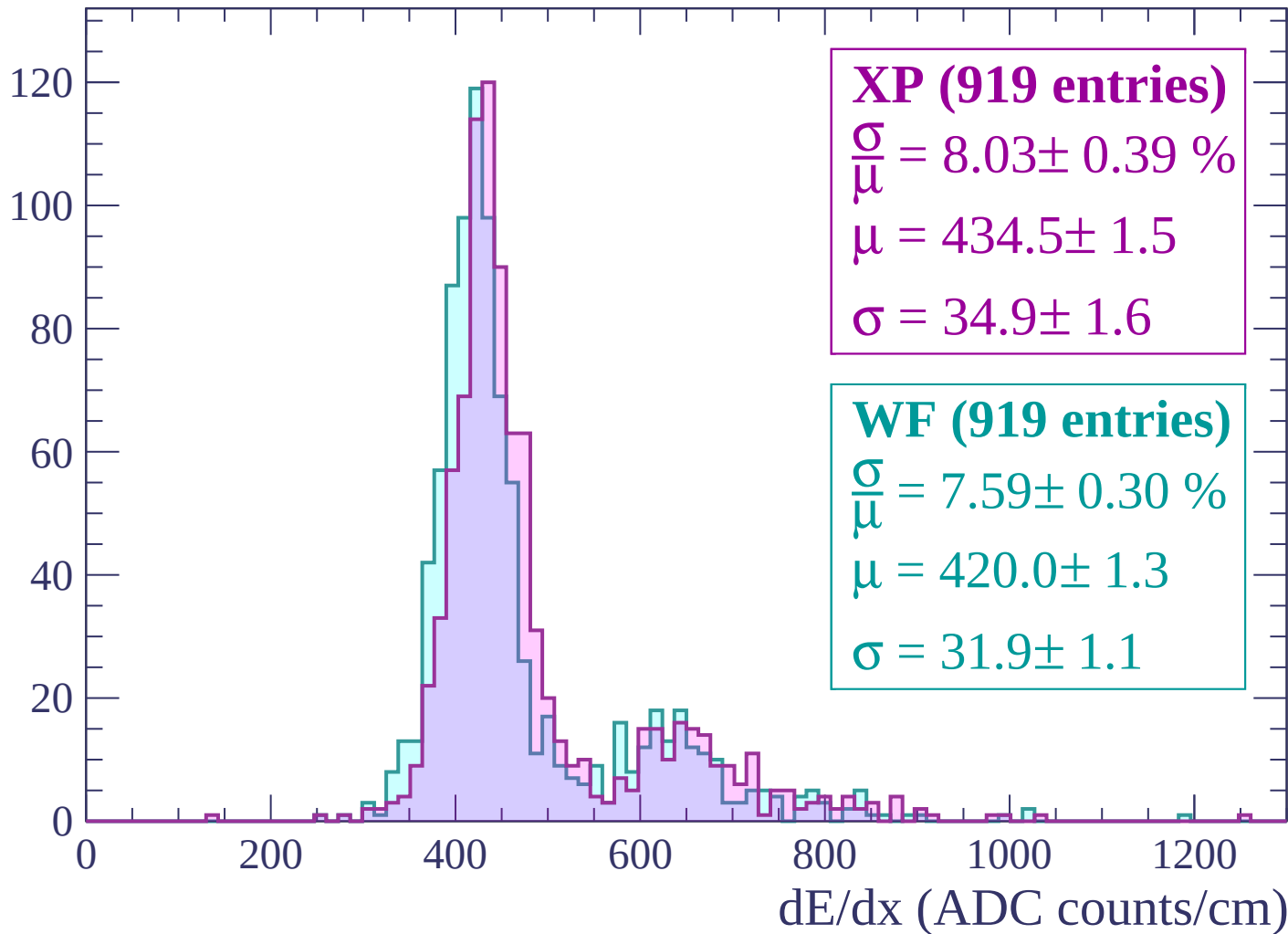
$$\mu = 419.2 \pm 1.7$$

$$\sigma = 34.2 \pm 1.7$$

dE/dx (ADC counts/cm)

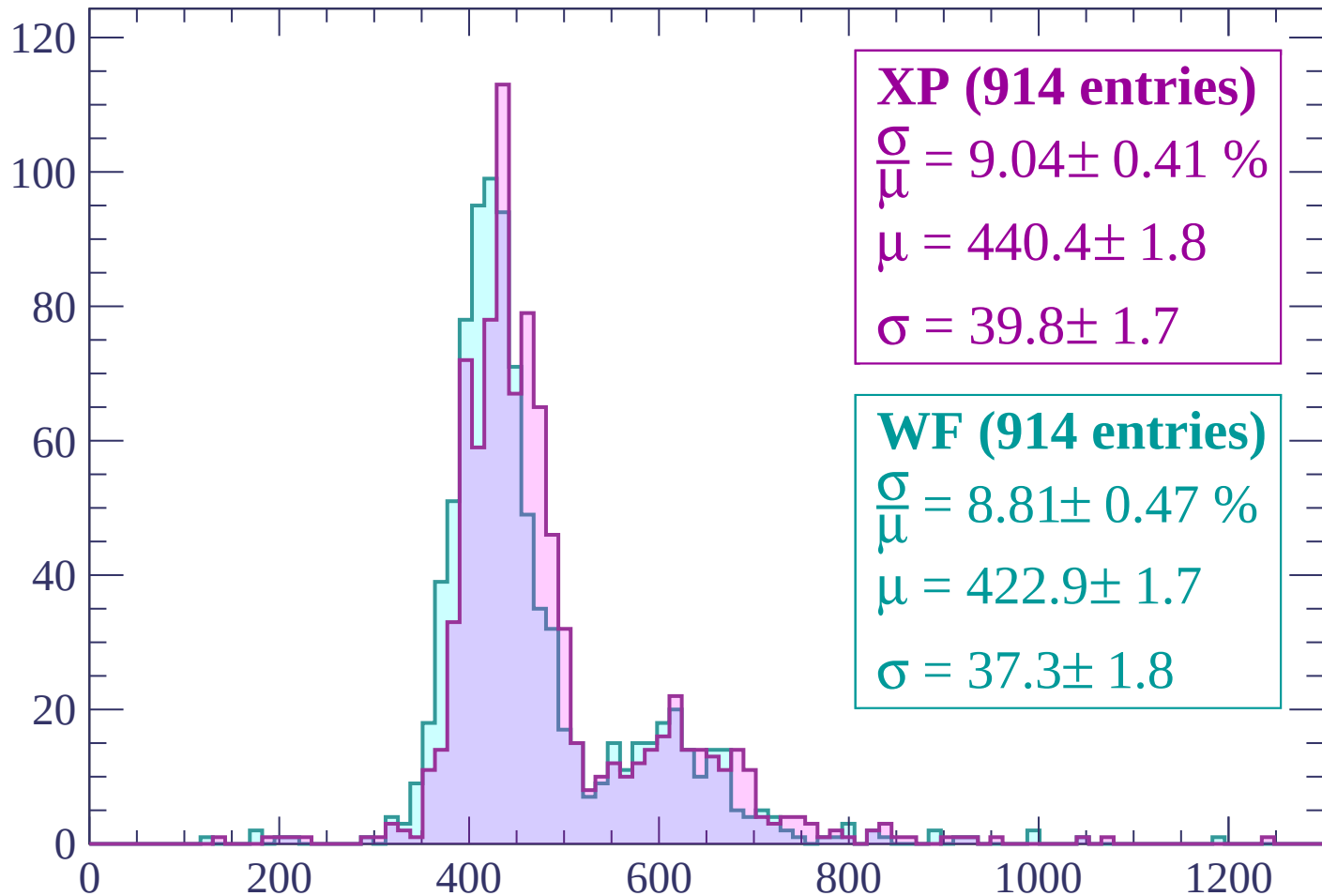
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP (914 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.41 \%$

$\mu = 440.4 \pm 1.8$

$\sigma = 39.8 \pm 1.7$

WF (914 entries)

$\frac{\sigma}{\mu} = 8.81 \pm 0.47 \%$

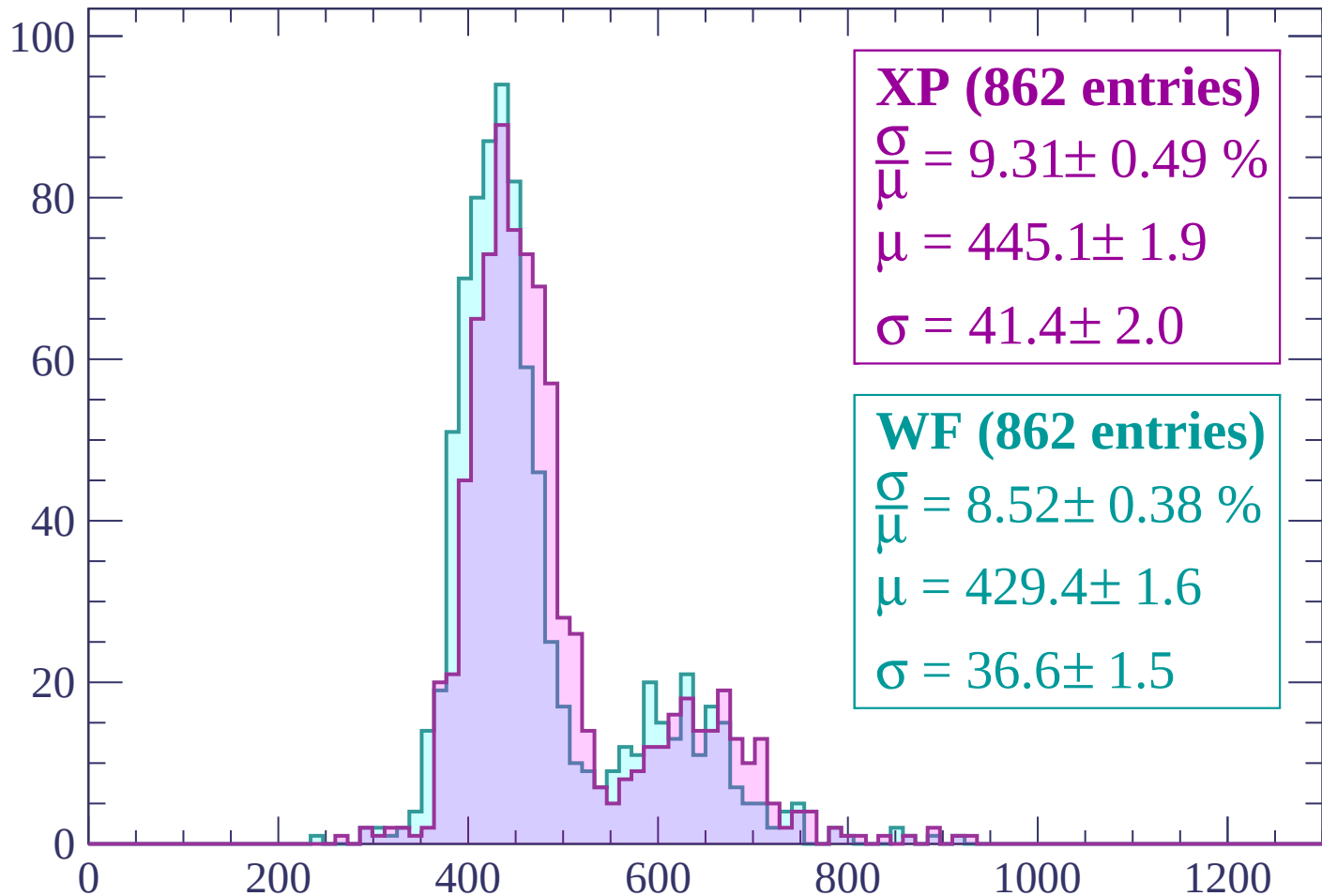
$\mu = 422.9 \pm 1.7$

$\sigma = 37.3 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (862 entries)

$\frac{\sigma}{\mu} = 9.31 \pm 0.49 \%$

$\mu = 445.1 \pm 1.9$

$\sigma = 41.4 \pm 2.0$

WF (862 entries)

$\frac{\sigma}{\mu} = 8.52 \pm 0.38 \%$

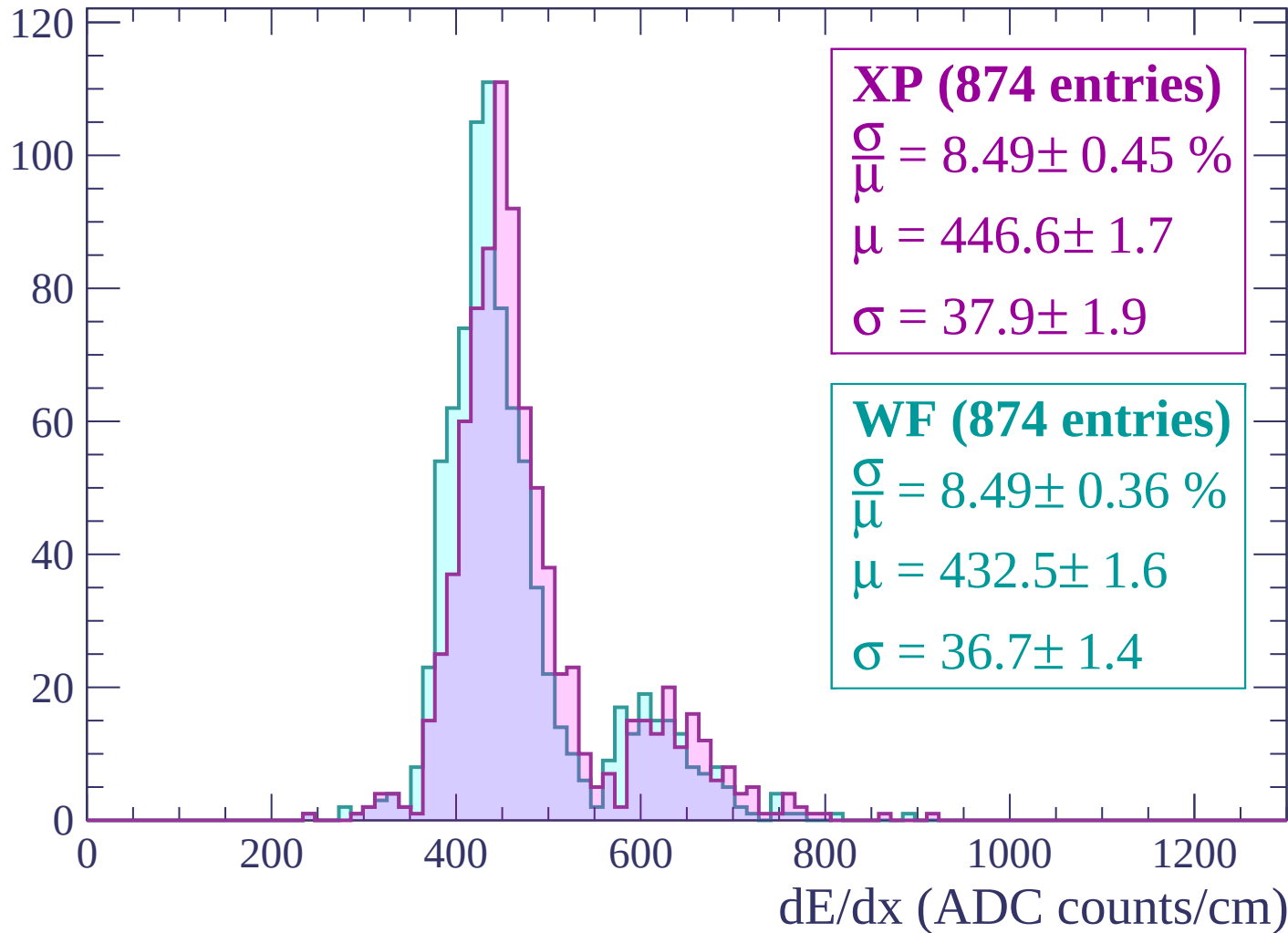
$\mu = 429.4 \pm 1.6$

$\sigma = 36.6 \pm 1.5$

dE/dx (ADC counts/cm)

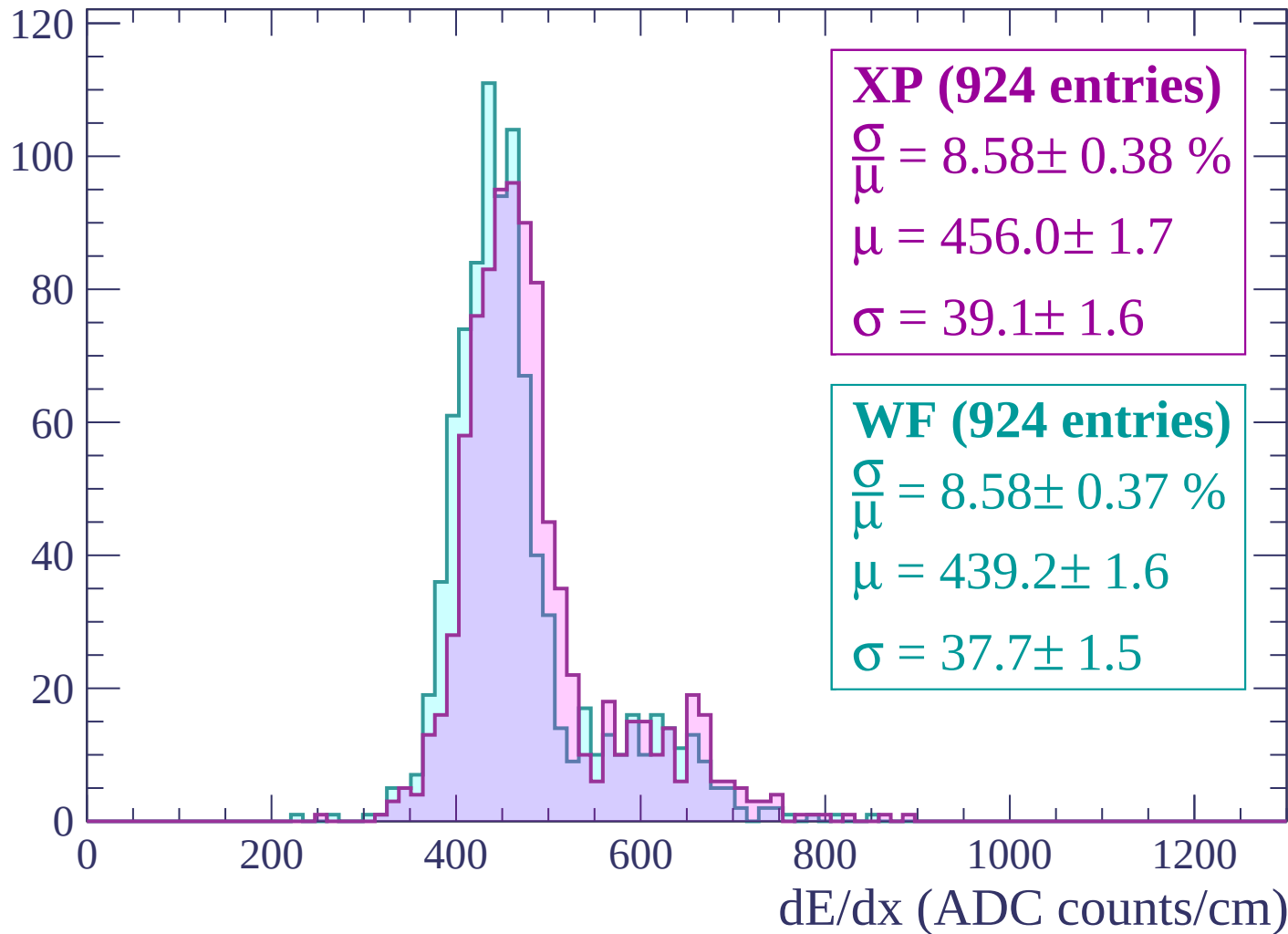
Energy loss | $960 < p < 1020$

Count



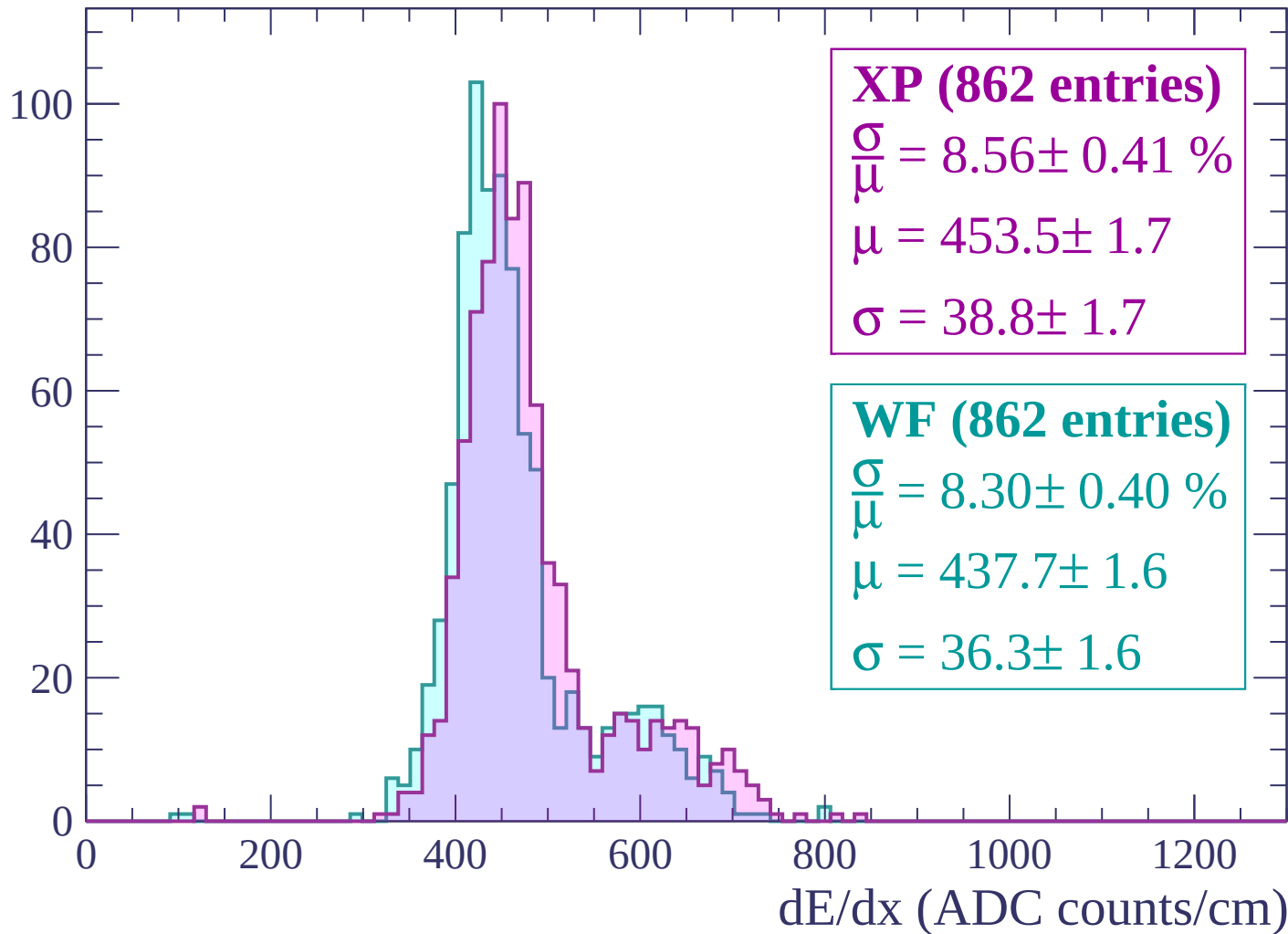
Energy loss | $1020 < p < 1080$

Count



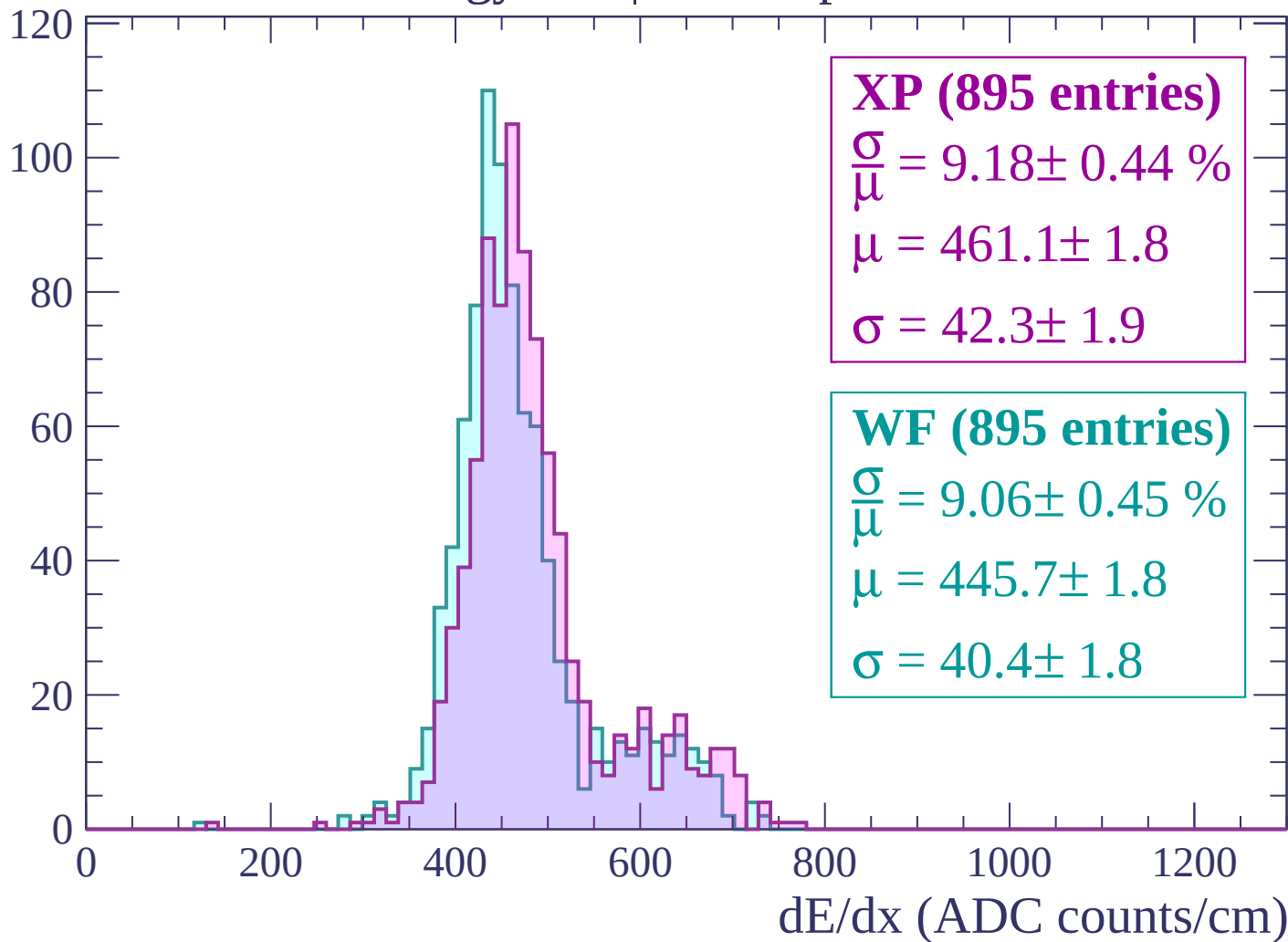
Energy loss | $1080 < p < 1140$

Count



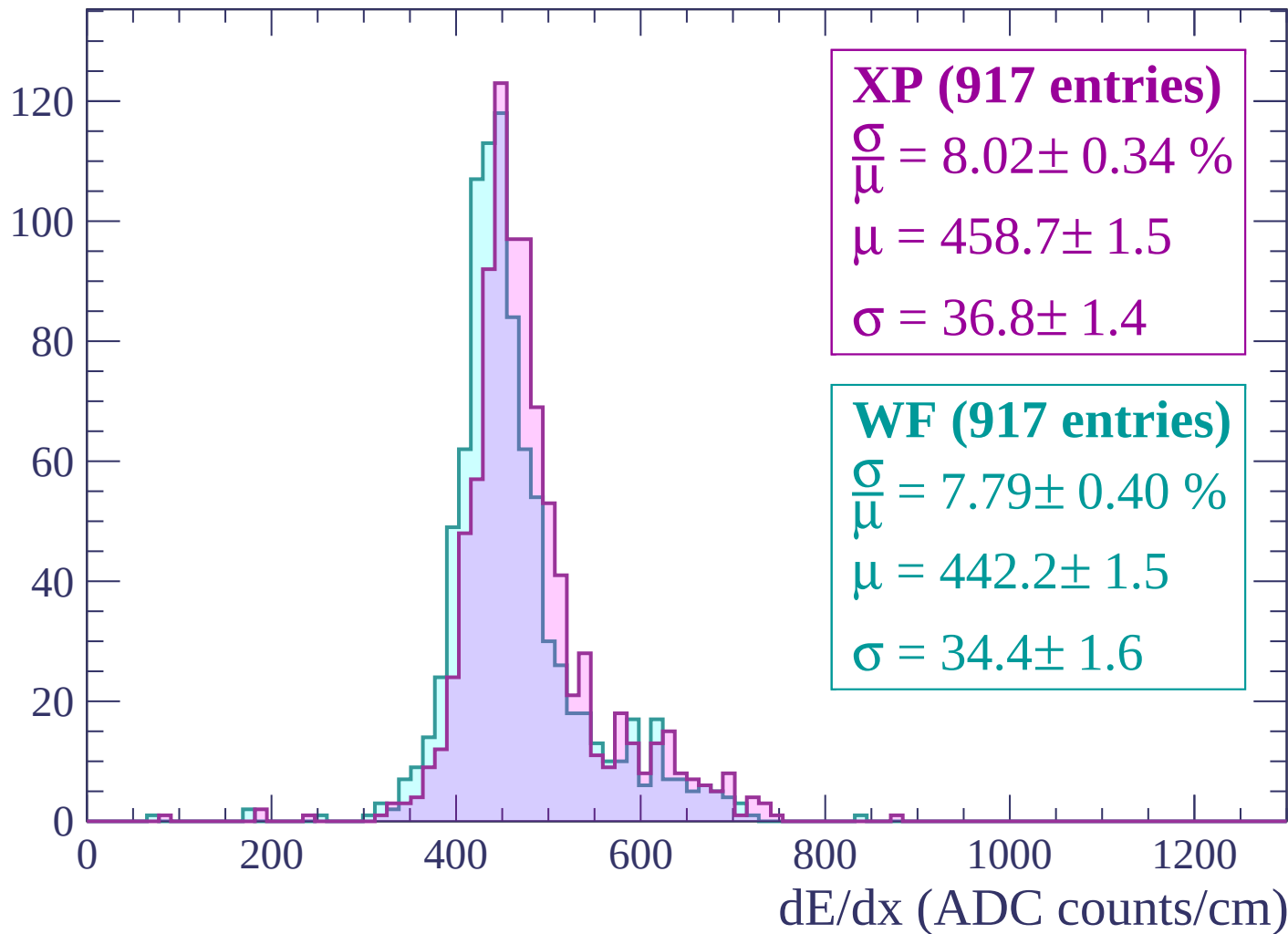
Energy loss | $1140 < p < 1200$

Count



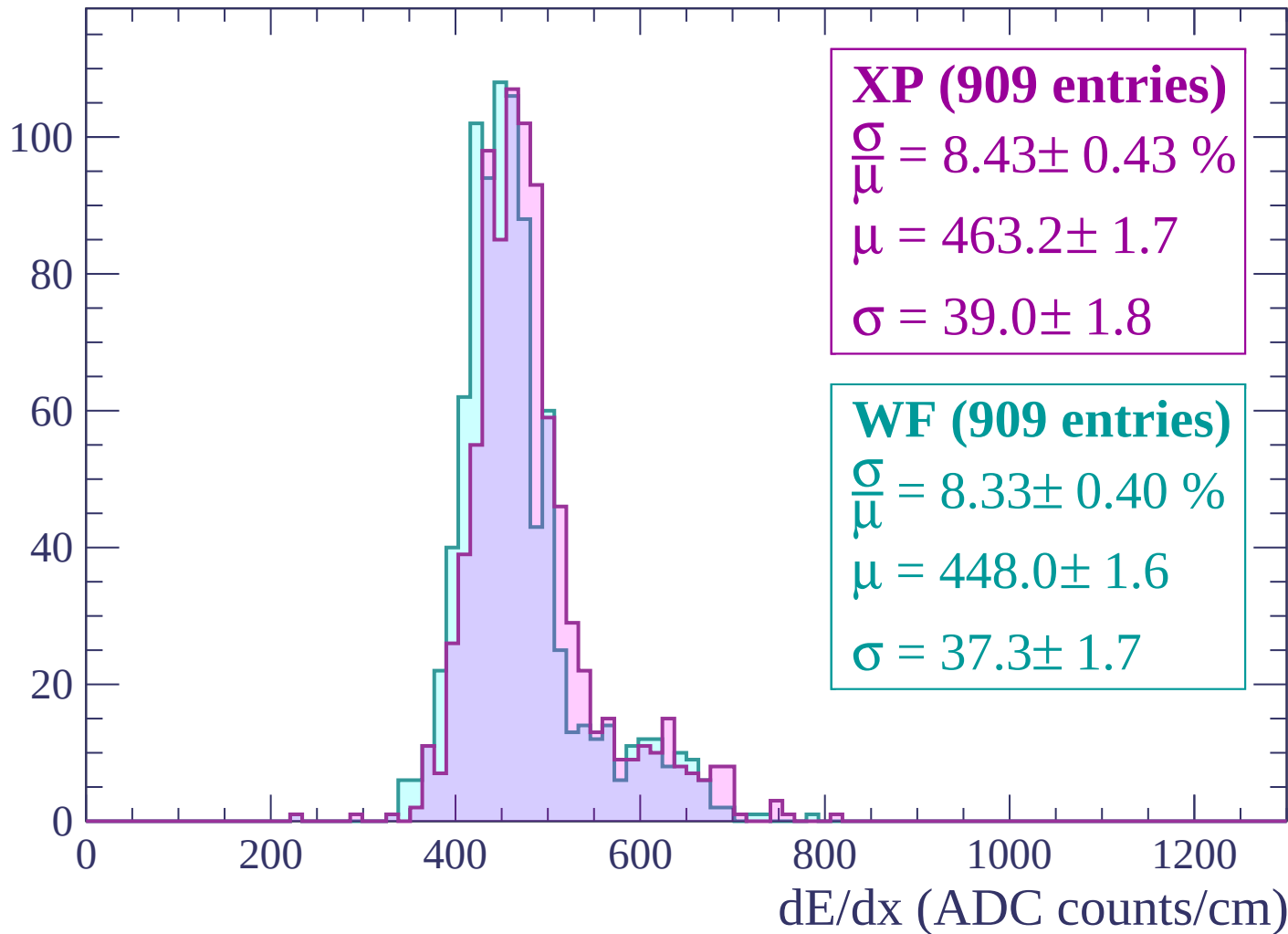
Energy loss | $1200 < p < 1260$

Count



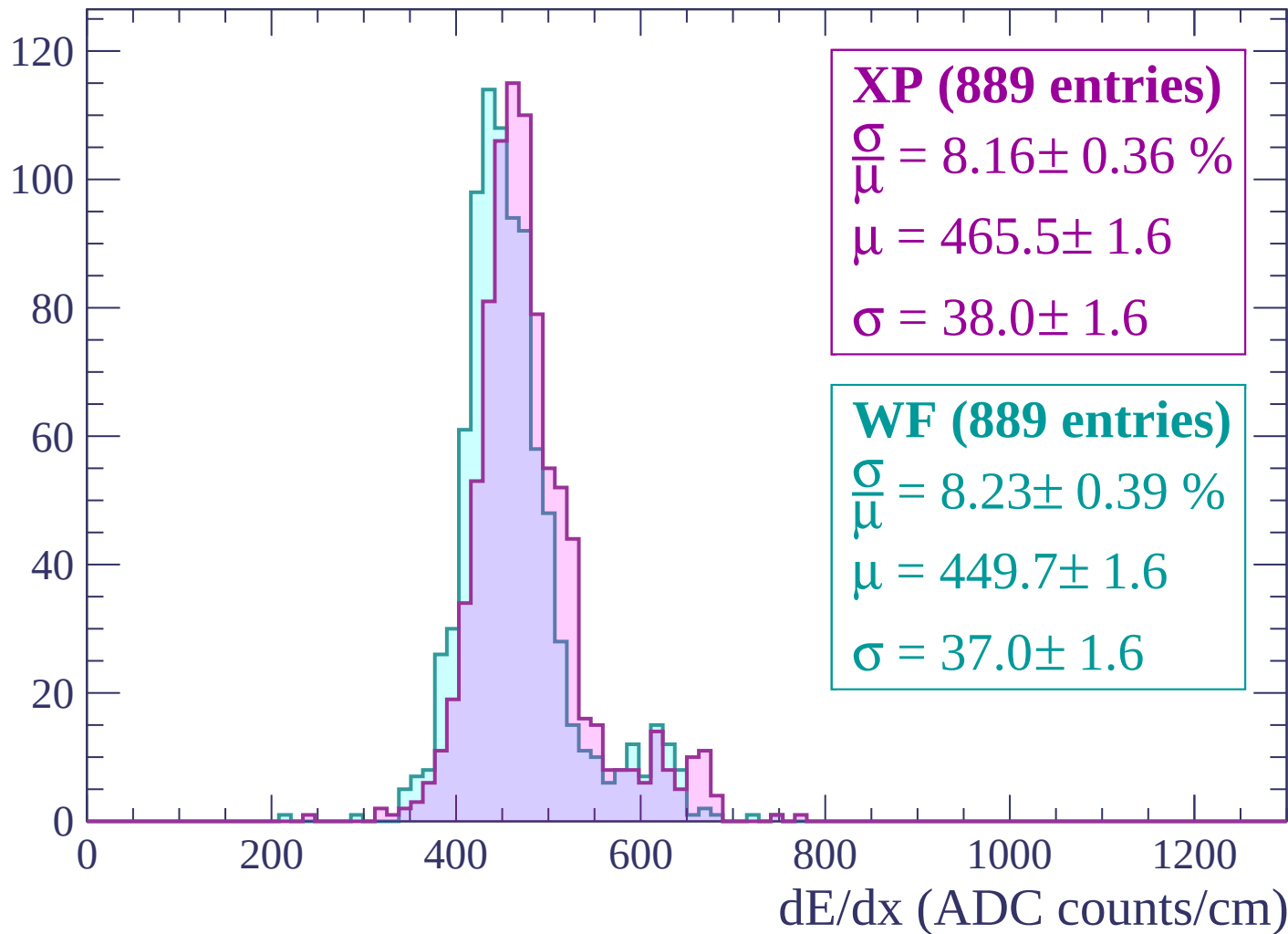
Energy loss | $1260 < p < 1320$

Count



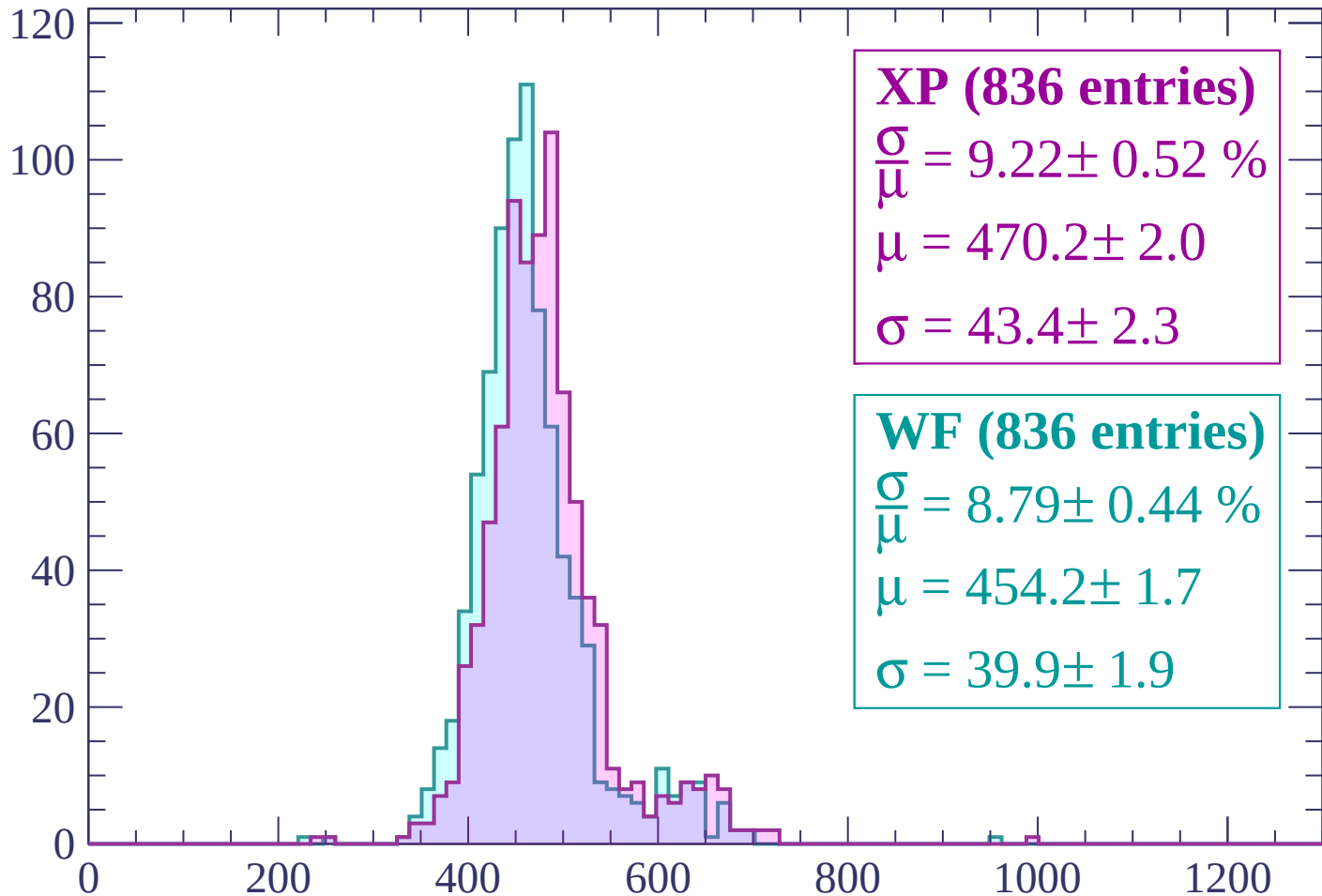
Energy loss | $1320 < p < 1380$

Count



Energy loss | $1380 < p < 1440$

Count



XP (836 entries)

$$\frac{\sigma}{\mu} = 9.22 \pm 0.52 \%$$

$$\mu = 470.2 \pm 2.0$$

$$\sigma = 43.4 \pm 2.3$$

WF (836 entries)

$$\frac{\sigma}{\mu} = 8.79 \pm 0.44 \%$$

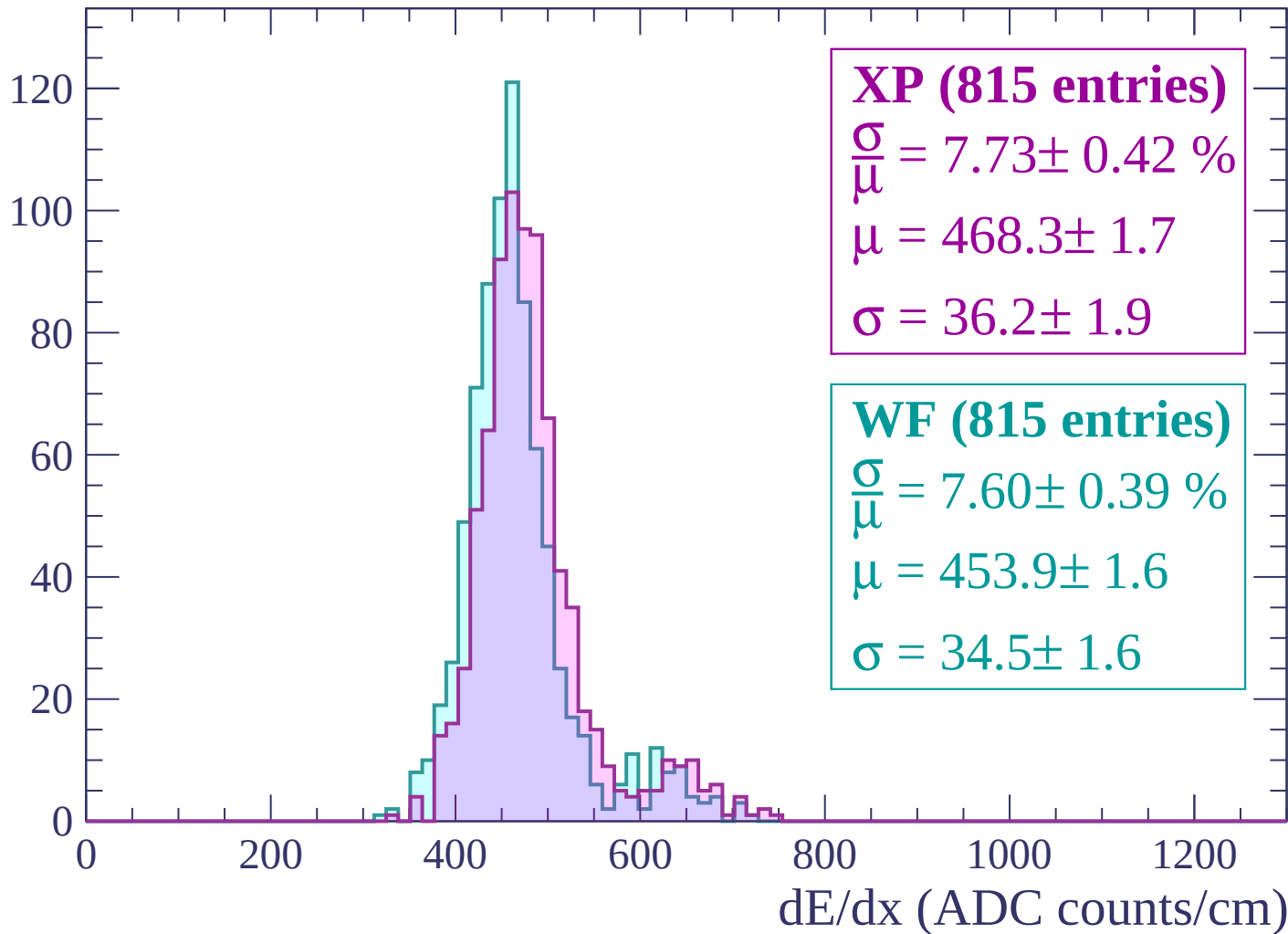
$$\mu = 454.2 \pm 1.7$$

$$\sigma = 39.9 \pm 1.9$$

dE/dx (ADC counts/cm)

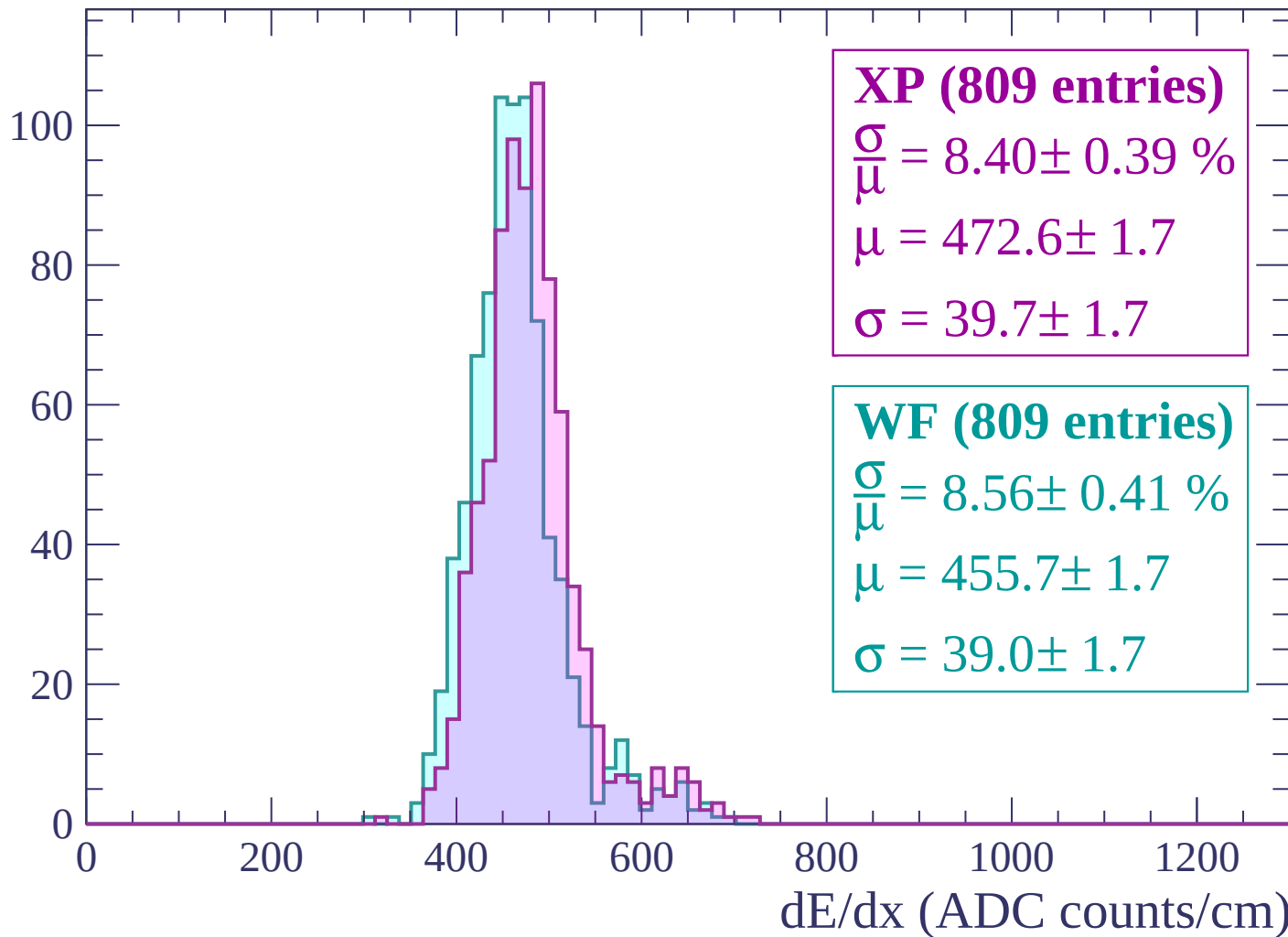
Energy loss | $1440 < p < 1500$

Count



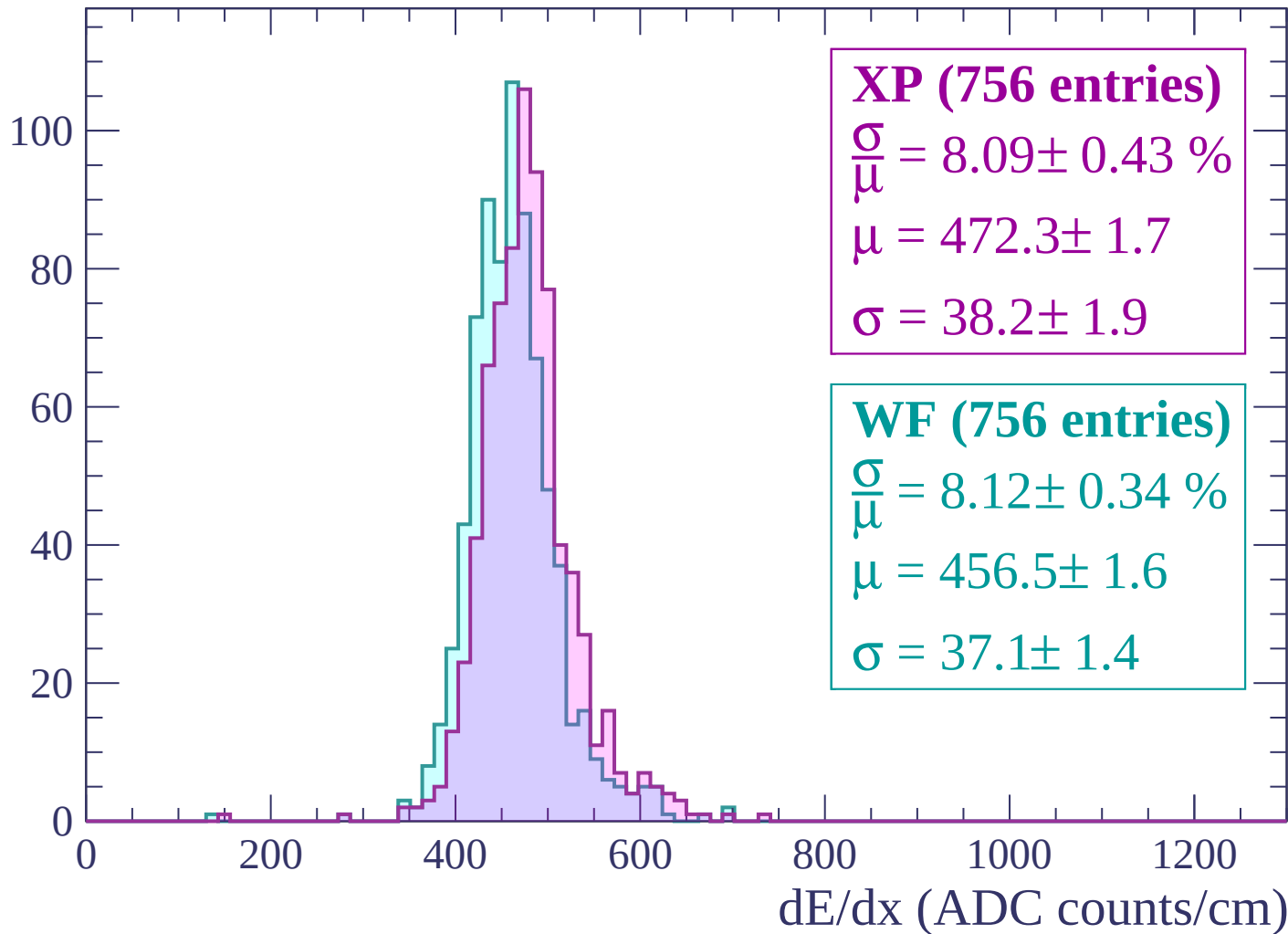
Energy loss | $1500 < p < 1560$

Count



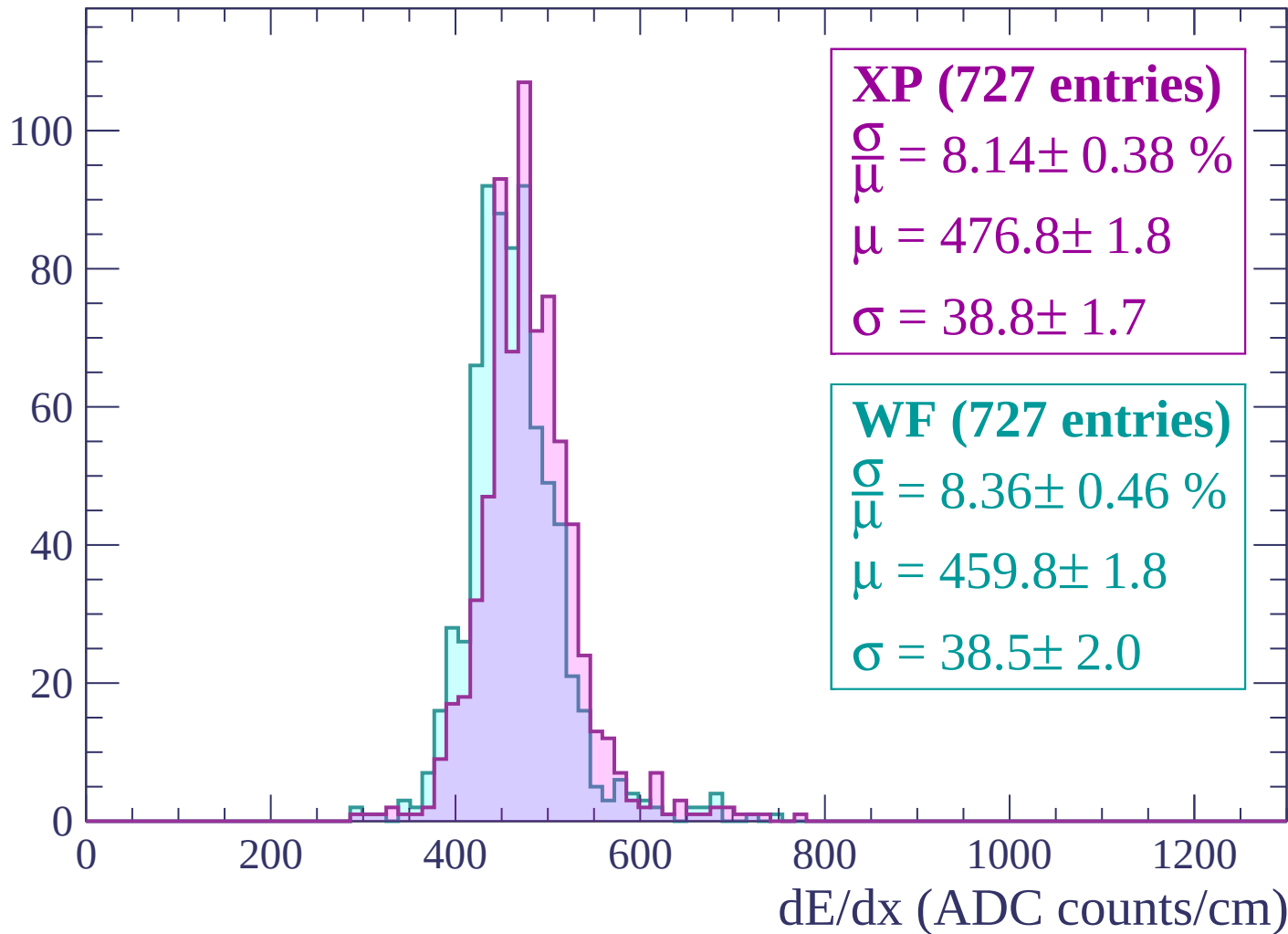
Energy loss | $1560 < p < 1620$

Count



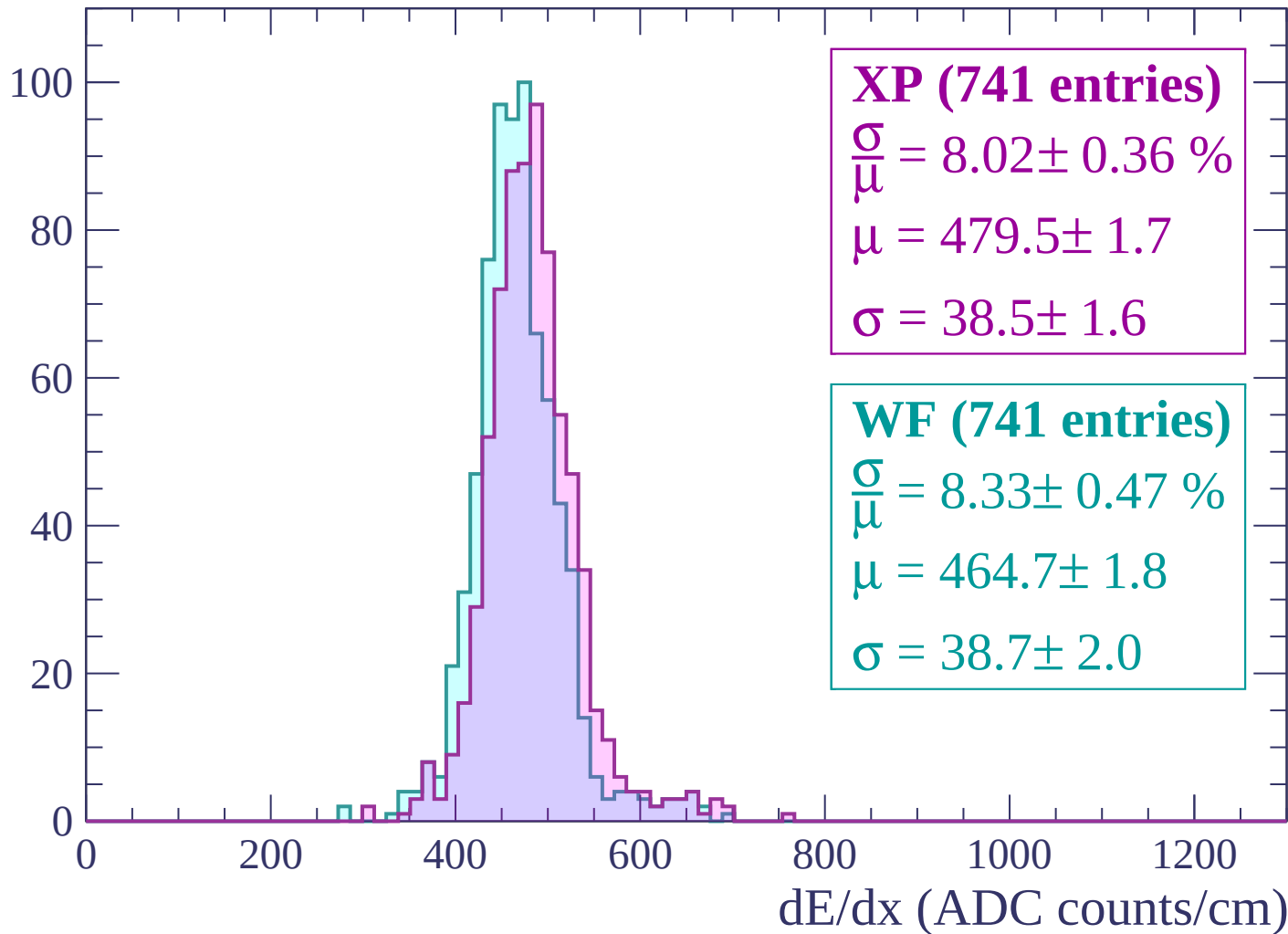
Energy loss | $1620 < p < 1680$

Count



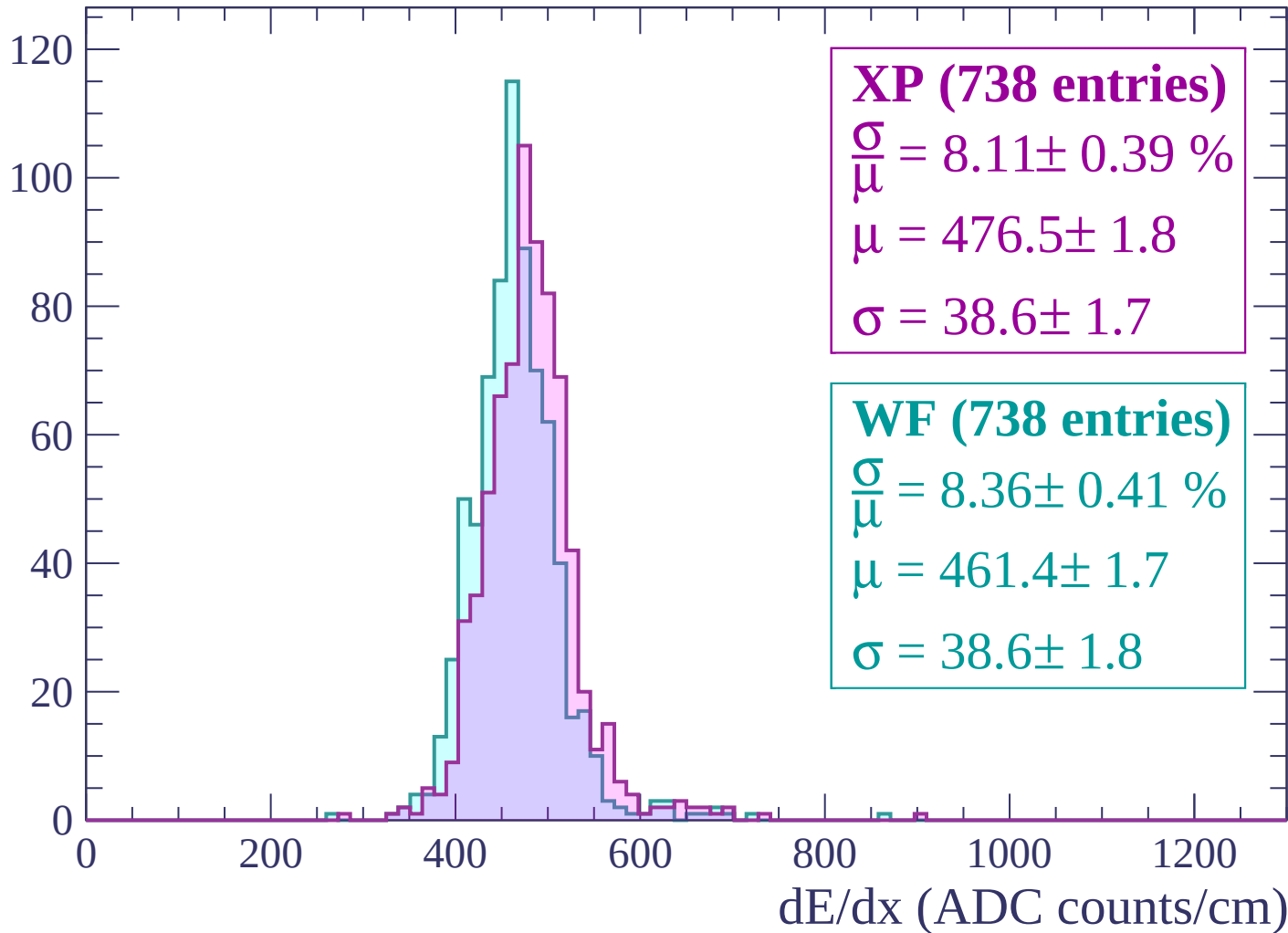
Energy loss | $1680 < p < 1740$

Count



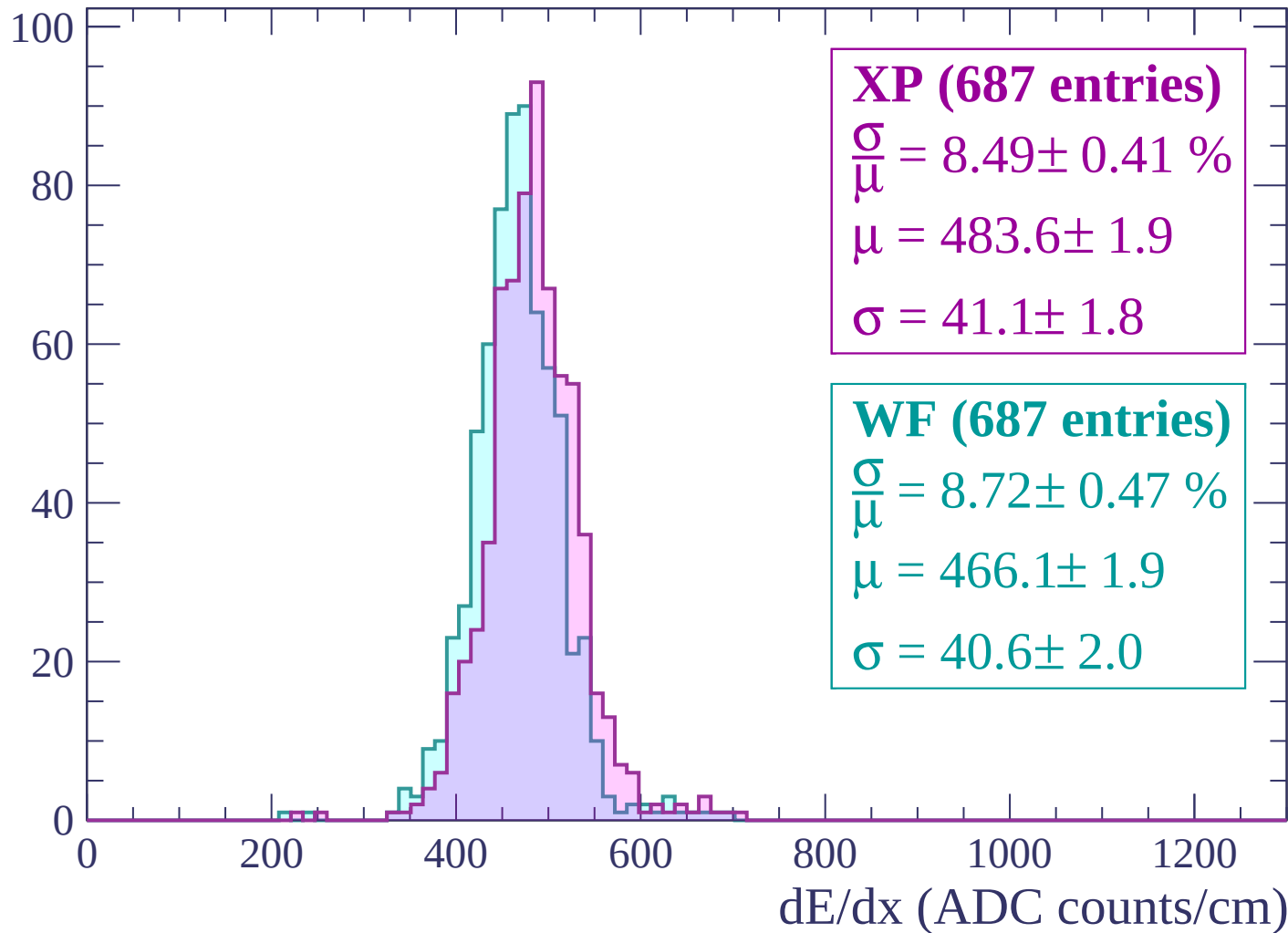
Energy loss | $1740 < p < 1800$

Count



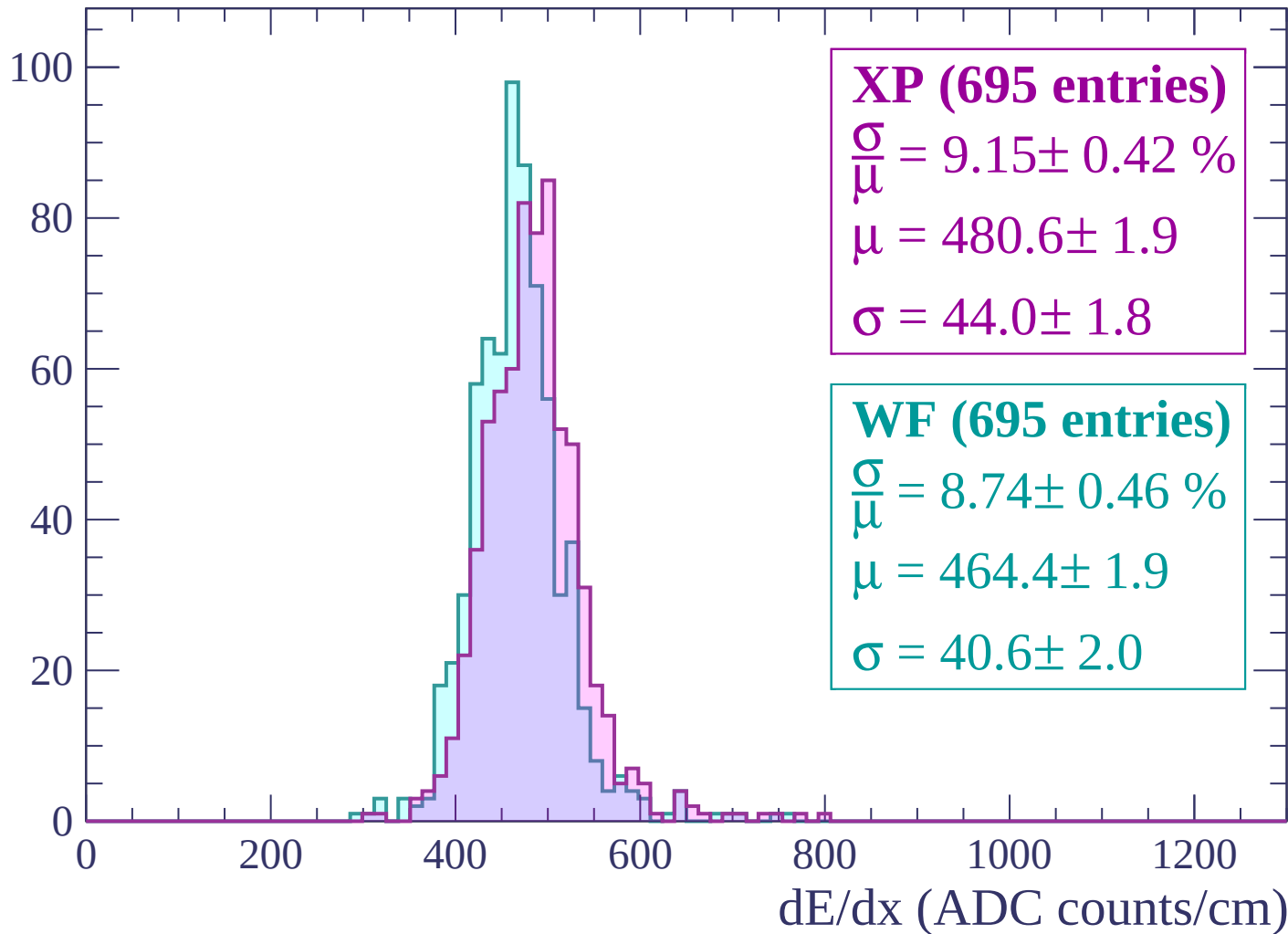
Energy loss | $1800 < p < 1860$

Count



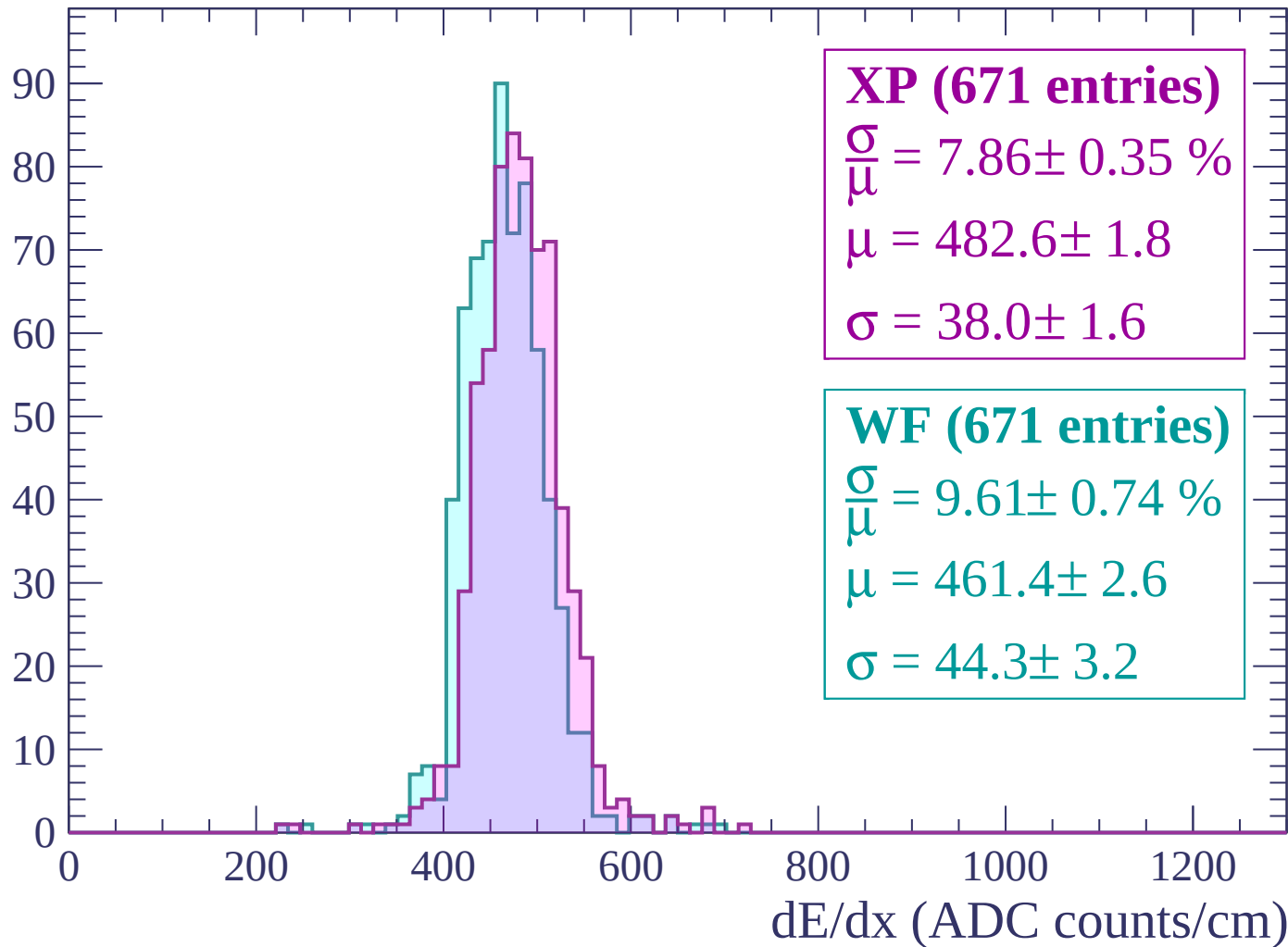
Energy loss | $1860 < p < 1920$

Count



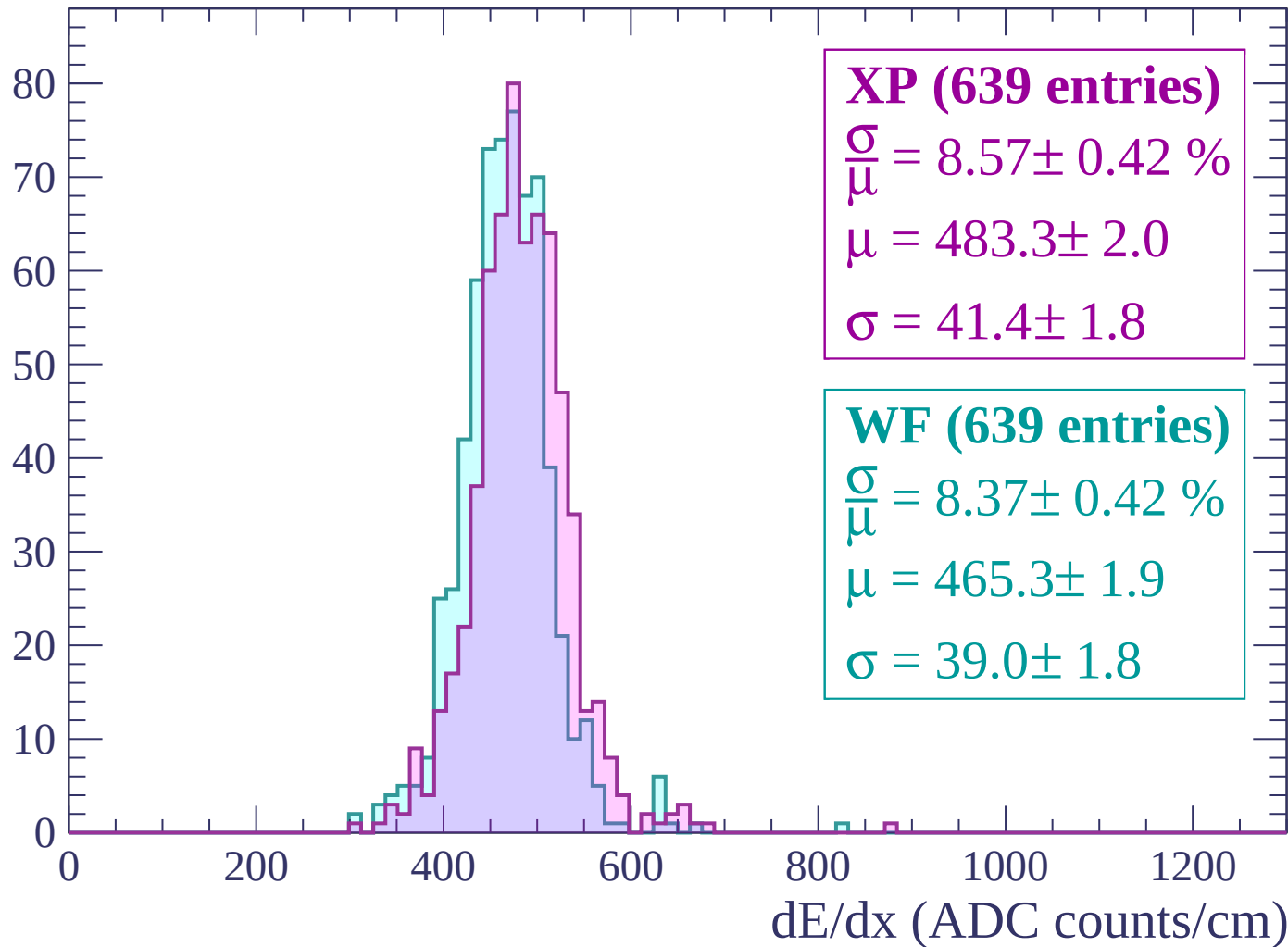
Energy loss | $1920 < p < 1980$

Count



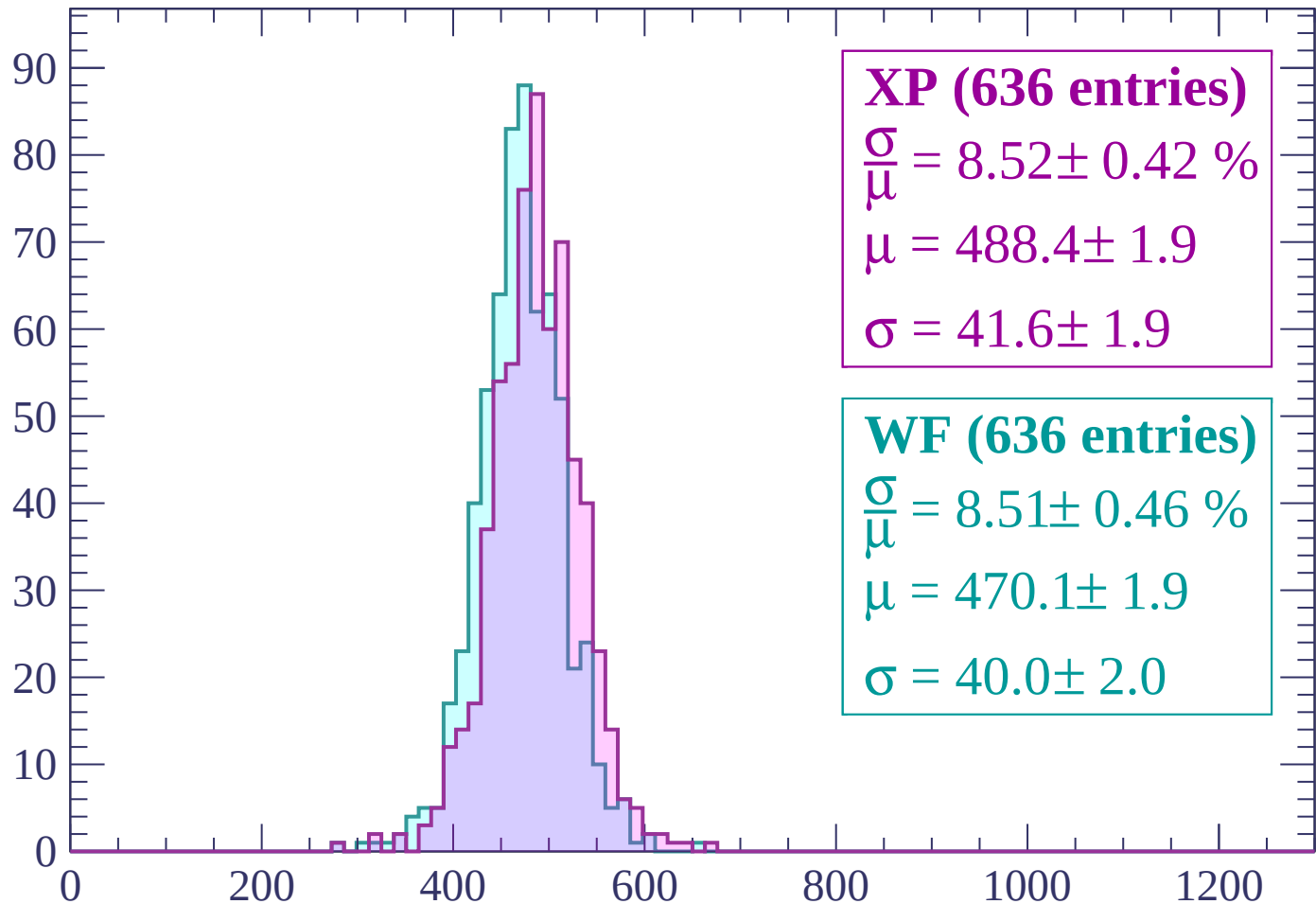
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (636 entries)

$\frac{\sigma}{\mu} = 8.52 \pm 0.42 \%$

$\mu = 488.4 \pm 1.9$

$\sigma = 41.6 \pm 1.9$

WF (636 entries)

$\frac{\sigma}{\mu} = 8.51 \pm 0.46 \%$

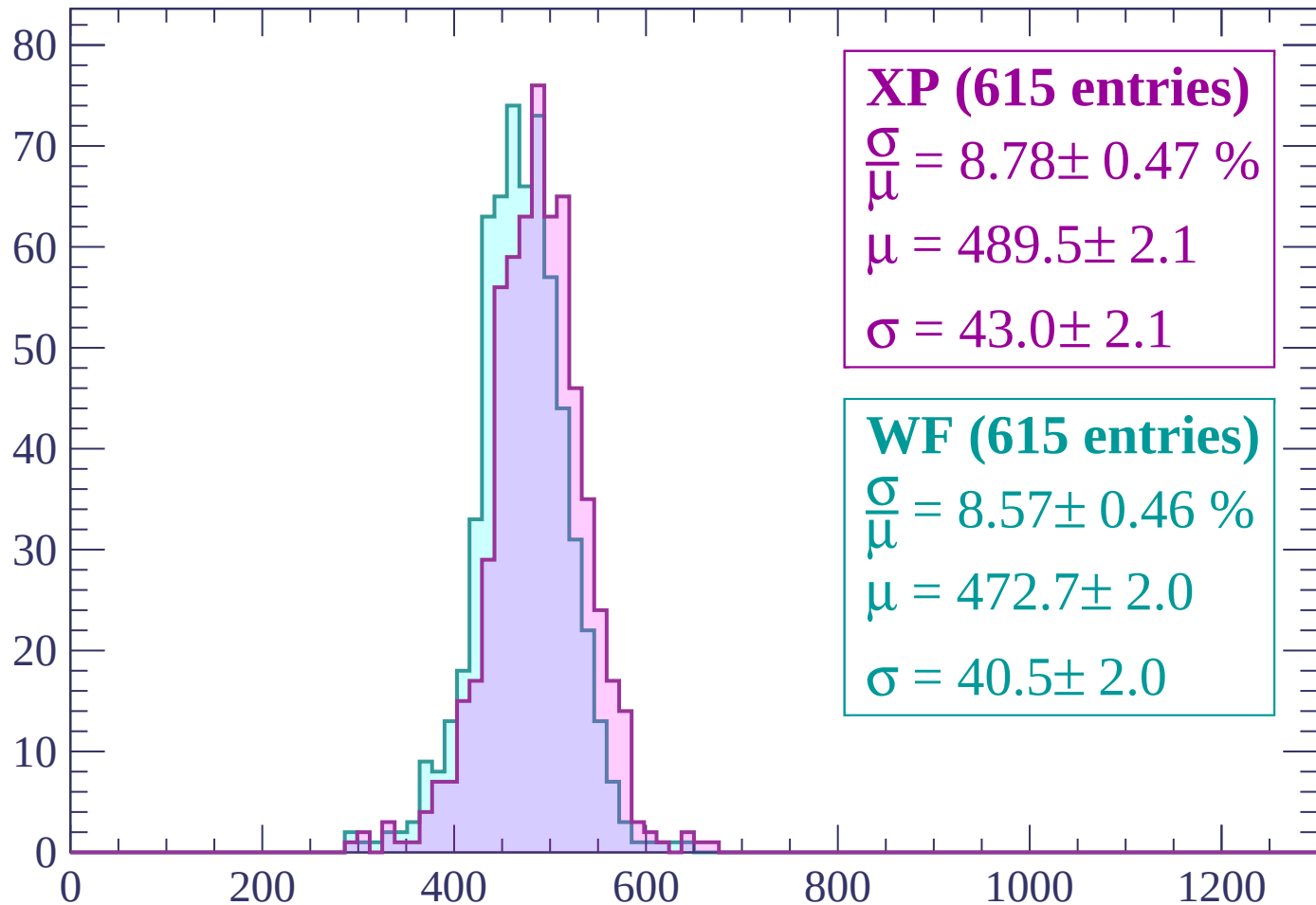
$\mu = 470.1 \pm 1.9$

$\sigma = 40.0 \pm 2.0$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (615 entries)

$\frac{\sigma}{\mu} = 8.78 \pm 0.47 \%$

$\mu = 489.5 \pm 2.1$

$\sigma = 43.0 \pm 2.1$

WF (615 entries)

$\frac{\sigma}{\mu} = 8.57 \pm 0.46 \%$

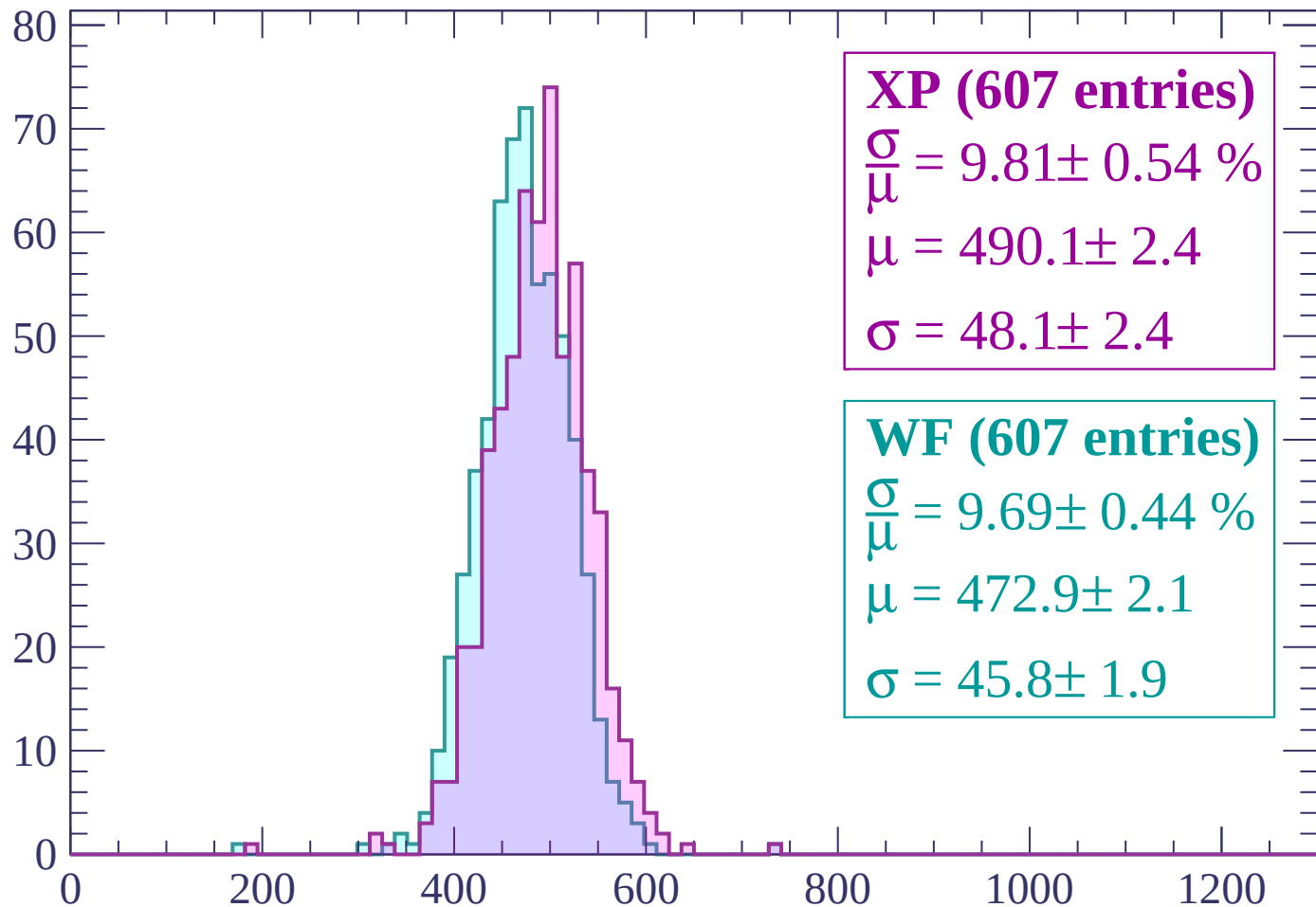
$\mu = 472.7 \pm 2.0$

$\sigma = 40.5 \pm 2.0$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (607 entries)

$\frac{\sigma}{\mu} = 9.81 \pm 0.54 \%$

$\mu = 490.1 \pm 2.4$

$\sigma = 48.1 \pm 2.4$

WF (607 entries)

$\frac{\sigma}{\mu} = 9.69 \pm 0.44 \%$

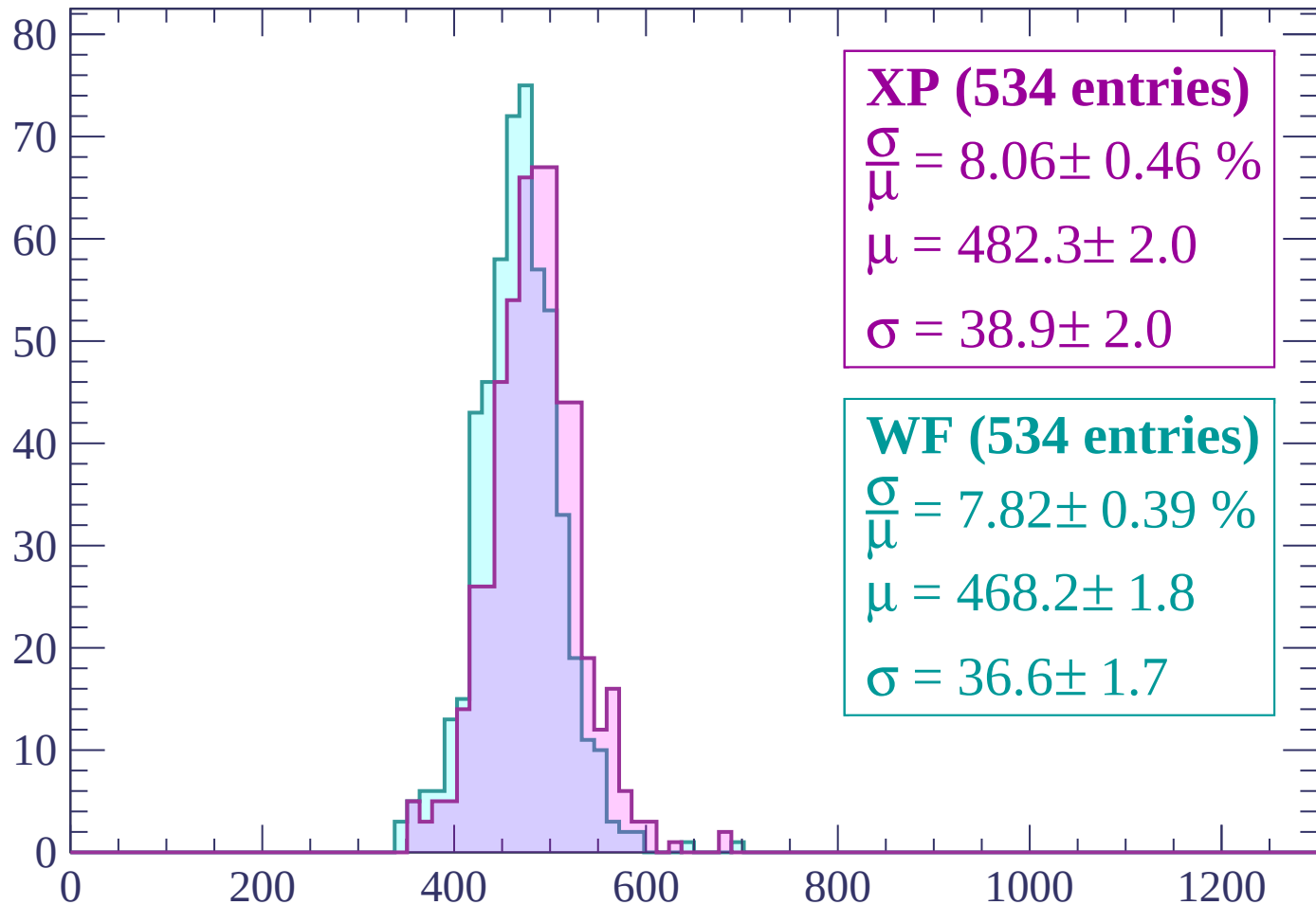
$\mu = 472.9 \pm 2.1$

$\sigma = 45.8 \pm 1.9$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (534 entries)

$\frac{\sigma}{\mu} = 8.06 \pm 0.46 \%$

$\mu = 482.3 \pm 2.0$

$\sigma = 38.9 \pm 2.0$

WF (534 entries)

$\frac{\sigma}{\mu} = 7.82 \pm 0.39 \%$

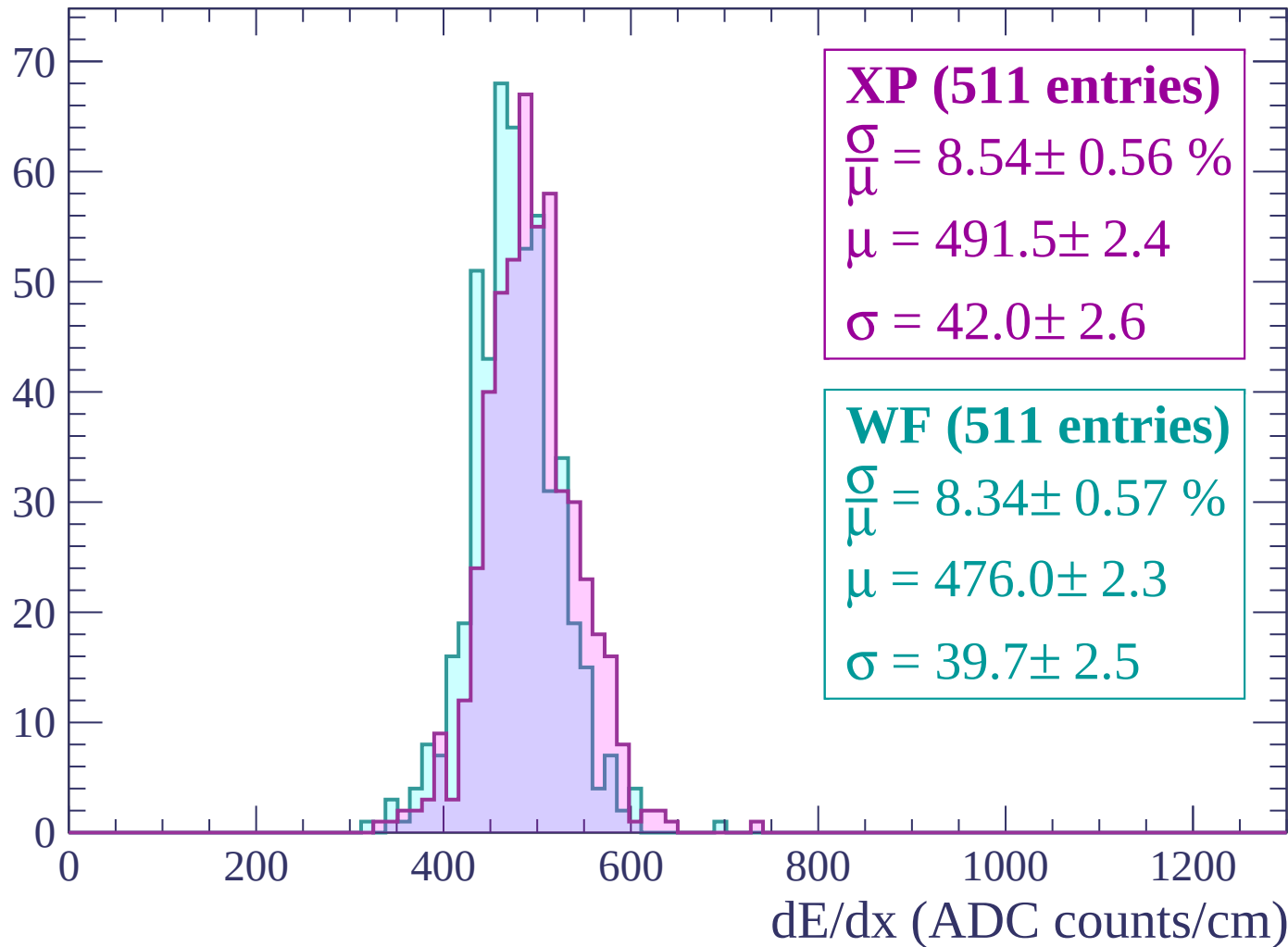
$\mu = 468.2 \pm 1.8$

$\sigma = 36.6 \pm 1.7$

dE/dx (ADC counts/cm)

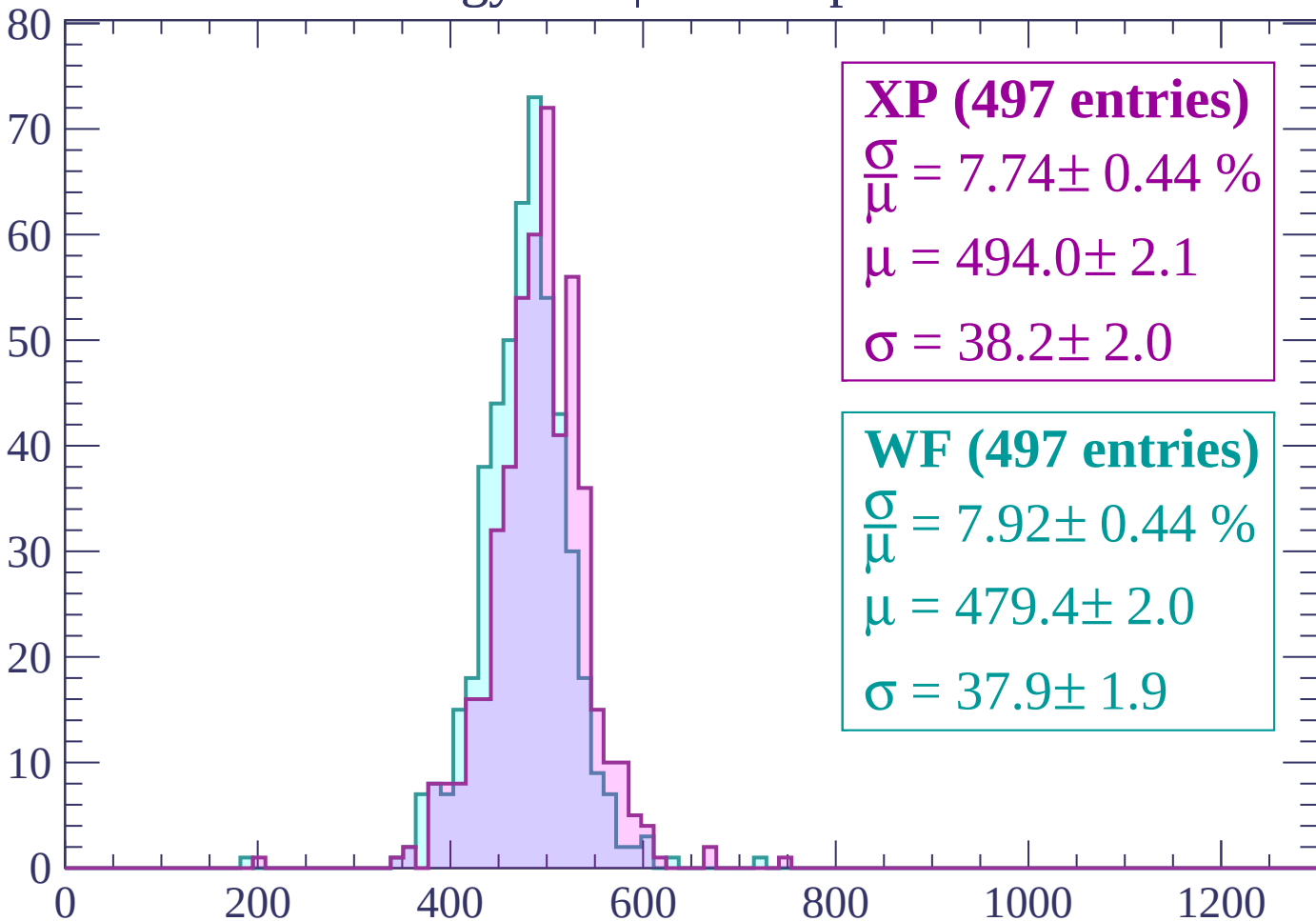
Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count



XP (497 entries)

$\frac{\sigma}{\mu} = 7.74 \pm 0.44 \%$

$\mu = 494.0 \pm 2.1$

$\sigma = 38.2 \pm 2.0$

WF (497 entries)

$\frac{\sigma}{\mu} = 7.92 \pm 0.44 \%$

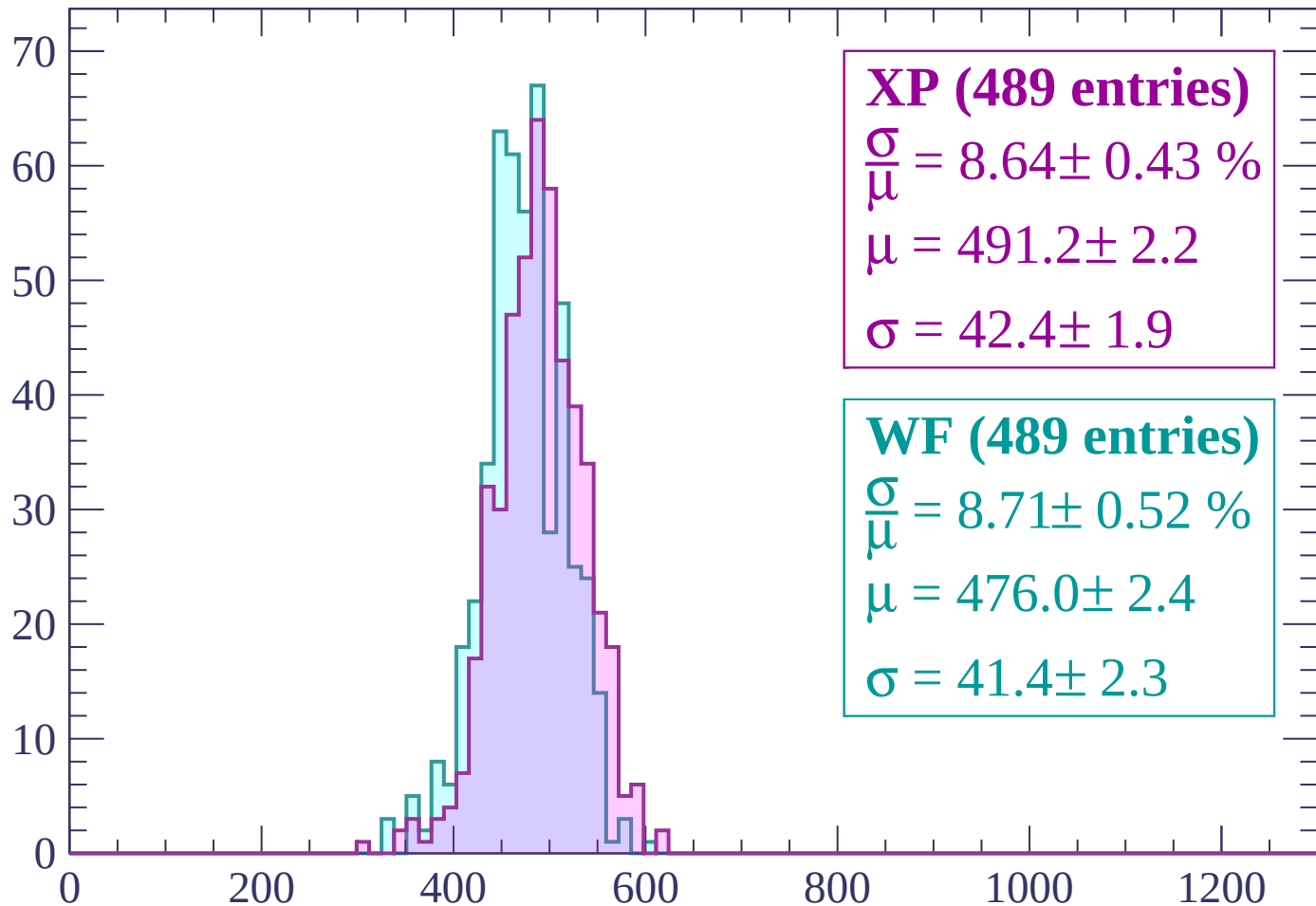
$\mu = 479.4 \pm 2.0$

$\sigma = 37.9 \pm 1.9$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (489 entries)

$\frac{\sigma}{\mu} = 8.64 \pm 0.43 \%$

$\mu = 491.2 \pm 2.2$

$\sigma = 42.4 \pm 1.9$

WF (489 entries)

$\frac{\sigma}{\mu} = 8.71 \pm 0.52 \%$

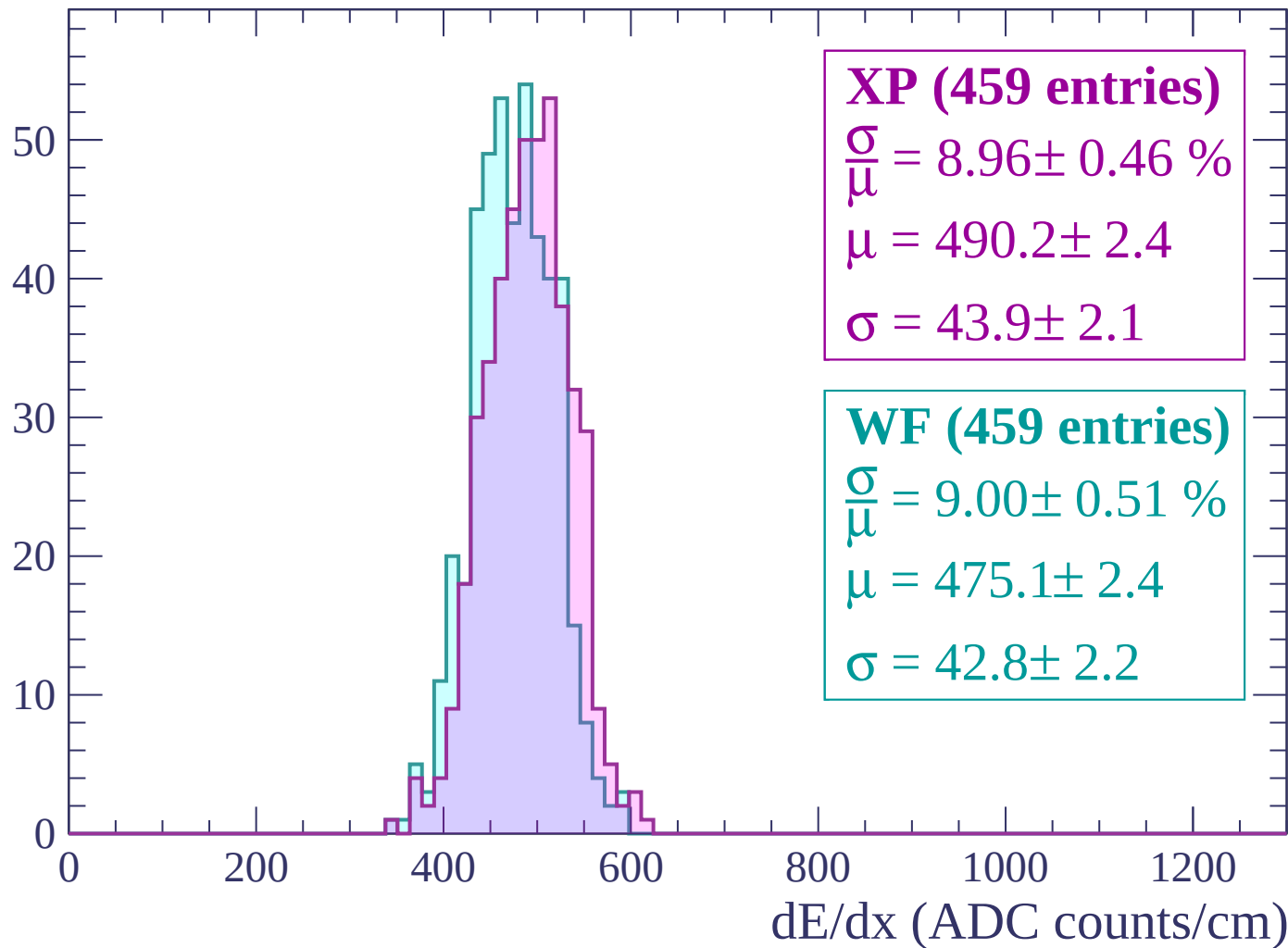
$\mu = 476.0 \pm 2.4$

$\sigma = 41.4 \pm 2.3$

dE/dx (ADC counts/cm)

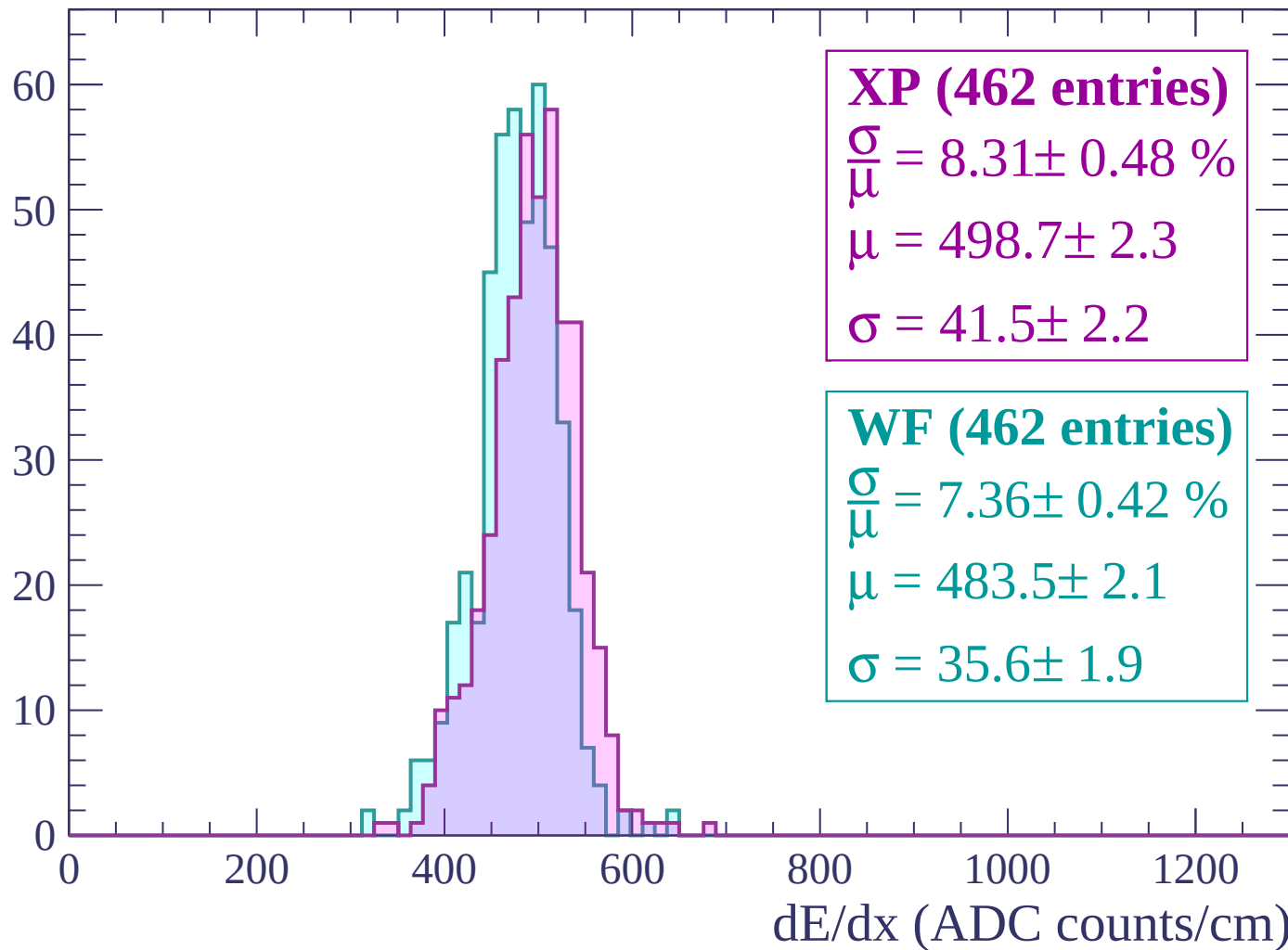
Energy loss | $2460 < p < 2520$

Count



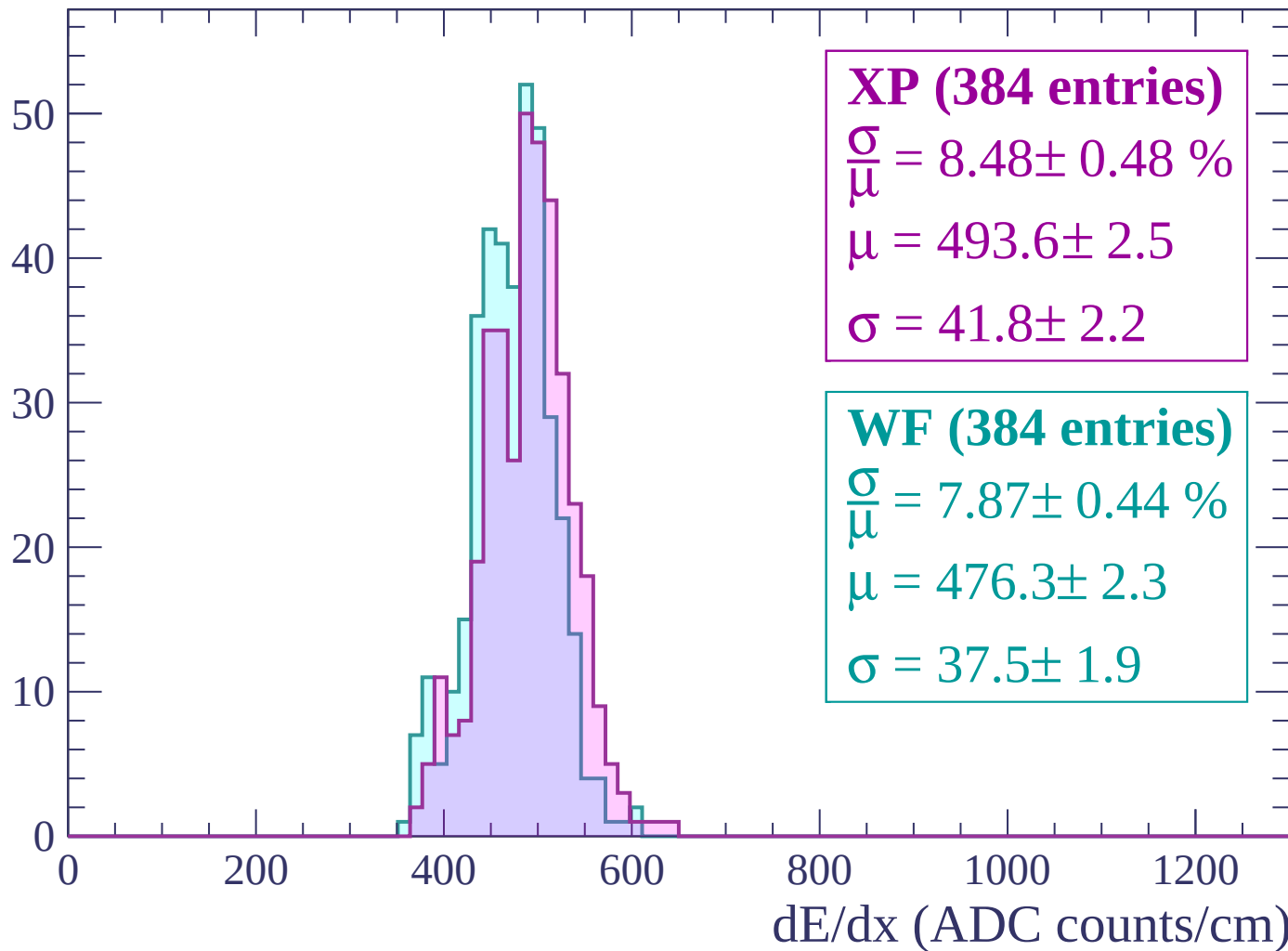
Energy loss | $2520 < p < 2580$

Count



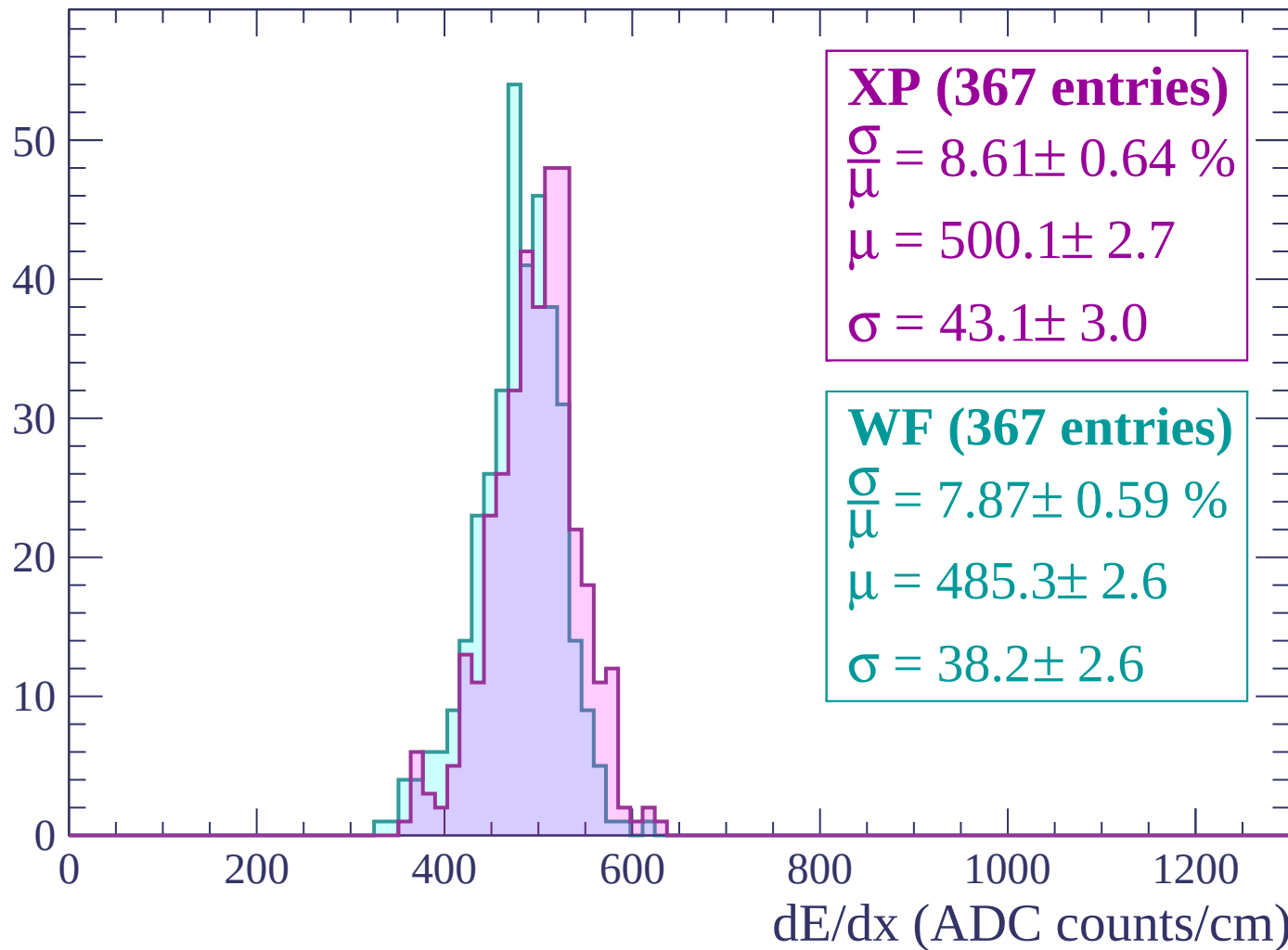
Energy loss | $2580 < p < 2640$

Count



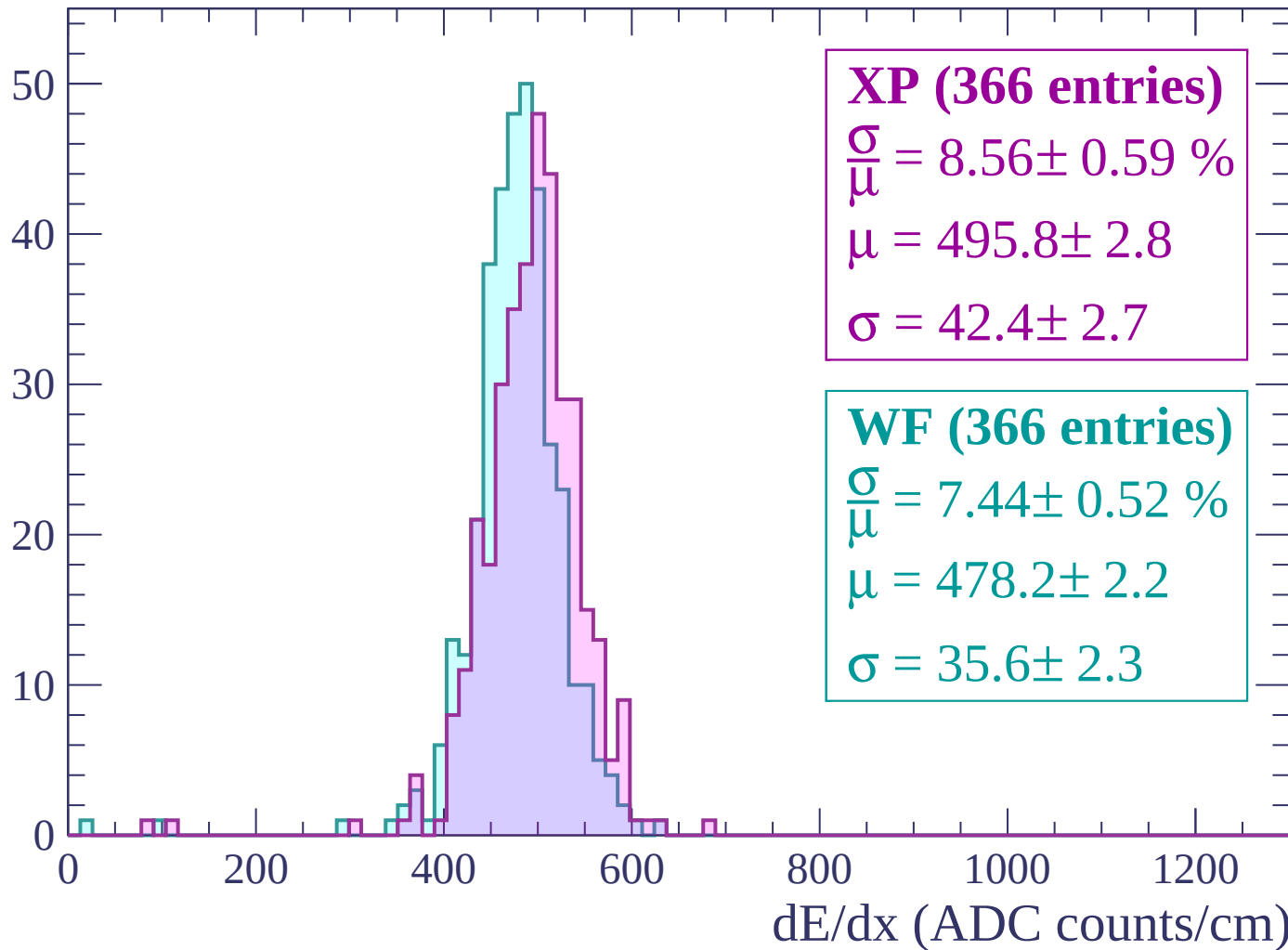
Energy loss | $2640 < p < 2700$

Count



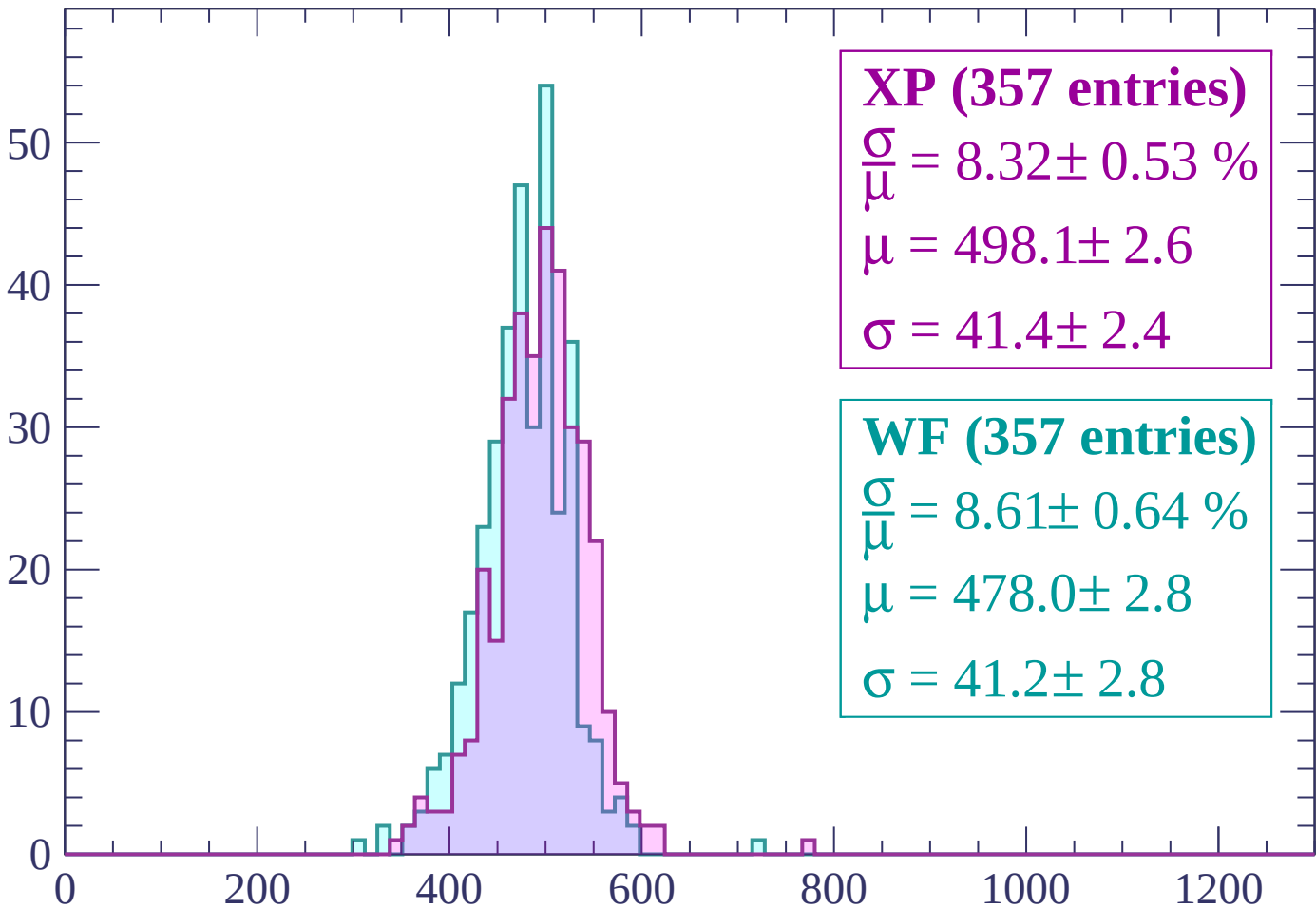
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP (357 entries)

$\frac{\sigma}{\mu} = 8.32 \pm 0.53 \%$

$\mu = 498.1 \pm 2.6$

$\sigma = 41.4 \pm 2.4$

WF (357 entries)

$\frac{\sigma}{\mu} = 8.61 \pm 0.64 \%$

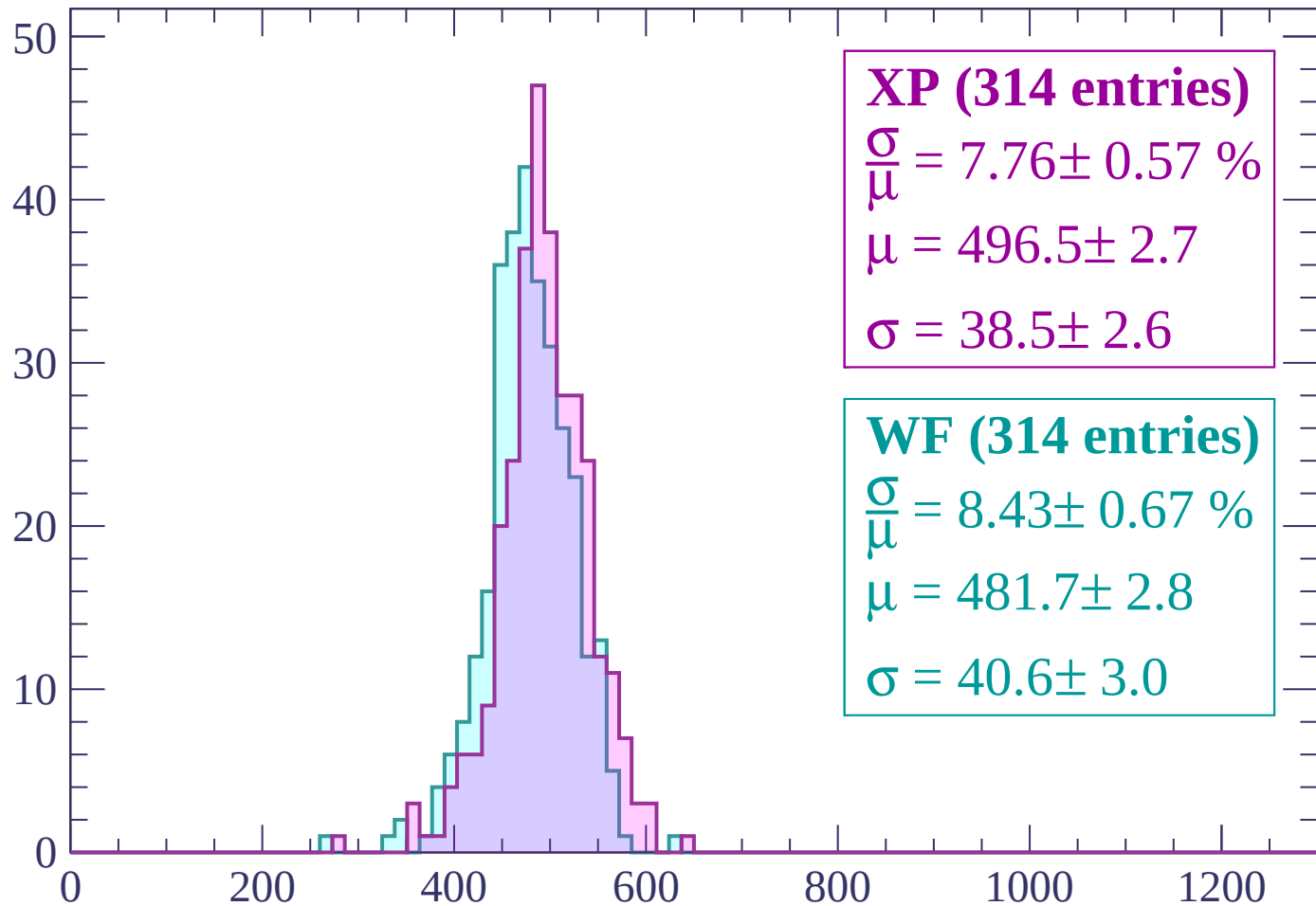
$\mu = 478.0 \pm 2.8$

$\sigma = 41.2 \pm 2.8$

dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



XP (314 entries)

$$\frac{\sigma}{\mu} = 7.76 \pm 0.57 \%$$

$$\mu = 496.5 \pm 2.7$$

$$\sigma = 38.5 \pm 2.6$$

WF (314 entries)

$$\frac{\sigma}{\mu} = 8.43 \pm 0.67 \%$$

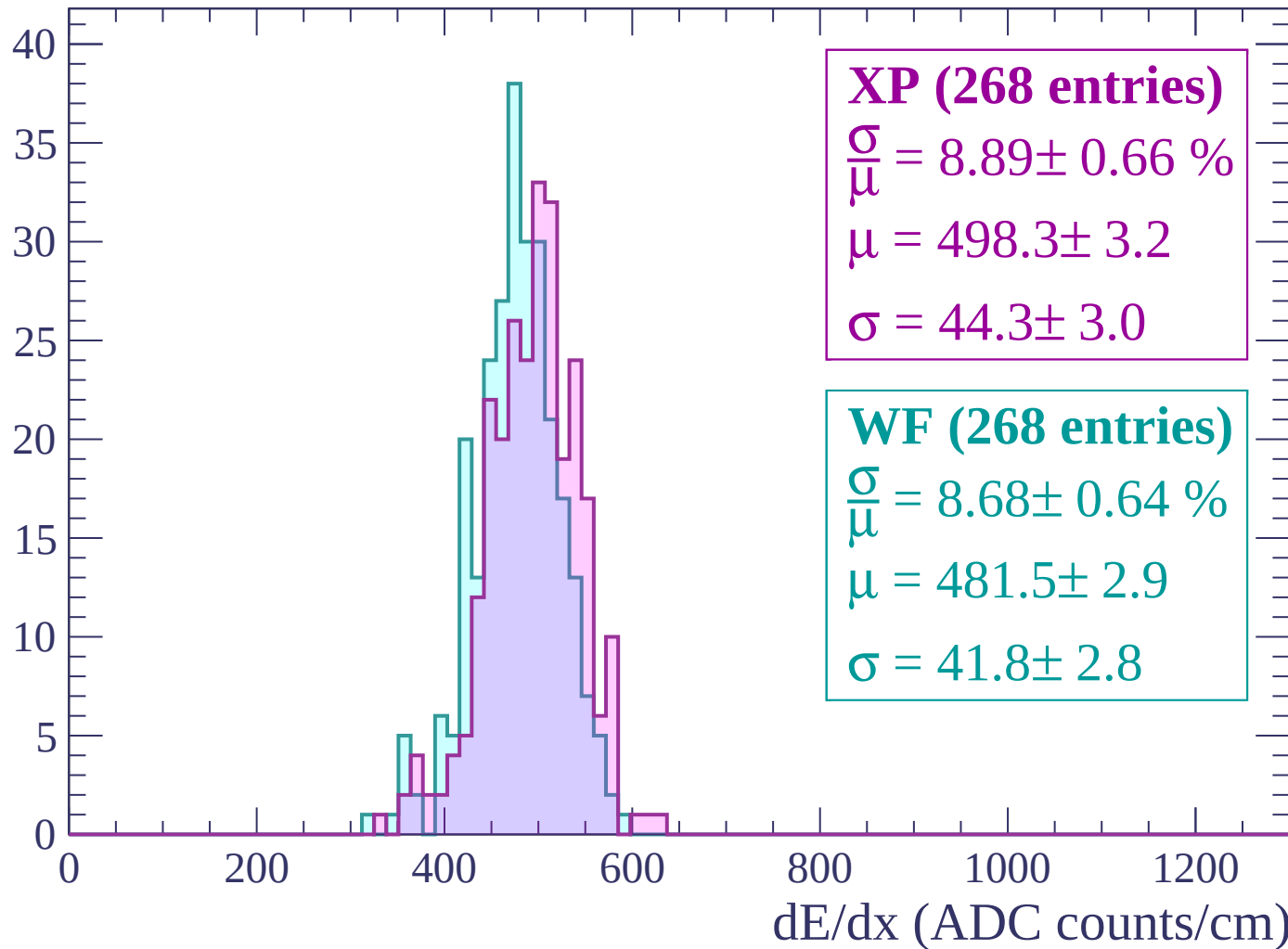
$$\mu = 481.7 \pm 2.8$$

$$\sigma = 40.6 \pm 3.0$$

dE/dx (ADC counts/cm)

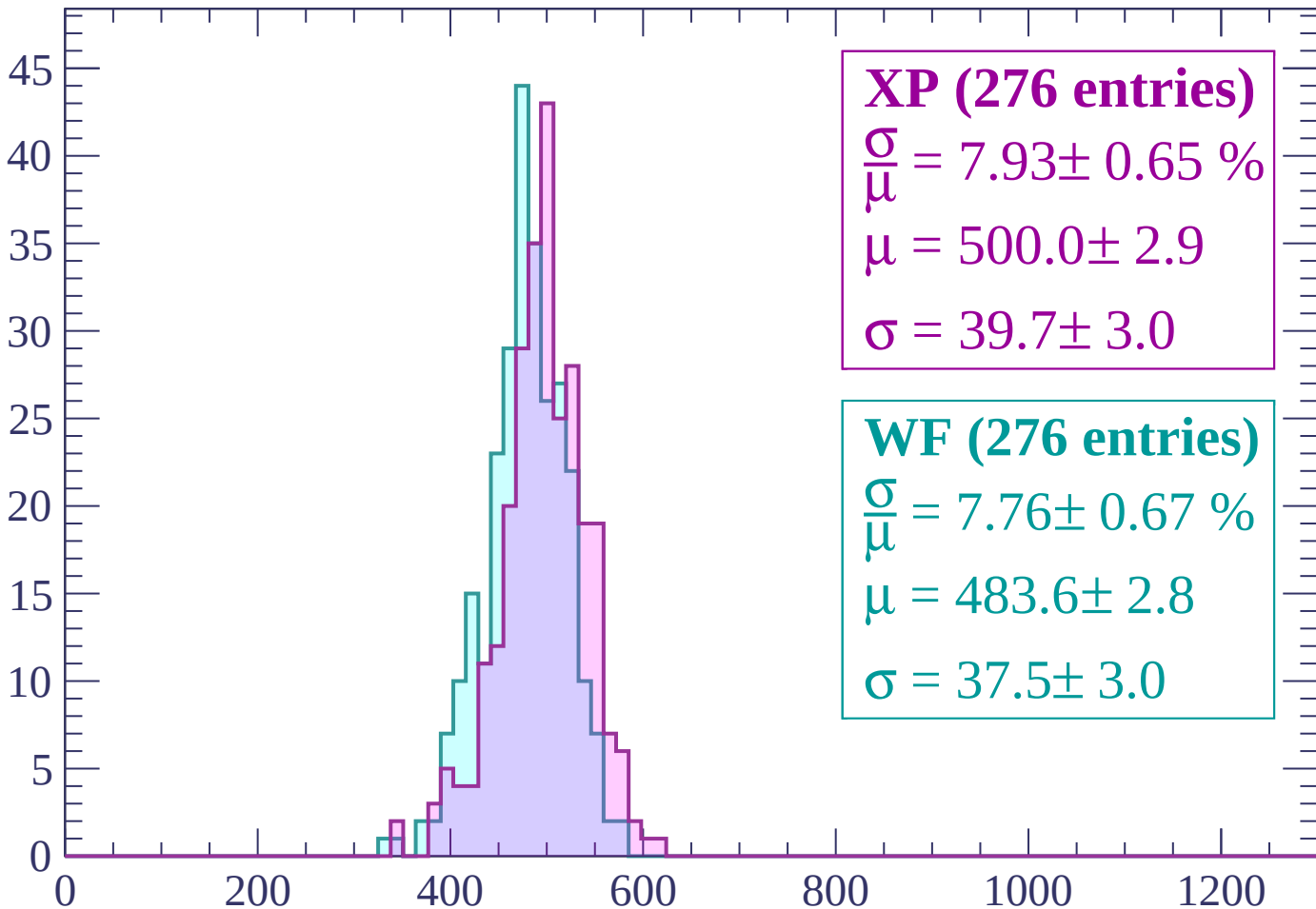
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP (276 entries)

$\frac{\sigma}{\mu} = 7.93 \pm 0.65 \%$

$\mu = 500.0 \pm 2.9$

$\sigma = 39.7 \pm 3.0$

WF (276 entries)

$\frac{\sigma}{\mu} = 7.76 \pm 0.67 \%$

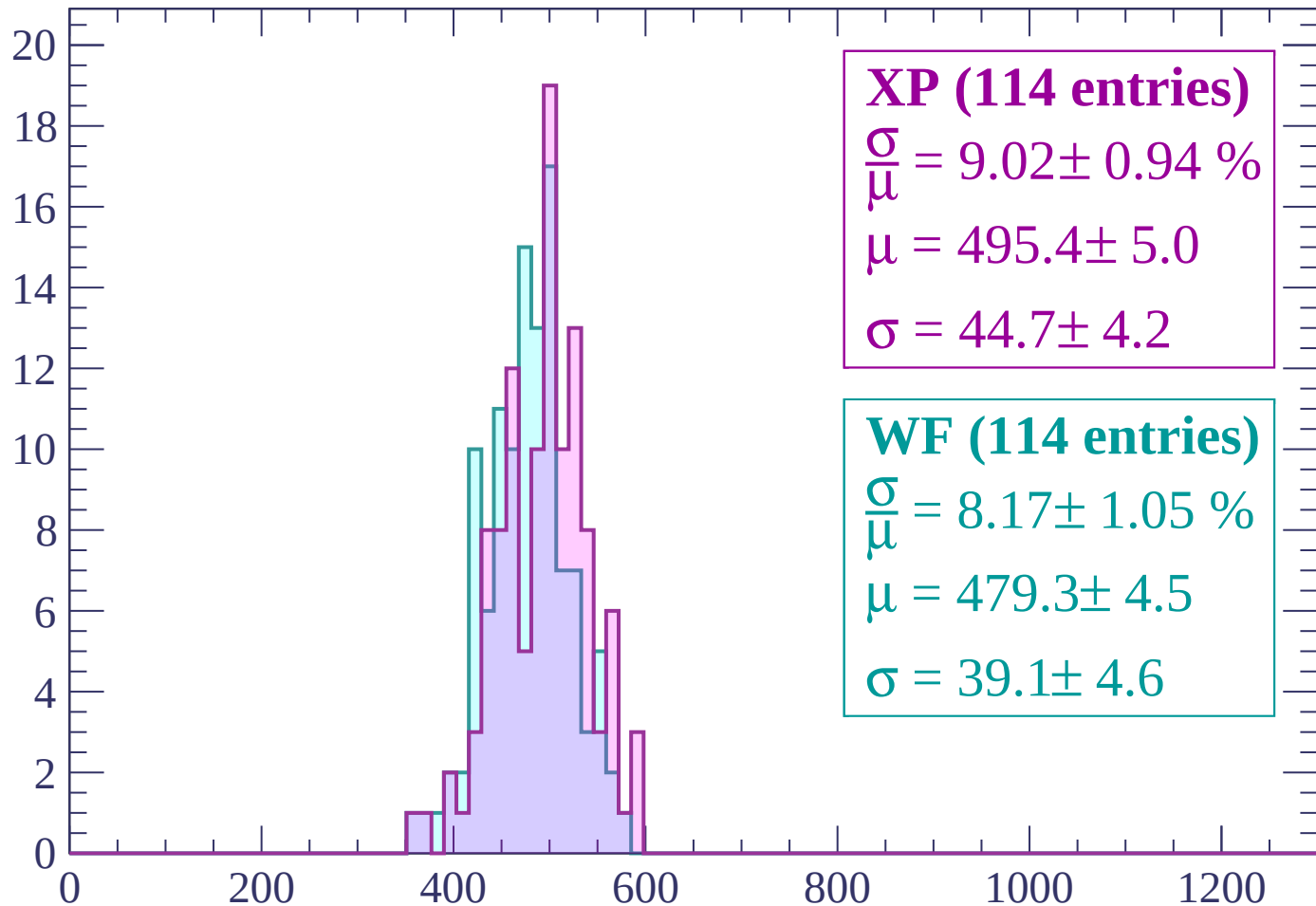
$\mu = 483.6 \pm 2.8$

$\sigma = 37.5 \pm 3.0$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



XP (114 entries)

$\frac{\sigma}{\mu} = 9.02 \pm 0.94 \%$

$\mu = 495.4 \pm 5.0$

$\sigma = 44.7 \pm 4.2$

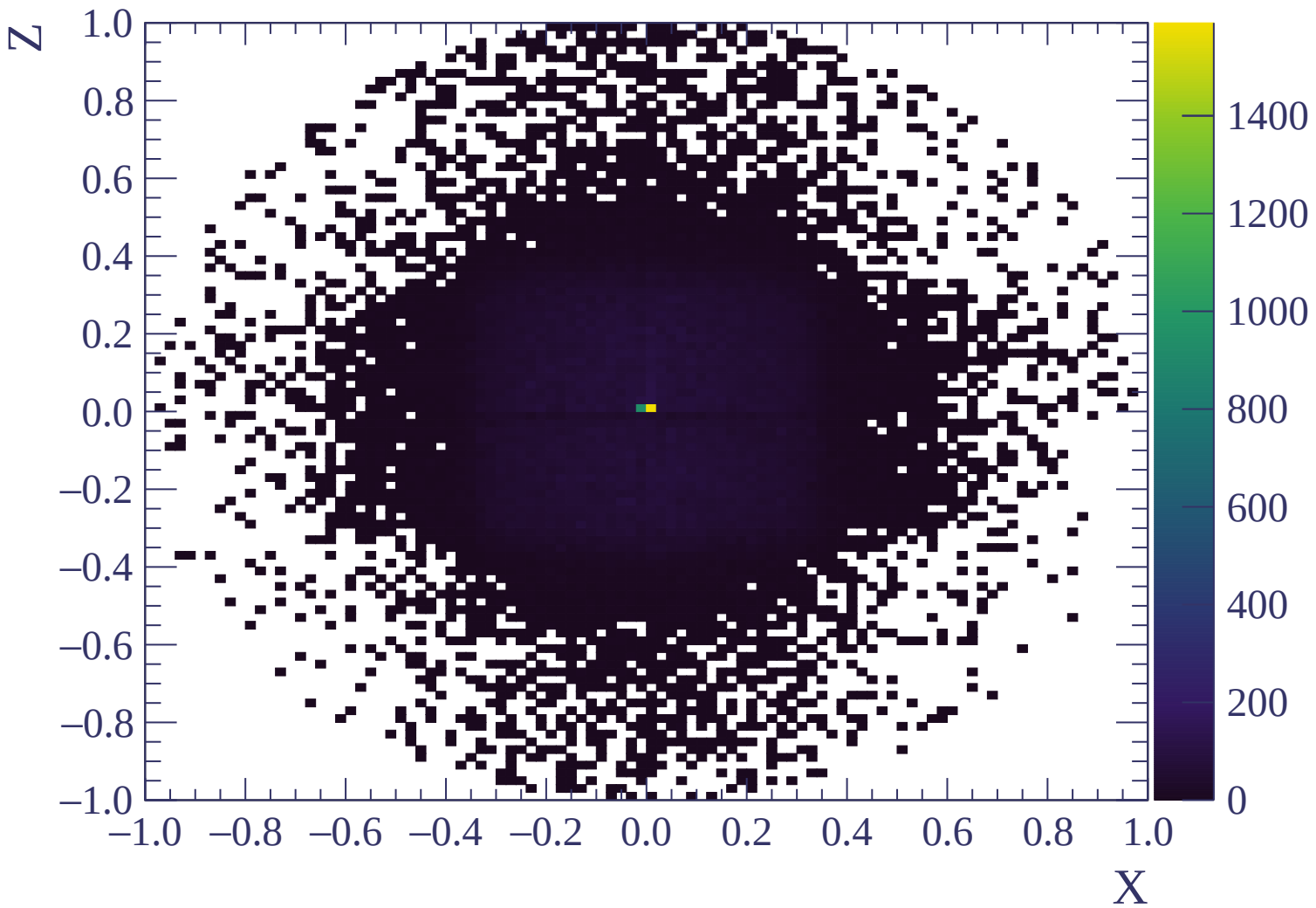
WF (114 entries)

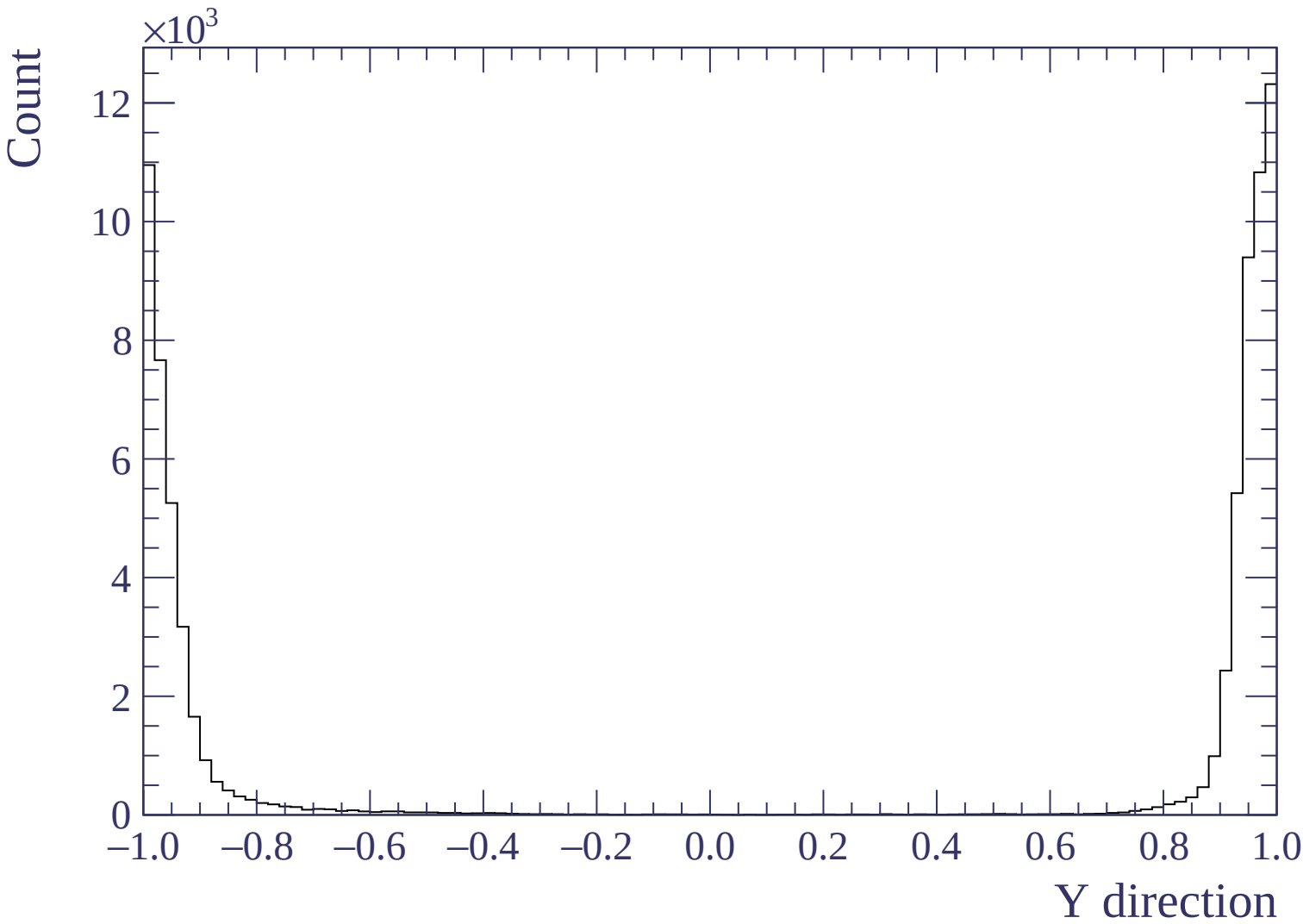
$\frac{\sigma}{\mu} = 8.17 \pm 1.05 \%$

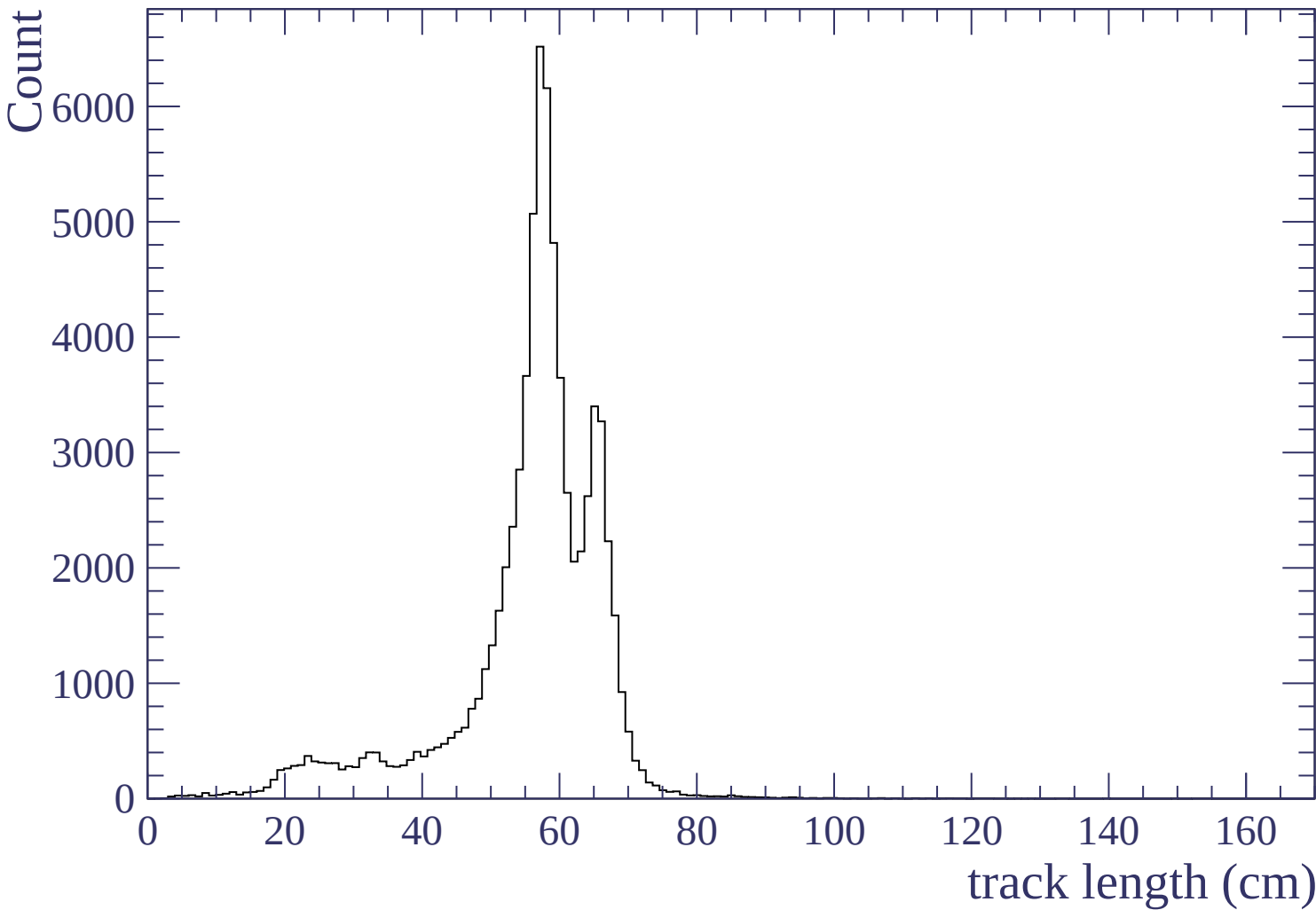
$\mu = 479.3 \pm 4.5$

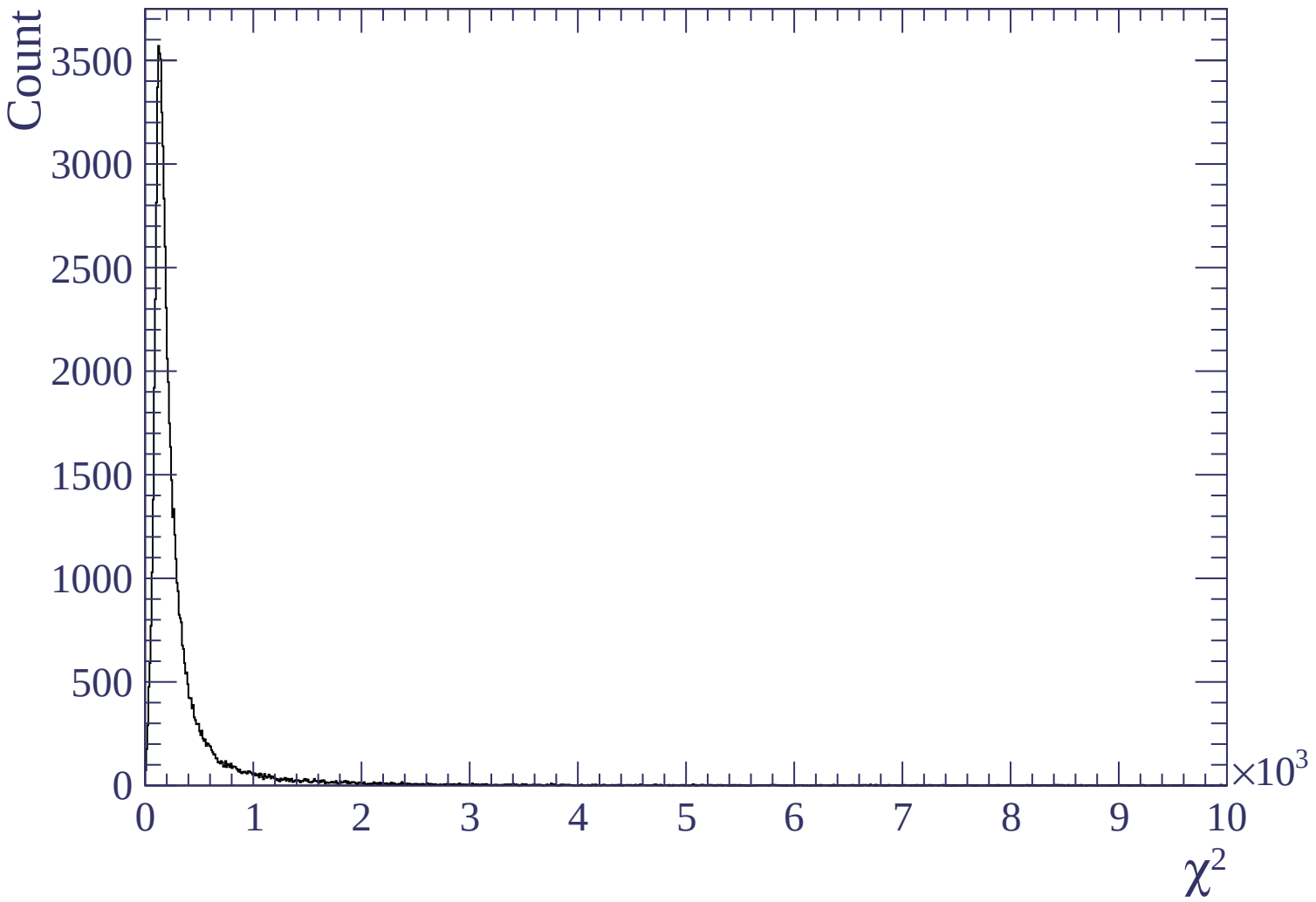
$\sigma = 39.1 \pm 4.6$

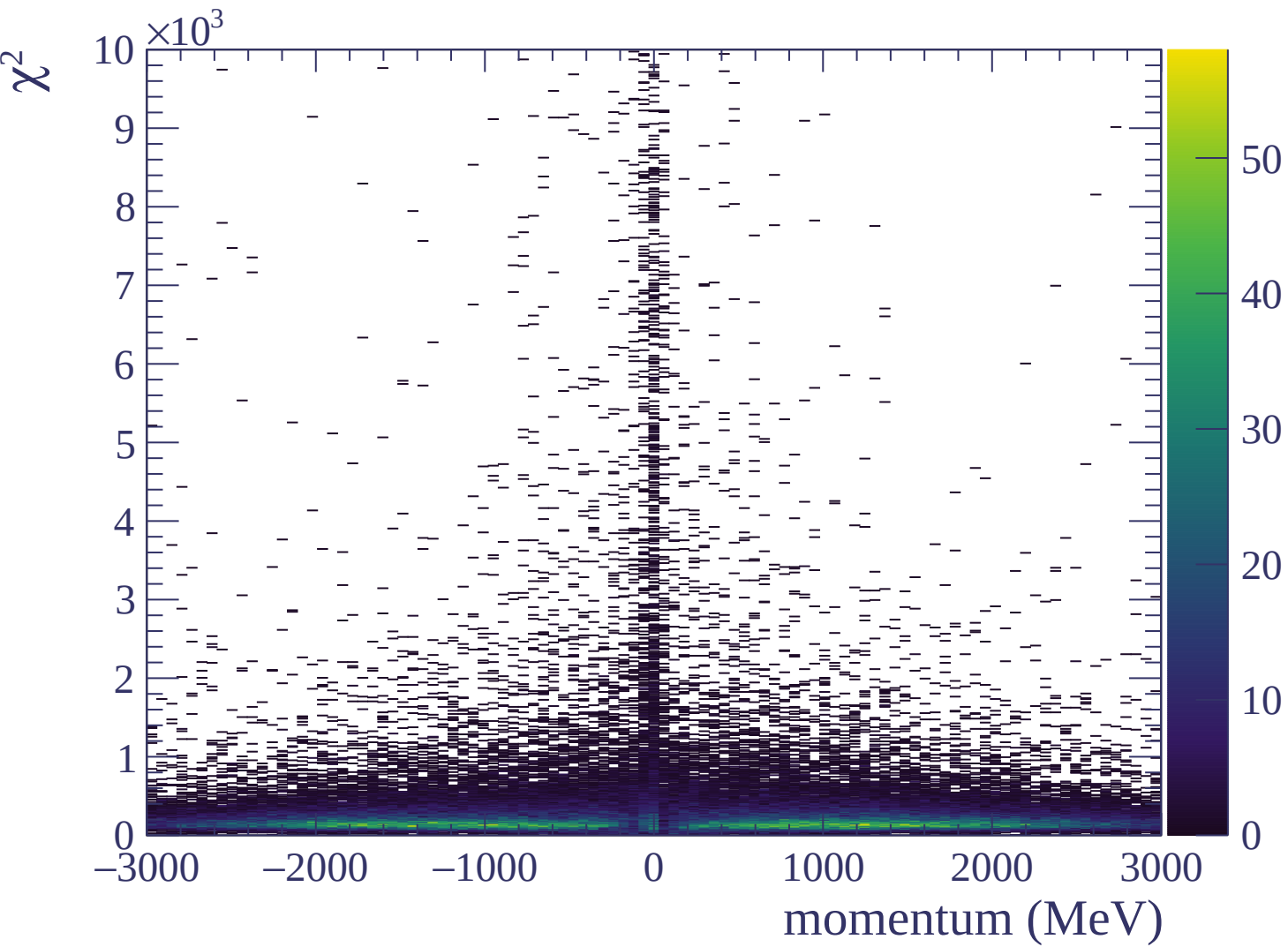
dE/dx (ADC counts/cm)

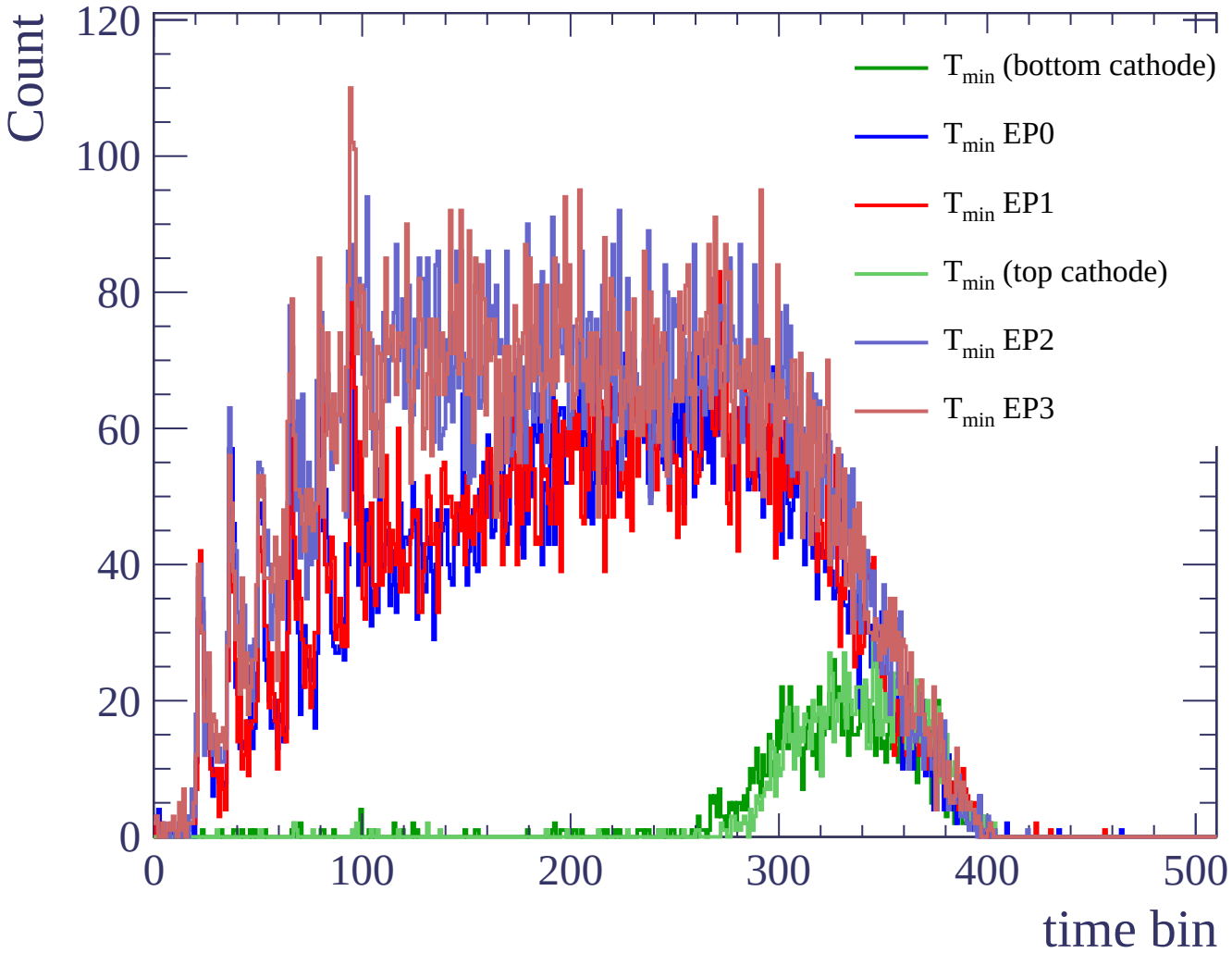












Count

